

REVIEWED INTEGRATED EDITION

Decision Theory for Rates E-Trading

Quoting, inventory, execution,
and evaluation

From information available at a decision to the value of a complete trading policy.

Twelve chapters with mathematical derivations,
numerical experiments, and source notes.

September 2026

Contents

How to read this book	vii
One decision chain, several explicit models	viii
Notation and units	ix
Experiments and source evidence	x
I Market data and estimation	1
1 RFQ, Streaming, Order Books, and Available Information	2
1.1 A protocol defines the meaning of an action	2
1.2 Fix side, quantity, and execution conventions	3
1.3 An RFQ is a sequence with more than two outcomes	4
1.4 Streaming adds quote versions and an execution contract	5
1.5 A book contains quantities and priority, not just a midpoint	6
1.6 An event has several clocks	7
1.7 Reconstruct the state the engine could actually use	8
1.8 Pending cancellation leaves a position to explain	11
1.9 Missing book messages and missing order detail are different problems	12
1.10 Count the observed population before estimating a win rate	13
1.11 Build a decision record, then reconcile what happened	15
1.12 The information contract carries through the book	16
Exercises	17
Source Notes	19
2 Reference Prices, Adverse Selection, and Markouts	20
2.1 A reference is an input with a time and a meaning	20
2.2 Reconcile one fill before averaging many	21
2.3 Selection enters the expected value of a quote	22
2.4 Learn from the direction of a trade	23
2.5 Competitive quotes reflect conditional values	24
2.6 An uninformed customer can select a stale quote	26

2.7	Preserve the reference used by the decision	28
2.8	State the benchmark clock and missing-data rule	29
2.9	A changing population can move the average	30
2.10	Benchmark choices change attribution	32
2.11	Interpret an adverse markout at the right level	33
2.12	Carry the conditioning into estimation	34
	Exercises	34
	Source Notes	36
3	Arrivals and Conditional Execution	37
3.1	Define the opportunity before estimating its probability	37
3.2	Estimate arrivals against exposure	38
3.3	From requests to executed flow	39
3.4	Keep outcome states and denominators separate	41
3.5	A label has an availability time	43
3.6	Response selection and quote support limit interpretation	44
3.7	Fit a conditional probability, with units attached	45
3.8	Separate fitting, selection, calibration, and testing in time	46
3.9	Score probabilities and inspect their calibration	47
3.10	Three reproducible experiments	48
3.11	Resting orders require a different event model	51
3.12	The probability contract passed to a quoting model	53
	Exercises	53
	Source Notes	55
II	Static quoting and risk	56
4	Optimizing a Single RFQ	57
4.1	The Decision and Its Inputs	57
4.2	Choosing the Denominator of Win Probability	58
4.3	Constructing Expected Profit and Loss	59
4.4	The Gain from a Wider Quote and the Opportunity It Sacrifices	60
4.5	Finding the Optimal Quote with a Logistic Win Probability	61
4.6	Reading a Single Request through Numbers	62
4.7	Changing Costs and the Competitive Environment	64
4.8	Non-Response and the Price Grid	65
4.9	Conditioning on Reference-Price Changes	67
4.10	Avoiding Double Counting in Markout and Costs	68

4.11	Connecting the Source Equations to This Chapter	69
4.12	Checks before Using the Policy	69
	Exercises	70
	Source Notes	71
5	Inventory Risk and Factor Representation	72
5.1	Positions, Trade Direction, and Supplied Sensitivities	72
5.2	Risk Comes from Joint Price Moves	73
5.3	Turning Variance into a Monetary Penalty	74
5.4	Charge the Trade for the Change in Risk	75
5.5	An Inventory-Aware Score for One RFQ	76
5.6	The Same Request from Three Starting Positions	77
5.7	Reservation Prices and Actual Optimal Quotes	80
5.8	Another Instrument Can Change This Quote	80
5.9	Factor Cancellation Does Not Eliminate Residual Risk	82
5.10	A Change of Factor Units Must Not Change the Trade	84
5.11	Limits Restrict Actions; Penalties Rank Them	84
5.12	What the Static Model Establishes	85
5.13	Forecast the Right Variance, on the Right Clock	86
5.14	Rolling Variance and Exponential Weighting	87
5.15	GARCH Adds a Fitted Long-Run Level	88
5.16	Realized Variance and a HAR-Style Forecast	89
5.17	Chronological Evaluation on a Common Variance Target	90
5.18	From Scalar Forecasts to a Portfolio Covariance	93
5.19	Keep Estimation and Decision Models Connected	94
	Exercises	94
	Source Notes	96
III	Dynamic quoting	97
6	Bellman and HJB	98
6.1	A Quote Changes the Next Decision	98
6.2	State, Requests, and Marked Wealth	99
6.3	What Is Being Maximized	100
6.4	The Value of an RFQ That Has Arrived	101
6.5	A Backward Bellman Recursion	102
6.6	From Bellman to HJB	103
6.7	Recovering the Static Model at the Horizon	104

6.8	A Five-State Numerical Model	106
6.9	Comparing Policies Under One Objective	109
6.10	Time Discretization and What the Checks Establish	111
6.11	What Must Change in a Richer Model	113
6.12	From an Assigned Penalty to a Computed Opportunity Value	113
	Exercises	114
	Source Notes	115
7	Multiple Instruments and Sizes	116
7.1	One Portfolio, Many Possible Requests	116
7.2	A Request Has an Instrument, a Direction, and a Size	117
7.3	The Same Bellman Logic with More Neighbors	118
7.4	Three Ways Size Can Change the Quote	120
7.5	Correlation Connects the Instrument Quotes	121
7.6	When Factor Inventory Is a Sufficient State	122
7.7	Residual Risk and Limits Test the Reduction	124
7.8	Two Instruments and Two Sizes on a Small Grid	125
7.9	An Exact Reduction on Reachable States	127
7.10	Turning a Compressed Model into a Feasible Policy	128
7.11	Accuracy and Computation Must Be Reported Together	130
7.12	What Carries Forward	132
	Exercises	133
	Source Notes	134
8	Flow Imbalance and Client Relationships	135
8.1	Four Reasons for a Quote to Move	135
8.2	Measure Incoming Flow Before Measuring Fills	136
8.3	Why an Arrival Rate Drops Out of a Single RFQ	137
8.4	Future Flow Changes Continuation Value	137
8.5	What an Exponential Benchmark Explains	139
8.6	A Controlled Flow Experiment	140
8.7	Access to Future Requests as a State	142
8.8	A Lost RFQ Can Change Value	143
8.9	Bellman Recursion with Client Feedback	145
8.10	Compare Policies Under the Same Feedback	146
8.11	Remove the Mechanism and Refine the Grid	148
8.12	What Must Be Observed Before Using the Model	150
	Exercises	150

Source Notes	152
IV Execution and evaluation	153
9 Internalization and External Hedging	154
9.1 Two Ways to Reduce the Same Inventory	154
9.2 A Hedge Is a Cash Transaction	154
9.3 Put Client Quotes and Hedges in One Objective	155
9.4 Specify the Clock Before Writing the Bellman Equation	156
9.5 The Region Where Waiting Wins	158
9.6 Evaluate a Stored Policy Independently	159
9.7 A Controlled Comparison of Five Policies	160
9.8 Why a Hedge Changes the Client Quote	162
9.9 Costs, Capacity, and Two Different Clock Experiments	164
9.10 When the External Instrument Leaves Residual Risk	166
9.11 Waiting for a Review Is Not Execution Latency	167
9.12 What the Experiment Establishes	168
Exercises	169
Source Notes	170
10 Optimal Execution	171
10.1 A Target, a Clock, and a Completion Rule	171
10.2 From Prices to a Cash Ledger	172
10.3 Which Costs Depend on the Schedule?	173
10.4 Choose a Risk-Cost Objective	175
10.5 Derive the Constant-Liquidity Schedule	175
10.6 A Complete Numerical Comparison	177
10.7 Read the Cost-Risk Frontier	178
10.8 Liquidity and Capacity Change the Answer	180
10.9 Refine Time Without Changing the Model Silently	182
10.10 A Terminal Penalty Is Not Completion	183
10.11 When Client Fills Change the Target	184
10.12 What an Execution Plan Commits To	185
Exercises	186
Source Notes	187
11 Transaction-Cost Analysis and Profit Attribution	188
11.1 Write down the clock before measuring performance	188

11.2	Derive the P&L bridge from cash	190
11.3	A second exact attribution uses a common terminal mark	191
11.4	Execution shortfall depends on its benchmark	192
11.5	Partial completion needs an explicit residual convention	193
11.6	Markouts measure a signed value at a stated horizon	195
11.7	Keep the cohort fixed when comparing horizons	196
11.8	Requests, quotes, and executions have different denominators	198
11.9	A reference-price change redistributes attribution	199
11.10	Observation times and units are part of the model	200
11.11	Make every metric auditable from its inputs	201
11.12	From reconciliation to evaluation	202
	Exercises	202
	Source Notes	204
12	Simulation and Evaluation	205
12.1	Define the object being evaluated	205
12.2	Reuse the cash ledger with an explicit event order	206
12.3	Start with policies simple enough to audit	208
12.4	Distinguish replay, generated paths, and action feedback	210
12.5	Use common random inputs without forcing common outcomes	211
12.6	Select on validation data and freeze before testing	211
12.7	Measure differences at the independent-episode level	213
12.8	Read the entire outcome distribution and the stress table	215
12.9	Historical logs need action support	217
12.10	A small example exposes selection and missing support	218
12.11	Do not transfer one-step guarantees to a sequential desk	220
12.12	Turn an experiment into a reviewable decision	221
	Exercises	222
	Source Notes	224
	Research and notebook guide	225

How to read this book

The decision at the center of this book is simple to state: an electronic dealer receives an opportunity, chooses whether and how to quote, manages the resulting position, and evaluates the decision using the information that was actually available. Each stage changes the next. A more generous quote can win a different population of trades; an inventory change can alter the value of waiting for future flow; and an external hedge can change both cash and future risk.

The chapters develop that decision in four parts. Chapters 1-3 establish observation, reference-based measurement, and the estimation of arrivals and conditional execution. Chapters 4-5 turn those objects into a static quote decision and portfolio risk charge, including volatility estimation. Chapters 6-8 introduce continuation value, multiple instruments, and client-flow states. Chapters 9-12 connect internalization, external execution, accounting, and policy evaluation.

Reference prices and instrument sensitivities are supplied inputs. We examine their timestamps, units, forecast uncertainty, and implications for decisions. The mathematical prerequisites are conditional probability, differentiation, matrix multiplication, and an understanding of expectation. Dynamic programming and the relevant stochastic-control equations are developed when the decision problem needs them.

The numerical examples are controlled teaching models. They deliberately vary one mechanism at a time and are not successive calibrations of one historical trading desk. The observed market studies motivate assumptions and identify limitations; their populations and empirical findings are kept separate from the book's simulations. An exact optimization result is always conditional on the stated model, action set, information, and objective.

For a first reading, follow the chapters in order and work through the small examples before the larger simulations. On a second reading, pair each chapter with its companion notebook. The main text supplies the economic argument; the notebook exposes inputs, numerical choices, and checks. A correct numerical answer matters less than understanding which observations and assumptions made that answer meaningful.

One decision chain, several explicit models

The book uses several models because different decisions require different state variables. Their connections are carried by accounting, information, and units. A new model must state which elements of the previous one it retains and which assumptions it changes.

Stage	Question	Connection to the next stage
Observe	What had arrived locally before the decision?	Establishes admissible features, pending exposure, and outcome cutoffs
Measure and estimate	What do the reference, arrival rate, probability, and variance forecast describe?	Supplies targets with explicit populations, units, and horizons
Quote	How does edge trade off against execution and inventory change?	Produces a distribution of cash and position changes
Control	What is the value of future flow and external execution?	Adds continuation value under a declared event clock
Evaluate	Did the actual policy improve the declared outcome?	Tests accounting, uncertainty, coverage, and model assumptions

A request intensity counts opportunities per unit of exposure. A conditional execution probability has no time unit. Their product describes expected completed outcomes indexed by request origin unless an instantaneous or stationary timing assumption connects it to physical executions. A size probability enters a marked arrival measure once. These conventions carry from [Chapter 3](#) into the Bellman models.

A covariance over a fixed horizon has different units from a covariance rate per unit time. A variance forecast has squared units; volatility is its square root. A chosen risk-aversion or penalty coefficient converts forecast risk into a preference-weighted objective. Estimating variance does not estimate that preference. [Chapter 5](#) develops the forecasting bridge, while the numerical control examples retain their declared fixed inputs.

A markout is also a contract: side, quantity weighting, fees, reference, horizon, and benchmark eligibility belong to its definition. Gross execution-price marked value and net dealer markout can be reconciled, but should not silently replace one another. Chapters 2, 4, and 11 make that reconciliation explicit. An accounting identity establishes consistency; it does not identify a causal market-impact effect.

Notation and units

Symbols are defined at first use. This concordance records recurring objects and source-inspired changes. Function arguments and subscripts distinguish fitted probabilities, positions, and prices.

Object	Working convention	Local specialization
Dealer trade direction	$\eta = +1$ for a dealer purchase; $\eta = -1$ for a sale	A customer buy has the opposite sign. Chapter 10 separately defines its temporary-cost coefficient η
Quantity and position	$z > 0$ is trade size; q is inventory	q_θ in Chapter 3 is a fitted probability function, not inventory
Quote edge	$\delta = \eta(M - p)$	Positive edge is favorable to the dealer against the stated reference; p_i is a transaction price in measurement chapters
Execution probability	$p(\delta \mid x)$ in decision models	Chapter 3 uses $q_\theta(a, x)$ for a fitted conditional probability; a valid quote differs from a decline message
Arrival intensity	λ or a marked intensity kernel	Units are requests per time. Chapters 10 and 12 define local λ risk weights with different units
Covariance and variance	Γ for a stated price-risk metric; σ^2 for a stated variance rate	Forecast horizon and coordinates are specified. σ is standard deviation, not variance
Cash and value	Currency units	Per-unit quote costs become cash costs after multiplying by executed quantity and adding any fixed costs
External trade	Chapter 9 $e > 0$ buys for the dealer	Chapter 10 $u > 0$ sells; Chapter 11 returns to the signed dealer inventory change

Examples generally use JPY thousands per normalized trade unit. Every forecast and decision also specifies its time unit: milliseconds, minutes, trading days, and simulation steps are distinct clocks. [Chapter 5](#) maps dimensionless daily return innovations into price-risk units.

K_m is a response fee, K a step count, $\ell(q)$ a terminal penalty, and ℓ_i a delay. The local λ weights have inverse-currency units in [Chapter 10](#) and currency per quantity squared per time in [Chapter 12](#).

Experiments and source evidence

Each chapter has a companion executable notebook, and [Chapter 5](#) adds a volatility-estimation supplement. The notebooks retain their declared synthetic populations, random seeds, parameter values, and numerical checks. Forecast fitting, model selection, probability calibration, and evaluation use explicit chronological cutoffs where relevant. New estimates should be evaluated on fresh data before replacing the supplied inputs of another chapter’s decision experiment.

The experiments illustrate mechanisms: quote selection, inventory geometry, finite-state control, model compression, feedback, execution schedules, attribution, and evaluation under policy support. They are not evidence of real-market profitability or universal parameter choices. A favorable simulated result can disappear when observation delays, invitation selection, costs, covariance dynamics, or other modeling assumptions change.

Source notes at the end of each chapter identify the saved editions, relevant passages, and the boundary between source results and the book’s own constructions. References in the text link to chapters, sections, equations, and those notes. The closing research guide provides direct source and notebook links. PDF bookmarks group the book by part and chapter, with colors matching the printed part dividers.

PART I

Market data and estimation

Observe the event, establish the reference, and estimate the population that a decision can reach.

- 1 RFQ, Streaming, Order Books, and Available Information
- 2 Reference Prices, Adverse Selection, and Markouts
- 3 Arrivals and Conditional Execution

Chapter 1

RFQ, Streaming, Order Books, and Available Information

A dealer sees a request, sends a price, and later learns whether a trade occurred. That description leaves out almost everything needed to model the decision. Who received the request? Was the price indicative or executable? Could it be changed? Did a cancellation take effect before a customer acted? Was the latest market update already available to the quoting engine, or merely present in a file downloaded afterward?

This book studies decisions made inside that sequence: which requests to answer, what prices to offer, how to manage the resulting inventory, and how to evaluate the outcome. Each numerical decision uses explicitly declared reference prices and risk inputs. We begin with trading protocols and observation, because a model cannot optimize a decision correctly if it gives the dealer the wrong information or treats a message as a trade.

The chapter develops three small examples: an RFQ log with an unresolved outcome, a streamed quote caught in a cancellation race, and a deliberately simple order book. They are synthetic constructions, not reconstructions of a particular venue. The accompanying notebook preserves both the events that occurred and the times at which the local system learned about them. That distinction will remain useful throughout the book.

1.1 A protocol defines the meaning of an action

A trading protocol specifies the permitted messages, their recipients, the conditions under which an instruction becomes effective, and the evidence of execution. An RFQ, a streamed price, and a resting order can all display a side, price, and quantity. Those fields do not make the three objects interchangeable.

In a request for quote, a requester asks one or more dealers for terms on a specified trade. A dealer may quote, decline, or fail to answer within the relevant time. A response can be valid for a limited period. A subsequent customer order, dealer acceptance, venue match, and execution report are different events; which ones are needed to form a trade depends on the protocol.

Streaming describes the repeated publication of prices. A bilateral stream can be addressed to a particular recipient or group. Its prices may be indicative, or they may be actionable under stated conditions. An aggregator combines available streams and

may route a request to a selected liquidity provider. The displayed best price is not by itself a promise that every recipient can execute every displayed quantity immediately.

In a central limit order book, participants submit orders to a matching mechanism. Resting orders specify prices and remaining quantities, and an incoming instruction can trade against eligible opposite-side orders. The mechanism determines priority, partial fills, cancellation, and any special phases. A price-time rule is one possible mechanism; the presence of a screen showing bids and offers does not prove that this rule applies.

Mechanism	A price refers to	Evidence needed to establish a trade
RFQ	A response to a particular request	The execution or acceptance sequence defined by that protocol
Bilateral stream or aggregator	A versioned price available to specified recipients	The relevant firm-quote match or subsequent acceptance and execution evidence
Central order book	An order or aggregated resting depth	Authoritative matching and execution records

The table classifies mechanisms, not products. An instrument can be available through more than one mechanism, and a venue can combine mechanisms or introduce special trading phases. The empirical sources discussed below describe particular historical settings. Their rules must be read as part of those settings.

1.2 Fix side, quantity, and execution conventions

We use the dealer's perspective throughout. Let $\eta = +1$ denote a dealer purchase and $\eta = -1$ a dealer sale. For an executed quantity $z > 0$, the signed change in inventory is

$$v = \eta z. \quad (1.1)$$

A client sell request therefore asks the dealer to buy. A client buy request asks the dealer to sell. Writing the perspective explicitly avoids the common mistake of using a client-side label as the sign of dealer inventory.

For a confirmed execution at monetary price p per unit, the elementary ledger update is

$$q' = q + v, \quad X' = X - vp - f. \quad (1.2)$$

Here q is inventory, X is cash, and f is a separately booked cash fee. A purchase increases inventory and spends cash. A sale does the reverse. Sending a quote, sending a cancellation request, or receiving a market-data update does not itself cause this execution update. Each synthetic account starts with $q_0 = X_0 = 0$, and the examples remain separate. Our opening examples have zero fees to isolate the event logic; later chapters introduce explicit costs.

With a supplied reference price M , marked wealth is

$$W = X + qM. \quad (1.3)$$

Marking inventory does not liquidate it. There is also a distinction between a complete execution ledger and a ledger containing only the confirmations available locally. If a trade has occurred but its report has not arrived, the locally known inventory can differ from the venue's executed inventory. We will keep both views in the example rather than silently assume immediate confirmation.

All numerical prices below are in thousands of JPY per normalized trading unit; cash and marked wealth are in thousands of JPY. These are supplied monetary prices. A source using a different quote variable, such as a yield, requires its own side and unit interpretation before its numbers can be compared. The notation does not turn unlike quote units into interchangeable prices.

1.3 An RFQ is a sequence with more than two outcomes

A useful RFQ record starts with a stable request identifier, the instrument, dealer-side direction, requested size, recipients known to the observer, and relevant deadlines. The dealer then records its own response separately from the request's eventual result. A decline is a response without a quote. No response is different from a decline. A quoted request with no observed result is different from a confirmed loss.

Consider the distinction between a response deadline and quote validity. The first determines how late a new response may be submitted. The second determines how long a particular price remains available under the specified terms. The requester can have received a quote before the response deadline while acting on it later. Neither deadline alone proves that an execution report has already reached the dealer.

Kakuma's saved English paper supplies a concrete historical example. Its Appendix describes Yensai as of end-June 2012: a requester specifies the issue, side, amount, delivery date, and invited dealers; the protocol separates the response deadline from the dealer's quote-validity period. The requester may act after receiving a response or cancel the inquiry. The paper does not establish every detail needed to implement a contemporary quoting system, including a general visibility rule for rival quotes or a universal acceptance-check mechanism.

Its empirical population also matters. The analysis selects inquiries sent to exactly five dealers with all five responses present, and includes canceled inquiries without trade agreement. Thus the analyzed records are neither all requests nor executed trades alone. A dataset selected for complete responses cannot directly reveal the frequency or causes of missing dealer responses in the broader population. The precise source pages and historical limits are retained in the reading note.

Our synthetic log uses one dealer's view. It does not invent the rival quotes, the requester's private valuation, or the number of competitors for a request when those

fields are absent. It records customer rejection only when the example explicitly supplies that notification. A silent outcome remains unresolved at the observation cutoff, even if a later file will eventually reveal what happened.

1.4 Streaming adds quote versions and an execution contract

A stream is a sequence of quote versions rather than a single enduring price. A version identifies the price, side, offered size, recipient scope, and validity or withdrawal conditions. If a request refers to a particular version, the response logic must retain that reference. Replacing a displayed price does not justify rewriting the version attached to an earlier request.

There is a material difference between an indicative stream with a later acceptance decision and a firm quote that can be matched under its terms. Oomen’s aggregator model concerns continuous indicative bilateral quotes followed by a routed deal request and acceptance or rejection by the selected liquidity provider. In that model, liquidity providers do not observe rival quotes directly. The aggregator’s consolidated view and a single provider’s observation are different objects. This is a model of a particular FX execution setting, not a universal rule for streamed trading.

The notebook’s cancellation example deliberately specifies a different contract: a firm ask quote for three units can execute while it remains active at the venue. The dealer sends a withdrawal request, but that instruction has not yet taken effect. One unit executes before the venue cancels the remaining two. The execution and cancellation confirmations then arrive locally at different times.

The lesson does not depend on describing either contract as preferable. A model must know which contract it is simulating. Under the indicative contract, selecting a displayed quote may still lead to rejection. Under the firm-quote example, a still-active quote can execute while a withdrawal message is in transit. A generic field called “quote status” should not conceal those different commitments.

Recipient scope is part of the same contract. A price visible to one client is not necessarily accessible to another client. Likewise, a consolidated display may combine prices that have different conditions, quantity limits, or availability. Any comparison of executable alternatives must first establish that the alternatives are available to the actor making the decision.

1.5 A book contains quantities and priority, not just a midpoint

For a nonempty two-sided book, let b be the highest resting bid price and a the lowest resting ask price. In the uncrossed book used here,

$$s = a - b, \quad m = \frac{a + b}{2}, \quad a \geq b. \quad (1.4)$$

The spread s and midpoint m summarize prices, not executable quantity. A missing side leaves these two-sided statistics unavailable. Averaging a valid bid with an absent ask coded as zero would manufacture a price.

Let r_o be an order's remaining displayed quantity. Aggregate ask depth at price p is

$$D^{\text{ask}}(p) = \sum_{o: \text{side}(o)=\text{ask}, p_o=p} r_o. \quad (1.5)$$

This sum discards order identities and priority within the price level. Gould and colleagues' survey distinguishes order-book mechanisms and the information available in different datasets. Its discussion of priority should not be read as saying that every market uses the same rule. The distinction between order-level events and aggregated depth is central to what can be reconstructed from a feed.

Our toy matching rule gives better prices priority and uses first-in-first-out priority among orders at the same price. An immediate buy instruction can walk upward through ask levels. If Q is the requested quantity and D_j is available depth at the j th ascending ask price, the filled amount at that level is

$$x_j = \min \left(D_j, \left[Q - \sum_{k < j} x_k \right]_+ \right), \quad [u]_+ = \max(u, 0). \quad (1.6)$$

Provided the filled quantity is positive, its volume-weighted execution price is

$$\bar{p} = \frac{\sum_j p_j x_j}{\sum_j x_j}. \quad (1.7)$$

The denominator is filled quantity. If the book cannot supply the full request, remaining quantity needs an explicit handling rule; it does not receive an imaginary execution at the last displayed price. A zero fill has no execution VWAP. The chapter's six-unit example has sufficient displayed depth, so it completes under the declared rule.

For the numerical walk, the initial best bid is 99.98 and best ask is 100.02, giving spread 0.04 and midpoint 100. The ask orders are ordered as follows under the toy price-time rule:

Ask order	Price	Starting quantity	Filled quantity
A1, first at this price	100.02	3	3
A2, second at this price	100.02	2	2
A3, next price level	100.04	4	1

The six-unit buy spends $3(100.02) + 2(100.02) + 1(100.04) = 600.14$, so its VWAP is 100.02333 when rounded to five decimals. Three ask units remain, all in A3 at 100.04. The bid orders are unchanged. This describes a match under the declared rule; it does not estimate a new fundamental value from the changed midpoint.

The historical BrokerTec study illustrates why mechanism details matter even within order-book markets. Its workup phase allows additional trading at a fixed transaction price after an initiating trade, with participation rules and observation conventions specific to that setting. The paper’s event aggregation and price observations cannot simply be substituted for a continuous FIFO matching engine. Our toy engine implements no workup phase and does not claim to reproduce BrokerTec.

1.6 An event has several clocks

An event can occur at a source, arrive at a gateway, and become usable by the decision engine at three different times. Denote those times by t^{src} , t^{recv} , and t^{ready} . In our synthetic examples all clocks are synchronized, so the delays can be written as

$$d^{\text{transport}} = t^{\text{recv}} - t^{\text{src}}, \quad d^{\text{processing}} = t^{\text{ready}} - t^{\text{recv}}. \quad (1.8)$$

These nonnegative differences are properties of the constructed log. Real source and receiver timestamps may use different clock domains, precisions, or synchronization quality. Their raw difference is not automatically a measured one-way delay. A local monotonic ordering remains useful even when cross-system clock differences cannot be interpreted safely.

Reception is not necessarily the point at which a field is usable. Decoding, validation, permission checks, or state reconstruction may still be pending. The notebook therefore uses a ready time together with a local ordering key. Event and decision occurrences sharing a timestamp receive distinct keys in a single total order. If an event and a decision share the same millisecond, that ordering key establishes whether the event was processed before the decision. A timestamp equality alone does not settle the issue.

Write $k_i = (t_i^{\text{ready}}, \ell_i)$ for the local event key and $d = (t_d, \ell_d)$ for the decision key, ordered lexicographically. The permitted history is

$$\mathcal{H}_d^{\text{local}} = \{e_i : k_i \preceq d, e_i \text{ is visible to this actor}\}. \quad (1.9)$$

Visibility is a restriction on the received information, not a prediction of what other parties probably saw. For example, a source file containing every provider's quotes does not establish that a particular dealer observed them during trading. A backtest needs an observation rule for the actor being evaluated.

A basic consistency requirement follows. If two complete event collections have the same permitted history at a decision, a deterministic policy with the same state initialization must make the same decision:

$$\mathcal{H}_d^{\text{local}} = \tilde{\mathcal{H}}_d^{\text{local}} \implies \pi(\mathcal{H}_d^{\text{local}}) = \pi(\tilde{\mathcal{H}}_d^{\text{local}}). \quad (1.10)$$

For a randomized policy, the comparison also holds its internal random input fixed. The notebook checks the observation side of this requirement by altering events that become ready only later and verifying unchanged earlier input snapshots. It replays specified actions; it does not fit or optimize a pricing policy. A previous decision must not gain information simply because the replay file has been supplemented with later records.

1.7 Reconstruct the state the engine could actually use

A decision rarely consumes a raw message in isolation. It uses a state reconstructed from earlier messages: the current request, the accepted quote version, confirmed inventory, or a view of the order book. Write this reconstruction abstractly as

$$\hat{S}_d = \text{Fold}(S_0, \mathcal{H}_d^{\text{local}}). \quad (1.11)$$

The fold applies the protocol's update rules in the declared local order. It includes duplicate handling and state-validity checks. It does not mean sorting the entire eventual dataset by source timestamp and pretending all of those events arrived instantly. A message with an early source timestamp can still be unavailable to the dealer at the decision time.

The feed example starts from a complete snapshot and applies ordered updates. An update exposes a missing sequence number, so the local book becomes untrusted. A replacement snapshot is already present in the complete replay file but is not ready locally for two more milliseconds. A hindsight selector that filters only on source time incorrectly uses that snapshot during the gap. The valid local reconstruction records the gap and waits for recovery.

The source and local clock values are retained for every feed message. All times in the following table are synthetic milliseconds. The last column is the ready-time key, not the venue sequence number.

Message	Source	Received	Ready key
Snapshot, sequence 100	90	100	(102, 1)
Update, sequence 101	95	103	(104, 2)
Update, sequence 103	97	105	(106, 3)
Complete snapshot, sequence 103	98	107	(108, 4)
Late update, sequence 102	96	109	(110, 5)

At local time 106, the stored book is untrusted because update 102 is missing. The recovery snapshot has source time 98, but it is not yet locally received or ready. Even at millisecond 108, decision key (108, 3) precedes recovery while (108, 5) follows it. The latter view has a trusted best ask of 100.03 with three displayed units. These are explicitly ordered synthetic occurrences, not an inference from millisecond timestamps alone.

Source age and time spent in the local cache also answer different questions. For the last accepted source-state update in this synchronized example,

$$\text{age}_{\text{source}}(d) = t_d - t_{\text{state}}^{\text{src}}, \quad \text{age}_{\text{cache}}(d) = t_d - t_{\text{state}}^{\text{ready}}. \quad (1.12)$$

A newly received message can contain an old source state. Conversely, an unchanged price can remain part of a valid newly confirmed snapshot. These ages describe timestamps and state provenance; neither by itself establishes the economic accuracy of the price. The example's maximum-age threshold is a declared operating rule, not an estimated optimal threshold.

At time 110, the recovered snapshot has source age $110 - 98 = 12$ milliseconds and cache age $110 - 108 = 2$ milliseconds. A maximum source-age rule of 15 milliseconds permits its use then. At time 124, the source age is 26 milliseconds and the same rule rejects it. The late update 102 does not refresh the state or its timestamp. The example requires both a trusted sequence and an acceptable age; freshness does not repair a gap.

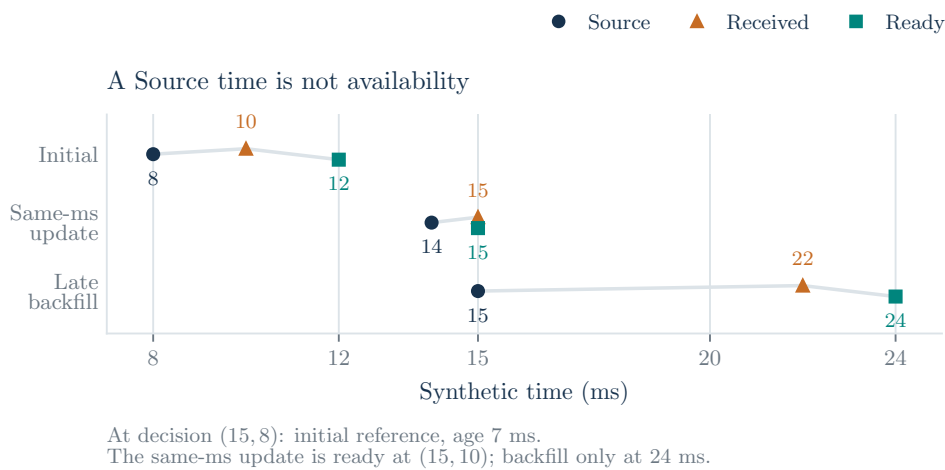
State reconstruction must also survive repeated delivery. For an identical event with the same stable identifier, an idempotent reducer satisfies

$$\mathcal{R}(\mathcal{R}(S, e), e) = \mathcal{R}(S, e). \quad (1.13)$$

This is especially important for execution messages: processing the same fill twice must not double inventory or cash. Identical identifiers with conflicting contents are not

ordinary duplicates; they require rejection or reconciliation. Venue sequence numbers, execution identifiers, and local ordering keys have separate roles and should not be substituted for one another.

A separate reference-price stream illustrates the same rule without book reconstruction. Its initial price 100 has source time 8, receipt time 10, and ready time 12. A replacement price 100.01, sourced at 14, becomes ready at key (15, 10). The observation at (15, 8) still uses 100; a later observation at (15, 11) can use 100.01. A further price 99.80 carries source time 15 but is received only at 22 and ready at 24. Selecting it for the earlier time-15 decision from the eventual file would introduce information the dealer did not have.



B Sequence trust and freshness are separate checks

	104	106	108	108	110	124
Local order	999	999	3	5	999	999
Source age (ms)	9	11	13	10	12	26
Sequence valid	Yes	No	No	Yes	Yes	Yes
Quote eligible	Yes	No	No	Yes	Yes	No

Columns: decision time (ms)

Panels use separate synthetic streams. Keys are (ready ms, local order).
Gap detected at 106; snapshot ready at (108, 4): before order 5, after order 3.
Age limit: 15 ms. At 124 ms the sequence is valid but its source age is 26 ms.

Figure 1.1: Two separate synthetic streams: reference-price source, receipt, and ready times; and sequenced-book validity, recovery, and source age. Local observations respect their ordering keys. A later reference or recovery snapshot present in the eventual file is not usable before it becomes ready.

1.8 Pending cancellation leaves a position to explain

An instruction to cancel expresses intent. A successful cancellation confirmation is evidence that specified remaining quantity has been removed under the protocol. Between the two, the order or quote may still execute. This interval cannot be erased by setting its remaining quantity to zero when the user or strategy first sends the cancellation.

For the stated firm-quote example, with fixed original quantity Q and no quantity-increasing amendments, the venue's quantity accounting is $Q = F + C + R$: filled quantity, canceled quantity, and quantity still live. This example permits only fills and cancellations as terminal quantity dispositions; other dispositions, such as expiry, would require additional terms. The local reconstruction instead knows confirmed fills and cancellations. Its not-yet-disposed quantity U obeys

$$Q = F^{\text{confirmed}} + C^{\text{confirmed}} + U, \quad U \geq 0. \quad (1.14)$$

U need not be the venue's actual resting quantity. Some of it may already have filled or been canceled without the confirmation being locally available. It is the quantity whose disposition remains unconfirmed. An amendment that adds quantity would require an additional term; the example does not silently absorb such amendments into the original Q .

In the firm-quote example, the ask quote offers three units at 100.06. The local withdrawal is sent at time 105. The venue fills one unit at 106 and cancels the remaining two at 107. The corresponding reports become ready locally at 112 and 114. At 105, all three units remain locally unresolved. At 112, one confirmed fill leaves two unresolved units. Only at 114 is the remaining disposition confirmed as canceled.

The venue and local views therefore disagree temporarily without either log being internally contradictory. A valid reconciliation retains both the source event time and local availability time. Reporting the trade at its true execution time for later analysis is appropriate; moving knowledge of that trade into an earlier decision is not.

One simple exposure bound makes the uncertainty tangible. Let $\hat{q}$ be confirmed local inventory, and let unresolved dealer-buy and dealer-sell instructions have remaining possible fill quantities U_i . If these are the only unreported inventory-changing events, each quantity can fill at most once, and every possible disposition is represented, then

$$\hat{q} - \sum_{i \in \text{sell}} U_i \leq q^{\text{venue}} \leq \hat{q} + \sum_{i \in \text{buy}} U_i. \quad (1.15)$$

This bound concerns execution uncertainty under a stated contract. It is not a probability interval and does not account for missing orders outside the recorded population. A strategy may use a more detailed exposure model, but using only confirmed inventory would ignore the possible fills represented by U_i .

Local expiry of a quote also does not prove that no trade occurred before expiry. If a report can arrive late, a still-unresolved prior execution remains possible after the displayed validity period ends. The RFQ example uses an explicit authoritative no-execution confirmation when it classifies an expired response as a known non-win. Time passing alone is insufficient for that label.

1.9 Missing book messages and missing order detail are different problems

Suppose a complete snapshot establishes sequence 100, and the next accepted update is 101. Receiving update 103 without 102 breaks the declared consecutive-update contract. A bid and ask may still be stored in memory, but they no longer establish a complete current book. The notebook keeps a validity flag separate from the numerical levels and declines to use the incomplete state.

Recovery requires a complete snapshot with a declared sequence baseline and a consistent continuation rule. The example receives a snapshot at sequence 103, restores a trusted state, and subsequently ignores an older update 102. Applying that late update merely because it arrived last would roll the reconstructed state backward. A real feed may have retransmission or snapshot-increment synchronization rules beyond this simple contract; they must be implemented explicitly.

Even a complete sequence of aggregated snapshots need not contain order-level information. For a price level, displayed depth changes can be decomposed conceptually as

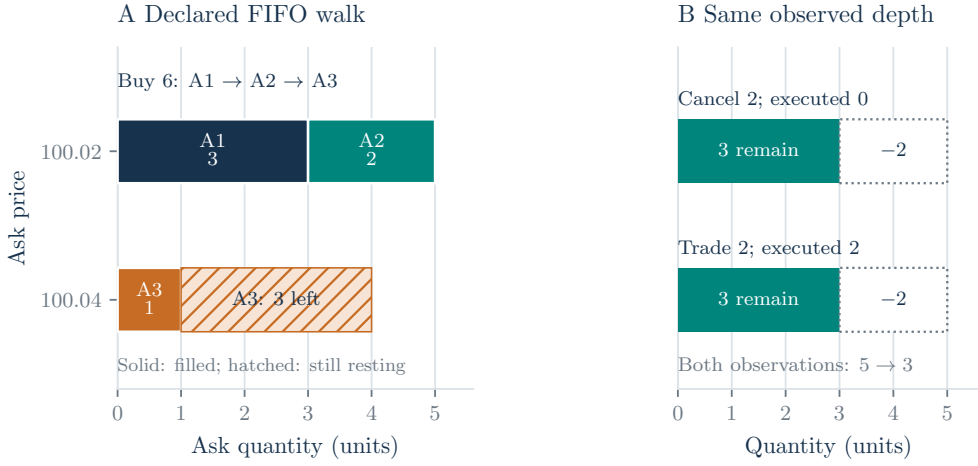
$$\Delta D = A - C - T, \quad (1.16)$$

where A is added displayed quantity, C removed quantity due to cancellation, and T removed quantity due to execution in the simplified example. Real feeds can have additional mechanisms, including hidden or refreshed quantity and representation changes. Equation (1.16) is an accounting description under its specified event vocabulary, not an identification formula from snapshots alone.

A depth change from five to three can result from canceling two units or executing two units. If only the initial and final aggregate depth are observed, those two histories produce the same data. They can have different trade volume and different implications for an individual resting order. Inventing a fill, a queue position, or a cancellation merely from the sign of the depth change would choose a history that the observation does not identify.

The synthetic order-level example is different because it explicitly supplies order identities and FIFO priority. At the best ask, order A1 has three units ahead of A2's two units; A3 offers four units at the next price. The incoming buy of six units can therefore be matched to particular orders. That full-information experiment demonstrates a

declared mechanism. It does not prove that a production depth feed contains the information needed to perform the same reconstruction.



A: declared order IDs and price-time FIFO; VWAP = 100.023333.
 Prices are JPY thousands per normalized unit. No fees or hidden quantity enter this toy walk.
 B: identical depth snapshots do not identify executed volume, order identity or queue position.

Figure 1.2: A declared FIFO book and two observationally identical depth changes. Order identities determine the toy six-unit execution, while aggregate depth alone does not distinguish canceled quantity from traded quantity.

1.10 Count the observed population before estimating a win rate

An RFQ request, a dealer response, a quote, an acceptance, an execution, and an execution report are different counting units. Multiple fills can belong to one request, and one fill can be delivered in repeated messages. Before estimating a frequency, specify the unit in both the numerator and denominator.

Our eight-request example has six dealer responses: five price quotes and one explicit decline. The remaining requests include a cancellation before quoting and a request receiving no response by its deadline. If N_R counts received requests, N_D dealer responses, and N_Q actual quotes, then

$$r_{\text{response}} = \frac{N_D}{N_R}, \quad r_{\text{quote}} = \frac{N_Q}{N_R}, \quad N_Q \leq N_D \leq N_R. \quad (1.17)$$

The request-level records at cutoff 100 are:

Request	Dealer response	Outcome known locally	Confirmed units
R1	Quote	Full fill	2
R2	Quote	Customer rejection	0
R3	Explicit decline	No dealer quote	0
R4	None	Customer canceled before quote	0
R5	Quote	Confirmed closure without execution	0
R6	Quote	Unresolved	0
R7	Quote	Full fill	1
R8	None	No response by deadline	0

The last column counts confirmed executed units, irrespective of side. Its zero for R6 is a statement about confirmations available locally, not proof of zero venue execution. R4's cancellation is ready at 46, before its planned response check at 47. R8's response deadline is 88. Neither request receives a quote in the retained log.

These inequalities use the example's one-response-per-request counting convention; quote replacements would not create new requests or additional responding requests. Message counts are retained separately. Here the response rate includes explicit declines. Chapter 3's valid-price response probability uses N_Q/N_R , corresponding to r_{quote} rather than r_{response} . Chapter 4 then conditions its win probability on those quoted requests. A zero denominator makes the corresponding rate unavailable rather than zero.

Among the five quoted requests, two are confirmed wins at the cutoff, two are confirmed non-wins, and one is unresolved. Let those counts be W , L , and U_Q , respectively, so $N_Q = W + L + U_Q$. The resolved-sample rate and the possible full-cohort eventual-win fraction satisfy

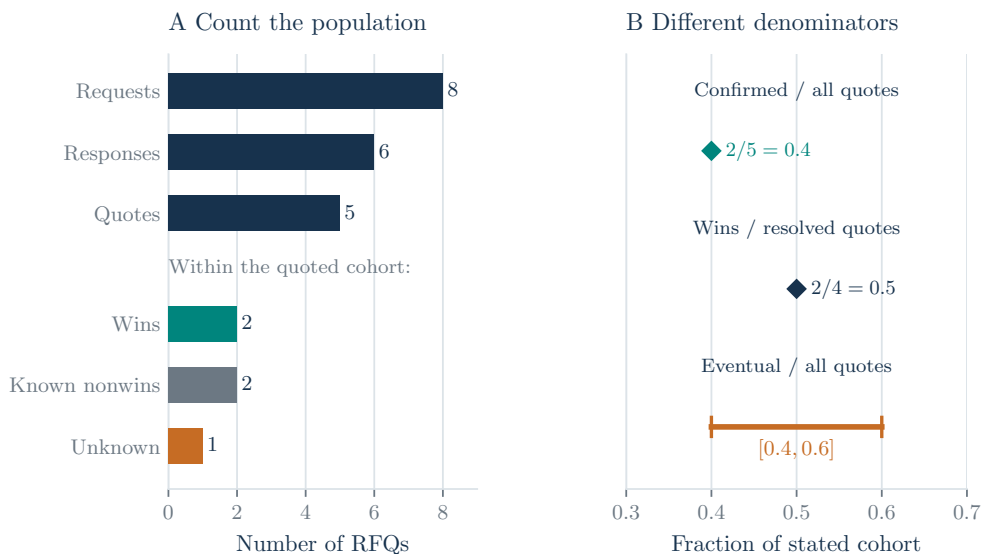
$$r_{\text{resolved}} = \frac{W}{W + L}, \quad \frac{W}{N_Q} \leq r_{\text{eventual}} \leq \frac{W + U_Q}{N_Q}. \quad (1.18)$$

The bound assumes each quoted request has exactly one eventual binary win/non-win outcome, the confirmed labels are final, and all unresolved cases are counted. It is an identification bound for this finite cohort, not a sampling confidence interval. Partial outcomes, reversals, or an incomplete request population would require another definition.

For this example, the response rate is 6/8, and the quote rate is 5/8. Confirmed wins divided by all quotes give 2/5, whereas restricting to resolved quotes gives 2/4. The eventual full-cohort fraction lies between 2/5 and 3/5. Neither replacing the unknown with a loss nor dropping it without changing the estimand preserves the

original question. Nor is a dealer's explicit decline evidence that the client failed to trade elsewhere.

These counts describe the recorded population at a cutoff. Chapter 3 uses such records to estimate arrival rates and execution probabilities, including the selection induced by earlier quote choices. Keeping the labels and denominators explicit is what makes that next step interpretable.



Cutoff $t = 100$ ms. Responses include one explicit decline; quotes exclude it.
 Five quotes partition into 2 confirmed wins, 2 known nonwins and 1 unresolved outcome.
 Each quoted RFQ has one possible full-size win or authoritative no-execution outcome.
 The orange segment is a finite-cohort logical bound, not a confidence interval.

Figure 1.3: Eight received requests and five quoted requests at a fixed observation cutoff. Dealer responses, quotes, confirmed outcomes, and unresolved outcomes have separate counts. The interval describes possible eventual labels of this finite cohort, not statistical sampling uncertainty.

1.11 Build a decision record, then reconcile what happened

A useful decision record captures the action together with the evidence used to produce it. It identifies the request or quote version, decision time and ordering key, admitted reference or book state, confirmed inventory, unresolved exposure, validity flags, and the selected action or reason for declining. Retaining only the final price is insufficient to explain why a quote was sent. The input snapshot ends at a decision key strictly before the quote-emission key, even when both occur in the same millisecond. The action output is not part of the evidence used to produce itself.

The notebook’s main RFQ cutoff is time 100. Only two fills are confirmed locally: a dealer purchase of two units at 99.98 and a sale of one unit at 100.04. The confirmed local ledger is therefore $\hat{q} = 1$ and $\hat{X} = -99.92$, with marked wealth 0.08 at the supplied reference 100.

The unresolved request R6 is a one-unit dealer-buy quote at 99.97. Its privileged source trace shows that it executes at 95, but its report is received at 130 and becomes ready at 132. Thus the actual executed inventory at source-time cutoff 100 is two units, cash is -199.89 , and marked wealth is 0.11. Those are ex-post facts supplied by the complete synthetic trace. They are not observations available to the dealer at the earlier cutoff.

At time 100, the one-unit unresolved buy places possible executed inventory in $[1, 2]$ under the example’s full-fill contract. The local view cannot select the realized endpoint until the execution is known. The later report updates the confirmed ledger and resolves the outcome label. It does not change the reference, quote version, or inventory information that the dealer had used to make the earlier decision.

This distinction also explains why “as of” needs a qualifier. An execution-time report can describe trades occurring before a market-time cutoff using information learned afterward. An as-known report restricts the data to what had become available locally by its observation cutoff. Both are useful, but they answer different questions. A reproducible analysis names the cutoff and the information convention instead of moving between them implicitly.

All three examples remain separate. The order-book walk is not added to the RFQ account, and the firm-stream cancellation race is not treated as another RFQ observation. Combining them would require a shared instrument, account, protocol, and time contract that this introductory experiment has not supplied.

1.12 The information contract carries through the book

Before a quoting model is used, its observation rules should answer several concrete questions. Which messages are visible to this actor? When are they ready? What makes a quote active? Which event creates an execution? What proves a cancellation? How are partial quantities, repeated reports, sequence gaps, and unresolved outcomes represented? Which actions are still possible when the system’s information is incomplete?

Those questions define the inputs to subsequent economic decisions. A reference price belongs to a particular observation time and source. Inventory comes from executions and confirmations, with an explicit treatment of unresolved exposure. A measured win rate belongs to a defined request population. A backtest must reconstruct the information available to the policy, while an ex-post ledger can use later evidence to reconcile what actually occurred.

The executable example preserves these distinctions through small, inspectable event traces. Its checks establish that the stated toy rules and arithmetic agree; they do not certify a venue implementation, feed handler, production deadline, or empirical trading result. Such claims would require the applicable protocol specifications and independent observations.

Chapter 2 next examines supplied reference prices, adverse selection, and markouts. Chapter 3 will then turn request and outcome records into arrival and conditional-execution models. Chapters 4 through 12 use those objects for quoting, inventory control, client-flow decisions, hedging, execution, attribution, and evaluation. The purpose of this opening chapter is to ensure that every such decision refers to an identifiable event and an honest information set.

Exercises

1. **Sides and confirmed executions.** Reconstruct the two fills known at cutoff 100. Explain the client and dealer sides, verify signed quantities, and reconcile confirmed inventory, cash, and marked wealth. Explain why a quote or cancellation request does not create a fill.
2. **What was available.** Reconstruct the feed state during the missing-update interval using ready-time order. Compare it with a hindsight selector that uses only source timestamps. At the snapshot's ready millisecond, show why the local tie-breaking order changes availability. Alter a later event and verify that an earlier decision is unchanged.
3. **Cancellation race.** Reconcile the three-unit firm quote at local times 105, 112, and 114. Distinguish confirmed filled quantity, confirmed canceled quantity, and unresolved quantity. Compare the local view with the privileged venue trace and explain the temporary difference.
4. **Book execution and observability.** Match the six-unit buy against the stated FIFO asks. Report fills by order, VWAP, and residual depth. Construct cancellation and execution histories that both change aggregate depth from five to three, and state which missing information prevents distinguishing them.
5. **State integrity.** Show how duplicate delivery, a conflicting identifier, a sequence gap, and a late old update should be handled under the example's declared rules. Recover from the complete snapshot, then apply the source-age threshold. Explain why a nonempty book is not necessarily usable.
6. **Requests, responses, and outcomes.** Reproduce the eight-request counts. Compute response rate, quote rate, confirmed wins per quote, resolved-sample win

rate, and the bound on eventual full-cohort wins. Explain the assumptions behind each denominator and why an unresolved request is not automatically a loss.

7. **Late confirmation and exposure.** Compare R6's execution time with its receipt and ready times. Reconcile the local confirmed ledger at 100, the ex-post venue ledger at that source-time cutoff, and the local ledger after confirmation. Derive the interval $[1, 2]$ for possible inventory before the report becomes ready.
8. **A reviewable information contract.** Specify the observation fields and execution evidence for the synthetic RFQ, firm stream, and FIFO book. Identify one source-specific convention that must not be treated as universal. Explain how to retain an eventual outcome label without leaking it into a prior decision record.

Source Notes

Kakuma. The saved English translation of *Liquidity Structure and Execution Cost in the Japanese Government Bond Market* describes a historical JGB inquiry platform and a selected empirical sample. Its request fields, response and quote-validity distinction, possible cancellation, and all-five-response filter inform Sections 1.3 and 1.10. We do not infer current platform rules, unseen rival quotes, or an unconditional dealer-response probability. The [official PDF](#) and [reading note](#) identify the edition and inspected passages.

Oomen. The saved June 2016 accepted manuscript of *Execution in an aggregator* distinguishes indicative bilateral streams, the aggregator’s selection, routed requests, and subsequent acceptance. It supplies a useful example of observer-specific information and execution timing. The chapter’s firm-quote cancellation race is an independent construction with a different explicit contract. See the [repository record](#) and [reading note](#).

Gould and colleagues. The saved arXiv 1012.0349v4 of *Limit Order Books* provides the source discussion of order-book mechanisms, quantities, priority, and data representations. Our small FIFO book and paired depth histories are constructed examples. No queue position or execution is inferred from absent order-level records. See the [source record](#) and [reading note](#).

Fleming, Mizrach, and Nguyen. The saved March 2017 revision of Staff Report 381, *The Microstructure of a U.S. Treasury ECN: The BrokerTec Platform*, describes a particular historical order book, workup mechanism, and data sample. It illustrates why matching phases and event aggregation belong to the observation contract. Its venue mechanics and measurements are not imported into our synthetic FIFO example. See the [New York Fed source record](#) and [reading note](#).

Construction and validation scope. The eight RFQs, firm-stream race, FIFO walk, timestamp keys, missing-update scenario, freshness rule, and uncertainty bounds are independent educational examples. Original source passages support the stated conceptual distinctions; the companion notebook supplies the constructed event records and reproduces their numerical results.

Chapter 2

Reference Prices, Adverse Selection, and Markouts

A quote can look attractive against the price used to construct it and still be expensive when it executes. The reference may have become old. The person choosing to trade may know something the dealer does not. A router may select the dealer precisely when its price is behind the market. These mechanisms can leave similar marks in an executed-trade data set, although they imply different models and different remedies.

Chapter 1 established the events and information that a dealer could observe. This chapter asks what a price difference means within that information set. We first reconcile a filled trade against its decision reference and later benchmarks. We then derive two small selection models: one with informed customers, and another with completely uninformed customers choosing among quotes. Finally, a six-fill experiment shows how reference delivery, observation gaps, weighting, and horizon eligibility affect measured markouts.

The reference-price series is supplied. Its economic interpretation and measurement rules must still be stated. All numerical examples are synthetic; prices are in thousands of JPY per normalized trading unit, and cash amounts are in thousands of JPY. The examples share notation but have separate populations and assumptions. None supplies an estimate of real-market information asymmetry or trading performance.

2.1 A reference is an input with a time and a meaning

Let a dealer decide at time d_i , execute at time t_i , and trade $z_i > 0$ units at monetary price p_i . Retain the dealer-side convention from Chapter 1: $\eta_i = +1$ means the dealer buys, and $\eta_i = -1$ means it sells. The signed inventory change is $v_i = \eta_i z_i$. A customer buy therefore corresponds to a negative dealer inventory change.

Write $M_{d_i}^{\text{obs}}$ for the reference actually available to the quoting process. The quote's signed edge against that input is

$$\delta_i = \eta_i (M_{d_i}^{\text{obs}} - p_i). \quad (2.1)$$

A dealer bid below its reference has positive edge. An ask above its reference also has positive edge, because the dealer's sign is negative. A favorable-looking price difference

does not establish expected profit: it is a comparison against a particular supplied number.

Several objects may be called a reference. A contemporaneous midpoint summarizes two quotes. An internal price may combine observable inputs. A later evaluation benchmark may be reconstructed after delayed messages arrive. In a theoretical model, a conditional expected terminal value may serve as the dealer's valuation. These objects can coincide under an explicit model; their labels do not make them interchangeable.

In particular, an unobserved terminal value V is not a field recovered by renaming a later midpoint. We will introduce V only in the two-state model, where its possible values and the dealer's uncertainty are declared. In the event-record experiment, the benchmark is simply a supplied timestamped price series. Its future values are evaluation observations, not evidence that the dealer knew the correct price when quoting.

The distinction matters even when all price fields use the same units. Replacing an old decision reference with a more recent reconstruction changes the apparent quote edge. It can make a deliberately ordinary quote look unusually good or bad after the fact. The original observation record is therefore part of the economic explanation, rather than incidental metadata.

2.2 Reconcile one fill before averaging many

Let $B(u)$ denote the declared evaluation benchmark at source time u . For now assume that the required benchmarks exist and meet the measurement rules. Allocate the actual fee f_i once to the fill. Define the net dealer markout per executed unit at elapsed horizon $h \geq 0$ by

$$\text{MO}_i(h) = \eta_i(B(t_i + h) - p_i) - \frac{f_i}{z_i}. \quad (2.2)$$

Positive markout is favorable to the dealer. A purchase benefits from a higher later benchmark, and a sale from a lower one. This definition includes the initial execution-price difference and the fee. Some systems use the word markout only for subsequent price movement; a report must carry its formula with its label. Our definition agrees with Chapter 11's dealer markout when the same benchmarks, fees, and horizons are used.

There are two distinct intervals between the decision input and the later benchmark. Define the signed decision-to-execution reference change and the signed post-execution reference change as

$$\begin{aligned} L_i &= \eta_i(B(t_i) - M_{d_i}^{\text{obs}}), \\ D_i(h) &= \eta_i(B(t_i + h) - B(t_i)). \end{aligned} \quad (2.3)$$

Adding and subtracting the two intermediate references in equation (2.2) gives the exact bridge

$$\text{MO}_i(h) = \delta_i + L_i + D_i(h) - \frac{f_i}{z_i}. \quad (2.4)$$

This is an algebraic identity, not a causal decomposition. The term L_i may include market movement while the response was in flight, stale information already present at the decision, or a difference between the two reference constructions. Calling the entire term a latency cost would assign a cause that the identity has not identified. Both L_i and $D_i(h)$ can be positive or negative.

For example, a dealer buys at 99.97 after using reference 100.00. The execution benchmark is 100.02, and the five-millisecond benchmark is 100.04. A fee of 0.001 per unit gives a net markout of 0.069: edge 0.030, decision-to-execution change 0.020, later change 0.020, and fee 0.001. The arithmetic is the same whether the original reference was economically well chosen or poorly chosen.

Multiplying the markout by z_i gives a monetary marked value allocated to that fill. It is not necessarily the realized profit of an independently held position. The dealer may have offset the inventory before $t_i + h$, and another trade records the offset. Chapter 11 reconciles those trades to the full account. Adding markouts at several horizons for the same fill would repeatedly count overlapping price movement.

2.3 Selection enters the expected value of a quote

A quoting decision concerns a possible fill, while a markout data set normally contains actual fills. Conditioning connects those objects. Consider a valid quote for one full-size RFQ with observed state x , size $z > 0$, chosen edge δ , and per-unit execution fee k . Let $F = 1$ denote its execution. Define the conditional signed reference cost

$$\begin{aligned} A(\delta, x, h) &= -\mathbb{E}\left[\eta(B(t+h) - M_d^{\text{obs}}) \mid F = 1, \delta, x\right], \\ \mathbb{E}[\text{MO}(h) \mid F = 1, \delta, x] &= \delta - A(\delta, x, h) - k. \end{aligned} \quad (2.5)$$

The state x includes the stated side, size, protocol, and relevant timing information. Equation (2.5) is defined only where the conditioning event has positive probability and the stated conditional expectation exists. Positive A means that the conditional reference movement is adverse to the dealer. It includes the full interval from the decision reference to the evaluation horizon. It is a measurement target under the declared benchmark; by itself it does not identify informed customers, the dealer's own impact, or a technical delay.

Suppose sending this quote has an incremental, avoidable cost c_{resp} in currency units, paid whether or not it executes. Its expected marked reward is

$$r(\delta, x, h) = \Pr(F = 1 \mid \delta, x) z[\delta - A(\delta, x, h) - k] - c_{\text{resp}}. \quad (2.6)$$

The request has already arrived, and equation (2.6) evaluates the action of sending this quote. Its execution probability is conditional on that action; it is not multiplied by the historical frequency with which the dealer chose to respond. Chapter 3 separately models response coverage over incoming invitations. The expression assumes a binary full-size execution, the stated fee allocation, and no additional inventory or hedging terms. At zero execution probability, define the execution-reward contribution directly as zero, giving $r = -c_{\text{resp}}$; there is no conditional A to estimate on that impossible event. Partial executions require their quantity distribution. Inventory preferences and future trading opportunities enter the later chapters through explicit value functions. They should not be hidden inside a measured markout and then charged again.

Changing the quote can change both its execution probability and the conditional reference movement when it executes. The dealer may win more often by improving its price, yet attract a different set of trades. A fixed average markout estimated from yesterday's fills is therefore not automatically the correct $A(\delta, x, h)$ for every possible new quote.

Nor does an unconditional reference forecast answer the conditional question. An average future movement of zero across incoming opportunities is compatible with a negative average signed movement among the opportunities that become this dealer's fills. The next two examples derive that difference through two separate mechanisms.

Observed horizon marks introduce another condition. An estimate made only from fills with valid benchmarks describes the benchmark-eligible subset. It does not automatically estimate the all-fill expectation in equation (2.5). Extending that estimate requires complete coverage or an explicit assumption about missing outcomes, in addition to the side, state, and quote conditioning. The final experiment retains these eligibility differences rather than silently treating its observed subset as all fills.

2.4 Learn from the direction of a trade

Consider a one-unit market with a terminal value V taking two values $V_H > V_L$. Before observing the next customer direction, the dealer assigns probability π to V_H . A customer is informed with probability μ , independently of the value. An informed customer knows the state and buys in the high state or sells in the low state. An uninformed customer buys with probability ρ , independently of V , and otherwise sells.

Take $0 < \pi < 1$, $0 < \rho < 1$, and $0 \leq \mu < 1$ for the interior calculation. The customer trade rule is an explicit part of this illustration. The resulting conditional-value quotes lie inside the two terminal values, so informed buying in the high state and selling in the low state are consistent with strictly positive informed gains when $\mu > 0$. We are

not deriving the full arrival and demand structure of the original Glosten–Milgrom model.

Let **B** mean a customer buy and **S** a customer sell. Splitting each direction into informed and uninformed cases gives

$$\begin{aligned}\Pr(\mathbf{B}) &= \mu\pi + (1 - \mu)\rho, \\ \Pr(\mathbf{S}) &= \mu(1 - \pi) + (1 - \mu)(1 - \rho).\end{aligned}\tag{2.7}$$

The numerator for a high value and a customer buy consists of all informed high-state customers plus the buying fraction of uninformed high-state customers. Dividing that joint probability by the first line of [equation \(2.7\)](#) yields

$$\pi_{\mathbf{B}} := \Pr(V = V_H \mid \mathbf{B}) = \frac{\pi[\mu + (1 - \mu)\rho]}{\mu\pi + (1 - \mu)\rho}.\tag{2.8}$$

In the high state, a seller must be uninformed under the stipulated trade rule. The corresponding posterior is

$$\pi_{\mathbf{S}} := \Pr(V = V_H \mid \mathbf{S}) = \frac{\pi(1 - \mu)(1 - \rho)}{\mu(1 - \pi) + (1 - \mu)(1 - \rho)}.\tag{2.9}$$

With $\pi = \rho = 1/2$ and $\mu = 0.2$, a buy raises the high-state probability from 0.5 to 0.6; a sell lowers it to 0.4. The buy is informative without revealing the customer's type. Informed customers and uninformed buyers both remain possible after observing it.

This conditional calculation is available before the next trade. The dealer can ask what a buy would reveal and choose its ask accordingly. It does not need to observe the future trade in advance. Confusing an ex-ante conditional valuation with the use of a future realized label would erase the economic content of the model.

The notebook evaluates the probabilities by both these formulas and an independent enumeration of value, customer type, and direction. Conditioning on an event with zero probability is reported as unavailable. A zero denominator is not a posterior of zero.

2.5 Competitive quotes reflect conditional values

Assume a risk-neutral competitive dealer, no fees, no inventory holding cost, and no binding position constraint. Conditional zero expected profit for a customer buy and a customer sell gives the ask and bid

$$a = \pi_{\mathbf{B}}V_H + (1 - \pi_{\mathbf{B}})V_L, \quad b = \pi_{\mathbf{S}}V_H + (1 - \pi_{\mathbf{S}})V_L.\tag{2.10}$$

The unconditional valuation before observing the direction is $M = \pi V_H + (1 - \pi)V_L$. In the symmetric case $\pi = \rho = 1/2$ and $V_H = M + d$, $V_L = M - d$, the formulas simplify to

$$a = M + \mu d, \quad b = M - \mu d, \quad a - b = 2\mu d. \quad (2.11)$$

For $M = 100$, $d = 0.10$, and $\mu = 0.2$, the ask is 100.02 and the bid is 99.98. The full spread is 0.04. At the ask, the dealer initially captures 0.02 relative to the unconditional mark. Conditional on that ask being taken, however, the expected terminal value has also risen by 0.02. The dealer-side reference movement is adverse by the same amount. On the bid side the signs reverse, with the same cancellation:

$$\mathbb{E}[a - V \mid \mathbf{B}] = 0, \quad \mathbb{E}[V - b \mid \mathbf{S}] = 0. \quad (2.12)$$

The spread here compensates the dealer for trading against an informative direction. It is not a guaranteed profit per round trip. Pairing an ask execution with an imagined bid execution at the original quotes ignores how the dealer's information and quotes change after the first trade.

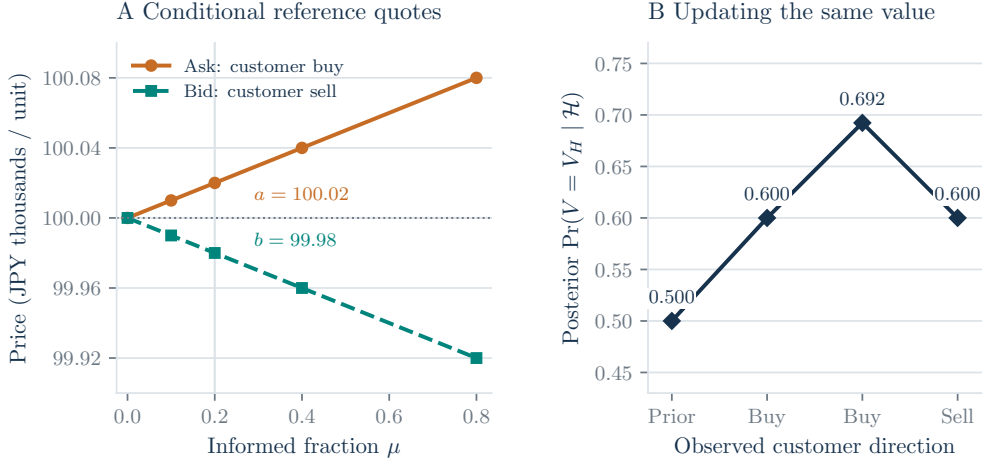
For repeated observations in this particular binary illustration, suppose the same terminal state persists, while customer types and liquidity directions are independently redrawn conditional on that state. Posterior odds then update by the likelihood ratio of the observed direction Y_n :

$$\frac{\pi_n}{1 - \pi_n} = \frac{\pi_{n-1}}{1 - \pi_{n-1}} \frac{\Pr(Y_n \mid V_H)}{\Pr(Y_n \mid V_L)}. \quad (2.13)$$

At the baseline parameters, a buy multiplies the odds by 1.5, and a sell by 2/3. Starting from probability 0.5, the sequence buy, buy, sell gives probabilities 0.6, 0.6923077, and 0.6. The next quote uses the newly updated prior. The symmetric shortcut in equation (2.11) cannot be reused unchanged once that prior is no longer one half; equations (2.8)-(2.10) remain the relevant calculation.

At $\mu = 0$ the direction conveys no value information, so both conditional quotes equal the prior mean. As μ approaches one, the conditional quotes approach the corresponding terminal values. At the endpoint an informed trader can become indifferent to execution. We treat that endpoint as a limit, rather than silently imposing a strict trading incentive that has disappeared.

Glosten and Milgrom provide the source argument for a competitive specialist pricing against the information revealed by a trade. The inspected saved scan explicitly states the risk-neutrality and cost assumptions. Our finite two-state calculation isolates that argument with a declared customer rule. Real quotes can additionally reflect inventory, operating costs, competition, and constraints, so an observed spread cannot be inverted into μ without establishing this model and its inputs.



$V = 100 \pm 0.10$; initial $\pi = 0.5$; uninformed buy probability $\rho = 0.5$.
 B: $\mu = 0.2$, fixed latent value and independent arrival types conditional on that value.
 At baseline, $\Pr(V_H | B) = 0.6$ and $\Pr(V_H | S) = 0.4$. Constructed conditional-value example.

Figure 2.1: The binary information model: conditional bid and ask values and posterior updating under the declared customer-direction likelihoods. The baseline has value uncertainty of 0.10 on each side of 100 and an informed fraction of 0.20. These are constructed model inputs, not estimates from market data.

2.6 An uninformed customer can select a stale quote

Now remove customer information entirely. The customer's direction is independent of a common value change J , which equals $+j$ or $-j$ with equal probability. A dealer still uses reference M , while a competing dealer has already received the update to $M + J$. Both quote a half-spread c . Their prices are

$$\begin{aligned} (b_{\text{stale}}, a_{\text{stale}}) &= (M - c, M + c), \\ (b_{\text{fresh}}, a_{\text{fresh}}) &= (M + J - c, M + J + c). \end{aligned} \quad (2.14)$$

The customer buys at the lower ask or sells at the higher bid. The quotes are firm for the stipulated one-unit trade; there is no subsequent acceptance option. With $j > 0$, there is no price tie between stale and fresh quotes. Using the dealer sign η , the stale dealer is selected exactly when

$$F_{\text{stale}} = 1 \iff \eta J < 0. \quad (2.15)$$

After an upward update, the stale ask is cheaper, so an uninformed customer buy reaches the stale dealer. After a downward update, its stale bid is higher, so an uninformed customer sell reaches it. The router needs only the quoted prices. The customer need not know the value update or have private information.

Set $M = 100$, $j = 0.06$, and $c = 0.02$. There are four equally likely combinations of customer direction and update. The stale dealer wins two of them. Its signed update is unfavorable by 0.06 in each winning case, although the signed update averages to zero across all opportunities:

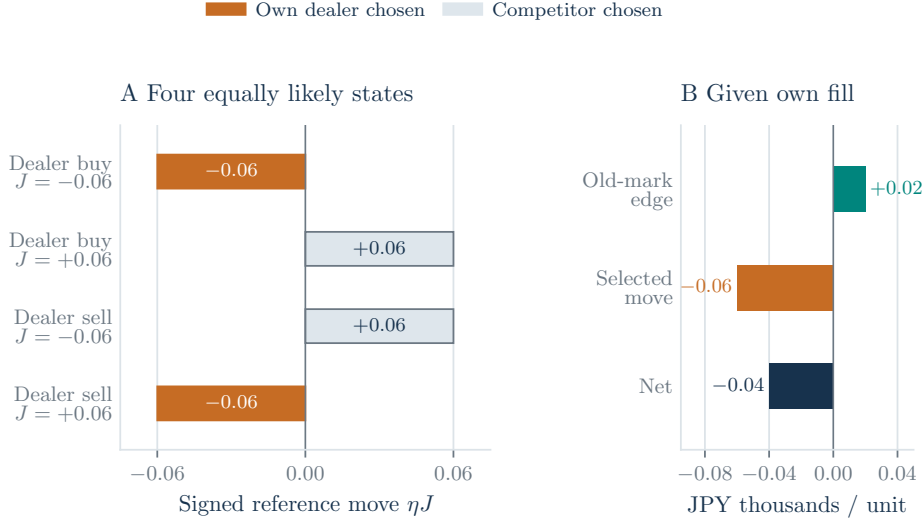
$$\begin{aligned}\mathbb{E}[\eta J] &= 0, & \Pr(F_{\text{stale}} = 1) &= \frac{1}{2}, \\ \mathbb{E}[\eta J \mid F_{\text{stale}} = 1] &= -j, \\ \mathbb{E}[\text{MO} \mid F_{\text{stale}} = 1] &= c - j = -0.04.\end{aligned}\tag{2.16}$$

This example has no fees. Expected monetary marked reward per one-unit opportunity is -0.02 , because the conditional loss is incurred only on the selected half of opportunities. The unconditional zero drift and the conditional negative drift answer different questions.

For a controlled comparison, let this dealer receive the update before quoting as well. Both dealers then quote the same prices around $M + J$. Declare an independent fair tie-breaker. Each wins half of opportunities and earns $c = 0.02$ per selected unit against the common value. The half-spread is a fixed toy input, not a competitive zero-profit solution. This comparison therefore cannot be combined with the zero-profit model in the previous section as though both used one equilibrium.

The event timing is explicit. The stale reference is ready at time zero, the competing dealer receives the update by time four, and the customer trades against quotes selected at time five. The stale dealer does not receive its update until time ten. The complete experiment contains J for evaluating outcomes; its earlier quote input contains only $M = 100$. Giving it the later update while keeping its stale price would create an inconsistent decision record.

Oomen supplies a related but distinct source result. In the accepted aggregator manuscript, uninformed client demand can produce unfavorable conditional changes in the winning liquidity provider's own midpoint. Selection among competing quotes is sufficient in that model. The benchmark is the provider's own midpoint, with the source's quote-error dynamics and conditioning. Our two-dealer common-value example is an independent construction, not a reproduction of Oomen's formula or a claim that every adverse midpoint move reveals the same mechanism.



Dealer sign: $\eta = +1$ buy, -1 sell. Customer direction is independent of J .
 Stale quotes: 100 ± 0.02 ; competing quotes: $100 + J \pm 0.02$. Own selection probability: 0.50.
 Across all opportunities, $\mathbb{E}[\eta J] = 0.00$; given own fill, old-mark edge + move = net. No fees.
 Independent stale-quote construction. Customers have no private information; this is not an Oomen replication.

Figure 2.2: Four equally likely opportunities with uninformed customer direction and an independent value update. Routing selects the stale dealer in the two states with adverse signed updates. Conditioning changes the population: zero mean signed movement over opportunities becomes a loss of 0.04 per selected unit after the 0.02 quote edge.

2.7 Preserve the reference used by the decision

The third experiment uses six actual synthetic fills and a shared reference stream. It does not use the binary-state population or the routing probabilities above. Each fill comes from a quote with intended edge 0.03 and execution fee 0.001 per unit. Two additional requests produce no fill: one quoted request is rejected and one is declined by the dealer. Thus eight requests produce seven quotes, six fills, and thirteen executed units.

Each reference record k has source time s_k , a local ready key r_k , and price P_k . A decision occurrence has its own key $D_i = (d_i, 8)$ and precedes its quote-emission key $(d_i, 9)$. For this simple reference stream, select the latest source observation among the locally admitted records:

$$\mathcal{J}_i = \{k : r_k \leq D_i\}, \quad k_i^* = \arg \max_{k \in \mathcal{J}_i} s_k, \quad M_{d_i}^{\text{obs}} = P_{k_i^*}. \quad (2.17)$$

The example has unique source times and deterministic local ordering. An empty admitted set means the reference is unavailable. A sequenced order book would require

its own reconstruction rule, as Chapter 1 explains; this maximum is not a general replacement for a book-state reducer.

Ordinary reference updates become ready one millisecond after their source time. The update sourced at 35, with price 99.96, is received at 38 and becomes ready at 39. The fourth quote's decision at (38, 8) therefore still uses the source-30 reference of 100.00. Its ask is 100.03. The eventual file contains the source-35 update, but a source-time-only join must not insert that update into the earlier decision.

The six recorded fills have the following inputs. Prices use the chapter's stated currency and quantity units; decision and execution times are synthetic milliseconds.

Fill and dealer side	Size; decision/execution time	Recorded decision reference	Execution price
F1: buy	1; 8 / 10	100.00	99.97
F2: sell	4; 18 / 20	100.04	100.07
F3: buy	2; 28 / 30	99.98	99.95
F4: sell	1; 38 / 40	100.00	100.03
F5: buy	3; 48 / 50	99.97	99.94
F6: sell	2; 58 / 60	100.02	100.05

Fill confirmations become ready two milliseconds after execution. The final analysis cutoff is 72, by which all six fills and all retained reference records are known. Later confirmation supplies an outcome label; it does not rewrite the earlier decision reference or quote price. Duplicate confirmations are likewise prevented from creating additional economic fills.

The baseline deliberately uses its available decision references without a quote-age filter. Some are old. The freshness rule introduced next governs evaluation benchmarks only. Applying that rule to quote eligibility would define a different experiment with different responses and fills, rather than a harmless cleanup of this one.

2.8 State the benchmark clock and missing-data rule

The evaluation stream ends at source time $H = 70$. For a target $u \leq H$, choose the latest reference sourced no later than u and delivered by the final analysis cutoff $T_{\text{eval}} = 72$:

$$\begin{aligned}\kappa(u) &= \arg \max_{k: s_k \leq u, r_k \leq (T_{\text{eval}}, 999)} s_k, \\ B(u) &= P_{\kappa(u)}, \quad \text{age}(u) = u - s_{\kappa(u)}.\end{aligned}\tag{2.18}$$

This is a retrospective source-time benchmark with a declared observation cutoff. It is not the reference available to the dealer at time u . In particular, the source-

time benchmark at each fill becomes ready after that fill. It is legitimate as a later performance label under this convention, and illegitimate as a field in the earlier quote input. An as-known-at- u report would use a different ready-time cutoff and would answer a different measurement question.

Use elapsed horizons $\mathcal{H} = \{0, 5, 10, 20\}$ and require benchmark source age no greater than $\bar{a} = 3$ milliseconds. A target beyond H is outside the observation window even if an old mark exists. A target within the window can also lack a sufficiently fresh mark. These are separate missingness reasons.

There is no source-65 reference record. F6's five-millisecond target is therefore 65, whose last source observation is at 60: age five fails the rule. Its twenty-millisecond target is 80, beyond the data window. The first is a stale benchmark inside the window; the second is an unavailable future horizon. Neither becomes a zero markout, and the last observed price at 70 is not a valid twenty-millisecond label for F6.

Let $\mathcal{E}_h$ contain fills confirmed by the final analysis cutoff for which the requested horizon and benchmark pass these rules. A fixed common cohort over the requested grid is

$$\begin{aligned}\mathcal{E}_h &= \{i : \text{fill known, } t_i + h \leq H, \kappa(t_i + h) \text{ exists, age}(t_i + h) \leq \bar{a}\}, \\ \mathcal{C} &= \bigcap_{h \in \mathcal{H}} \mathcal{E}_h.\end{aligned}\tag{2.19}$$

The intersection is over every requested horizon, not merely the longest one. Here the same fill happens to fail both an intermediate and the longest horizon. In a less convenient data set, an intermediate quote gap could remove a fill whose longest-horizon mark is present.

The available cohorts contain six, five, six, and five fills at horizons zero, five, ten, and twenty. Their quantities are thirteen, eleven, thirteen, and eleven units. The common cohort consists of F1 through F5, with eleven units at every horizon. Counts can recover at a later horizon when an intermediate gap ends; eligibility need not decrease monotonically with elapsed time.

2.9 A changing population can move the average

For a nonempty eligible set $\mathcal{A}$, define its quantity-weighted mean markout by

$$\overline{\text{MO}}_{\mathcal{A}}(h) = \frac{\sum_{i \in \mathcal{A}} z_i \text{MO}_i(h)}{\sum_{i \in \mathcal{A}} z_i}.\tag{2.20}$$

The numerator is the sum of per-fill marked currency values at their respective elapsed horizons. The denominator is executed quantity. This answers a per-unit question. Giving every fill equal weight answers a per-fill question, and can change when one execution is split into several reporting records. Splitting a fill into proportional pieces with apportioned fees leaves the currency total and quantity-weighted mean unchanged.

The computed means are:

Horizon (ms)	Available fills / units	Available mean	Common-cohort mean
0	6 / 13	0.047462	0.056273
5	5 / 11	0.067182	0.067182
10	6 / 13	0.058231	0.059909
20	5 / 11	0.070818	0.070818

The common cohort always contains five fills and eleven units. All entries are net dealer markouts in thousands of JPY per executed unit. They include the quote edge and fee, rather than measuring only subsequent reference drift.

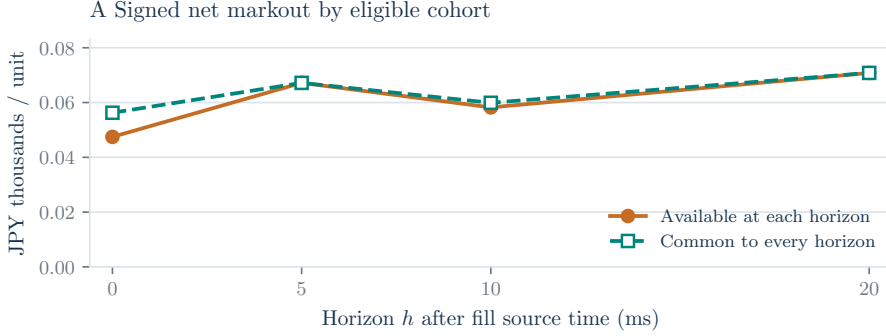
For any two displayed horizons whose common cohort is nonempty, inserting and subtracting its two means gives an exact comparison:

$$\begin{aligned}
 \overline{\text{MO}}_{\mathcal{E}_{h_2}}(h_2) - \overline{\text{MO}}_{\mathcal{E}_{h_1}}(h_1) &= \overline{\text{MO}}_{\mathcal{C}}(h_2) - \overline{\text{MO}}_{\mathcal{C}}(h_1) \\
 &\quad + [\overline{\text{MO}}_{\mathcal{E}_{h_2}}(h_2) - \overline{\text{MO}}_{\mathcal{C}}(h_2)] \\
 &\quad - [\overline{\text{MO}}_{\mathcal{E}_{h_1}}(h_1) - \overline{\text{MO}}_{\mathcal{C}}(h_1)].
 \end{aligned} \tag{2.21}$$

The first line compares the same fills. The remaining lines expose differences associated with sample composition at each horizon. This identity does not impute the missing markouts. The common cohort is itself selected, and its trajectory need not represent the late fills that are absent from it.

Weighting also changes the conclusion drawn from these six fills. Equal-fill available means at horizons zero, five, ten, and twenty are 0.049, 0.061, 0.049, and 0.047. The quantity-weighted twenty-millisecond mean is approximately 0.070818. Both calculations are correct for their denominators: the larger trades and smaller trades have different outcomes in this constructed data set.

There are also two nonexecuted requests. Their actual executed quantities are zero, and they have no realized fill markout. Assigning them zero and averaging over all eight requests would mix execution coverage with fill quality. A separately defined per-request reward can legitimately include zero execution revenue on those requests, but it must state response costs and any hypothetical outcomes separately, as [equation \(2.6\)](#) does.



B Keep the denominators visible

	$h = 0$	$h = 5$	$h = 10$	$h = 20$
Available: fills	6	5	6	5
Available: units	13	11	13	11
Common: fills	5	5	5	5
Common: units	11	11	11	11

Quantity-weighted means after fees; synthetic synchronized milliseconds. Offline cutoff: 72 ms.
Source window ends at 70 ms; maximum source age: 3 ms.
F6 is stale at target 65, available again at 70, and censored at 80. Missing marks are not zero.
The common cohort is an ex post intersection. These curves are neither causal impact estimates nor additive PnL.

Figure 2.3: Net markouts with available and common cohorts at four elapsed horizons. Counts and executed quantities accompany the averages. F6 has a stale five-millisecond benchmark and an out-of-window twenty-millisecond target; neither missing value is replaced with zero or a carried-forward endpoint.

2.10 Benchmark choices change attribution

A measurement can be exact and still depend on its benchmark. Hold each actual execution price, quantity, and fee fixed. If an alternative final benchmark differs by $b_i(h)$, then

$$\widetilde{\text{MO}}_i(h) - \text{MO}_i(h) = \eta_i b_i(h), \quad b_i(h) = \widetilde{B}(t_i + h) - B(t_i + h). \quad (2.22)$$

Adding 0.01 to the endpoint benchmark increases a dealer purchase's per-unit markout by 0.01 and decreases a sale's by 0.01. This sensitivity does not establish that either benchmark is preferable. It identifies what the reported number depends on. A benchmark that systematically differs by side or client population can create a systematic apparent difference in measured quality.

Changing only the decision reference has a different effect. If $M_{d_i}^{\text{obs}}$ is shifted upward by b in a diagnostic calculation, the quoted-edge component increases by $\eta_i b$ and the decision-to-execution component decreases by the same amount. Equation (2.4)'s net markout is unchanged when the actual price and endpoint benchmark stay fixed. The

component bridge exposes a change in explanation that a single headline number would hide.

F4 provides a concrete example. Its actual decision used 100.00 and produced an ask of 100.03, giving edge 0.03. A retrospective source-time join at decision time 38 instead selects 99.96, the update not ready until 39. Against that reconstructed reference, the same actual ask appears to have edge 0.07. Its execution benchmark is 99.94, so the signed decision-to-execution component falls from 0.06 to 0.02. The total remains unchanged. Recomputing an ask of 99.99 would describe a hypothetical quote, not recover the one that was sent.

The notebook tests both sensitivities and the complete component identity at every valid fill-horizon pair. It also checks invariance when a fill is split with proportional fees, rejects zero or negative executed size, and keeps unknown benchmarks unavailable. These checks establish arithmetic and information consistency. They do not certify that a supplied benchmark is unbiased or executable.

2.11 Interpret an adverse markout at the right level

The binary model and the stale-quote model can both produce adverse conditional reference movement. In the first, customer direction reveals information about a terminal state. In the second, customer direction contains no such information, but routing selects the stale dealer when its price is unfavorable relative to an update. A negative markout alone cannot choose between these explanations.

Other mechanisms can contribute as well. A trade may move prices, public information may arrive during the horizon, hedging may interact with the market, and a reference may change because its own sampling or construction changes. An observed association between execution and a later mark is not a decomposition of those effects.

A causal claim needs a defined intervention and a defensible comparison for what would have happened under that intervention's absence. The post-trade benchmark path for an executed request does not reveal the path that would have occurred without execution. Unexecuted requests generally differ in prices, timing, states, and customer choices. Their observed future marks are therefore not automatically the missing comparison for the executed group.

The BrokerTec study illustrates a more structured empirical object. Its transaction observations aggregate an initiating trade with its workup and use the midpoint at the end of the aggregated transaction. The inspected specification models price and order flow jointly, with a structural ordering and a cumulative response measured over fifty transactions. Those assumptions and that transaction clock travel with the result. It is neither our fixed-millisecond fill markout nor a randomized estimate of the cash cost caused by an arbitrary individual fill.

The notebook performs no empirical causal estimation. Its constructed populations are useful because every mechanism and observation rule is stated. They show what a

particular assumption can generate, and what a markout calculation can distinguish. They do not establish which mechanism dominates an actual customer's trading.

2.12 Carry the conditioning into estimation

Before using a measured markout in a quote model, specify the dealer side, monetary units, executed quantity, decision reference, benchmark source, availability cutoff, horizon, fee coverage, sample, and weighting. Retain the input record and the later outcome record separately. A benchmark that arrives later can improve an evaluation label without becoming an earlier observation.

The principal economic connection is [equation \(2.6\)](#): the probability of execution and the value conditional on execution jointly determine a quote's expected reward. They can both depend on its price and the information available when it is sent. An attractive conditional outcome with almost no executions and a high execution rate with adverse selected outcomes are different situations, neither described adequately by one unqualified average.

[Chapter 3](#) turns request and outcome records into arrival and conditional-execution models, with explicit observation populations and validation splits. [Chapter 4](#) combines those probabilities with the conditional value in [equation \(2.6\)](#) to choose a quote. Both steps retain this chapter's dealer signs, benchmark conventions, and separation of decision inputs from later outcome labels.

Exercises

1. **Bayesian direction information.** Enumerate value state, customer type, and direction at the baseline parameters. Recover both posteriors and both quotes independently of the closed-form implementation. Repeat for priors 0.2 and 0.8, and explain why the symmetric spread shortcut cannot simply be applied to those priors.
2. **Conditional compensation and updating.** Reconcile the dealer's positive initial edge with zero expected terminal gain conditional on each direction. Update the prior through buy, buy, sell. Explain the no-information case, the limit as the informed fraction approaches one, and the handling of an impossible conditioning event.
3. **Selection without informed customers.** Enumerate all four stale-quote routing states. Compute the unconditional signed update, conditional signed update, fill probability, conditional marked reward, and reward per opportunity. Repeat the reward calculation when both dealers quote around the updated value with the stipulated tie-breaker.

4. **The recorded decision reference.** Reconstruct F4's admitted reference at decision time 38, its actual ask, and the naive source-time-only alternative. Alter the later delivered update and verify that the earlier decision input is unchanged. Distinguish a retrospective explanation of an actual quote from a hypothetical requote.
5. **Signs and the component bridge.** Calculate every valid fill-horizon markout directly from price and benchmark, then independently from decision edge, decision-to-execution change, later movement, and fee. Show the calculations for a purchase and a sale. Explain why adding several horizons does not produce account P&L.
6. **Horizon eligibility.** Reproduce the stale and out-of-window missingness reasons, the available cohorts, and the intersection over all four horizons. Recalculate the means and verify equation (2.21). Explain why availability can recover at ten milliseconds after failing at five.
7. **Weighting and benchmark sensitivity.** Compare per-fill and per-unit averages. Split a fill with proportional fees and verify which totals and averages are invariant. Shift only the decision reference, then only the final benchmark, and reconcile the two different effects. Keep the two unexecuted requests out of realized fill averages.
8. **What the experiment identifies.** State the observation and execution contracts of all three examples. Explain why an adverse selected markout does not identify customer private information or causal price impact. Name one source-specific benchmark or clock that must accompany a result when it is discussed outside the original paper.

Source Notes

Glosten and Milgrom. The saved Columbia copy of *Bid, Ask and Transaction Prices in a Specialist Market with Heterogeneously Informed Traders* provides the competitive conditional-valuation argument. The inspected opening model discussion makes the specialist risk neutral and removes transaction and inventory-holding costs. Equation (4) on saved-copy page 9 gives the conditional bid and ask; pages 10-11 interpret the information revealed by trade and retain participation and bracketing qualifications. These visible copy folios match the inspected PDF pages. The binary value and prescribed customer-direction population in this chapter are our own finite illustration. We do not claim to reproduce the paper’s full model or infer its parameters from observed spreads. See the [author-hosted original](#) and [chapter reading note](#).

Oomen. The local source is the June 2016 accepted manuscript of *Execution in an Aggregator*, distinct from the journal publication date. Printed page 9, source equations (6)-(8), measures conditional marked value using the winning provider’s own midpoint. Uninformed client demand does not prevent adverse selection through quote competition. The source’s full effective spread, quote dynamics, and request/acceptance conditioning are not our per-unit level-price cash formula or the common-value routing example. See the [LSE source record](#) and [chapter reading note](#).

Fleming, Mizrach, and Nguyen. The saved March 2017 revision of Staff Report 381, *The Microstructure of a U.S. Treasury ECN: The BrokerTec Platform*, supplies the empirical discussion of transaction aggregation, price discovery, and fitted responses to order-flow innovations. The inspected transaction-clock and structural-model conventions qualify the interpretation of the fifty-transaction impact measure. The synthetic millisecond example does not reproduce that data set or estimate the study’s responses. See the [New York Fed source record](#) and [chapter reading note](#).

Construction and validation scope. All value states, customer-type probabilities, quote comparisons, reference records, requests, fees, and quantities in the notebook are educational constructions. The executable examples verify probability calculations, signed identities, observation boundaries, and measurement populations. The three original sources support the stated conceptual distinctions; their selected passages do not establish real-market performance for these examples. The companion notebook preserves the inputs and checks needed to reproduce those examples.

Chapter 3

Arrivals and Conditional Execution

A dealer can receive many requests and execute very few trades. It can also win a large fraction of the requests it answers while serving only a narrow part of the incoming flow. These observations describe different stages of the trading process. A model that combines them into a single historical fill rate loses the distinction between access to opportunities, the choice to respond, and the customer's subsequent decision.

Chapters 1 and 2 established the information available when quoting and the meaning of a trade's conditional marked value. This chapter supplies the probability models that connect those objects. We count requests against an explicit exposure clock, separate response and execution probabilities, preserve unresolved outcomes, and estimate a small probability model using chronological data. We also explain why a resting order needs a separate model of queue events and cancellation.

All numerical examples are synthetic. The reference price and risk inputs are supplied, and no market calibration is claimed. An RFQ is one invitation delivered to this dealer, identified once across retransmissions. Unless stated otherwise, a response can produce at most one full-size execution. The examples concern this observation population; they do not estimate requests that the dealer never received or opportunities hidden during a feed outage.

The arrival and competing-event examples measure time in minutes. The explicit request log and probability-fitting experiment use synchronized synthetic milliseconds. Quantities are normalized trading units, and the fitted quote offsets are in thousands of JPY per unit. These scales are teaching conventions, not empirical specifications of a trading venue.

3.1 Define the opportunity before estimating its probability

Let request i arrive locally at r_i , and let the dealer send a quote at $d_i \geq r_i$. Its request state x_i contains information available at the stated decision stage: instrument, dealer side, requested size, market state, and protocol. Retain $\eta_i = +1$ for a dealer purchase and $\eta_i = -1$ for a dealer sale. With supplied reference $M_{d_i}^{\text{obs}}$ and quote price p_i , define

$$\delta_i = \eta_i(M_{d_i}^{\text{obs}} - p_i), \quad \ell_i = d_i - r_i. \quad (3.1)$$

Larger positive edge means a less attractive price for the customer on either side. The elapsed response time ℓ_i is known when the quote is sent. The time until a future execution report is received is a different variable, unavailable at that decision. A model used before choosing when to respond must use a planned delay or integrate over future delay, rather than quietly inserting the realized value.

Write $R_i = 1$ when a valid dealer response is sent, and $Y_i = 1$ when that response ultimately produces the declared full-size execution. Here a valid response means a usable price quote. An explicit decline counts as a dealer reply in [Chapter 1](#), but it has $R_i = 0$ in this chapter. An invitation with no response has no actual execution under this protocol. It still belongs in request counts and service-coverage calculations. Its hypothetical outcome under a quote that was never sent has not been observed.

Let $N^{\text{req}}(t)$ count distinct received requests, and let $H(t)$ indicate declared observation exposure. The predictable request intensity satisfies

$$\mathbb{E}[dN^{\text{req}}(t) \mid \mathcal{F}_{t-}] = H(t)\lambda^{\text{req}}(t) dt. \quad (3.2)$$

Here $\mathcal{F}_{t-}$ is the information just before the event. A statement of requests per minute needs both the event definition and the minutes being counted. A trading break, a known recorder outage, and a quiet but healthy connection have different meanings. Quiet observed minutes contribute exposure with zero requests. Unobserved minutes cannot be filled with invented zero counts.

The target may deliberately be the dealer's observed and eligible population. Extending that target to the entire market requires more information about invitation routing and unavailable periods. In particular, making the observation indicator depend on whether activity looks interesting can select the sample. Declaring an exposure clock documents that selection; it does not remove it.

3.2 Estimate arrivals against exposure

Begin with a deliberately simple model: within each declared state, requests follow a Poisson process with constant rate λ . For observed counts n_j in exposure intervals E_j , the part of the log likelihood depending on the rate is

$$\mathcal{L}(\lambda) = \sum_j (n_j \log \lambda - \lambda E_j) + \text{constant}. \quad (3.3)$$

The remaining count-likelihood terms depend on E_j and n_j , but not on λ . Differentiating gives $\sum_j n_j / \lambda - \sum_j E_j = 0$. With positive total exposure, the maximum-likelihood estimate is

$$\hat{\lambda} = \frac{\sum_j n_j}{\sum_j E_j}. \quad (3.4)$$

This is a ratio of totals, not an equally weighted average of rates from intervals of unequal length. If no requests are observed during positive exposure, the maximum lies at zero. If exposure is zero, there is no rate estimate from that sample. A positive expected number of future requests must then come from a prior or another stated model, not from division by zero.

Different rates may apply to different observed states. Estimate each state against its own exposure and reconstruct an aggregate using exposure weights. Pooling a quiet state and a busy state can make count variation look like event clustering. Conversely, fitting a different rate after every observed burst can conceal genuine failure of the Poisson assumption.

The first experiment observes 40 requests in 10 minutes and another 40 in 20 minutes. The interval estimates are 4 and 2 requests per minute. A single pooled constant-rate fit gives $80/30 = 8/3$ requests per minute. Averaging 4 and 2 would instead give 3, assigning the shorter interval too much exposure weight. The pooled fit is useful arithmetic; the unequal interval rates remain a reason to examine the constant-rate assumption.

A constant-rate Poisson model implies independent increments and equality between the mean and variance of counts in equal exposure windows. Those are model restrictions. Inspect chronological residuals, dispersion within comparable states, time-of-day patterns, and reporting bursts before relying on them. A Hawkes or other history-dependent intensity is a possible extension, but a more elaborate formula does not repair a broken exposure clock or duplicated events.

3.3 From requests to executed flow

Let $m = (\eta, z)$ denote a request mark containing dealer side and requested size. For a fixed state, write the request intensity measure as

$$\Lambda^{\text{req}}(dm) = \lambda^{\text{req}} \nu(dm), \quad \int \nu(dm) = 1. \quad (3.5)$$

The probability measure ν allocates requests across marks. The total rate λ^{req} has units of requests per time. If a supplied measure already gives the request rate for each size, it is already Λ^{req} ; multiplying it by the size probabilities again changes the model.

For one request population, the probability chain rule gives

$$\Pr(Y = 1 \mid x, m) = \rho(x, m) \Pr(Y = 1 \mid R = 1, x, m), \quad \rho(x, m) = \Pr(R = 1 \mid x, m). \quad (3.6)$$

This identity does not assume independence between responding and winning. The conditional probability on the right describes the responders selected by the actual policy. It incorporates that policy's quote choices. When $\rho = 0$, define the product's

execution contribution directly as zero; a win probability conditional on an impossible response is not identified.

To make quote choices explicit, let a denote the response action, including edge and any timing choice, and let $\pi(da \mid R = 1, x, m)$ be its distribution among responders. Let $q(a, x, m)$ be the corresponding conditional execution probability. In this chapter q denotes a probability, not the inventory variable of Chapters 1 and 5. Chapter 4 writes this probability as $p(\delta \mid x)$ when optimizing the price offset. The expected completed fills originating from a unit of request-time exposure have marked rate

$$\begin{aligned}\Lambda^{\text{orig}}(dm) &= \lambda^{\text{req}} \nu(dm) \rho(x, m) \bar{q}(x, m), \\ \bar{q}(x, m) &= \int q(a, x, m) \pi(da \mid R = 1, x, m).\end{aligned}\tag{3.7}$$

All factors refer to the same state and population. If information arrives between invitation and response, the conditioning and averaging must also cover that information. Equation (3.7) is a request-cohort expectation. It is not, in general, the instantaneous rate of physical executions on the wall clock. A request that arrives now can execute later.

For illustration, suppose request and outcome mechanisms are specified exogenously, and a successful request originating at s has execution-delay density $g(u \mid s)$. In the next expression, $\lambda^{\text{req}}(s)$ denotes the rate already gated to the tracked request population by $H(s)$. This defines completions originating in that population; it does not set unobserved whole-market activity to zero. Suppressing marks, the expected physical execution rate is

$$\bar{\Lambda}^{\text{exec}}(t) = \int_{-\infty}^t \lambda^{\text{req}}(s) \rho(s) \bar{q}(s) g(t - s \mid s) ds.\tag{3.8}$$

This is a first-moment convolution under the stated model, not a general conditional-intensity theorem for interacting queues. A deterministic delay is handled by a point mass instead of a density. Equal rates follow in a stationary setting with a proper time-homogeneous delay distribution, or in an instantaneous-execution approximation. They need not agree during a market opening, an outage, or a burst of requests.

Start with no outstanding requests. Four requests per minute during the first minute, zero during the second, and an execution probability of one quarter produce one expected eventual fill from the first cohort. If every successful execution occurs one minute after its request, the expected physical counts in the first and second minutes are zero and one. The notebook checks this shift explicitly.

For full-size executions, the expected gross quantity and signed inventory contributions attached to request cohorts are

$$v^{\text{gross}} = \int z \Lambda^{\text{orig}}(dm), \quad v^{\text{signed}} = \int \eta z \Lambda^{\text{orig}}(dm).\tag{3.9}$$

The first is nonnegative; the second can cancel between buys and sells. Neither is the number of trades. Quantities and size probabilities each enter once. For a model with partial execution, replace the full requested size by the conditional executed-quantity distribution.

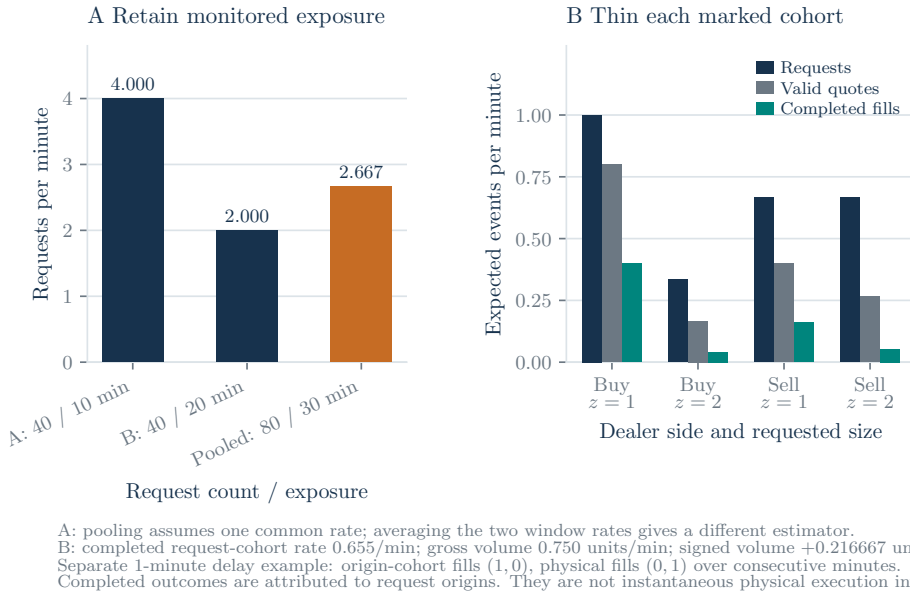


Figure 3.1: Request exposure, response selection, and conditional execution produce different rates. The synthetic calculations attach completed outcomes to request cohorts; physical execution times require a delay model.

3.4 Keep outcome states and denominators separate

A request log should retain more detail than a binary training column. A terminal cancellation can be a known non-execution for the declared full-size RFQ. A missing outcome report is not a cancellation. A dealer decision to decline an invitation is not a competitive loss. A response rejected as invalid also needs a recorded state and an explicit choice of response denominator.

Recorded state	Role in the actual process	Use in a conditional win sample
No valid response	Received invitation, no executable dealer quote	Coverage denominator; no observed hypothetical quote outcome
Confirmed win	Full-size dealer execution	Known positive label
Confirmed competitive loss	Terminal result without this dealer's execution	Known negative label
Confirmed cancellation	Terminal request cancellation before any fill	Known negative label for the declared target
Unresolved	Terminal result unavailable at the cutoff	Unknown label; retain separately

Let N be received invitations, n_R valid responses, and let those responses contain w known wins, l known losses, c known cancellations, and u unresolved results. Thus $n_R = w + l + c + u$. Two descriptive ratios are

$$\hat{q}_{\text{known}} = \frac{w}{w + l + c}, \quad \hat{\rho} = \frac{n_R}{N}. \quad (3.10)$$

Both ratios require positive denominators. The first conditions on having a known result as well as a response. It estimates neither all-responder probability nor a price-response curve merely by being called a win rate. Excluding cancellations would produce a different target, conditional on reaching the remaining outcome states. An explicit trade/no-trade factorization can be useful, but both stages must then be modeled and multiplied consistently, as in Chapter 4.

Without assumptions about the unresolved responses, the eventual realized win fraction among these n_R responses lies in the sharp bounds

$$\frac{w}{n_R} \leq q_{\text{cohort}} \leq \frac{w + u}{n_R}. \quad (3.11)$$

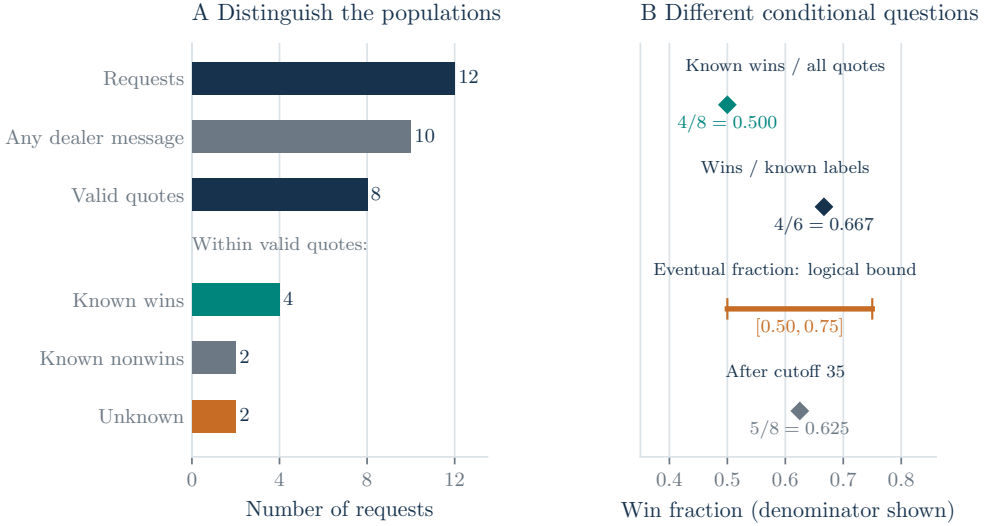
These bounds concern the finite cohort. They are not statistical confidence intervals for a population parameter. The lower endpoint assigns every unresolved response a non-win; the upper endpoint assigns every unresolved response a win. If all responses are unresolved, the known-outcome fraction is undefined but the bounds remain $[0, 1]$. With no response, even the finite-cohort conditional fraction is undefined.

The same logic applies to actual wins per invitation, using N as the denominator and retaining no-response invitations as actual non-executions under this protocol. It does not identify how often the dealer would have won if it had quoted every invitation. A single log can support several useful ratios, provided each is labeled with its population and its observation cutoff.

In the second experiment, 12 invitations produce eight valid quotes, two explicit declines, one cancellation before quoting, and one timeout. Counting any dealer message gives ten responses, but the executable-quote coverage used by R is $8/12$. At cutoff 25, the eight quotes have four confirmed wins, one confirmed competitive loss, one

confirmed cancellation, and two unresolved outcomes. The known-outcome fraction is $4/6$, while the eventual responder-cohort bounds are $[4/8, 6/8] = [0.50, 0.75]$. Actual wins per invitation are bounded by $[4/12, 6/12]$.

At cutoff 35, both missing results are available: one is a win and one a non-win. The eventual quoted-cohort fraction is $5/8 = 0.625$. This later knowledge does not belong in a model fitted at cutoff 25. The missing results were unknown observations at the earlier cutoff, rather than errors to be silently replaced with their final database values.



At cutoff 25 ms: valid-quote fraction $8/12$; any dealer-message fraction $10/12$ includes 2 declines. The 8 quoted outcomes are 4 known wins, 2 known nonwins and 2 unresolved labels. The bound assumes one eventual full-size binary result per quote; it is not a confidence interval. Cutoff 35 is later knowledge. It must not be backfilled into a fit performed at cutoff 25.

Figure 3.2: A synthetic request log preserves no-response invitations, terminal outcomes, and unresolved quoted outcomes. Known-outcome ratios and bounds on the eventual cohort fraction answer different questions.

3.5 A label has an availability time

Let b_i be the time at which the terminal outcome becomes available to the model-building process. A source timestamp alone does not establish that the result could have been used at an earlier fit. For a model fitted at origin T , define label availability by

$$O_i(T) = \mathbf{1}\{b_i \leq T \text{ and the terminal outcome is verified}\}. \quad (3.12)$$

The feature row must separately reproduce information available at the quote decision. A feature can be old yet available; a freshly reconstructed feature may have

been unavailable. Feature validity and label maturity are two checks with different clocks.

Dropping unavailable labels gives an observed-data target

$$q_{\text{obs}}(a, x; T) = \Pr(Y = 1 \mid R = 1, A = a, X = x, O(T) = 1). \quad (3.13)$$

It equals the all-responder conditional probability only under an additional condition, such as complete outcome coverage or conditional independence between outcome availability and the outcome given the modeled state and action. Fast wins and slow losses can violate that condition. Waiting for a fixed horizon helps only if the protocol and reporting guarantees make the selected labels complete or the remaining missingness assumption defensible.

A cancellation acknowledgement can also race with a fill. The negative label must follow the protocol's terminal resolution, rather than the first locally observed cancellation message. Likewise, a fixed-horizon target such as execution within two seconds requires a complete observation window through those two seconds before assigning a zero. It is a different target from eventual execution of the response.

A historical training table built from the final database often contains labels that were not yet known at the historical fit origin. Sorting its rows by request time does not remove that leak. Reconstruct the eligible set at each origin, freeze it, and record which recent requests were excluded for unavailable outcomes. This links the event discipline of Chapter 1 to the evaluation discipline of Chapter 12.

3.6 Response selection and quote support limit interpretation

Historical data describe the quotes a particular policy chose to send. Inventory, client access, risk limits, market state, and human intervention can influence both response decisions and quote prices. A regression may therefore learn useful predictions under that policy without identifying the result of changing the quote.

Let $Y(a)$ denote the potential execution outcome if the dealer responds using a well-defined action a . Under consistency, the logged conditional probability is

$$q_{\text{log}}(a, x) = \mathbb{E}[Y(a) \mid R = 1, A = a, X = x], \quad q_{\text{target}}(a, x) = \mathbb{E}[Y(a) \mid X = x]. \quad (3.14)$$

The second target refers to responding with that action in the specified invitation population. Equality requires adequate assumptions about selection and confounding, in addition to correct outcome measurement. Conditional exchangeability of the potential outcome with response and action assignment, together with consistency and positivity, is one sufficient route. It is not established by fitting a smooth curve.

The observable action support is

$$\mathcal{A}_{\log}(x) = \text{supp}(A \mid R = 1, X = x). \quad (3.15)$$

For a continuous quote, support means positive probability in relevant neighborhoods, not observing the exact same decimal price repeatedly. A model can output a probability outside this set because its formula extrapolates. The output is then a modeling assumption rather than evidence supplied by nearby logged actions. Sparse support within the range can also make interpolation unreliable.

Similarly, a request stratum to which the logging policy never responded contains no observed quote outcomes for that stratum. Coding all its invitations as losses teaches the model the old decision to decline. It does not reveal the latent outcome under a newly offered quote. Service coverage and response propensity are useful companion reports precisely because a good responder-only score can coexist with poor coverage of incoming requests.

Kakuma’s historical JGB inquiry study makes the population issue concrete. Its retained inquiries invited exactly five dealers and obtained rates from all five; cancellations were retained. That design supports analysis within the selected sample. It does not provide this dealer’s unconditional response probability across every invitation, or exposure for an all-market request-arrival estimator. The relevant source passages and exclusions are recorded in the reading notes.

3.7 Fit a conditional probability, with units attached

For a small descriptive model, let $\phi(a, x)$ be a vector of decision-time features with an intercept. It can contain the dealer’s edge, requested size, dealer side, observed market state, and elapsed response time at quote submission. A logistic specification is

$$q_{\theta}(a, x) = \sigma\left(\theta^{\top} \phi(a, x)\right), \quad \sigma(v) = \frac{1}{1 + e^{-v}}. \quad (3.16)$$

Feature definitions include their scales. An edge in monetary price units differs from an edge divided by a fixed tick-like scale. A latency coefficient per millisecond differs by a factor of one thousand from a coefficient per second. Normalize using a declared physical scale or training-only statistics; avoid normalization estimated from the future test sample.

For eligible training responses $\mathcal{I}_{\text{tr}}$, estimate parameters by minimizing a penalized Bernoulli loss. One convention is

$$\begin{aligned} \hat{\theta} = \arg \min_{\theta} \Bigg\{ & -\frac{1}{|\mathcal{I}_{\text{tr}}|} \sum_{i \in \mathcal{I}_{\text{tr}}} [y_i \log q_{\theta i} + (1 - y_i) \log(1 - q_{\theta i})] \\ & + \frac{\kappa}{2} \sum_{j > 0} \theta_j^2 \Bigg\}. \end{aligned} \quad (3.17)$$

The intercept is unpenalized here, and the numerical experiment uses this mean-loss convention. A penalty added to a summed loss has a different strength from the same number added to a mean loss, so the convention belongs with the parameter. Regularization can stabilize separated or sparse samples. It does not supply missing action support, correct mislabeled outcomes, or make correlated observations independent.

If the edge feature is $(\delta - \delta_0)/s_\delta$ with fixed center δ_0 and scale $s_\delta > 0$, and no interactions involving that edge, then

$$\frac{\partial q_\theta}{\partial \delta} = \frac{\theta_\delta}{s_\delta} q_\theta (1 - q_\theta). \quad (3.18)$$

A negative coefficient makes the fitted curve decrease as dealer edge rises while the other features are held fixed. This is a property of the specification. A negative descriptive coefficient alone is not a causal price elasticity. If interactions or shape constraints are added, derive and check the resulting derivative instead of borrowing equation (3.18).

The object entering a decision model must have the same target as the fitted probability. A model trained only on known competitive wins and losses omits cancellations. A model trained only on a chosen client segment describes that segment. A probability measured per incoming request includes the response policy. Each may be useful, but inserting any one of them into an optimization formula expecting another changes expected reward.

3.8 Separate fitting, selection, calibration, and testing in time

Use four consecutive decision-time blocks: training estimates candidate parameters, validation selects among declared candidates, calibration estimates a probability mapping for the selected candidate, and test evaluates the frozen pipeline. If block k spans $[T_k, T_{k+1})$ and has a declared label cutoff $C_k \geq T_{k+1}$, its eligible scored responses are

$$\mathcal{I}_k = \{i : T_k \leq d_i < T_{k+1}, R_i = 1, O_i(C_k) = 1\}. \quad (3.19)$$

In the experiment, the first three label cutoffs equal their block boundaries. The training model is fitted at the first boundary, candidate selection occurs at the second, and calibration is fitted at the third. Test predictions use the frozen coefficients and mapping. The test label cutoff includes an additional reporting window; those later labels can be used for scoring, but not to alter the already evaluated pipeline.

Equation (3.19) is a clear operational cutoff, not a universal cure for outcome-dependent reporting. The synthetic experiment uses bounded reporting delay that is independent of the simulated outcome, conditional on its declared generation process. Its excluded boundary rows are counted. In live data, compare outcome maturity and

missingness across states and label types; a complete follow-up window or a suitable observation model may be necessary.

Do not refit the selected model on additional blocks halfway through this experiment. That would change the prediction scale on which the calibration mapping was fitted. Refitting can be a reasonable deployment procedure, but it requires a newly specified training and calibration sequence followed by fresh evaluation.

Chronological splitting protects the direction of information flow. It does not make adjacent outcomes independent or guarantee a stable market. Dependence affects uncertainty estimates, while structural change affects transport to future periods. Repeated evaluation should preserve the full fitting sequence at each rolling origin, with uncertainty calculations that respect the dependence structure.

3.9 Score probabilities and inspect their calibration

On an eligible scored set of size n , the mean log loss is

$$\text{LL} = -\frac{1}{n} \sum_{i=1}^n [y_i \log \hat{q}_i + (1 - y_i) \log(1 - \hat{q}_i)]. \quad (3.20)$$

The Brier score is the mean squared probability error. For a fixed prediction q and a Bernoulli outcome with success probability p , its expectation decomposes as

$$\text{BS} = \frac{1}{n} \sum_i (\hat{q}_i - y_i)^2, \quad \mathbb{E}[(q - Y)^2] = (q - p)^2 + p(1 - p). \quad (3.21)$$

The second expression is minimized at $q = p$, making the score sensitive to the probability level, not just the ranking of quotes. Log loss has the same population optimum and penalizes confident errors strongly. Probabilities exactly zero or one yield infinite log loss when contradicted. Use numerically stable log-sigmoid calculations or declare any clipping used for reporting; do not hide bad predictions with an unexplained floor.

A simple calibration mapping applies an intercept and slope to the selected model's logit:

$$\tilde{q} = \sigma(\alpha + \beta \text{logit } \hat{q}), \quad \text{logit } q = \log \frac{q}{1 - q}. \quad (3.22)$$

Fit α and β using only calibration responses. Identity calibration is $\alpha = 0$, $\beta = 1$. An intercept shifts the overall probability level, while the slope changes how extreme predictions are. A positive slope preserves ranking. A slope constraint must be stated if ranking preservation is required; the unconstrained formula itself does not guarantee it.

For bins B_b fixed before inspecting test labels, record the test count, average prediction, and observed fraction:

$$n_b = |B_b|, \quad \bar{q}_b = \frac{1}{n_b} \sum_{i \in B_b} \tilde{q}_i, \quad \bar{y}_b = \frac{1}{n_b} \sum_{i \in B_b} y_i. \quad (3.23)$$

Empty bins have no observed calibration fraction. Plotting zero for an empty bin would invent evidence. Wide bins can conceal opposite errors in different states; narrow bins can be too sparse to interpret. A reliability plot should therefore show counts and support, and should be supplemented by state and time slices chosen for an economic reason.

On a fixed set of opportunities with correctly specified conditional probabilities, expected wins add even when outcomes are dependent:

$$\mathbb{E} \left[\sum_i Y_i \mid \mathcal{X} \right] = \sum_i q_i, \quad q_i = \mathbb{E}[Y_i \mid \mathcal{X}]. \quad (3.24)$$

Here $\mathcal{X}$ denotes the common conditioning information for that statement. Predictive variance, unlike the mean, also depends on covariances. Aggregate expected wins matching aggregate observed wins is a useful check, but can hide severe offsetting errors across prices, sides, or client groups. Calibration is one component of model adequacy, alongside resolution, stability, coverage, and economic consequences.

3.10 Three reproducible experiments

The companion notebook separates three populations. Its first experiment estimates request rates and carries normalized size marks through response and execution probabilities. Its second uses a small explicit event log to expose denominator and reporting-cutoff errors. Its third fits and evaluates probabilities on a seeded synthetic chronological sample. These examples share definitions; they are not pooled into one empirical estimate.

The first experiment allocates its 80 requests to dealer-buy sizes one and two with counts 30 and 10, and dealer-sell sizes one and two with counts 20 and 20. Dividing by 30 minutes gives the request rates below. Response and conditional win probabilities are supplied assumptions, not estimates from these request counts.

Request mark	Requests per minute	Response times conditional win	Completed fills per minute of request exposure
Dealer buy, $z = 1$	1	0.8×0.5	0.400000
Dealer buy, $z = 2$	1/3	0.5×0.25	0.041667
Dealer sell, $z = 1$	2/3	0.6×0.4	0.160000
Dealer sell, $z = 2$	2/3	0.4×0.2	0.053333

The total completed-fill cohort rate is 0.655 per minute. Multiplying each mark's contribution by size gives gross executed quantity 0.750 units per minute of request exposure; also applying the dealer sign gives 0.216667 units per minute of expected inventory increase. The incoming requested gross quantity rate is $110/30 = 11/3$. These quantities differ because the response and win probabilities select both side and size.

The third experiment generates 12,000 invitations, ten milliseconds apart, with 4,000 in training, 2,000 in validation, 2,000 in calibration, and 4,000 in test. Its feature vector uses normalized edge $(\delta - 0.15)/0.05$, a size-two indicator, dealer side, a public state indicator, elapsed quote-submission latency divided by three milliseconds, and a side-state interaction. These normalizing constants are fixed before fitting. The generating mechanism is specified in the companion notebook. Hidden potential outcomes and true probabilities are retained only as synthetic oracle fields; they never enter fitting or prediction.

The logging policy has a structural no-quote region: size two, public state one, and the six-millisecond latency choice. A responder-only evaluation cannot validate quote probabilities there. Logged edges otherwise span the declared range $[0.05, 0.25]$ in synthetic monetary price units. Outputs beyond that range also require extrapolation. These limits remain present even if a full-feature logistic model matches the generating form.

The training candidates combine two feature sets, edge-only and full-state, with penalties $\kappa \in \{0.001, 0.01, 0.1\}$. Validation chooses once by mean log loss. The chosen coefficients then remain fixed while the calibration block fits equation (3.22). The calibration and raw predictions are both reported on the same supported test responses, with no second model choice based on that test.

Label cutoffs are 40,000, 60,000, and 80,000 milliseconds for the first three blocks, and 120,500 milliseconds for test scoring. The last test invitation arrives at 119,990 milliseconds. Quote submission occurs one, three, or six milliseconds after each invitation, so the request and quote-decision block memberships agree in this construction. The final reporting window makes all quoted test outcomes observable.

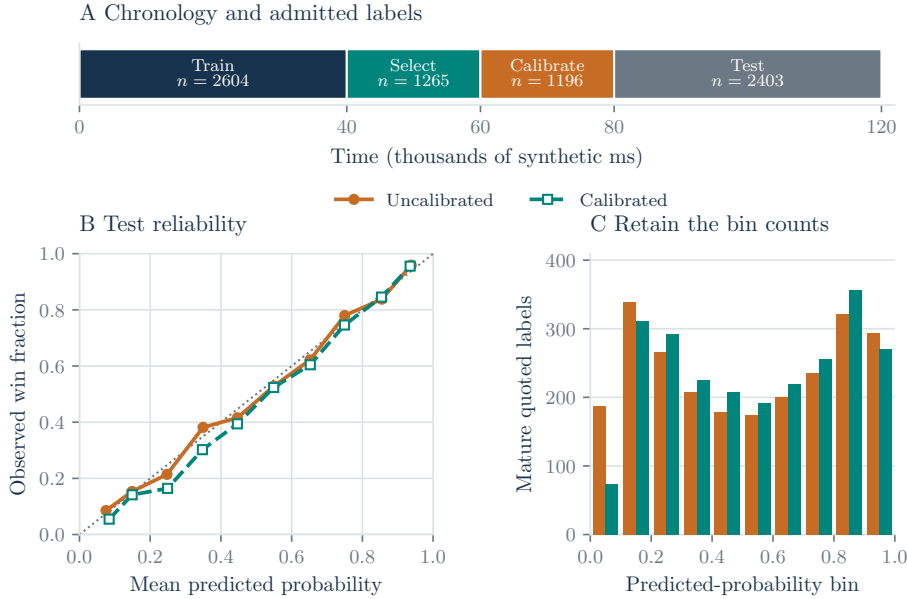
Chronological block	Valid quoted responses	Mature labels at cutoff	Quoted labels still unavailable
Training	2,614	2,604	10
Validation	1,276	1,265	11
Calibration	1,205	1,196	9
Test	2,403	2,403	0

Validation selects the full-state candidate with $\kappa = 0.001$. The fitted calibration parameters are $\alpha = 0.143031$ and $\beta = 0.908739$. The slope happens to be positive; no positivity constraint was imposed. Scores use probability clipping at 10^{-12} and $1 - 10^{-12}$ for numerical reporting, while likelihood fitting uses stable log-sigmoid arithmetic. The test predictions already lie strictly inside those clipping limits.

Frozen test predictor	Mean log loss	Brier score
Selected raw model	0.485516	0.158973
Selected model with calibration	0.489009	0.160312

There are 1,230 wins among the 2,403 test responses, an observed fraction of 0.511860. Mean predicted probabilities are 0.514407 before calibration and 0.538536 after it. The calibration mapping worsens both test scores in this realization. This result is retained, together with the ten equal-width reliability bins and their counts. No parameter, seed, or candidate is changed to reverse that finding.

The companion notebook reproduces these results from the recorded experiment inputs, split boundaries, frozen model parameters, and eligibility rules. Its figures use the same calculations as the tables, and its checks retain the cutoff and support restrictions.



Test: 2403 mature quoted labels, known by 120500 ms. The model and calibration are frozen before test evaluation.
 Test log loss: 0.485516 $\rightarrow$ 0.489009; Brier: 0.158973 $\rightarrow$ 0.160312 (higher is worse).
 Calibration worsens this synthetic test. Fixed bins are descriptive; they are not confidence intervals.
 Only observed quoted populations are evaluated. A structural no-quote region remains unsupported.

Figure 3.3: Training, model selection, calibration, and testing occur in chronological blocks. Test reliability compares average predictions with observed fractions and retains bin counts; the synthetic logging policy limits the supported population.

The test set is reserved for evaluation of the declared procedure. An improvement in one finite synthetic test is neither a guaranteed benefit from calibration nor a real-market performance estimate. Repeatedly altering the procedure after looking at its test score would make that score part of model development. A new claim would then need a fresh evaluation sample.

3.11 Resting orders require a different event model

An RFQ response is evaluated under a request-specific contract. A resting order sits in a priority system while other orders arrive, trade, amend, and cancel. Its execution probability depends on quantity ahead, the allocation rule, price changes, the dealer's cancellation policy, and the chosen time horizon. Aggregate displayed depth need not identify the dealer's position in that queue.

As a deliberately reduced example, suppose a fully executable resting order faces an independent execution clock of rate $a > 0$ and a terminal cancellation clock of rate $c \geq 0$. Both rates are constant, there are no partial fills, and the first event terminates the order. The probability of execution by horizon h is

$$\Pr(T_F \leq h, T_F < T_C) = \int_0^h a e^{-(a+c)t} dt = \frac{a}{a+c} (1 - e^{-(a+c)h}). \quad (3.25)$$

The density integrates the execution hazard against survival of both clocks. Its eventual value is $a/(a+c)$, whereas the probability of any terminal event by h is $1 - e^{-(a+c)h}$. Treating a cancellation as an ordinary still-live observation, or using the latter expression as a fill probability, changes the target. This toy is a competing-event illustration, not a reconstructed queue model or an estimate of a venue's hazards.

With $a = 2$ per minute, $c = 1$ per minute, and $h = 0.5$ minutes, the fill probability is $\frac{2}{3}(1 - e^{-1.5}) \approx 0.517913$. Ignoring cancellation instead gives $1 - e^{-1} \approx 0.632121$. The difference follows from the declared competing clocks, not from a fitted empirical cancellation cost.

For a real partially executable order, let $V_h \in [0, z]$ be quantity executed by horizon h . Three distinct targets are

$$\Pr(V_h > 0), \quad \Pr(V_h = z), \quad \mathbb{E}[V_h]/z. \quad (3.26)$$

The first measures any fill, the second completion, and the third expected participation in the requested quantity. They generally differ. A single binary RFQ label cannot encode all three. A realistic queue experiment must explicitly update priority after the relevant events and represent the actual venue's allocation rules.

For example, requested quantity two and executed quantities zero, one, and two with probabilities 0.4, 0.4, and 0.2 give any-fill probability 0.6, completion probability 0.2, and expected filled fraction 0.4. Reporting the first as a completion rate would overstate what the model actually predicts.

Gould and coauthors distinguish price-time, pro-rata, and other priority mechanisms in their survey. The implication for this book is to state the matching rule and observable state before choosing a fill model. A FIFO example in one chapter does not turn FIFO into a universal market rule. Similarly, the survey's discussions of clustering and queue models motivate diagnostic questions; they do not provide fitted hazards for this dealer's current trading population.

3.12 The probability contract passed to a quoting model

The output of this chapter is a documented probability contract. It names the request population and exposure clock, the response event, the execution target and horizon, the available features, the treatment of terminal cancellations and unresolved labels, and the quote support on which evidence exists. It also names the fitting origin, calibration procedure, and evaluation population.

A dynamic model can use request arrivals and conditional outcomes as separate transitions, as in Chapters 6 through 9. An execution-intensity formulation can compress those stages under its stated timing approximation. Both representations can be coherent. Their inputs must use compatible units, marks, and populations; multiplying a conditional probability into a rate that already contains it counts the same selection twice.

Chapter 4 takes the next decision step: choosing a quote for a request that has already arrived. Its win probability is conditional on sending a valid quote, and its trading value uses the execution conditioning developed in Chapter 2. Arrival rates and historical response coverage are therefore not extra multipliers in that single-action objective. An optimizer can be numerically exact and still act on the wrong economic object if the probability's denominator, clock, or support has been lost.

Exercises

1. Derive equation (3.4) from the Poisson log likelihood. Explain why zero observed requests with positive exposure differs from zero exposure, and reproduce the notebook's pooled and state-specific rates.
2. Reproduce the marked request, response, and completed-fill rates in the first experiment. Compute gross and signed quantity rates, then show the error caused by multiplying the size probabilities twice.
3. Reproduce the one-minute deterministic-delay example. Explain when request-cohort completed-fill rates can equal physical execution rates and why that equality fails at the start of the example.
4. Reconstruct every denominator in the small request log. Compute the known-outcome conditional fraction and the sharp bounds for the eventual responder and invitation cohorts. State why these are not confidence intervals.
5. Apply each fitting cutoff to label availability, and list the recent responses excluded at a boundary. Explain the bias that could arise if wins were reported faster than losses, even after rows were sorted chronologically.

6. Differentiate the logistic model with respect to monetary edge, retaining its scale. Explain why the derivative does not identify an intervention effect, and identify the notebook's invitation stratum with no response support.
7. Reproduce the frozen test log loss, Brier score, and nonempty reliability bins. Verify the Brier identity in [equation \(3.21\)](#), and explain why calibrating on the test outcomes would invalidate this evaluation.
8. Derive the competing-event probability in [equation \(3.25\)](#), check the notebook's numerical example, and construct a partial-fill distribution for which the three targets in [equation \(3.26\)](#) differ.

Source Notes

Kakuma, historical JGB inquiry data. The saved English translation of the September 2012 study describes its sampling conditions on PDF/printed page 4, Section 3; page 5, footnote 8 records the exclusions. The sample covers inquiries from 2003 through 2011 that invited exactly five dealers and received all five rates, including canceled inquiries, with 117,360 retained observations. We use those restrictions to motivate explicit populations and denominators. The source is not used to estimate an unconditional response probability or to calibrate the synthetic logistic experiment. See the [Chapter 3 Kakuma notes](#).

Bergault and Guéant, size-dependent execution intensities. The saved arXiv version 5, revised 21 September 2022, gives marked execution kernels in Section 2.1 on PDF/printed page 3. Its Remark 1 interprets the scalar execution-intensity function as a constant RFQ arrival rate multiplied by the conditional probability of transaction; the size measure is a probability measure. This is the direct theoretical connection to equation (3.7), under compatible response and instantaneous-execution conventions. Our separate response stage, exposure estimator, delayed-cohort calculation, missing-label analysis, and statistical experiment are independent constructions. The source's execution kernel must not be relabeled as an unthinned request rate. See the [Chapter 3 Bergault-Guéant notes](#).

Gould and coauthors, limit order books. The saved arXiv version 4, revised 1 April 2013, distinguishes order queues, cancellation, and sampling clocks on PDF/printed pages 3-5 and priority rules on page 7. We use those distinctions to explain why resting-order fill targets require their own event state and allocation rule. The competing-event and partial-quantity examples above are original teaching examples, rather than reproductions of the survey's empirical results or claims about a current venue. See the [Chapter 3 Gould notes](#).

PART II

Static quoting and risk

Connect conditional execution to quote economics, inventory geometry, and forecasts of physical variance.

- 4 Optimizing a Single RFQ
- 5 Inventory Risk and Factor Representation

Chapter 4

Optimizing a Single RFQ

A client sends a request. The reference price is available, and the quantity the dealer can quote is already fixed. The remaining decision is how far from that reference price to quote. A wider price offset increases the revenue from a successful trade, but it may reduce the probability that the client selects the dealer. This chapter starts by putting both effects into the same objective function.

Chapter 2 identified value conditional on execution; Chapter 3 separated that event's probability from request arrivals and the decision to respond. We now combine them to maximize expected incremental trade P&L for one request. Transaction costs, conditional reference changes, and the option not to quote enter the same comparison. Chapter 5 adds an explicit inventory-risk penalty, and Chapter 6 adds the value of future opportunities. By the end of this chapter, the reader should be able to choose a quote for a given win-probability curve and explain which assumptions support that choice.

4.1 The Decision and Its Inputs

Consider an RFQ in which the client specifies the trade direction and quantity, and the dealer responds with a single price. There are no quote revisions, partial fills, or repeated negotiations. If the client trades at that quote, the entire requested quantity executes; otherwise, nothing executes. These restrictions make it possible to trace the connection between the chosen price and the resulting profit and loss.

Let S_0 be the reference price and $z > 0$ the requested quantity. The reference price is available at the time of the decision and enters this chapter as an input. Define the price offset δ as the distance from the reference price in the dealer's favor. A positive δ means that the dealer buys below the reference price or sells above it. This convention expresses the potential trading margin consistently on both sides.

Set $\eta = +1$ when the dealer buys and $\eta = -1$ when the dealer sells. The quoted execution price Q and the inventory change Δq , conditional on execution, are

$$Q = S_0 - \eta\delta, \quad \Delta q = \eta z. \quad (4.1)$$

On a client sell request, the dealer buys: $\eta = +1$ and $Q = S_0 - \delta$. On a client buy request, the dealer sells: $\eta = -1$ and $Q = S_0 + \delta$. Writing "buy" without specifying

whose trade direction is meant can reverse these signs. Throughout this book, inventory changes and profit and loss are described from the dealer's perspective.

Collect the information available at the decision time in x . It may include the instrument, direction, quantity, client attributes, recent order flow, and quote freshness. It must exclude information that becomes available only after the dealer responds. Since we are studying one decision in a fixed context, we suppress x where doing so simplifies the notation. Here S_0 is the observed decision reference M_d^{obs} of Chapters 2 and 3, and Q is the quoted price previously written as p_i . We reserve $p(\delta \mid x)$ for execution probability and retain q for inventory.

Symbol	Meaning	Unit in the numerical example
δ	Price offset in the dealer's favor	JPY thousands per reference trade unit
z	Requested quantity	Reference trade units
$p(\delta \mid x)$	Probability that the dealer executes this request	Dimensionless
$C(z)$	Total transaction cost incurred if the trade executes	JPY thousands
$c(z) = C(z)/z$	Transaction cost per unit	JPY thousands per reference trade unit
K	Cost incurred on responding, whether the dealer wins or loses	JPY thousands per request

The "reference trade unit" is a teaching convention for expressing quantity after monetary conversion. It does not represent a particular instrument's face value or market quotation convention. Stating the units explicitly prevents probabilities, price offsets, and monetary profit and loss from being added together incorrectly.

4.2 Choosing the Denominator of Win Probability

Let F denote the event that this request executes at the dealer's quote. As in Chapter 3, responding here means sending a valid price quote; an explicit decline belongs to non-response in this decision problem. Conditional on that action, define the win probability as

$$p(\delta \mid x) = \Pr(F \mid \delta, x, \text{respond}). \quad (4.2)$$

The denominator must include requests to which the dealer responded but on which the client ultimately traded with no one. Suppose the dealer sends valid quotes to 100 requests. The client trades with some dealer on 80 of them, and our dealer wins 40. The ratio $40/80 = 50\%$ is a win probability conditional on the client trading. The probability to multiply by the profit from a trade when evaluating one request is instead $40/100 = 40\%$.

The distinction can also be expressed as a product of probabilities. Let T be the event that the client trades with any counterparty on this request. Since F is a subset of T ,

$$p(\delta \mid x) = \Pr(T \mid \delta, x, \text{respond}) \times \Pr(F \mid T, \delta, x, \text{respond}). \quad (4.3)$$

This decomposition does not require independence. Both the client's decision to trade and the choice of dealer may depend on the quoted price and the observed state. In his discussion of market-making intensities, Guéant notes that a hit ratio calculated only over RFQs that lead to a trade is insufficient: RFQ arrival and the client's willingness to trade must also be considered. [G, printed p. 231, PDF p. 256, footnote 25]

An RFQ arrival intensity λ has units of requests per unit of time; the win probability p for one request does not. In the instantaneous-execution approximation of Chapter 3, homogeneous Poisson requests with independently marked outcomes across requests give execution intensity $\lambda p p$, where ρ is valid-quote coverage. If every request is quoted, $\rho = 1$ and this becomes λp . The corresponding probability of at least one execution during an interval Δt is $1 - \exp(-\lambda p p \Delta t)$. Delayed executions require the clock treatment in Section 3.3. None of these time-interval probabilities is the probability of winning the individual request now being considered.

Cartea and coauthors distinguish the arrival rate of market orders from the probability that an arriving order reaches the dealer's quoted price and executes. [C, Section 10.1, printed p. 246, PDF p. 263] Here the request has already arrived. There is therefore no need to multiply the objective for this request by the RFQ arrival rate again.

4.3 Constructing Expected Profit and Loss

Start by assuming that the reference price is also the valuation price, with no subsequent price change. If the trade executes, the dealer earns the quantity z times the price offset δ , giving $z\delta$, and pays the transaction cost $C(z)$. Neither arises if the dealer loses the request. By definition, the response cost K is incurred whenever the dealer responds, regardless of the outcome. It is an incremental, avoidable cost of responding; it excludes fixed costs already incurred.

The profit and loss Π from responding, and its expectation J , are therefore

$$\Pi = \mathbf{1}_F \{z\delta - C(z)\} - K, \quad (4.4)$$

$$\begin{aligned} J(\delta \mid x) &= p(\delta \mid x) \{z\delta - C(z)\} - K \\ &= z p(\delta \mid x) \{\delta - c(z)\} - K. \end{aligned} \quad (4.5)$$

This is the incremental expected profit and loss from responding to this request, rather than the profit and loss of the dealer's entire existing inventory. Including the

mark-to-market change of existing inventory would mix components affected by this decision with components that are not. When inventory effects are introduced in the next chapter, they will likewise enter as the difference between the values before and after the trade.

The equation should make the conditions under which each cost arises visible. A cost paid only on execution contributes pC in expectation; a cost paid whenever the dealer responds contributes K . Putting K inside the execution-dependent brackets would remove the response cost on lost requests. Conversely, deducting an execution cost regardless of whether a trade occurs would charge a fictitious execution expense to a lost request.

4.4 The Gain from a Wider Quote and the Opportunity It Sacrifices

Hold quantity fixed and assume that the unit cost c does not depend on the price offset. If p is differentiable, the derivative of expected profit and loss with respect to the offset is

$$J'(\delta) = z [p(\delta) + (\delta - c)p'(\delta)]. \quad (4.6)$$

The first term is the gain from earning an additional price increment on requests that execute. The second captures the change in profit opportunities caused by a change in win probability. If win probability falls as the offset widens, then $p' < 0$, so this second term is negative for a quote with a positive net trading margin.

At an interior solution with $p > 0$, rearranging $J' = 0$ gives

$$(\delta - c) \left(-\frac{p'(\delta)}{p(\delta)} \right) = 1. \quad (4.7)$$

The one on the right corresponds to the direct increase in profit per unit from increasing the offset by one unit. The left side multiplies the profit per executed unit, $\delta - c$, by the relative rate at which win probability declines. Where win probability drops sharply as the quote widens, maintaining a large margin is difficult. Where win probability declines more gradually, a wider offset becomes a candidate.

Writing $h(\delta) = -p'(\delta)/p(\delta)$, the interior condition becomes $\delta - c = 1/h(\delta)$. The quantity h measures the local relative decline of the win-probability curve and has units of inverse price offset. Its form resembles a distribution's hazard rate, but here we first define it directly from the win-probability curve.

This condition identifies candidates at differentiable interior points. If quotes have upper or lower bounds, or must lie on a price grid, a boundary may be better. A general estimated win-probability curve may also produce several local extrema. The final decision must compare feasible quotes and non-response on the same basis.

4.5 Finding the Optimal Quote with a Logistic Win Probability

For a concrete calculation, assume a logistic win-probability curve:

$$p(\delta) = \frac{1}{1 + \exp((\delta - a)/b)}, \quad b > 0. \quad (4.8)$$

The parameter a is the offset at which win probability is 50%. The parameter b controls the width of the transition: a smaller b makes win probability change more sharply near a . This is a teaching assumption for this chapter, not a claim that the logistic function always fits actual RFQs. With other features fixed, a negative linear edge coefficient in [Chapter 3](#)'s logistic model gives a curve of this form. Estimation still requires that chapter's label, support, and validation rules. The parameters used below are separate synthetic inputs, not the fitted coefficients of its experiment.

Differentiation gives $p' = -p(1 - p)/b$, so the optimality condition is

$$\frac{(\delta - c)(1 - p(\delta))}{b} = 1. \quad (4.9)$$

We should also verify why solving this equation gives a maximum. For $\delta \leq c$, profit is non-positive when $K = 0$. For $\delta > c$, let $u(\delta)$ denote the left side of (4.9). Then

$$u'(\delta) = \frac{1 - p}{b} + \frac{(\delta - c)p(1 - p)}{b^2} > 0. \quad (4.10)$$

The function u is zero at $\delta = c$ and increases without bound as δ increases. There is consequently exactly one point at which $u = 1$. Before it, $J' > 0$; after it, $J' < 0$. The model without quote bounds therefore has a unique maximizer. This conclusion is specific to the assumed curve and is not being extended to every smooth win-probability function.

[Equation \(4.9\)](#) can be solved reliably by bisection. For an analytical check, introduce the function $W(y)$ defined by $w \exp(w) = y$. The solution can then be written as

$$\delta^* = c + b \left[1 + W \left(\exp \left(\frac{a - c}{b} - 1 \right) \right) \right]. \quad (4.11)$$

To obtain it, set $w = (\delta - c)/b - 1$ and rearrange [\(4.9\)](#) as $w \exp(w) = \exp((a - c)/b - 1)$. The calculations in this chapter use bisection and price-grid search, so they do not depend on a special-function implementation. The analytical expression provides a separate check on the computed solution.

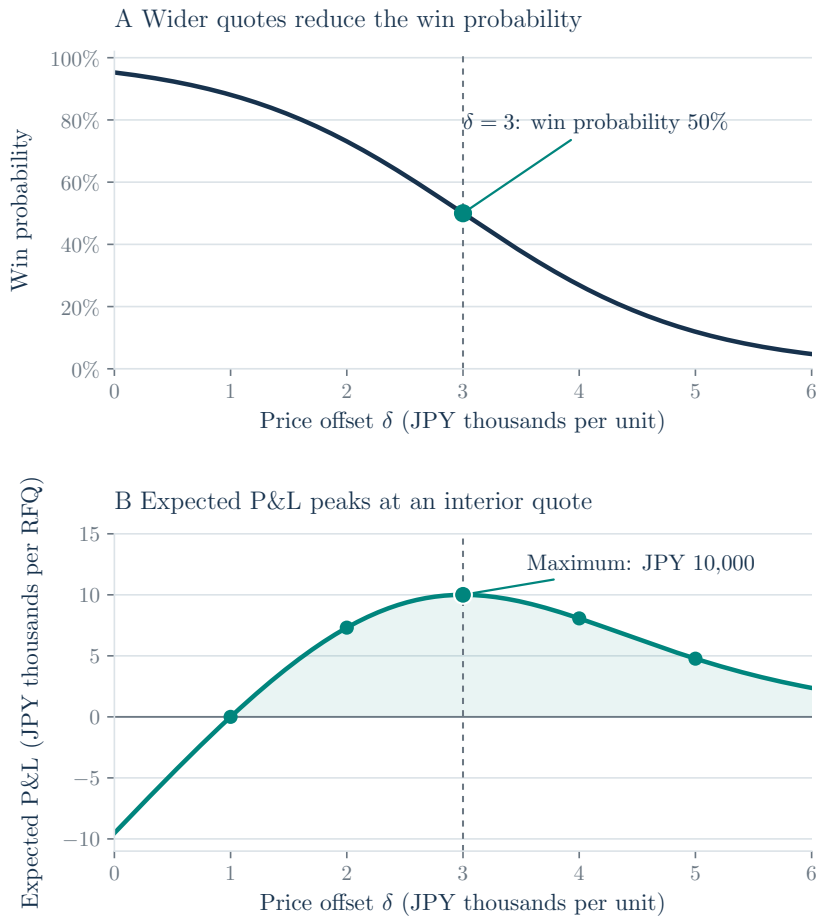
4.6 Reading a Single Request through Numbers

For a hypothetical request, set $a = 3$, $b = 1$, $c = 1$, $z = 10$, and $K = 0$. Price offsets and unit costs are measured in JPY thousands per reference trade unit. Quantity is 10 units, and expected profit and loss is measured in JPY thousands. Feasible offsets run from 0 to 6 in increments of 0.25. These values are not estimates from a live market.

Substituting $\delta = 3$ into (4.9) gives $p = 0.5$ and $(3 - 1)(1 - 0.5) = 1$. Both the continuous solution and the grid solution are therefore $\delta = 3$. If the trade executes, it earns JPY 20 thousand after transaction costs. That outcome occurs with probability 50%, giving expected profit of JPY 10 thousand per request.

Price offset δ	Win probability	Profit after costs, if executed	Expected profit per request
1	88.08%	JPY 0 thousand	JPY 0.000 thousand
2	73.11%	JPY 10 thousand	JPY 7.311 thousand
3	50.00%	JPY 20 thousand	JPY 10.000 thousand
4	26.89%	JPY 30 thousand	JPY 8.068 thousand
5	11.92%	JPY 40 thousand	JPY 4.768 thousand

Choosing only for a high win probability can produce a quote that earns nothing even when it executes. At the other extreme, a large profit conditional on execution can still produce little expected profit per request if the quote is rarely selected. The table is a minimal example of why these two quantities should be examined separately.



Synthetic inputs: $a = 3$, $b = 1$, $c = 1$, $z = 10$, $K = A = 0$.
Dots match the quotes in the chapter table. No market data are used.

Figure 4.1: Win probability and expected profit for a hypothetical RFQ

The upper panel of Figure 4.1 shows win probability declining with the offset; the lower panel shows expected profit after costs. Win probability is not maximized at the optimal quote. What is being compared is the product of profit conditional on execution and the probability of that event.

4.7 Changing Costs and the Competitive Environment

Now change only the unit cost c . In general, the optimal offset does not move by simply adding the change in cost, because raising the offset also changes win probability. In the logistic model, the interior continuous solution satisfies

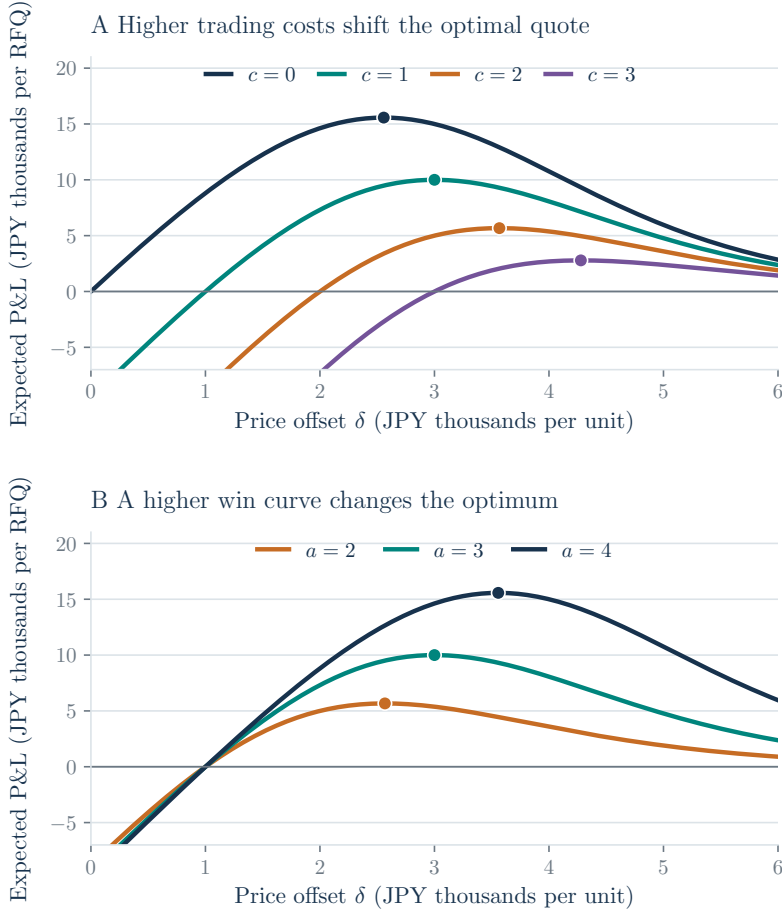
$$\frac{\partial \delta^*}{\partial c} = 1 - p(\delta^*), \quad \frac{\partial \delta^*}{\partial a} = p(\delta^*). \quad (4.12)$$

To derive these results, differentiate $u(\delta, c, a) = 1$ implicitly. At the optimum, the partial derivatives of u with respect to δ , c , and a are $1/b$, $-(1-p)/b$, and $-p/b$, respectively, which gives (4.12). A small increase in unit cost is therefore not necessarily passed through in full to the quoted offset. The opportunity to win the client's trade remains part of the comparison.

In the base example, win probability at the optimum is 50%, so the local rate at which the optimal offset increases with c is also 0.5. An increase in a , meanwhile, shifts the curve toward a higher win probability at the same offset. These are local sensitivities of the interior solution. Large changes, price increments, and binding quote bounds require a fresh numerical calculation.

The role of quantity z also needs to be separated from other effects. In the simple model used here, holding p and the unit cost c fixed means that z only scales expected profit and loss. It does not change the optimal offset conditional on responding. The same optimal quote for different sizes is a consequence of these assumptions.

To obtain size-dependent quotes, specify which of $p(\delta, z | x)$, $C(z)$, subsequent price changes, or the value of inventory changes depends on size. Merely multiplying the objective by quantity does not create a size-dependent quote shape. With a fixed response cost K , however, quantity can affect whether the dealer responds at all.



Synthetic inputs; dots mark continuous optima. $b = 1$, $z = 10$, $K = A = 0$.
 Panel A holds $a = 3$ fixed; panel B holds $c = 1$ fixed.

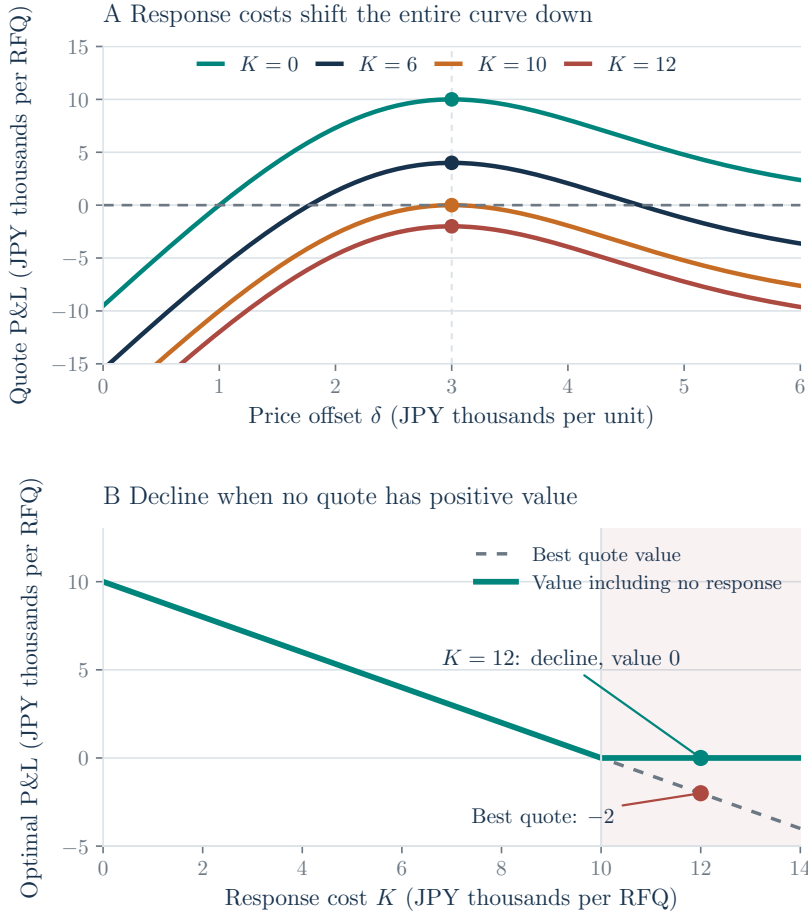
Figure 4.2: Sensitivity to unit cost and the location of the win-probability curve

4.8 Non-Response and the Price Grid

Set the incremental profit and loss from not responding to this request to zero. If $\mathcal{D}$ is the set of feasible price offsets, the optimal value is

$$J^* = \max \left\{ 0, \max_{\delta \in \mathcal{D}} J(\delta) \right\}. \quad (4.13)$$

In the base example, maximum expected profit before the response cost is JPY 10 thousand. If K is JPY 12 thousand, even the best quote has expected profit of JPY -2 thousand, below the zero value of non-response. If K is JPY 10 thousand, the two choices have equal value. The implementation in this chapter breaks that tie in favor of non-response. A tie-breaking convention is part of the policy.



Synthetic inputs: $a = 3$, $b = 1$, $c = 1$, $z = 10$, $A = 0$.
 K is incurred for every response. At a tie ($K = 10$), decline.

Figure 4.3: Response cost and the decision to respond

If K is independent of the offset, it does not change the optimal quote conditional on responding. It changes the boundary between responding and not responding. Combining these two decisions too early obscures which costs affect the price and which affect participation.

In an implementation, evaluate J on the actual set of feasible prices rather than simply rounding a continuous solution. If the candidate set is small, exhaustive search is sufficient. For the single-peaked logistic objective in this chapter, grid points adjacent to the continuous optimum are natural candidates, but a general estimated curve offers no such guarantee. Record the boundary cases, maximum expected profit, and comparison with non-response.

Restricting offsets to nonnegative values is itself a teaching constraint. Once later chapters introduce the value of reducing inventory, trading at a price unfavorable

relative to the reference price may become a candidate. Here that inventory value is set to zero so that the economics of one request can be examined first.

4.9 Conditioning on Reference-Price Changes

So far, the reference used to mark the incremental position has been held fixed. Let S_H be the supplied benchmark at a valuation time H at or after execution. For a fill at time t and an elapsed horizon h , the convention $H = t + h$ and $S_H = B(t + h)$ recovers Chapter 2's endpoint; the interval from S_0 still starts at the quote decision. Isolating this trade, and adding the cash change at execution to the value of the resulting position at H , gives

$$\begin{aligned} & -\eta zQ + \eta zS_H - C(z) \\ & = z\delta + \eta z(S_H - S_0) - C(z). \end{aligned} \tag{4.14}$$

If the dealer buys and the price subsequently falls, $\eta = +1$ makes the price change an additional loss. If the dealer sells and the price subsequently rises, $\eta = -1$ again produces a loss. Expressing both directions in one equation also provides a useful implementation check on the signs.

When taking expectations, retain the condition that the trade executed. An unconditional mean price change of zero does not imply a zero mean conditional on the dealer having been selected. Define the conditional adverse-selection cost A as

$$A(\delta, x) = -\eta \mathbb{E}[S_H - S_0 \mid F, \delta, x, \text{respond}]. \tag{4.15}$$

As in equation (2.5), this conditional expectation requires positive execution probability and a finite conditional mean. At zero execution probability, define the execution contribution in (4.16) directly as zero, so $J = -K$; there is no conditional A to estimate on that null event. A positive A denotes an expected price change unfavorable to the dealer; a negative A denotes a favorable one. Calling it a "cost" does not impose nonnegativity. Like win probability, it may depend on the chosen offset, direction, client, size, and valuation horizon.

The quantity A is an accounting summary of price changes conditional on execution. It does not identify only the causal loss from asymmetric information. Predictable drift or price impact may also contribute. Separating those causes requires an empirical identification strategy in addition to a definition of profit and loss.

$$J(\delta) = z p(\delta) \{\delta - c - A(\delta)\} - K. \tag{4.16}$$

If A is constant, the calculation is equivalent to replacing c with $c + A$. For example, adding $A = 0.75$ to the base case gives the same optimization as an increase in unit cost of that amount. This example treats the conditional price change as a known input; it does not guarantee the accuracy of a forecast used to supply it.

If A varies with the offset, its derivative cannot be ignored. The interior condition becomes

$$1 - A'(\delta) = \left(-\frac{p'(\delta)}{p(\delta)} \right) \{ \delta - c - A(\delta) \}. \quad (4.17)$$

A wider offset may change not only win probability but also the mix of counterparties on winning requests and the conditional distribution of price changes. The term A' captures that effect. Adding the same historical average markout to every request's profit and loss may fail to represent this selection effect.

Cartea and coauthors show how expected price changes affect quotes in an adverse-selection model that links market orders to price jumps. [C, Section 10.4.1, printed pp. 262-264, PDF pp. 279-281] The definition of A here is a conditional profit-and-loss definition and is not restricted to that particular jump model.

4.10 Avoiding Double Counting in Markout and Costs

The gross markout from execution price Q to benchmark S_H is $\eta(S_H - Q) = \delta + \eta(S_H - S_0)$ per unit. It already contains the quoted offset. Chapter 2's net markout also deducts the allocated execution fee; it matches (4.14) divided by z when that fee equals $C(z)$. A transaction-cost definition containing additional items needs a corresponding cost reconciliation. Adding z times either complete per-unit markout to $z\delta$ would count the quoted edge twice, and subtracting an already included fee would count that cost twice.

The price change used in A is measured from the decision-time reference price S_0 to the valuation time H . If a data field instead measures markout from the execution price, check which components it includes and reconcile them with the terms of the objective. Two columns with the same name need not be interchangeable if their markout definitions differ.

The same caution applies to the cost C . If C already includes the full cost of an actual hedge measured against the decision-time price, do not deduct the price-change component again through A . Fix the valuation time, hedging policy, and cost benchmark, and verify that the decomposition reconciles to a single realized profit-and-loss amount.

In this chapter, the incremental position after execution is valued at H , while C is a separately defined transaction cost. Chapter 9 introduces the decision of when to hedge; Chapter 11 reconciles benchmark changes and actual execution costs against one account ledger. Those extensions preserve the requirement to book each price change and cash cost once.

4.11 Connecting the Source Equations to This Chapter

In Section 10.2.1, Cartea and coauthors maximize a term of the form $\delta \exp(-\kappa\delta)$ and obtain the optimal offset $1/\kappa$. [C, printed pp. 253-254, PDF pp. 270-271, equations (10.15)-(10.16)] Equation (4.7) gives the same result when costs are zero and $p(\delta) = \exp(-\kappa\delta)$, since $-p'/p = \kappa$. If this exponential function is used as a probability, its domain must ensure that it does not exceed one; $\delta \geq 0$ is one such restriction.

This correspondence establishes the local structure of optimizing the product of a trading margin and the likelihood of execution. It does not make the source's arrival model or dynamic control problem identical to a single client RFQ. Here we evaluate one outcome conditional on the request having arrived.

Guéant's equation (11.26) contains, within a dynamic problem with CARA utility, maximization terms that combine execution intensity with a nonlinear function of the trading margin and the change in inventory value. [G, printed p. 234, PDF p. 259] Simply replacing such a term with $p(\delta)(\delta - c)$ would discard the treatment of utility, time, and inventory. The expected-profit objective in this chapter is instead derived independently from the single-decision accounting in (4.4).

This distinction also determines how the book proceeds. Chapter 5 adds the change in a specified inventory-risk penalty, using covariance inputs over a declared horizon. Estimating that covariance and choosing risk aversion are separate tasks. Chapter 6 then introduces future request arrivals and continuation value. We will connect models only after aligning their decision horizons and objectives, rather than joining equations merely because their notation looks similar.

4.12 Checks before Using the Policy

First, the observed response policy affects the estimated win-probability curve. In regions where the dealer has rarely quoted, p may be estimated poorly. If historical quote choices vary by client or market state, selection also affects the apparent relationship between price and win probability. This chapter solves the optimization problem with p as an input. The validity of the estimate supplying that input must be assessed separately.

Second, maximizing expected profit does not determine the outcome of an individual request. At the optimal quote in the base example, the dealer loses the request half the time. Once price changes are included, profit and loss can also vary conditional on execution. Recording expected profit, win probability, profit conditional on execution, and price risk separately prepares the ground for later policy comparisons.

Third, the present model assigns zero value to any effect of today's response on future RFQs or client relationships. The zero value assigned to non-response rests on the same assumption. If lost future requests matter, their effect belongs in continuation value; it should not be hidden inside an arbitrary fixed cost.

The accompanying notebook checks the base solution, sensitivities to costs and the location of the win-probability curve, the price grid, the non-response boundary, and the signs for both trade directions. It compares the continuous solution with a search over a wide price grid, checking that a point satisfying the derivative condition has not been mistaken for the desired maximum. All inputs and numerical results are saved, and the chapter's tables and figures are generated from the same example.

Exercises

1. If requests on which the client trades with no one are removed from the win-probability denominator, which quantity in (4.5) changes? Compare expected profit using the example of 100 responses, 80 requests that lead to a trade, and 40 wins.
2. Increase c from 1 to 2 in the base example. Find the continuous solution and the solution on a grid with increments of 0.25. Use (4.12) to explain why the optimal offset does not increase by exactly 1.
3. If only z changes from 10 to 5, what happens to the optimal offset and maximum expected profit when $K = 0$? What is the response decision when $K = 6$?
4. Suppose $S_H - S_0 = -0.5$ after a dealer purchase and $S_H - S_0 = +0.5$ after a dealer sale. Use η to verify that both produce the same loss from an unfavorable price change.
5. Consider an implementation that adds markout measured from the execution price to the quoted offset δ . Use (4.14) to identify the component counted twice.

Source Notes

[G] Olivier Guéant, *The Financial Mathematics of Market Liquidity: From Optimal Execution to Market Making*, CRC Press, 2016. The supplied PDF has 302 pages. The passages used are the model setup in Section 11.2.1 (printed pp. 222-223; PDF pp. 247-248), the discussion of RFQs and intensities in Section 11.3.1, including footnote 25 (printed p. 231; PDF p. 256), and equation (11.26) in Section 11.3.2 (printed p. 234; PDF p. 259). PDF edition ISBN: 9781498725484.

[C] Álvaro Cartea, Sebastian Jaimungal, and José Penalva, *Algorithmic and High-Frequency Trading*, Cambridge University Press, 2015. The supplied PDF has 360 pages. The passages used are Section 10.1 (printed p. 246; PDF p. 263), Section 10.2.1 and equations (10.15)-(10.16) (printed pp. 253-254; PDF pp. 270-271), and Section 10.4.1 and equations (10.26)-(10.27) (printed pp. 262-264; PDF pp. 279-281). ISBN: 9781107091146.

The logistic win-probability curve, numerical inputs, and comparisons involving response costs are teaching assumptions introduced for this book. They are kept distinct from the markets, utility functions, and arrival processes in the sources. The accompanying reading notes record which source passages were checked and how they connect to the derivations in this chapter.

Chapter 5

Inventory Risk and Factor Representation

Consider the same request as in Chapter 4. Its size, reference price, client information, and win-probability curve have not changed. The dealer's portfolio has. A purchase that adds to an existing position and a purchase that offsets an existing position need not have the same value. To distinguish them, we need a measure of the risk of the whole portfolio and a way to charge the request for the change it creates.

This chapter builds that connection in stages. We map positions into factor exposures, turn their joint price uncertainty into a monetary penalty, and add the change in that penalty to the single-RFQ objective. The resulting rule has a useful interpretation as an effective transaction cost. It also exposes two common mistakes: treating a flat position in one instrument as the absence of portfolio risk, and treating cancellation of a few factors as the absence of all risk.

The model remains a static decision about one request. Future client arrivals and the choice of when to hedge will enter later. Reference prices and sensitivities are supplied inputs. The first twelve sections keep covariance inputs fixed; Sections 5.13-5.19 add a separate, chronological variance-estimation exercise before reconnecting those estimates to the inventory charge.

5.1 Positions, Trade Direction, and Supplied Sensitivities

Let $q = (q_1, \dots, q_n)^\top$ be the dealer's signed positions in n instruments. A positive component denotes a long position; a negative component denotes a short position. Each component is measured in the reference trade units used for that instrument. Let e_i be the vector with a one in component i and zeros elsewhere.

An RFQ requests a trade of size $z > 0$ in instrument i . We retain the direction convention from Chapter 4: $\eta = +1$ when the dealer buys and $\eta = -1$ when the dealer sells. Conditional on execution, the position changes by

$$d = \eta z e_i, \quad q_{\text{after}} = q + d. \quad (5.1)$$

If the request does not execute, the position remains q . A single request therefore changes one instrument position, but its economic effect can depend on every component of the portfolio.

Fix a common valuation horizon H . Represent changes in the vector of reference prices by

$$\Delta S = D \Delta f + \varepsilon. \quad (5.2)$$

Here $\Delta S = S_H - S_0$ is an n -vector of price changes, Δf is a k -vector of factor changes, and D is an n -by- k matrix of supplied sensitivities. The residual ε collects price changes not represented by these factors. Typically k is smaller than n .

The sign and units of each sensitivity are part of the input specification. If a factor is measured in basis points, a row of D describes the signed price change per trade unit for a one-basis-point factor move. We use those supplied numbers directly. A label such as DV01 is not enough to determine whether a stored sensitivity is signed or reported as a positive magnitude.

Quantity	Meaning	Units
q_i, z	Position and requested size	Reference trade units
ΔS_i	Change in price per trade unit	Currency per trade unit
D_{ik}	Supplied sensitivity to factor k	Currency per trade unit per factor unit
Δf_k	Factor change to horizon H	Factor units
γ	Coefficient converting variance into a penalty	Inverse currency

All numerical examples below measure currency in JPY thousands. Sensitivities, covariance matrices, and risk tolerances are synthetic inputs. Their role is to make the mechanism visible, rather than to estimate a market risk number.

5.2 Risk Comes from Joint Price Moves

Condition on information available when the request arrives. Write the factor and residual covariance matrices over the chosen horizon as

$$\Omega_H = \text{Cov}(\Delta f), \quad \Psi_H = \text{Cov}(\varepsilon), \quad \text{Cov}(\Delta f, \varepsilon) = 0. \quad (5.3)$$

The last equality is an assumption of this representation. Without it, cross-covariance terms must also be included. A residual covariance matrix need not be diagonal: two instruments can share risk that the chosen factors do not explain.

Under (5.3), the covariance of reference-price changes is

$$\Gamma_H = D \Omega_H D^\top + \Psi_H. \quad (5.4)$$

The matrices Ω_H and Ψ_H must be symmetric and positive semidefinite. Their construction then makes Γ_H positive semidefinite as well. This ensures that the

variance assigned to every portfolio is nonnegative. A matrix that produces a negative variance is an invalid input to this model, not an unusually attractive portfolio.

The change in value of a fixed portfolio is $q^\top \Delta S$. Define the factor exposure $r = D^\top q$. Its variance can be written either in instrument coordinates or in factor coordinates:

$$\begin{aligned} r &= D^\top q, \\ V_H(q) &= q^\top \Gamma_H q = r^\top \Omega_H r + q^\top \Psi_H q. \end{aligned} \tag{5.5}$$

The factor exposure is a sensitivity of portfolio value, not a position in an instrument. For example, its component for a basis-point factor has units of currency per basis point. Multiplying exposures by the factor covariance gives a variance in currency squared.

Correlation enters through cross terms. With two instruments,

$$V_H(q) = q_1^2 \Gamma_{11} + 2q_1 q_2 \Gamma_{12} + q_2^2 \Gamma_{22}. \tag{5.6}$$

We suppress the subscript H on matrix entries where the horizon is fixed. If the covariance is positive, opposite positions reduce the cross term; positions of the same sign increase it. An instrument-by-instrument sum of squared positions cannot reproduce this effect.

Guéant's multi-asset discussion motivates precisely this portfolio perspective and specifies correlated reference-price dynamics. [G, printed pp. 231-232, PDF pp. 256-257] The covariance matrix in those continuous-time dynamics is a covariance rate. In this chapter, Γ_H is already a covariance over a holding horizon. With the decision time set to zero, a constant covariance rate Σ gives horizon covariance $H\Sigma$ under that model. Multiplying by the horizon again would count the exposure time twice.

5.3 Turning Variance into a Monetary Penalty

Variance and profit have different units. We cannot subtract a variance directly from the expected profit in Chapter 4. Introduce a coefficient $\gamma > 0$ and define

$$R(q) = \frac{\gamma}{2} V_H(q) = \frac{\gamma}{2} q^\top \Gamma_H q. \tag{5.7}$$

Since V_H is measured in currency squared, γ has units of inverse currency and R has units of currency. The factor $1/2$ simplifies the derivative; it does not change the economic interpretation. A larger γ means that the decision rule attaches a larger penalty to the same price uncertainty.

There is a familiar justification for this quadratic form. For a fixed position q with a zero-mean Gaussian price shock, exponential utility gives the certainty equivalent

$$-\frac{1}{\gamma} \log \mathbb{E} \left[\exp \{ -\gamma q^\top \Delta S \} \right] = -\frac{\gamma}{2} q^\top \Gamma_H q = -R(q). \quad (5.8)$$

To obtain (5.8), note that $q^\top \Delta S$ is a scalar Gaussian variable with variance $V_H(q)$. Its exponential moment is $\exp \{ \gamma^2 V_H(q)/2 \}$. Taking the logarithm and multiplying by $-1/\gamma$ gives the result. A nonzero mean would add the expected portfolio price change to the certainty equivalent.

This calculation justifies the penalty for a fixed holding under those distributional assumptions. It does not make every decision rule using that penalty an exact exponential-utility solution. An RFQ has a random execution event, a random resulting position, and possibly selection in subsequent price changes. Its exact exponential-utility objective involves an expectation over those outcomes before taking a logarithm.

Our rule instead subtracts the expected change in a chosen post-trade inventory penalty. It is not the mean-variance objective of the entire random RFQ payoff either: that payoff also has uncertainty about execution itself. The model will be stated explicitly in Section 5.5.

Guéant distinguishes a CARA formulation from a formulation based on expected marked-to-market wealth minus running inventory penalties. [M, Section 2.2, p. 6; multi-asset formulation, pp. 25-26] This distinction matters here. Our fixed-horizon state penalty is a separate, static construction; it is not the time integral of the penalty in that paper.

The covariance is a description of price uncertainty. The coefficient γ is a decision preference or control parameter. Fitting one from price data does not identify the other. They should be recorded separately, with a common horizon and a clear currency convention.

5.4 Charge the Trade for the Change in Risk

The existing portfolio already carries risk before the request arrives. Charging the request for all of $R(q + d)$ would make it pay for risk that is also present when the dealer declines. The relevant quantity is the difference

$$\begin{aligned} \Delta R(q, d) &= R(q + d) - R(q) \\ &= \gamma d^\top \Gamma_H q + \frac{\gamma}{2} d^\top \Gamma_H d. \end{aligned} \quad (5.9)$$

Expand $(q + d)^\top \Gamma_H (q + d)$, use symmetry of Γ_H , and subtract $q^\top \Gamma_H q$. The first term measures how the trade interacts with the current portfolio. The second measures the risk of the trade itself.

For a single-instrument request, divide by z to obtain a charge per trade unit:

$$u_i^\eta(q, z) = \frac{\Delta R(q, \eta z e_i)}{z} = \eta \gamma (\Gamma_H q)_i + \frac{\gamma z}{2} \Gamma_{ii}. \quad (5.10)$$

This is an exact finite-trade difference for the quadratic penalty. A marginal derivative alone would retain only the first term and miss the second. The missing term matters even when the portfolio starts at zero.

In a one-instrument example with $\Gamma_{11} = 1$, $\gamma = 0.2$, and $z = 10$, the charge is $u^\eta(q, 10) = 0.2\eta q + 1$. At $q = 10$, a dealer purchase has charge 3 per unit; a dealer sale has charge -1 per unit. Selling all 10 units removes an inventory penalty of JPY 10 thousand, so its total change is $\Delta R = -10$. A negative charge is a benefit under the risk criterion, not a cash payment received from the market.

The finite-size term also explains why "reducing inventory" should be checked on the final position. In one dimension, a sale from a long position reduces q^2 only while it does not overshoot too far into a short position. If $q > 0$, selling size z reduces the penalty when $0 < z < 2q$. At $z = 2q$ the penalty is unchanged, and larger sales increase it. Trade direction alone does not determine risk reduction.

5.5 An Inventory-Aware Score for One RFQ

Retain the definitions of price offset δ , win probability $p(\delta)$, transaction cost per unit c , response cost K , and conditional price-change cost $A(\delta)$ from Chapter 4. A response incurs the inventory change only if it executes. Define its expected score by

$$J_\gamma(\delta; q) = p(\delta) [z\{\delta - c - A(\delta)\} - \Delta R(q, \eta z e_i)] - K. \quad (5.11)$$

One way to read (5.11) is to subtract the inventory-penalty change from the executed trade's marked-to-market payoff and then take the expectation. If the request is lost, neither the trade's cash flows nor its inventory-penalty change occurs. The response cost still occurs. Thus ΔR belongs inside the term multiplied by p , whereas K remains outside.

Declining has zero incremental score under the present assumptions. That does not mean the existing portfolio becomes riskless: its penalty is present in both alternatives and has been subtracted as the baseline. Similarly, the expected marked-to-market change of existing positions cancels when its unconditional law is unaffected by the response. We do not assume that every realized outcome of the existing book is unchanged.

The matrix Γ_H and the penalty function are fixed at this decision state. This is a modeling choice. If the chosen quote changes the appropriate conditional covariance, or if the trade changes future prices through its own impact, a fixed state penalty cannot capture all those effects. Chapter 4's A continues to describe the conditional

mean price change of the incremental trade; the quadratic penalty measures a specified price-risk exposure. Keep their roles distinct.

For a constant A , define an effective unit cost

$$\begin{aligned} c_{\text{eff}} &= c + A + u_i^\eta(q, z), \\ J_\gamma(\delta; q) &= z p(\delta)(\delta - c_{\text{eff}}) - K. \end{aligned} \quad (5.12)$$

We can now reuse the optimization in Chapter 4. With the logistic win curve and fixed a , b , and c_{eff} , the interior condition is

$$\frac{(\delta - c_{\text{eff}})(1 - p(\delta))}{b} = 1. \quad (5.13)$$

The unrestricted continuous optimum is

$$\delta_\gamma^* = c_{\text{eff}} + b \left[1 + W \left(\exp \left\{ \frac{a - c_{\text{eff}}}{b} - 1 \right\} \right) \right]. \quad (5.14)$$

The same one-dimensional bisection and grid search remain available. The inventory layer changes the effective cost; it does not require a new optimizer for this particular model. As before, price boundaries and the option not to respond must be checked explicitly.

An informative sensitivity follows from Chapter 4's derivative with respect to cost. Holding the win curve, A , size, and covariance fixed, an interior continuous solution satisfies

$$\frac{\partial \delta_\gamma^*}{\partial q_j} = \{1 - p(\delta_\gamma^*)\} \eta \gamma \Gamma_{ij}. \quad (5.15)$$

This equation shows both the portfolio connection and the incomplete pass-through of a risk charge into the quote. Correlated holdings enter through Γ_{ij} . The factor $1 - p$ reflects the remaining trade-off against execution probability. Formula (5.15) is not a derivative of a discrete, limit-constrained policy, and it assumes that inventory does not separately change the estimated win curve.

5.6 The Same Request from Three Starting Positions

Use the Chapter 4 inputs $a = 3$, $b = 1$, $c = 1$, $z = 10$, $K = 0$, and $A = 0$. Add $\gamma = 0.2$ and a one-instrument horizon covariance $\Gamma_{11} = 1$. Feasible offsets now run from -6 to 10 , in increments of 0.25 . Allowing negative offsets lets us represent willingness to pay for inventory relief; this is still a synthetic teaching range.

First compare a flat position with a long position of 10 units. By symmetry, the short-position case swaps the buy and sell results.

Starting position and dealer side	Effective cost	Continuous optimum	Grid optimum
$q = 0$, buy or sell	2	3.5671	3.50
$q = 10$, buy	4	5.1200	5.00
$q = 10$, sell	0	2.5571	2.50
$q = -10$, buy	0	2.5571	2.50
$q = -10$, sell	4	5.1200	5.00

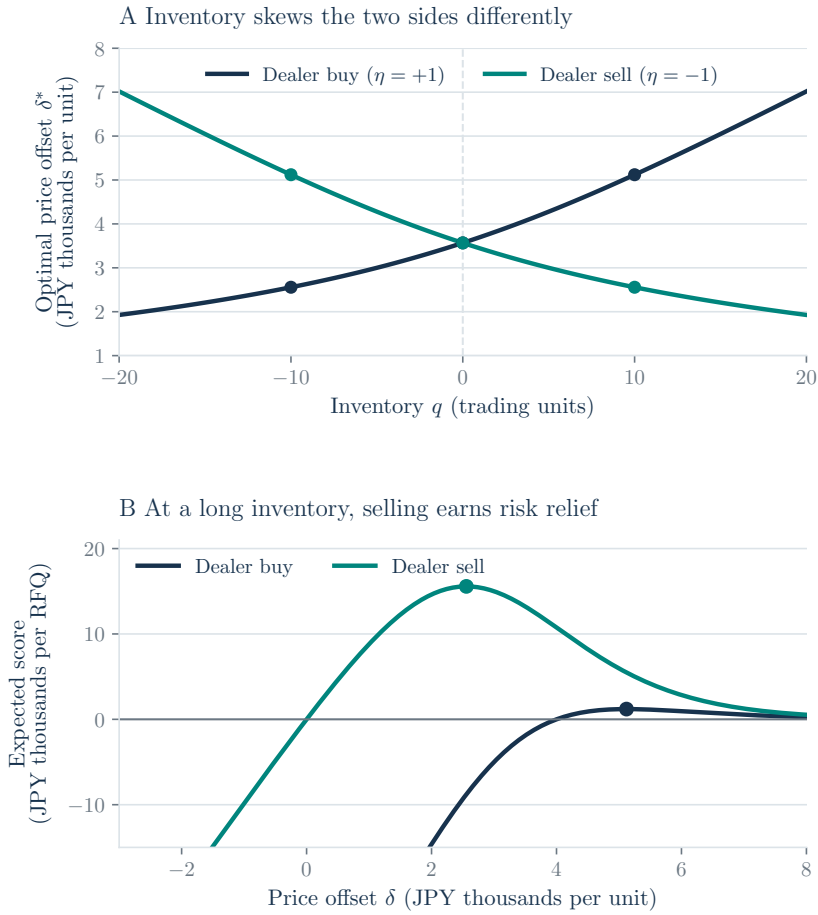
For the long dealer, a purchase receives a wider offset. Since the purchase price is $S_0 - \delta$, the dealer bids less aggressively. A sale receives a narrower offset. Since the sale price is $S_0 + \delta$, the dealer offers more aggressively. Both changes tend to protect or reduce the long position.

Even the flat dealer quotes differently from Chapter 4. Starting with no inventory does not remove the risk of a new 10-unit position. That position contributes the finite-size charge of 1 per unit, taking the effective cost from 1 to 2. Setting $\gamma = 0$, rather than merely setting $q = 0$, recovers Chapter 4.

Keep the financial payoff separate from the score:

Case at its grid optimum	Win probability	Expected trade P&L	Expected score
Flat dealer, either side	37.75%	9.439	5.663
Long dealer, buying	11.92%	4.768	1.192
Long dealer, selling	62.25%	9.337	15.561

The last two columns are JPY thousands per request. The expected trade P&L includes transaction costs and, in this example, zero conditional price change. The score additionally subtracts the expected inventory-penalty change. Selling has a score larger than its cash-related P&L because it relieves inventory risk. That extra amount must not be booked as trading revenue.



Synthetic inputs: $a = 3$, $b = c = 1$, $z = 10$, $\gamma = 0.2$, $\Gamma_{11} = 1$.
 $K = A = 0$. Dots mark continuous optima; score is not cash P&L.

Figure 5.1: Optimal offsets and expected scores for different starting inventories

Quantity now changes the quote even if p and c are held fixed: the finite-size term in (5.10) depends on z . Realistic size effects can also enter the win curve, transaction cost, and conditional price change. The quadratic term isolates one source of size dependence without claiming to describe all of them.

For a deliberately extreme illustration, take $q = 150$ and consider selling 10 units. The penalty falls from 2250 to 1960, a change of -290 , giving $c_{\text{eff}} = -28$. A negative offset can then maximize the score even though the expected trade P&L is negative. The model is willing to incur a trading loss to reduce a much larger specified penalty. Whether that is an acceptable policy depends on the risk criterion and controls; the optimizer does not establish it by itself.

5.7 Reservation Prices and Actual Optimal Quotes

The portfolio term suggests a marginal reservation price for instrument i :

$$\bar{S}_i(q) = S_{0,i} - \gamma(\Gamma_H q)_i. \quad (5.16)$$

This is an inventory-dependent indifference reference. It is not a replacement market-price estimate. The observable reference price $S_{0,i}$ is unchanged; the dealer's willingness to add or remove risk has changed.

For a finite trade of size z , ignore transaction costs and expected price changes for a moment. Set the executed trade's margin minus its penalty change to zero, before any response cost K . The highest purchase price and the lowest sale price giving zero executed-trade value on this basis are

$$\begin{aligned} Q_{\text{indiff}}^{\text{buy}} &= \bar{S}_i(q) - \frac{\gamma z}{2} \Gamma_{ii}, \\ Q_{\text{indiff}}^{\text{sell}} &= \bar{S}_i(q) + \frac{\gamma z}{2} \Gamma_{ii}. \end{aligned} \quad (5.17)$$

For a purchase, the break-even condition is $\delta = u_i^+$ and $Q = S_{0,i} - \delta$. For a sale, it is $\delta = u_i^-$ and $Q = S_{0,i} + \delta$. Substituting (5.10) gives (5.17). The midpoint is the marginal reservation price; the gap is a finite-size risk allowance of $\gamma z \Gamma_{ii}$.

These are indifference levels for an executed trade. They are not the optimal bid and offer. Choosing an actual quote still requires the probability of winning at that price, transaction costs, expected price changes, response costs, and feasible-price constraints. In the logistic example, those elements are combined through (5.12)-(5.14).

Nor does a customer request normally invite the dealer to execute both sides at once. Comparing buy and sell requests of the same size and context is a diagnostic for understanding inventory effects. It does not assume simultaneous execution or identical real-world win curves on the two sides.

5.8 Another Instrument Can Change This Quote

The vector appearing in the risk charge has the factor decomposition

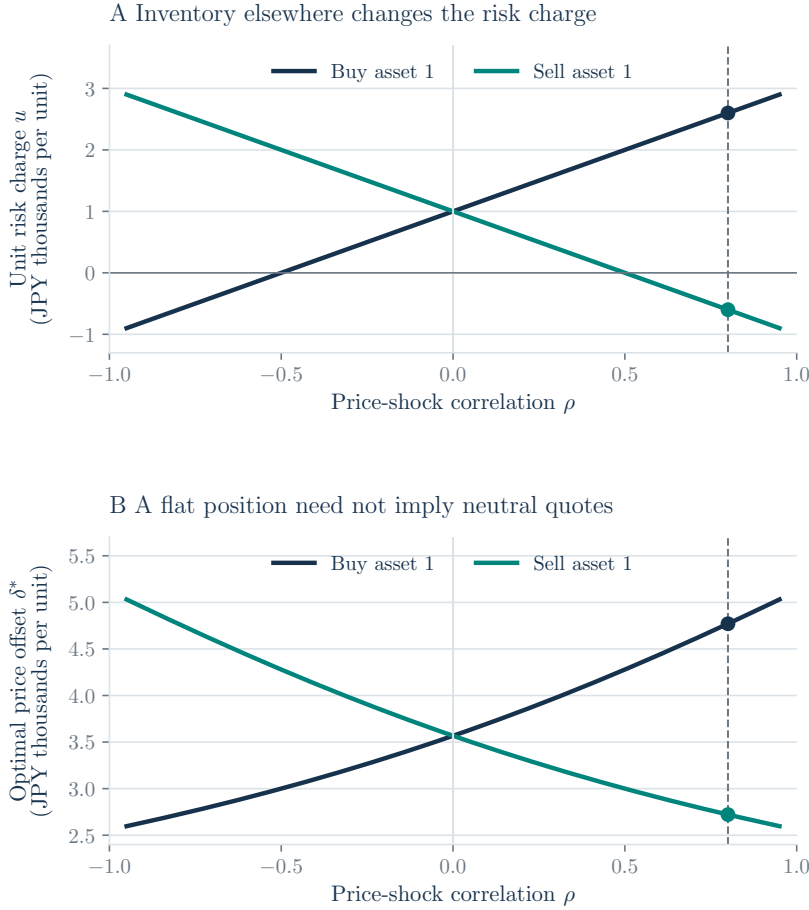
$$\Gamma_H q = D\Omega_H r + \Psi_H q, \quad r = D^\top q. \quad (5.18)$$

The first term prices the trade's interaction with the portfolio's factor exposures. The second adds its interaction with residual instrument risk. Both can matter when $q_i = 0$.

For a two-instrument illustration, let both individual price variances be 1 and their covariance be ρ . Start with $q = (0, 10)^\top$ and receive a 10-unit request in instrument 1. With $\gamma = 0.2$, the inventory charges per unit are $1 + 2\rho$ for a purchase and $1 - 2\rho$ for a sale.

At $\rho = 0.8$, the purchase charge is 2.6 and the sale charge is -0.6 . The corresponding effective costs are 3.6 and 0.4. Selling instrument 1 can reduce portfolio variance even though it creates a short position in an instrument the dealer did not previously hold. The new short offsets part of the risk of the existing long position in instrument 2.

At zero covariance, the two directions have the same charge of 1. With negative covariance, the preferred direction reverses. The quantities in (5.10) and (5.15), rather than an instrument's standalone position, identify the portfolio effect.



Synthetic inputs: $q = (0, 10)$, $\Gamma_{11} = \Gamma_{22} = 1$, $\gamma = 0.2$, $z = 10$.
 $a = 3$, $b = c = 1$, $K = A = 0$. At $\rho = 0.8$, buy $u = 2.6$; sell $u = -0.6$.

Figure 5.2: Correlation changes the quote even when the requested instrument has zero inventory

This mechanism does not imply that any historical correlation is a reliable hedge. The covariance horizon, estimation period, stressed dependence, and residual risk affect

the comparison. Here those inputs are fixed so that the portfolio arithmetic can be examined separately from their estimation.

5.9 Factor Cancellation Does Not Eliminate Residual Risk

A low-dimensional factor representation can greatly reduce the information required for a decision. If $\Psi_H = 0$, the quadratic penalty depends on q only through $r = D^\top q$. If residual risk is present, that statement is generally false.

Bergault and Guéant write a covariance decomposition into factor and residual components, then set the residual component to zero for their initial dimension-reduction result. Their factor inventory is the transpose of the loading matrix times instrument inventory. [B, Section 3, p. 11, unnumbered decomposition; Section 3.1, pp. 11-12, equations (3)-(4)] Their discussion also warns that ignoring residual risk can make some combinations of instruments appear riskless. [B, Section 3.2, p. 12] These observations concern their dynamic model; we use the same risk geometry in a separate static example.

Take two instruments with a single common factor. Each has loading 1, the factor variance is 1, and the residual covariance is diagonal with entries 0.09. For $q = (10, -10)^\top$, the common-factor exposure is zero. Factor-only variance is therefore zero, but residual variance is $0.09(10^2 + (-10)^2) = 18$. At $\gamma = 0.2$, the full penalty is 1.8.

The flat portfolio and this long-short portfolio have the same factor exposure, but different risk. They also have different marginal residual charges. For the long-short portfolio, a purchase of 10 units of instrument 1 has full risk charge 1.27 per unit, while the same-sized purchase of instrument 2 has charge 0.91. Dropping the residual covariance assigns charge 1 to both requests.

The second purchase has a smaller full charge because it reduces an existing residual short exposure. This is an important directional point: dropping a positive semidefinite residual covariance always understates the level of the penalty, but it does not always understate the change in the penalty.

Let $R_{\text{factor}}(q) = \gamma r^\top \Omega_H r / 2$. The omitted finite-trade charge is exactly

$$\Delta R - \Delta R_{\text{factor}} = \eta \gamma z (\Psi_H q)_i + \frac{\gamma z^2}{2} \Psi_{ii}. \quad (5.19)$$

The first term can have either sign. In the example, the omitted total charge is 2.7 for the first purchase and -0.9 for the second. A factor-only decision rule cannot infer those two corrections from the common-factor exposure of zero.

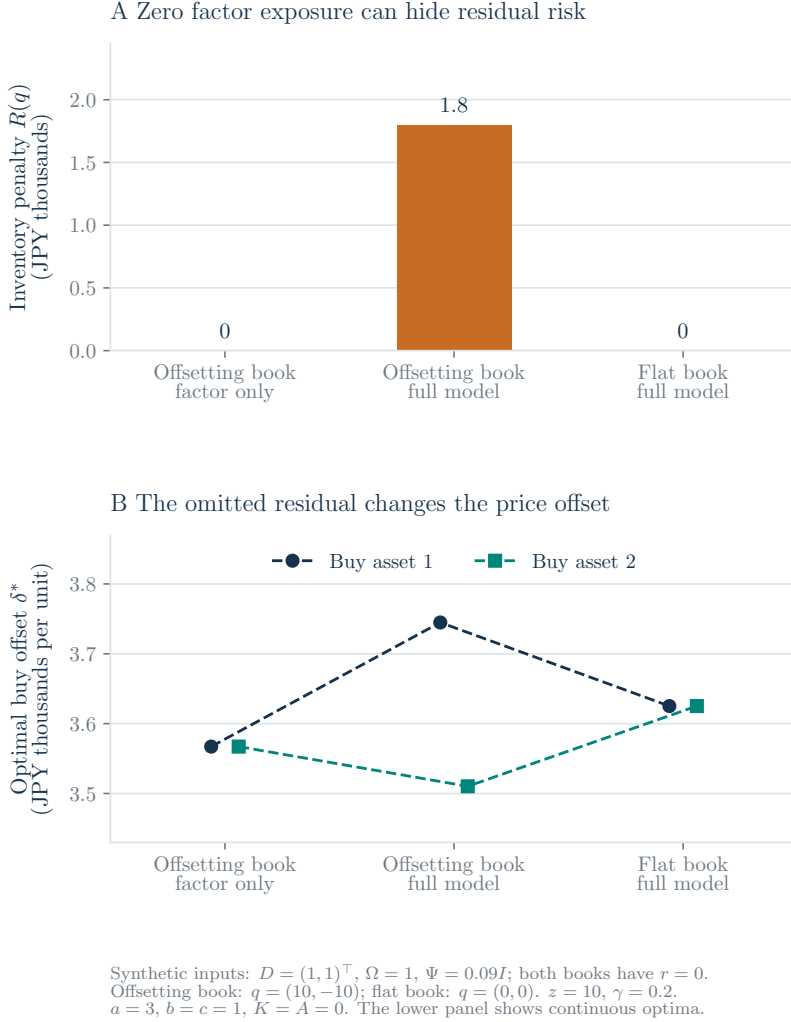


Figure 5.3: Residual risk distinguishes portfolios and RFQs with the same factor exposure

For a fixed request, fixed win curve, and unchanged feasible-action set, the difference in the score assigned to a particular quote obeys

$$|J_\gamma(\delta; q) - J_{\text{factor}}(\delta; q)| = p(\delta) |\Delta R - \Delta R_{\text{factor}}| \leq |\Delta R - \Delta R_{\text{factor}}|. \quad (5.20)$$

This is a direct accounting comparison, not a bound on the error of an estimated covariance. It also does not say that the optimal quote or response decision is unchanged. A small score correction can switch a decision near a tie. If residual risk enters a hard limit, omitting it may change the feasible-action set as well.

Even with zero residual covariance, factor inventory is sufficient only for the parts of a model that depend on inventory through those factors. Instrument limits, inventory-dependent client-flow models, or inventory-dependent execution costs may require

additional state. Exact reduction of a quadratic penalty is not by itself exact reduction of an entire trading system.

5.10 A Change of Factor Units Must Not Change the Trade

The risk assigned to a fixed portfolio should not depend on the coordinates used to report the same factor model. Suppose T is an invertible k -by- k matrix and define new factor changes by $\Delta f' = T\Delta f$. Consistent transformed inputs are

$$\begin{aligned} D' &= DT^{-1}, & \Omega'_H &= T\Omega_H T^\top, & r' &= T^{-\top} r, \\ D'\Omega'_H D'^\top &= D\Omega_H D^\top, & r'^\top \Omega'_H r' &= r^\top \Omega_H r. \end{aligned} \quad (5.21)$$

Substitution cancels T and T^{-1} , leaving the same covariance in instrument coordinates and the same portfolio variance. With the residual covariance unchanged, Γ_H , the inventory charge, and the optimal quote are unchanged.

Changing a factor from basis points to decimal units is a simple rescaling of this kind. The sensitivity and covariance must both change consistently. Rescaling only one of them changes the economic input rather than merely its representation.

The invariance statement applies to the risk geometry. Coordinate-wise factor limits do not automatically remain the same under a general rotation: the limit set itself must be transformed if it is meant to represent the same economic restriction. A rotated box of limits is generally not another axis-aligned box.

Before using a supplied covariance, check its symmetry, units, horizon, and positive semidefiniteness. Rounding can create very small negative eigenvalues; a materially indefinite estimate requires an explicit correction or rejection. Such a repair changes the input and should be recorded, rather than concealed inside the optimizer.

5.11 Limits Restrict Actions; Penalties Rank Them

A penalty makes an undesirable position less attractive. A hard limit prohibits it. One does not replace the other. For illustration, define a permitted position set by instrument caps, factor caps, and a variance cap:

$$\mathcal{Q} = \left\{ q : |q_i| \leq L_i \text{ for all } i, \quad |(D^\top q)_k| \leq B_k \text{ for all } k, \quad q^\top \Gamma_H q \leq V_{\max} \right\}. \quad (5.22)$$

Here L_i is measured in trade units, B_k in the units of factor exposure k , and $V_{\max}$ in currency squared. A variance cap and a monetary-penalty cap are different specifications, even though one can be converted into the other for a fixed positive γ .

For an all-or-nothing request, test the position conditional on a full execution. Under the strict post-trade feasibility rule used in the experiment, the best attainable score is

$$J_{\gamma}^*(q) = \max \left\{ 0, \max_{\substack{\delta \in \mathcal{D} \\ q + \eta z e_i \in \mathcal{Q}}} J_{\gamma}(\delta; q) \right\}. \quad (5.23)$$

An empty inner action set has value $-\infty$, so declining remains available with value zero. Since size and direction are fixed, feasibility in this example does not depend on δ . A wider quote can reduce the probability of a breach; it cannot make the position feasible if the quote executes. A low win probability therefore does not justify an otherwise prohibited trade.

For a scalar example with an absolute position limit of 20, a dealer at $q = 15$ cannot accept a 10-unit purchase under this rule, because the executed position would be 25. A 10-unit sale leads to 5 and passes that instrument limit. The score optimizer runs only after this feasibility check.

If the starting state is already outside a limit, the same strict rule can reject a trade that reduces the breach but does not eliminate it. A system may instead define an explicit exception for reducing an existing breach. That is a separate control rule, not something to infer silently from the sign of the risk charge. Our numerical experiment records the behavior of the strict rule.

5.12 What the Static Model Establishes

The chapter's experiment checks the exact quadratic difference against direct before-and-after penalties, compares numerical optima with a price-grid search, and separates trade P&L from inventory-adjusted score. It also checks directional symmetry, finite-size effects, covariance and factor-coordinate identities, residual-risk corrections, and post-trade feasibility.

The result is a decision rule conditional on supplied estimates and chosen controls. The examples do not calibrate γ , establish a reliable covariance horizon, or validate a real client's win curve. Those are distinct tasks. A quote can be mathematically optimal for this score and still depend on unsuitable inputs or an unsuitable objective.

There is also no model of waiting for future offsetting flow yet. Guéant's dynamic equations include differences in continuation value across neighboring inventory states; the CARA and expected-wealth formulations combine those differences with trade revenue differently. [G, equation (11.26), printed p. 234, PDF p. 259; M, equations (5.8) and (5.12), pp. 25-26] Our static construction uses the particular value assignment $-R(q)$.

This makes the connection to the next chapter explicit:

$$\Delta\theta(t, q; d) = \theta(t, q + d) - \theta(t, q), \quad \theta(t, q) = -R(q) \implies \Delta\theta = -\Delta R. \quad (5.24)$$

In Chapter 6, θ will be a value of future opportunities under a specified model, rather than a penalty inserted at the current decision. Before taking that step, the following sections examine how a risk estimate can be constructed and checked. The single-trade accounting and portfolio geometry remain unchanged.

5.13 Forecast the Right Variance, on the Right Clock

The preceding inventory examples supply a covariance matrix. A desk that estimates this input faces a second modeling problem: deciding which price changes to measure, how to forecast their uncertainty, and how to translate the result into the units of the inventory penalty. A variance estimator cannot repair an inconsistent reference series or identify the dealer's risk tolerance.

For this section, let y_t denote a dimensionless daily return innovation, with conditional mean zero. It is distinct from the factor exposure vector $r = D^\top q$ used earlier. Write the next-day conditional variance and its forecast as

$$v_{t+1|t} = \mathbb{E}[y_{t+1}^2 \mid \mathcal{F}_t], \quad \hat{v}_{t+1|t} > 0. \quad (5.25)$$

Information $\mathcal{F}_t$ contains only observations available when the forecast is made. Day $t + 1$ is the forecast target, not an input. If a nonzero conditional mean is modeled, subtract its forecast before defining the innovation. A rolling second moment about zero and a sample variance about an estimated mean answer slightly different questions; the notebook explicitly adopts the zero-mean convention.

Daily return variance is dimensionless squared return. Its square root is daily return volatility. Annualized return volatility, often a useful reporting convention, is

$$\hat{\sigma}_{\text{ann},t} = \sqrt{252\hat{v}_{t+1|t}}, \quad \hat{\Gamma}_{t,1d}^{\text{price}} \simeq S_t^2 \hat{v}_{t+1|t}. \quad (5.26)$$

The second relation uses the local approximation $\Delta S \simeq S_t y$ and a supplied reference level. Its output has squared price units. An annualized percentage cannot be inserted directly into $q^\top \Gamma_H q$. Nor is the first formula automatically the forecast standard deviation of the next calendar year's return: that interpretation requires assumptions about future variance and cross-time covariances. Here 252 is a declared display convention.

For example, daily return variance 10^{-4} gives daily volatility 1% and annualized volatility about 15.87%. At reference level 100 JPY thousands per trade unit, the local one-day price-change variance is 1 in squared price units. The same input written in

percent-return units has numerical variance 1 rather than 10^{-4} . Both the data and every variance parameter must transform by the square of the return-unit conversion.

The horizon is equally important. A fixed-horizon covariance $\Gamma_{t,H}$ already includes exposure over H . A covariance rate C_t has squared price units per unit time. The approximation $\Gamma_{t,H} \simeq HC_t$ needs a specified common time unit and a local constant-rate assumption. It is not a license to multiply an already accumulated covariance by time again. When volatility is predictable and changes over the horizon, cumulative risk should be built from the corresponding sequence of forecasts.

5.14 Rolling Variance and Exponential Weighting

A rolling estimator places equal weight on the last L completed innovations:

$$\hat{v}_{t+1|t}^{\text{roll}} = \frac{1}{L} \sum_{j=0}^{L-1} y_{t-j}^2. \quad (5.27)$$

This is a transparent baseline. It is easy to reproduce and makes its effective observation window explicit. A short window responds rapidly but estimates from fewer observations. A long window smooths noise but can retain an old regime. An unusually large return loses all its weight on the day it leaves the window, so a sharp forecast move need not indicate new information.

The exponential weighted moving average, or EWMA, replaces that cutoff with a recursive weight:

$$\hat{v}_{t+1|t}^{\text{EWMA}} = \lambda \hat{v}_{t|t-1}^{\text{EWMA}} + (1 - \lambda)y_t^2, \quad 0 < \lambda < 1. \quad (5.28)$$

After n updates, the starting variance retains weight λ^n and the observed squared innovations receive weights $(1 - \lambda)\lambda^j$. The complete weights sum to one:

$$\lambda^n + \sum_{j=0}^{n-1} (1 - \lambda)\lambda^j = 1, \quad h_{1/2} = \frac{\log(1/2)}{\log \lambda}. \quad (5.29)$$

The half-life measures the decay of an observation's weight; it is not a forecast horizon. At $\lambda = 0.94$ it is about 11.2 daily updates. The initialization must also be recorded. The notebook initializes EWMA after 60 completed days using their mean squared return, so no value from the target day enters that starting state.

RiskMetrics' fourth-edition document gives the recursive variance and covariance constructions and reports its historical daily decay choice of 0.94. This is a source convention, not a universal estimate for present trading. [V-R, printed pp. 82-83 and 100; PDF pp. 94-95 and 112] Our experiment compares the declared candidates 0.90, 0.94, and 0.97 on a separate validation block.

Both rolling variance and EWMA are estimators whose lag weights are specified directly. The basic EWMA recursion has no explicit long-run variance level. If the

EWMA recursion itself is assumed to specify conditional variance in a compatible martingale-difference return model, its forecast of future daily variance remains at its current level until new observations arrive. The martingale-difference property alone does not imply a flat variance forecast. This differs from a stationary GARCH model's conditional mean reversion. More responsive is not synonymous with more accurate: a rapid response can follow measurement noise as easily as a true change in uncertainty.

5.15 GARCH Adds a Fitted Long-Run Level

A GARCH(1,1) recursion uses the last innovation and the last conditional variance:

$$\begin{aligned} y_t &= \sqrt{v_{t|t-1}} \xi_t, \\ v_{t+1|t} &= \omega + \alpha y_t^2 + \beta v_{t|t-1}, \\ \omega &> 0, \quad \alpha, \beta \geq 0, \quad \alpha + \beta < 1. \end{aligned} \tag{5.30}$$

The standardized innovation has conditional mean zero and unit variance. Conditional Gaussianity is an additional likelihood assumption in the example. The restriction $\alpha + \beta < 1$ supports a finite unconditional second moment under the stationary model, with level $\bar{v} = \omega / (1 - \alpha - \beta)$. It does not establish every higher moment or validate a fitted model on a new regime. Bollerslev's original construction, stationarity result, and GARCH(1,1) specialization provide the relevant source definitions. [V-G, printed pp. 309-311; PDF pp. 3-5]

The notebook fits the mean Gaussian negative log likelihood, dropping constants independent of the parameters:

$$\mathcal{L}(\omega, \alpha, \beta) = \frac{1}{2n} \sum_{t \in \mathcal{I}_{\text{train}}} \left(\log v_{t|t-1} + \frac{y_t^2}{v_{t|t-1}} \right). \tag{5.31}$$

The conditional recursion and its initialization are part of that criterion. To improve numerical scaling, the implementation fits percent returns, then converts the variance intercept and initialization back to decimal-return units. The initial filter variance uses all training returns: early training filter values are therefore in-sample initialized quantities, not historical out-of-sample forecasts. Fitting and initialization are complete before validation begins. The implementation uses three predetermined starts, an analytic gradient, explicit parameter constraints, and an independent transformed-parameter optimizer check. Convergence is evidence about the specified finite optimization; it is not evidence that GARCH generated a real dataset.

Once coefficients are fitted and frozen, new completed returns update the variance state without refitting the coefficients. These are different operations. Fitting all coefficients again after seeing a test period would change the evaluation procedure. In a live adaptive implementation, refit dates and eligible historical data would have to be part of the information contract.

Let $p = \alpha + \beta$ and take the current one-day forecast as given. Under the fixed-parameter model, successive conditional daily forecasts satisfy

$$\begin{aligned}\hat{v}_{t+j|t} &= \bar{v} + p^{j-1}(\hat{v}_{t+1|t} - \bar{v}), \\ \text{Var}_t \left(\sum_{j=1}^H y_{t+j} \right) &= \sum_{j=1}^H \hat{v}_{t+j|t}.\end{aligned}\tag{5.32}$$

The second equality uses zero cross-time conditional covariances of the return innovations. Conditional heteroskedasticity does not by itself create nonzero covariance between martingale-difference returns. The sum generally differs from $H\hat{v}_{t+1|t}$ because future daily variances move toward the fitted long-run level. Serially correlated returns, a changing mean, or a different accumulation rule require additional terms.

5.16 Realized Variance and a HAR-Style Forecast

Daily squared returns are noisy observations of daily variance. If a complete day is partitioned into m intraday intervals with return innovations $y_{t,j}$, define realized variance by

$$\text{RV}_t = \sum_{j=1}^m y_{t,j}^2, \quad y_t = \sum_{j=1}^m y_{t,j}.\tag{5.33}$$

The sum of squares differs from the square of the sum. Under the notebook's independent Gaussian intraday increments, constant conditional variance within a day, and no jumps or observation noise, $\mathbb{E}[\text{RV}_t | \mathcal{F}_{t-1}] = v_{t|t-1}$. Its conditional error variance is $2v_{t|t-1}^2/m$, compared with $2v_{t|t-1}^2$ for the squared daily return. These are properties of this declared construction. [V-P, Section 2.2, printed p. 249; PDF p. 4]

An observed realized variance becomes available after its measurement window. It is an ex-post proxy for that day's uncertainty, not an ex-ante forecast of the same day. High-frequency data do not automatically make it accurate: bid-ask effects, asynchronous observations, stale marks, missing intervals, price jumps, and overnight exposure change the measurement problem. A denser grid can accumulate observation noise. A trading-day RV that omits the overnight interval does not measure a full close-to-close daily target without further treatment.

A heterogeneous autoregressive, or HAR-style, forecast summarizes completed daily RV at several lag scales. Our positive specification predicts log variance using the log of arithmetic averages:

$$\begin{aligned}\overline{\text{RV}}_{t,L} &= \frac{1}{L} \sum_{j=0}^{L-1} \text{RV}_{t-j}, \\ \log \text{RV}_{t+1} &= b_0 + b_1 \log \text{RV}_t + b_5 \log \overline{\text{RV}}_{t,5} \\ &\quad + b_{22} \log \overline{\text{RV}}_{t,22} + e_{t+1}.\end{aligned}\tag{5.34}$$

Daily, weekly, and roughly monthly lags provide a compact way to combine persistence at several scales. The log of an average is not the average of logs. The exact transformation and lag construction therefore belong with the fitted coefficients.

Corsi's published presentation uses realized volatility, the square root of summed intraday squares, and averages that quantity over the longer horizons. It explicitly permits variance and logarithmic variants. Equation (5.34) is our stated log-variance variant, not a literal reproduction of the source's volatility-level equation. [V-H, printed pp. 177, 179, and 181; PDF pp. 4, 6, and 8]

Exponentiating a fitted conditional log value does not generally give the conditional arithmetic mean of RV. The notebook uses an empirical smearing correction estimated only from training residuals:

$$\hat{v}_{t+1|t}^{\text{HAR}} = \exp(\hat{b}^\top X_t) \hat{s}, \quad \hat{s} = \frac{1}{n_{\text{train}}} \sum_{j \in \mathcal{I}_{\text{train}}} \exp(\hat{e}_j). \quad (5.35)$$

This corrects retransformation using the training residual distribution. A constant correction is justified as a conditional-mean estimate only under suitable stability and conditional residual assumptions. It is not an exact guarantee for an arbitrary new state. Even the uncorrected exponential is a conditional median only under an appropriate zero-median residual condition. We therefore state the conversion rather than attaching an automatic mean or median interpretation.

HAR uses intraday information that the daily-return rolling, EWMA, and GARCH forecasts do not use. Their comparison evaluates complete data-and-model procedures, not equal-information algorithms. A lower loss can reflect richer measurements as well as a better dynamic specification.

5.17 Chronological Evaluation on a Common Variance Target

The new companion experiment generates 1,800 retained synthetic days after a 500-day burn-in. Each day has 48 independent Gaussian intraday innovations conditional on that day's variance. The daily variance follows a supplied GARCH recursion with $\omega = 2 \times 10^{-6}$, $\alpha = 0.08$, and $\beta = 0.90$. This deliberately transparent mechanism makes the latent daily variance available to an audit. It is never used as a fitting feature or a validation label.

The first 900 days fit GARCH and log-HAR parameters. The next 300 choose the rolling window from 20 and 60 days and the EWMA decay from 0.90, 0.94, and 0.97. The final 600 evaluate all four frozen procedures. Each validation or test forecast for day t uses completed days strictly before t and parameters fitted before validation. Early training filter values remain in-sample quantities. During testing, variance recursions and lag features update with newly completed observations, while fitted coefficients and selected settings remain fixed.

Every method is scored against the same strictly positive next-day RV proxy. We report variance MSE and the nonnegative QLIKE convention

$$L_{\text{QLIKE}}(y, f) = \frac{y}{f} - \log \frac{y}{f} - 1, \quad y > 0, f > 0. \quad (5.36)$$

Here y is the realized variance proxy, not a signed daily return. The ratio is dimensionless. Compared with $\log f + y/f$, this expression subtracts $\log y + 1$, which is common to all forecasts on the same scored observations. Thus pairwise loss differences agree. This convention rejects zero or invalid proxies rather than silently inserting a numerical floor; a zero-permitting loss convention or an eligibility rule would need to be stated separately.

When both forecasts are measurable on $\mathcal{F}_t$ and $\mathbb{E}[\text{RV}_{t+1} \mid \mathcal{F}_t] = v_{t+1|t}$, QLIKE's expected ranking can be preserved when variance is replaced by that noisy proxy, under the required integrability conditions. That statement concerns expected loss differences, not equality of realized finite-sample rankings or immunity to measurement bias. Patton defines that robustness criterion and the QLIKE comparison; he does not equate it with resistance to arbitrary outliers. [V-P, printed p. 248; PDF p. 3]

Procedure	Validation choice	Test mean QLIKE	Test variance MSE
Rolling	20 days	0.058334	6.9613×10^{-10}
EWMA	Decay 0.94	0.034323	4.8264×10^{-10}
GARCH(1,1)	Training fit frozen	0.026043	4.2525×10^{-10}
Log-variance HAR	Training fit and smearing frozen	0.039926	6.4074×10^{-10}

The GARCH forecast has the lowest realized loss in this sample generated by a GARCH mechanism. This is not a recommendation that it must win on other assets, horizons, or regimes. The notebook retains the fixed seed, all raw intraday increments, forecast arrays, choices, and scores. It makes no statistical-significance claim from one dependent path. Dependence-aware uncertainty and genuinely new evaluation periods would be needed for a broader performance statement.

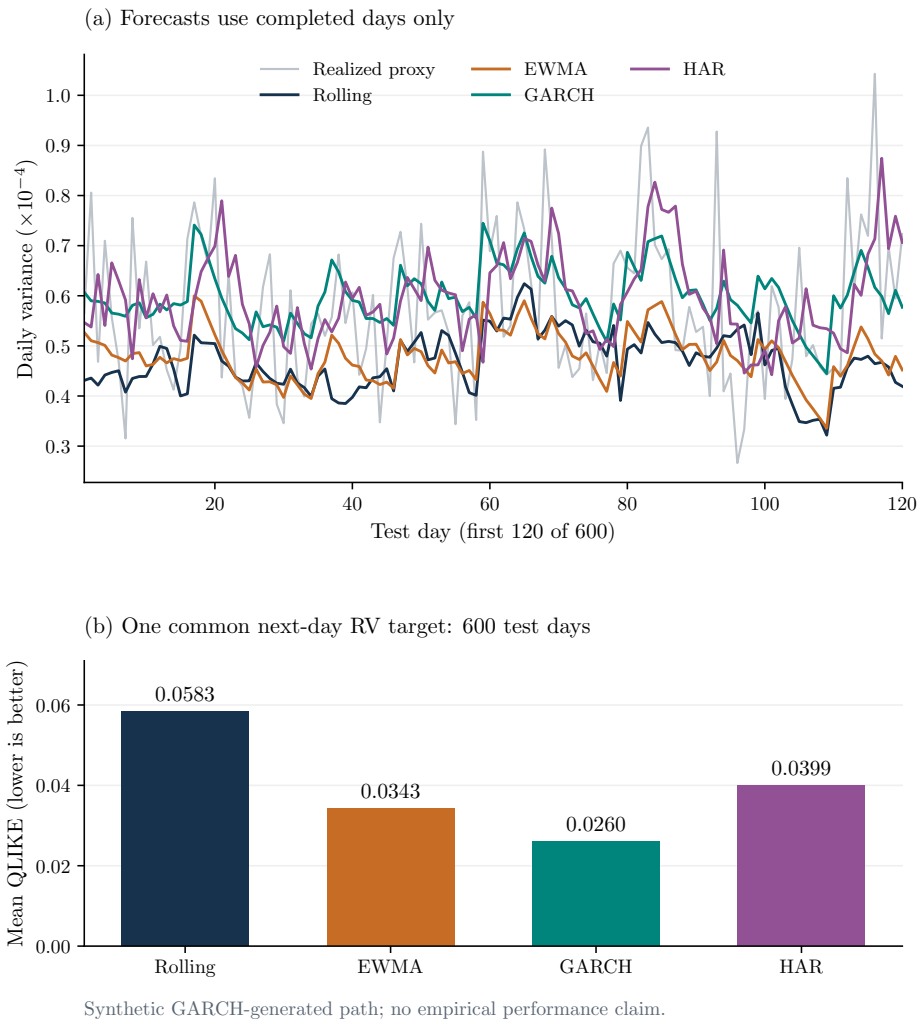


Figure 5.4: Next-day daily variance forecasts use completed observations, while realized variance is an ex-post proxy. The four procedures share the same 600-day test cohort; the upper panel shows its first 120 days and the lower panel reports full-test QLIKE. Data are synthetic and generated by a GARCH mechanism.

The 27 checks include independent optimizer and gradient comparisons, chronological perturbations, lag identities, training-only retransformation, positive forecasts, common-target score arithmetic, consistent unit changes, and covariance geometry. They verify this implementation and its declared assumptions. They do not verify that a market data feed is complete or that a forecast will remain calibrated after a structural change.

5.18 From Scalar Forecasts to a Portfolio Covariance

A set of separate volatility forecasts is not yet a portfolio covariance. Cross-instrument dependence, alignment, and residual risk remain to be specified. For aligned zero-mean vector innovations x_t , a common-weight update preserves positive semidefiniteness:

$$\begin{aligned}\widehat{V}_{t+1}^x &= \lambda \widehat{V}_t^x + (1 - \lambda) x_t x_t^\top, \\ \widehat{\Gamma}_{t,H} &= D \widehat{\Omega}_{t,H} D^\top + \widehat{\Psi}_{t,H}.\end{aligned}\tag{5.37}$$

In the first line, $\widehat{V}_t^x$ is a covariance for one specified observation interval. To convert it to a covariance rate C_t , divide by that interval's duration under the relevant accumulation assumptions. Each update is a nonnegative weighted sum of positive semidefinite matrices. In the second line, factor and residual estimates must refer to the same target horizon and consistent units, with the uncorrelated-factor/residual representation assumed in [Section 5.2](#).

Independently estimating every correlation on a different pairwise-missing sample can lose positive semidefiniteness. Choosing different entrywise decay weights can do the same. One coherent alternative combines positive marginal variance forecasts with a positive semidefinite correlation matrix, while recording how that correlation was estimated. Shrinkage or an eigenvalue correction changes the input; its amount and purpose should be retained. The optimizer should not discover an invalid covariance by exploiting negative portfolio variance.

The experiment maps two supplied factor-return variances through sensitivities D and adds a strictly nonnegative residual covariance. It verifies the total against factor-plus-residual risk, changes factor units consistently, and tests the full finite trade difference. The incremental quote charge remains

$$u_{i,\eta,z}(t, H) = \eta \gamma (\widehat{\Gamma}_{t,H} q)_i + \frac{\gamma z}{2} (\widehat{\Gamma}_{t,H})_{ii}.\tag{5.38}$$

This is [equation \(5.10\)](#)'s geometry with an explicitly dated estimate. Both the inventory exposure term and the finite-lot term change when the covariance estimate changes. A factor-neutral portfolio can still carry residual risk, and the incremental residual charge can be negative for a residual-risk-reducing trade even though the residual covariance is positive semidefinite.

No volatility estimator chooses γ , supplies a client win curve, or estimates execution-conditioned adverse selection automatically. Statistical forecast uncertainty is also different from the future price innovation whose variance is being forecast. A control rule may allow for estimation uncertainty, but doing so requires an explicit additional objective or state assumption.

5.19 Keep Estimation and Decision Models Connected

The original one-RFQ experiment remains a fixed-input illustration with its original matrices, currency units, penalties, and numerical outputs. The new notebook is a separate forecasting exercise. Its estimates are not silently substituted into the old tables. That separation makes it possible to verify the optimization before asking whether a particular estimate is a suitable live input.

A usable risk input carries an observation definition, information cutoff, forecast horizon, units, model and parameter version, and any covariance repair or residual treatment. Its timestamp does not alone show that all data used in fitting were available then. The chronological discipline of Chapters 1 and 3 applies to variance forecasts just as it applies to references and execution labels.

Chapter 6 now replaces the assigned static penalty with a computed continuation value. Its numerical model holds its risk inputs fixed across the decision horizon. An adaptive covariance forecast would introduce additional state and transition rules, or a stated sequence of model updates. Reusing a newly estimated number without specifying that change would alter the decision problem being solved.

Exercises

1. In the one-instrument example, calculate $R(q)$ and $R(q + d)$ directly for $q = 10$, $z = 10$, and each trade direction. Reconcile the total changes with the per-unit charges in (5.10).
2. Starting at $q = 0$, halve the trade size from 10 to 5 while holding the win curve and unit transaction cost fixed. Find the continuous and grid-optimal offsets. Explain why this differs from the quantity-invariance result in Chapter 4.
3. For the two-instrument example with $q = (0, 10)^\top$, find the covariance ρ at which a 10-unit sale of instrument 1 leaves the variance unchanged. Verify that the answer concerns a finite trade rather than only a marginal derivative.
4. Reproduce the factor-only and full risk charges of 1, 1.27, and 0.91 in Section 5.9. Explain why a positive semidefinite residual covariance can produce a negative correction to a trade's incremental charge.
5. Change the units of a factor through an invertible diagonal matrix T . Transform the sensitivities, covariance, and exposures consistently, then check that the risk charge and optimal quote do not change.
6. Under an absolute position cap of 20, compare a 10-unit purchase and a 10-unit sale from $q = 15$. Then start at $q = 25$ and consider selling only 2 units. Explain what

the strict feasibility rule does, and why a breach-reduction exception would be a different policy.

7. For the sale from $q = 150$ in Section 5.6, find the best offset on the stated grid. Report expected trade P&L and expected score separately. Explain why a positive score does not imply positive expected trading P&L.
8. In the separate volatility notebook, reconcile daily variance, annualized return volatility, and price-change variance. Compare the five-day fixed-parameter GARCH sum with five times its one-day forecast; explain why they differ.
9. Reproduce the chronological four-model comparison and the log-HAR smearing correction. State which observations each procedure may use, why RV is an ex-post target, and which conditional-unbiasedness assumption supports the QLIKE proxy comparison.

Source Notes

[G] Olivier Guéant, *The Financial Mathematics of Market Liquidity: From Optimal Execution to Market Making*, CRC Press, 2016. The supplied PDF has 302 pages. Chapter 5 uses the portfolio discussion and covariance setup in Sections 11.3.1-11.3.2.1 (printed pp. 231-232; PDF pp. 256-257), and the dynamic inventory-value terms in equation (11.26) and the following discussion (printed pp. 234-235; PDF pp. 259-260). PDF edition ISBN: 9781498725484.

[B] Philippe Bergault and Olivier Guéant, *Size matters for OTC market makers: general results and dimensionality reduction techniques*, arXiv:1907.01225v5, 21 September 2022, supplied 42-page PDF. Printed and PDF page numbers agree. The passages used are the penalty formulation and Remark 2 on p. 4; the unnumbered covariance decomposition in Section 3, p. 11; the zero-residual factor reduction in Section 3.1, pp. 11-12, equations (3)-(4); and the residual-risk discussion in Section 3.2, pp. 12-14. The dynamic reduction and its formal residual correction are not claimed as results for our static RFQ example.

[M] Olivier Guéant, *Optimal market making*, arXiv:1605.01862v5, 2017, supplied 43-page PDF. Printed and PDF page numbers agree. Section 2.2, p. 6, distinguishes CARA utility from expected marked-to-market wealth with a running inventory penalty. The multi-asset formulations on pp. 25-26, including equations (5.8) and (5.12), clarify the different roles of continuation-value increments.

The static penalty, its finite-trade accounting, the numerical matrices, and the RFQ experiments in this chapter are educational constructions. The primary sources support the portfolio-risk geometry and the distinctions between model formulations. None of the numerical examples is presented as a fit to JGB trading data.

[V-G] Tim Bollerslev, *Generalized autoregressive conditional heteroskedasticity*, Journal of Econometrics 31 (1986), 307-327. Author-hosted 21-page PDF; the definitions and second-moment/GARCH(1,1) discussion inspected here are printed pp. 309-311, PDF pp. 3-5.

[V-R] J.P. Morgan/Reuters, *RiskMetrics Technical Document*, Fourth Edition, 1996. Official institutional 296-page PDF; recursive variance and covariance formulas are printed pp. 82-83, PDF pp. 94-95; the daily decay convention is summarized on printed p. 100, PDF p. 112. Our validation grid and synthetic results are independent constructions.

[V-H] Fulvio Corsi, *A Simple Approximate Long-Memory Model of Realized Volatility*, Journal of Financial Econometrics 7 (2009), 174-196. The institutional 23-page copy distinguishes its volatility-level construction from admissible variance/logarithmic variants: printed pp. 177, 179, and 181; PDF pp. 4, 6, and 8. The notebook uses a declared log-variance variant with training-only smearing.

[V-P] Andrew J. Patton, *Volatility forecast comparison using imperfect volatility proxies*, Journal of Econometrics 160 (2011), 246-256. Author-hosted 11-page PDF; the expected-ranking definition and QLIKE are on printed p. 248, PDF p. 3; the simplified intraday RV construction is on printed p. 249, PDF p. 4. No empirical findings are reproduced.

PART III

Dynamic quoting

Value future decisions across time, instruments, and changing client-flow states.

- 6 Bellman and HJB
- 7 Multiple Instruments and Sizes
- 8 Flow Imbalance and Client Relationships

Chapter 6

Bellman and HJB

6.1 A Quote Changes the Next Decision

The first RFQ model valued the trade in front of us. Chapter 5 added the change in a chosen inventory penalty. Both calculations can be useful even when another request will arrive later. Neither calculation, by itself, assigns a value to the decisions that those later requests will create.

Consider a dealer with a long position. Selling now earns a quote edge and reduces inventory. It also gives up the opportunity to sell that same inventory to a later client on potentially different terms. Buying now can earn an edge while using capacity that a later, more attractive purchase would need. Waiting preserves the current position and the opportunities attached to it. The relevant comparison therefore includes what happens after the present RFQ is resolved.

This chapter makes that comparison precise. We begin with a small model in which the state, possible requests, admissible quotes, cash flows, and terminal condition are explicit. We then write a backward recursion over short time intervals. Its continuous-time limit is a Hamilton-Jacobi-Bellman equation, or HJB equation. In our finite inventory model, this is a coupled system of ordinary differential equations.

The purpose of the derivation is to make each term accountable. An arrival rate describes how often a decision becomes available. A win probability describes the outcome of responding to a particular request. A continuation value describes what the resulting position allows us to do afterward. Confusing these three objects produces equations that may look familiar while representing the wrong decision.

The chapter's numerical example uses the same quantity, quote offsets, and transaction-cost convention as Chapters 4 and 5. It compares a dynamic policy with myopic and static-penalty policies under a common objective. All inputs are synthetic. Solving the model accurately establishes what its assumptions imply; it does not establish that those assumptions describe a particular trading desk.

Chapter 5's added forecasting exercise concerns how a covariance input may be estimated. The present model instead holds its price and risk specifications fixed while computing a value of future RFQs. Updating a covariance estimate during the horizon would require an enlarged state or an explicitly scheduled model update; it is not hidden inside the recursion below.

6.2 State, Requests, and Marked Wealth

Begin with one instrument and a fixed positive trade size z . Let q denote dealer inventory, X cash, and S the supplied reference price. A request has a dealer direction η : $+1$ means the dealer buys and -1 means the dealer sells. Conditional on execution, inventory becomes

$$q^\eta = q + \eta z, \quad \eta \in \{+1, -1\}, \quad q \in \mathcal{Q}. \quad (6.1)$$

The finite set $\mathcal{Q}$ is the permitted inventory grid. A response is admissible only if the full executed position q^η belongs to that set. A request may arrive even when its direction is blocked; the dealer then declines it. Feasibility does not depend on the chance that the quote will win.

Assume the two request streams are independent Poisson processes, with constant rates ν_+ and ν_- . These are rates of RFQs presented to the dealer, including requests that the dealer declines or loses. When the dealer responds with offset δ , the conditional probability of execution is $p_\eta(\delta)$. Over a sufficiently short interval h ,

$$\begin{aligned} \Pr(\text{one direction-}\eta \text{ request in } h) &= \nu_\eta h + O(h^2), \\ \lambda_\eta(t, q; \delta) &= \nu_\eta p_\eta(\delta) \quad \text{when the dealer responds.} \end{aligned} \quad (6.2)$$

The first line counts requests. The second is the execution intensity under the stated response. Declining sets the execution intensity to zero. Multiplying a win probability by a time step without an arrival rate would confuse a probability conditional on a request with a probability of an event occurring during time.

Let the transaction price be $Q = S - \eta\delta$. Thus a positive offset places a dealer purchase below the reference price or a dealer sale above it. The unit execution cost is c , and K is a response cost paid whenever the dealer responds. Let I be the fill indicator, equal to one on a win and zero otherwise. For a response to an RFQ, holding the reference price fixed across the execution itself, the accounting is

$$\begin{aligned} \Delta X &= -I\eta z Q - Izc - K, \\ \Delta q &= I\eta z, \\ \Delta(X + qS) &= Iz(\delta - c) - K. \end{aligned} \quad (6.3)$$

The final line follows by adding $S\Delta q$ to the cash change. The terms involving the reference price cancel on both trade directions. A lost response changes cash by $-K$ and leaves inventory unchanged. A declined request changes neither cash nor inventory and incurs no response cost.

This deliberately small model takes c , the win curve, and the request rates as supplied inputs. It assumes instantaneous resolution, full execution of a winning request, and no queue or response latency. The numerical example sets the adverse-selection adjustment

from Chapter 4 to zero. A predictive price signal, correlated request outcomes, or price impact would require a corresponding state and transition model.

6.3 What Is Being Maximized

Fix a terminal time T . Let $\psi(q)$ be a nonnegative holding penalty per unit time and $\ell(q)$ a nonnegative terminal inventory penalty. A policy π chooses whether to respond and, if so, which admissible quote to send using the information available at the decision. Its objective is expected terminal marked wealth less those two penalties:

$$V(t, x, s, q) = \sup_{\pi} \mathbb{E}_{t,x,s,q}^{\pi} \left[X_T + q_T S_T - \ell(q_T) - \int_t^T \psi(q_u) du \right]. \quad (6.4)$$

The response and execution costs are already included in cash. Subtracting them again outside cash would double-count them. The holding penalty is an accumulating preference or charge, while the terminal penalty is assessed once at the horizon. Neither is automatically an actual cash expense.

For the reduction used here, assume S is a zero-drift Brownian reference price, with constant volatility, independent of the request and fill innovations. Assume also that the win probabilities, costs, and admissible actions do not depend on the absolute levels of X and S . Inventory is bounded. Under these assumptions, adding cash or shifting the current reference price changes expected marked wealth linearly, so we can write

$$V(t, x, s, q) = x + qs + \theta(t, q), \quad \theta(T, q) = -\ell(q). \quad (6.5)$$

The continuation value θ is measured in currency. It is the optimal expected future trading contribution after accounting for the remaining holding and terminal penalties, beyond the current marked wealth. It can be positive at zero inventory because a flat dealer can earn future quote revenue.

The price risk has not become zero. Instead, its mean contribution vanishes under this particular expected-wealth objective and the martingale assumptions:

$$\mathbb{E}^{\pi} \left[\int_t^T q_u - dS_u \right] = 0. \quad (6.6)$$

Bounded predictable inventory makes the stochastic integral well behaved over the finite horizon. Its variance can be substantial even though its expectation is zero. The chosen function ψ expresses a preference about holding inventory; it is not generated automatically by maximizing expected wealth.

This distinction matters when comparing formulations in the literature. Exponential utility introduces a nonlinear transformation of execution gains and continuation values. An expected-wealth objective with a running penalty combines those quantities linearly. We use the latter formulation. Cartea, Jaimungal, and Penalva derive a related cash-plus-marked-price reduction in their market-making model; Guéant's Model B makes

the additive formulation explicit. [C, Section 10.2, equations (10.3)-(10.7), printed pp. 247-248, PDF pp. 264-265; M, Sections 2.2 and 3, pp. 6-7]

The terminal condition is a modeling choice, not a hidden instruction to liquidate instantly. A quadratic ℓ charges for residual inventory. It does not specify an external execution process or guarantee that the final position will be zero. If the task requires physical liquidation, that requirement must be represented explicitly.

6.4 The Value of an RFQ That Has Arrived

Suppose a direction- η RFQ has arrived. For the moment, let $w(q)$ be any assigned value of the state after resolving it. A response at δ has total value, relative to current marked wealth,

$$B_\eta(q, \delta; w) = -K + p_\eta(\delta) [z(\delta - c) + w(q^\eta)] + [1 - p_\eta(\delta)] w(q). \quad (6.7)$$

Declining has value $w(q)$. Subtract that common continuation value from the response value. The incremental advantage of responding is

$$g_\eta(q, \delta; w) = p_\eta(\delta) [z(\delta - c) + w(q^\eta) - w(q)] - K. \quad (6.8)$$

The continuation difference is multiplied by the win probability because inventory changes only when the quote executes. The cost K sits outside that probability because every response pays it. Both terms matter before applying the decision to decline.

Define the best incremental value of the arrived request as

$$\mathcal{H}_\eta(q; w) = \max \left\{ 0, \max_{\substack{\delta \in \mathcal{D} \\ q^\eta \in \mathcal{Q}}} g_\eta(q, \delta; w) \right\}. \quad (6.9)$$

Here $\mathcal{D}$ is the set of allowed quote offsets. If execution would leave the inventory grid, the inner action set is empty and the request contributes zero through decline. If the best quote has zero incremental value, our convention is to decline. Ties among positive-value quotes select the smallest offset.

The maximization is performed after observing the request direction. This is why the eventual HJB equation adds the separately optimized contributions from the two request types. It does not require the dealer to choose a single action before knowing whether the next request is a purchase or a sale.

The connection to [Chapter 4](#) is immediate. A continuation difference is equivalent to a state-dependent unit cost:

$$c_{\text{dyn}, \eta}(q; w) = c - \frac{w(q^\eta) - w(q)}{z}, \quad (6.10)$$

$$g_\eta = z p_\eta(\delta) [\delta - c_{\text{dyn}, \eta}(q; w)] - K.$$

A more valuable post-trade state lowers this effective cost. A less valuable state raises it. The value difference includes all modeled future opportunities and penalties. It should not be relabeled as cash received from reducing risk.

For a logistic win curve $p_\eta(\delta) = 1/[1 + \exp((\delta - a)/b)]$, with $b > 0$ and a fixed continuation assignment, an interior continuous quote optimum obeys

$$\frac{[\delta^* - c_{\text{dyn},\eta}] [1 - p_\eta(\delta^*)]}{b} = 1. \quad (6.11)$$

This is the same first-order condition as before, with a different effective cost. It is not an instruction to ignore the permitted price grid, its endpoints, or the decline action. In the numerical model we compare every allowed quote directly. The difficulty of the dynamic problem lies in finding the continuation values consistently, not in replacing a simple one-RFQ maximization with a different local objective.

6.5 A Backward Bellman Recursion

Divide the horizon into N intervals of length $h = T/N$, with times $t_n = nh$. Construct a discrete model in which an interval contains no request with probability $1 - \nu h$, one buy-direction request with probability $\nu_+ h$, or one sell-direction request with probability $\nu_- h$, where $\nu = \nu_+ + \nu_-$. Require $\nu h \leq 1$ so these are valid probabilities.

Each request's response and full-fill outcome are resolved inside its branch before the next state is entered. Thus this model has no pending execution or reporting-latency state. Chapter 3's distinction between request-origin completed fills and delayed physical execution counts still applies: identifying the two clocks here is an explicit immediate-resolution modeling assumption, not a general identity for live RFQ logs.

In this discrete model, the holding penalty is charged at the inventory at the beginning of the interval, and any request is resolved at the end. There can be at most one request in an interval. These are explicit timing conventions. They give an exact finite-step Bellman recursion for this categorical-arrival model, and an approximation to the continuous-time Poisson model.

Write $\theta_n(q)$ for its optimal continuation value. Conditioning on the next interval's event gives

$$\begin{aligned} \theta_n(q) = & -h\psi(q) + (1 - \nu h)\theta_{n+1}(q) \\ & + h \sum_{\eta \in \{+1, -1\}} \nu_\eta [\theta_{n+1}(q) + \mathcal{H}_\eta(q; \theta_{n+1})], \quad \theta_N(q) = -\ell(q). \end{aligned} \quad (6.12)$$

The no-request branch retains the same inventory. The bracketed request branch includes the unchanged-state continuation value and the best incremental response value. A declined or lost request can leave the state unchanged; these possibilities belong inside the request branch rather than being omitted from the event accounting.

The coefficients of the unchanged-state continuation value sum to one. Collecting those terms yields the more convenient recursion

$$\theta_n(q) = \theta_{n+1}(q) + h \left[-\psi(q) + \sum_{\eta} \nu_{\eta} \mathcal{H}_{\eta}(q; \theta_{n+1}) \right]. \quad (6.13)$$

Begin at the known terminal vector $-\ell$. For each earlier time, evaluate the possible responses using the next-time vector, select the best action for each state and request direction, and apply (6.13). Only after all states have been updated should the time index move backward. Updating some states using newly computed values and others using old values would implement a different numerical method.

This is the principle of optimal continuation in a concrete form. Once the next state is known, the policy used from that state onward must be the optimal remaining policy represented by θ_{n+1} . Otherwise an improvement in the continuation would improve the original decision as well.

The recursion also clarifies why the current holding charge does not appear inside the RFQ quote comparison. Under the discrete timing convention, $h\psi(q)$ is paid before any possible inventory change and is common to all actions in that interval. Future holding penalties do depend on the executed position and are already included in the continuation difference.

6.6 From Bellman to HJB

Rearrange (6.13), divide by h , and let the step tend to zero. The time difference becomes $\partial_t \theta$. The two possible request contributions become their arrival rates multiplied by their optimal per-arrival values:

$$0 = \partial_t \theta(t, q) - \psi(q) + \sum_{\eta \in \{+1, -1\}} \nu_{\eta} \mathcal{H}_{\eta}(q; \theta(t, \cdot)), \quad \theta(T, q) = -\ell(q). \quad (6.14)$$

The signs can be checked with a dealer who receives no further requests. Then $\partial_t \theta = \psi$, so moving closer to the terminal time removes future holding costs and raises the value. With a positive trading opportunity and no penalties, moving farther from the terminal time adds opportunities and raises the value. Both checks agree with (6.14).

Each term has units of currency per unit time. The derivative is a rate of change in value, ψ is a rate of holding penalty, and $\nu_{\eta} \mathcal{H}_{\eta}$ is a request rate times a currency value per request. The expression p_{η} alone has no time unit and cannot replace ν_{η} .

For computation it is often clearer to use time remaining, $\tau = T - t$, and write $v(\tau, q) = \theta(T - \tau, q)$. The terminal condition becomes an initial condition:

$$\partial_\tau v(\tau, q) = -\psi(q) + \sum_\eta \nu_\eta \mathcal{H}_\eta(q; v(\tau, \cdot)), \quad v(0, q) = -\ell(q). \quad (6.15)$$

The backward recursion is forward Euler applied to this equation in time remaining. The name HJB does not imply that a large partial differential equation must always be solved. Once cash and the martingale price have been removed, our only varying state is one of finitely many inventory levels. The unknown is a vector with one component per permitted position.

With a finite quote set and inventory set, this system is also mathematically well behaved. Let F be the right side of (6.15). A continuation-value perturbation changes a difference across two states by at most twice its largest absolute component. Probabilities are at most one, and taking a maximum preserves that bound:

$$\|F(v) - F(w)\|_\infty \leq 2(\nu_+ + \nu_-) \|v - w\|_\infty. \quad (6.16)$$

Thus the vector field is globally Lipschitz and the finite-dimensional initial-value problem has a unique solution on the horizon. The maximizing quote can change discontinuously when two actions exchange rank, even while the value vector remains continuous. Numerical value convergence and a visually smooth quote curve are different requirements.

The source treatments use dynamic programming with diffusion and counting-process generators to reach closely related equations. Our explicit request stream, conditional win probability, response cost, and decline action specify the adaptation to an RFQ decision. They should not be silently attributed to a model that instead starts directly from quote-dependent execution intensities. [C, Sections 5.4.1-5.4.2, printed pp. 115-117, PDF pp. 132-134; M, equations (3.5)-(3.8), p. 7]

6.7 Recovering the Static Model at the Horizon

At zero time remaining, continuation is exactly $-\ell(q)$. If the terminal penalty is $\ell(q) = \alpha_T q^2/2$, then the continuation cost of a finite trade is

$$-\frac{v(0, q + \eta z) - v(0, q)}{z} = \frac{\ell(q + \eta z) - \ell(q)}{z} = \eta \alpha_T q + \frac{\alpha_T z}{2}. \quad (6.17)$$

This is Chapter 5's finite-trade inventory charge when its quadratic penalty coefficient equals α_T . The linear-in-inventory term changes sign between buying and selling. The positive size term remains on both sides, including at zero inventory.

The statement concerns the limiting quote decision for a request resolved immediately before the horizon, or the final request branch of the discrete recursion. An RFQ at exactly the terminal instant has probability zero in a continuous Poisson model. The

terminal vector itself is a condition on remaining inventory, not an extra opportunity to trade afterward.

For $\alpha_T = 0.2$, $q = 10$, and $z = 10$, a purchase changes the terminal penalty from 10 to 40, while a sale changes it from 10 to zero. With $c = 1$, the effective costs are 4 and 0. On the price grid used in the example, the corresponding best offsets are 5 and 2.5. These are the [Chapter 5](#) terminal-penalty decisions.

There is another exact check. When both request rates are zero, there is no control that can change the position. For all remaining times,

$$\theta(t, q) = -\ell(q) - (T - t)\psi(q). \quad (6.18)$$

If $\ell(10) = 10$, $\psi(10) = 1$, and one unit of time remains, the continuation value is -11 . This is negative because the inventory incurs penalties, not because any trading loss has been realized.

For a short but nonzero remaining time, use the first derivative in (6.15) to obtain a local approximation:

$$v(\tau, q) = -\ell(q) + \tau \left[-\psi(q) + \sum_{\eta} \nu_{\eta} \mathcal{H}_{\eta}(q; -\ell) \right] + O(\tau^2). \quad (6.19)$$

This formula explains the first departure from the static assignment. It adds the immediate holding charge and the arrival-weighted opportunity values evaluated with terminal continuation. It is a local approximation as τ tends to zero, not a justification for using the same terminal penalty at every date.

With more time remaining, a position can influence several future RFQs. Capacity, offsetting trades, and holding costs all enter through the evolving value vector. There is no universal rule that a longer horizon must make every quote narrower or that it must make every inventory position more valuable.

Once continuation already includes the specified inventory costs, adding [Chapter 5](#)'s static trade penalty again would count the same preference twice. A separate penalty is legitimate only if it represents an additional term in the objective and that term is included consistently in the recursion and policy evaluation.

6.8 A Five-State Numerical Model

The experiment has five inventory levels, $\mathcal{Q} = \{-20, -10, 0, 10, 20\}$, and a fixed request size of 10. Quotes lie on the grid from -6 to 10 in steps of 0.25 . A response is compared with decline after the post-trade limit check.

Input	Value	Interpretation
a, b, c	3, 1, 1	Logistic location, scale, and unit execution cost
z, K	10, 0	Full trade size and response cost
ν_+, ν_-	1, 1	Requests per unit time in each direction
T	1	Total normalized time horizon
$\psi(q)$	$0.01q^2$	Holding penalty in currency per unit time
$\ell(q)$	$0.10q^2$	Terminal penalty in currency

Prices, offsets, and unit costs are in JPY thousands per normalized trading unit. Inventory is in normalized trading units. Values, response costs, and penalties are in JPY thousands. Time is a normalized unit; no empirical intraday request rate is being asserted.

The coefficient of the terminal penalty and the coefficient of the holding penalty have different time units. They are chosen directly for this example. Reusing a fixed-horizon covariance from [Chapter 5](#) as a running penalty without converting its horizon would create an inconsistent objective.

At the upper inventory boundary, a purchase is disabled. The value of holding 20 is still computed, and selling remains possible. We do not set the continuation value to zero at that boundary. At zero inventory, both directions are available, so setting the value there to zero would also incorrectly remove future opportunities.

The numerical solver keeps a complete next-time value vector, evaluates each allowed action, and stores the policy as well as the value. It repeats the calculation on finer time grids. The notebook recomputes these arrays in WSL and includes exact checks of the terminal condition, symmetry, event probabilities, limit handling, and the Bellman equation.

With $N = 800$, the initial continuation values and quotes are as follows. Values are in JPY thousands; offsets are in JPY thousands per trading unit.

Inventory q	Initial value	Buy offset	Sell offset
-20	-17.622008	2.00	Blocked
-10	5.293276	2.75	4.50
0	12.533055	3.50	3.50
10	5.293276	4.50	2.75
20	-17.622008	Blocked	2.00

At $q = 10$, the dynamic initial offsets differ from the static offsets of 5 and 2.5: the dynamic policy buys at 4.5 and sells at 2.75. In this example, future opportunities reduce the severity of the initial inventory skew. This conclusion is conditional on the specified horizon, arrival rates, penalties, and available capacity.

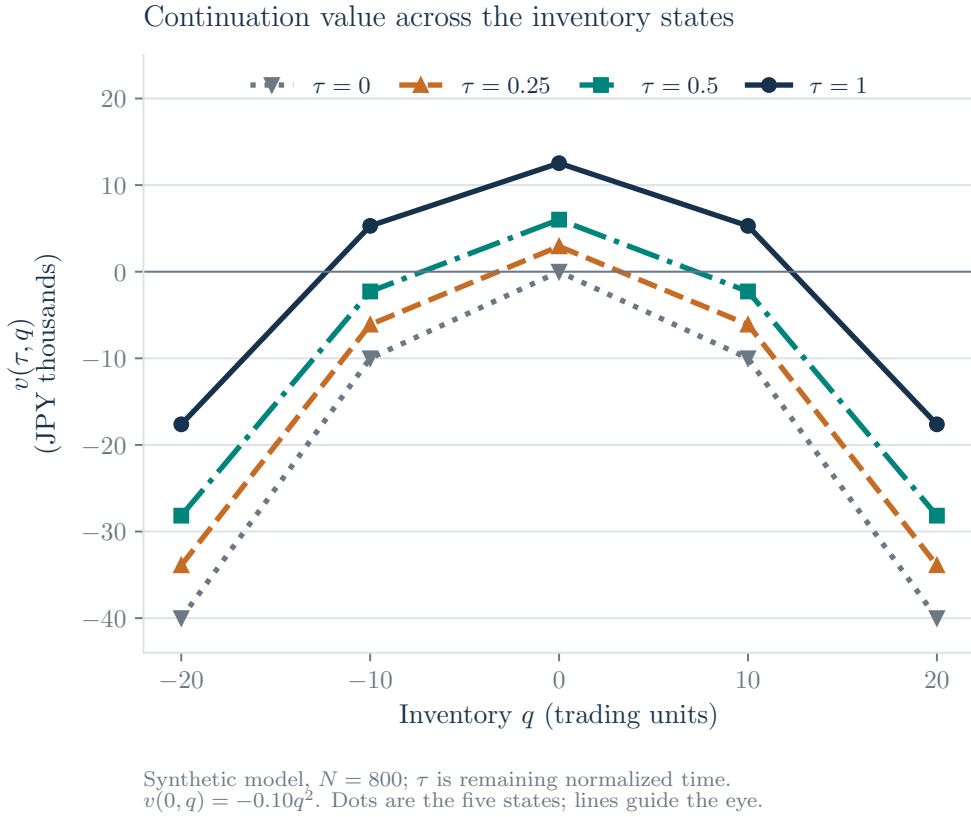


Figure 6.1: Continuation value evolves from the terminal penalty as time for future requests increases

The continuation values in Figure 6.1 include future expected trading gains as well as penalties. A common upward shift in all states would not change any quote: decisions use differences between neighboring states. It is therefore insufficient to look only at the absolute value of a flat position when explaining inventory skew.

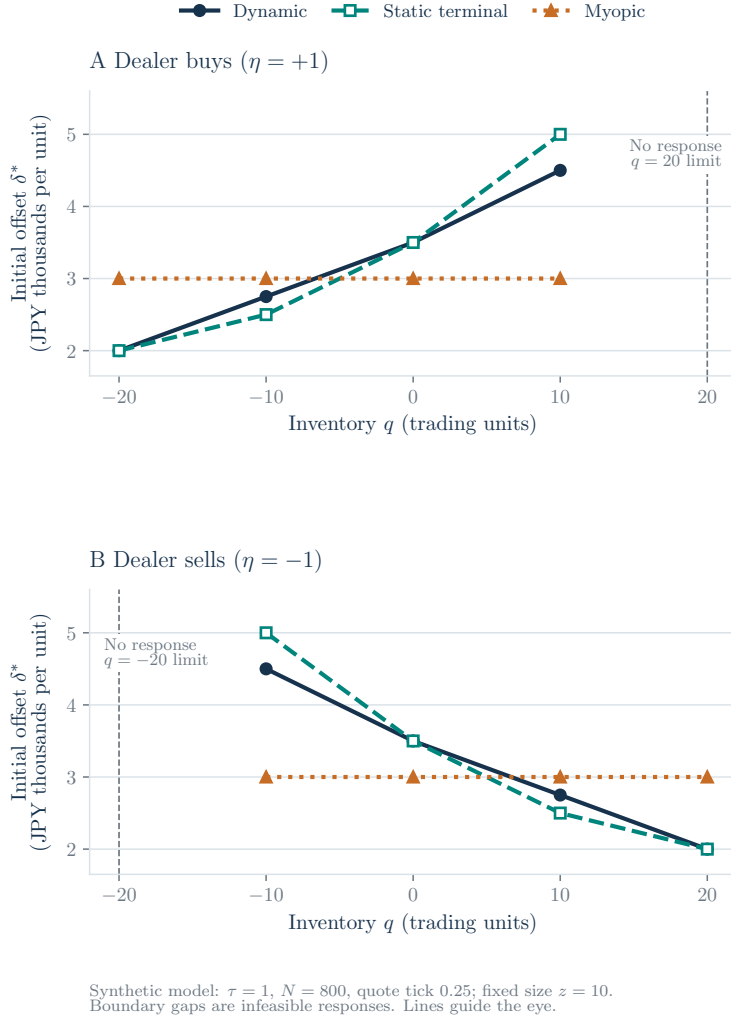


Figure 6.2: Dynamic quotes use continuation differences and respect the same inventory boundaries as the comparison policies

The quote grid introduces discrete steps. At symmetric request rates and symmetric costs and penalties, buying at inventory q mirrors selling at inventory $-q$. The implementation checks that identity directly. A blocked direction is shown as unavailable, rather than assigned an artificial extremely wide price.

6.9 Comparing Policies Under One Objective

A policy comparison is meaningful only if all policies face the same requests, costs, constraints, holding penalties, and terminal condition. We compare three rules. The myopic rule uses a zero continuation difference and maximizes current expected trading contribution. The static rule uses $-\ell(q)$ at every request. The dynamic rule uses the next-time vector from Bellman's recursion.

These are competing decision rules within the same model. The myopic rule is still evaluated with the actual holding and terminal penalties. We do not make it look favorable by scoring it with a different objective, and we do not treat the static penalty as an additional cash cost.

No Monte Carlo sampling is required for this small example. Fix any policy at time step n . Let $r_{n,\eta}(q)$ equal one if it responds and zero if it declines, and let $\delta_{n,\eta}(q)$ be its offset when responding. Its one-step inventory transition matrix has entries

$$\begin{aligned} P_n(q, q^\eta) &= h\nu_\eta r_{n,\eta}(q) p_\eta(\delta_{n,\eta}(q)), \\ P_n(q, q) &= 1 - \sum_{\eta} P_n(q, q^\eta). \end{aligned} \quad (6.20)$$

A declined or lost request contributes to the unchanged-state probability. For an unavailable direction the transition probability is zero. Under the request-step condition $\nu h \leq 1$, every entry is nonnegative and each row sums to one.

The expected immediate reward of the same policy, including the current holding penalty, is

$$b_n(q) = h \left[\sum_{\eta} \nu_\eta r_{n,\eta}(q) \{p_\eta(\delta_{n,\eta}(q)) z[\delta_{n,\eta}(q) - c] - K\} - \psi(q) \right]. \quad (6.21)$$

Where the rule declines, the corresponding contribution is defined as zero and no quote needs to be evaluated. The response cost again applies to every response, not just the responses that win.

Policy evaluation is then a matrix-vector recursion with no maximization:

$$U_n^\pi = b_n^\pi + P_n^\pi U_{n+1}^\pi, \quad U_N^\pi(q) = -\ell(q). \quad (6.22)$$

We also propagate expected trading contribution, holding penalty, and terminal penalty separately. Their reconciliation with U^π prevents a preference for less risk from being reported as extra cash income. The trading contribution includes quote revenue minus execution and response costs; under the martingale assumptions it is the expected contribution to marked wealth before inventory penalties.

Starting at zero inventory with $N = 800$, the results are:

Policy	Trading contribution	Total penalties	Objective
Dynamic	17.743546	5.210490	12.533055
Static penalty	17.615792	5.091100	12.524692
Myopic	19.531379	9.295164	10.236215

All entries are expected JPY thousands. For the dynamic policy, the total penalty consists of 0.286969 in holding penalties and 4.923521 at the terminal time. The corresponding components are 0.278324 and 4.812775 for the static policy, and 0.466717 and 8.828447 for the myopic policy. Calculations use unrounded values; the displayed columns can differ by a final rounding unit.

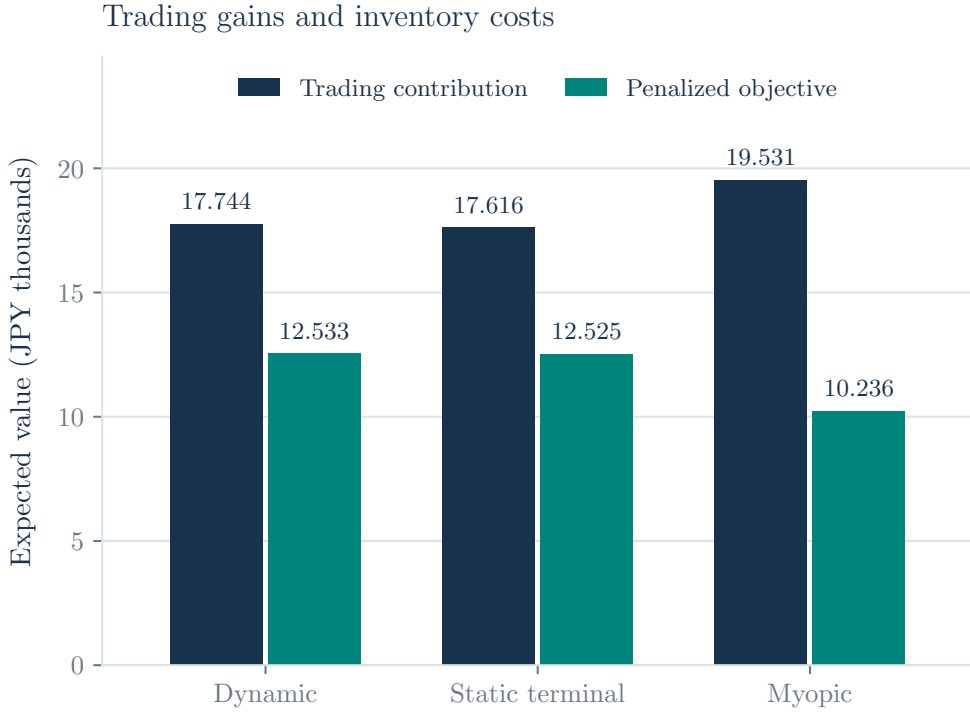
The myopic rule earns the largest expected trading contribution in this comparison. Its larger inventory penalties lower its value under the common objective. The dynamic rule improves that objective over the static rule by only about 0.008364 thousand yen in this particular calibration. Reporting that modest difference is more informative than treating every successful HJB solution as a large economic improvement.

The dynamic and static rules start with the same offsets at zero inventory. They still have different values because their decisions can differ after inventory changes and time passes. Comparing only the first quote would miss that policy difference.

Bellman's maximizing policy must dominate any feasible comparison policy under the same discrete model:

$$U_n^\pi(q) \leq \theta_n(q), \quad U_n^{\pi^*}(q) = \theta_n(q). \quad (6.23)$$

This follows by backward induction. The terminal vectors agree. If the inequality holds at the next time, nonnegative transition probabilities preserve it, and maximizing over the available actions cannot lower the value. Equality for the stored maximizing policy is a strong implementation check because policy evaluation takes a separate computational path.



Synthetic model: initial $q = 0$, $T = 1$, $N = 800$; deterministic expectations.
 All policies use the same running and terminal penalties in evaluation.
 Objective = trading contribution – running penalty – terminal penalty.

Figure 6.3: Expected trading contribution and the common penalized objective distinguish cash opportunities from inventory preferences

6.10 Time Discretization and What the Checks Establish

The categorical-arrival recursion is not the exact transition law of a Poisson process over a finite interval. With total rate ν , a Poisson interval can contain two or more requests. Their probability is $1 - e^{-\nu h}(1 + \nu h)$, which is of order h^2 . Those events are discarded by the one-request approximation.

For example, with $N = 200$, the step is 0.005 and $\nu h = 0.01$. The categorical model assigns probability 0.99 to no request and 0.005 to each request direction. The exact probability of two or more Poisson requests is about 4.97×10^{-5} per interval. A small per-interval difference is a reason to check refinement, not a reason to call the discrete law exact.

The experiment uses $N = 100, 200, 400, 800$ and compares initial value vectors on the same inventory and quote grids. A useful refinement statistic is

$$E_N = \left\| \theta_0^{(2N)} - \theta_0^{(N)} \right\|_{\infty}. \quad (6.24)$$

Steps N	Initial value at $q = 0$	E_N across all states
100	12.527707	0.009266
200	12.530767	0.004620
400	12.532293	0.002307
800	12.533055	Not computed

The largest differences occur at the boundary states, so inspecting only the flat-state value would understate this refinement statistic. The successive differences roughly halve. Initial quote choices happen to agree on all four time grids; that is an observed feature of this example, not a guarantee that every policy switch is resolved exactly.

Forward Euler has first-order time accuracy for this finite system. Successively smaller value differences are consistent with that accuracy. They do not quantify error caused by the chosen quote tick, fixed trade size, inventory cap, or model assumptions. Those are separate approximations or restrictions.

The time-step probability bound is also a numerical safeguard. The Bellman update can be expressed as a maximum of expectations with nonnegative event weights. If the step makes the no-request probability negative, that interpretation fails even if a numerical routine still returns a vector of plausible-looking values.

The checks therefore cover several distinct questions. Terminal and zero-arrival cases verify signs and timing. Direct comparisons of quadratic differences verify the connection to Chapter 5. Symmetry checks compare both trade directions. Transition rows verify probability accounting. Independent policy evaluation verifies the stored maximizing actions. Refinement compares time resolutions without changing the economic inputs.

Passing these checks supports the implementation of the stated finite model. It does not estimate uncertainty in ν_{η} , identify a client's win curve, justify the penalty coefficients, or establish that the inventory grid captures the positions a desk actually needs to manage.

6.11 What Must Change in a Richer Model

The reduction to inventory and time depends on what information predicts future rewards and transitions. If the arrival rate changes with a market regime, that regime belongs in the state. If a predictive signal changes expected reference-price moves, its transition law affects continuation value. If the client identity changes the win curve or future flow, a single pooled probability may be insufficient.

For multiple instruments, the state can be a vector of inventories, with a separate arrival type for each instrument, direction, and size. The continuation difference then moves between two portfolio states. The covariance and factor representation from Chapter 5 can help describe holding penalties, but they do not automatically make factor exposure sufficient for the entire dynamic decision.

Execution latency changes the event sequence. A quote may remain outstanding while other requests arrive or the reference price moves. Then pending quotes, cancellation rights, or stale-price exposure may matter. Modeling the response as instantaneously resolved would no longer represent the same control problem.

Likewise, an external hedge introduces another action and another cost process. It should not be represented merely by declaring unfavorable inventory states to have zero value. The model must say when hedging is possible, what changes in inventory it produces, and what the dealer pays for it.

Even within the simple model, boundaries deserve economic interpretation. A hard capacity of 20 is a binding restriction on execution. Increasing that cap changes the admissible strategies and may change interior values and quotes. It is not merely adding extra plotting points to the same problem.

These extensions make the state larger because they preserve information relevant to future decisions. Approximation should begin only after that full decision structure is clear. A lower-dimensional model can be useful, but the information it discards and the resulting restrictions on the policy should be stated.

6.12 From an Assigned Penalty to a Computed Opportunity Value

The progression across the three chapters can now be stated precisely. Chapter 4 uses the expected contribution of the current RFQ. Chapter 5 uses a particular continuation assignment, the negative of a static inventory penalty. Chapter 6 computes continuation from future arrivals, feasible responses, running penalties, and a terminal condition.

The local RFQ comparison retains the same accounting structure throughout. Execution earns a quote edge net of trading cost. Only a fill changes inventory. Every response pays its response cost. What changes is the value attached to the resulting position.

This is the practical meaning of Bellman and HJB in the present setting: a consistent way to assign values to future decision opportunities and to insert their differences into today's quote choice. The equations become useful only after the state, objective, timing, and constraints have been specified carefully.

The next chapter extends the decision across multiple instruments and sizes. It will retain the same separation between an arrived request and the stream of future requests, and between a modeled inventory preference and actual trading cash flows.

Exercises

1. For $N = 200$, $T = 1$, and $\nu_+ = \nu_- = 1$, calculate the categorical probabilities of no request and each request direction in an interval. Calculate the exact Poisson probability of two or more requests. Explain which probability law the Bellman recursion uses.
2. At terminal continuation, take $q = 10$, $z = 10$, $c = 1$, and $\ell(q) = 0.1q^2$. Calculate the continuation difference, effective cost, and grid-optimal quote for each direction. Reconcile the results with Chapter 5.
3. At $q = 0$ with terminal continuation, change K from zero to 6. Compute the best response advantage and the chosen action. Explain why replacing $-K$ with $-p(\delta)K$ solves a different problem.
4. At $q = 20$, identify which request direction is blocked. Describe the unchanged-state probability in the policy transition matrix and explain why the value at that inventory boundary is not set to zero.
5. Set both request rates to zero. Derive the exact value at every inventory state and compute the value at $q = 10$, $t = 0$. Use the result to check both the terminal condition and the sign of the holding term.
6. Starting at $q = 0$, evaluate the dynamic, static-penalty, and myopic policies under the same discrete model. Report expected trading contribution, expected holding and terminal penalties, and the resulting objective separately. Check equality between Bellman's value and independent evaluation of its stored policy.
7. Repeat the solution with $N = 100, 200, 400, 800$. Compute E_{100} , E_{200} , and E_{400} , compare the initial value vectors, and inspect any quote changes. Explain why convergence of values does not require quotes to vary smoothly.

Source Notes

[C] Alvaro Cartea, Sebastian Jaimungal, and Jose Penalva, *Algorithmic and High-Frequency Trading*, Cambridge University Press, 2015, user-supplied PDF. Sections 5.4.1-5.4.2 provide dynamic programming with counting processes and the HJB equation: equation (5.31), printed p. 115, PDF p. 132; equation (5.35), printed p. 117, PDF p. 134. Section 10.2, printed pp. 247-248, PDF pp. 264-265, provides an additive market-making objective and the reduction in equations (10.3)-(10.7). Section 10.3, printed pp. 260-261, PDF pp. 277-278, provides a separate exponential-utility formulation and a comparison under its own assumptions.

[M] Olivier Guéant, *Optimal market making*, arXiv:1605.01862v5, 6 May 2017, supplied 43-page PDF. Printed and PDF page numbers agree. Section 2.2, p. 6, distinguishes Model A's exponential utility from Model B's expected marked wealth with a running inventory penalty and a terminal penalty. Model B's equations (3.5)-(3.8), p. 7, are the direct reference for the additive continuation-value structure used here. Equations (3.9)-(3.10), p. 8, give the unified inventory ODE system and terminal condition. The paper is available from [arXiv](https://arxiv.org/abs/1605.01862v5).

[G] Olivier Guéant, *The Financial Mathematics of Market Liquidity: From Optimal Execution to Market Making*, CRC Press, 2016, user-supplied 302-page PDF. Sections 11.3.1-11.3.2 provide the RFQ and multi-instrument context. Equations (11.22)-(11.26), printed pp. 233-234, PDF pp. 258-259, use exponential utility; they are a comparison formulation, not the additive objective derived in this chapter.

The explicit request-arrival rates, conditional logistic win probabilities, response cost, decline action, categorical time discretization, five-state grid, and numerical comparisons are educational constructions for this book. The source passages support the dynamic-programming method and clarify differences between objectives. They do not constitute empirical validation of these inputs or policies.

Chapter 7

Multiple Instruments and Sizes

7.1 One Portfolio, Many Possible Requests

A multi-instrument dealer receives requests for particular instruments and quantities, while managing the inventory of the whole portfolio. A quote for one instrument can therefore depend on positions in other instruments. A request twice as large can change more than the cash earned on execution: it changes the position reached after execution, the capacity available for later requests, and potentially the probability of winning.

The accounting from [Chapter 6](#) survives this extension. The decision still compares an immediate trading contribution with a continuation-value difference. What changes is the set of neighboring portfolio states and the distribution of requests that can reach them.

This chapter develops that extension and then asks when a portfolio can be represented by a smaller factor inventory. The distinction is between a factor that summarizes part of the risk and a state that contains all the information needed for the trading decision. The first does not automatically provide the second.

We use two instruments and two trade sizes so that every state and action can be inspected. The full model is small enough to solve and evaluate directly. A compressed model provides a faster policy whose decisions can be checked against that full benchmark. A separate experiment shows exact factor reduction when the rewards, transitions, and constraints satisfy the required conditions.

Throughout, reference prices, covariances, costs, and win curves are supplied inputs. The numerical parameters are educational constructions. The goal is to make the connection between portfolio risk, trade size, state representation, and decision quality explicit. The variance estimators added in [Chapter 5](#) are a separate experiment: their forecasts do not replace the fixed matrices used in this chapter's verified tables.

7.2 A Request Has an Instrument, a Direction, and a Size

Let $q = (q_1, \dots, q_d)^\top$ be inventory across d instruments. A request carries a mark $m = (i, \eta, z)$: an instrument i , a dealer direction η , and a positive size z . The direction convention remains +1 for a dealer purchase and -1 for a dealer sale. The full execution moves inventory by

$$m = (i, \eta, z), \quad d_m = \eta z e_i, \quad q^m = q + d_m. \quad (7.1)$$

The vector e_i has a one in coordinate i and zeros elsewhere. The requested size is observed before quoting. It is not a quantity selected by the dealer in the example. A winning response executes the entire request; a loss or decline leaves inventory unchanged.

There are two equivalent ways to specify the request distribution. One can give an arrival rate $\lambda_{i,\eta}$ for an instrument and direction together with a conditional size distribution $\pi_{i,\eta}(z)$. Alternatively, one can give a separate arrival rate ν_m for every complete mark:

$$\nu_{i,\eta,z} = \lambda_{i,\eta} \pi_{i,\eta}(z), \quad \sum_{z \in \mathcal{Z}_{i,\eta}} \pi_{i,\eta}(z) = 1. \quad (7.2)$$

Once ν_m includes the size probability, multiplying it by that probability again would change the request model. The total request rate and the execution intensity of a quoted mark are

$$\nu = \sum_m \nu_m, \quad \lambda_m^{\text{exec}}(t, q) = \nu_m p_m(\delta_m(t, q)) \quad \text{when responding.} \quad (7.3)$$

Here p_m is a probability conditional on the complete arrived request and the chosen quote. Decline sets its execution intensity to zero. The mark rate counts requests, including requests that are subsequently lost or declined.

Trade size also matters when converting counts to quantities. For a fixed response policy, let r_m be its response indicator. Its expected rate of gross executed units and its expected inventory-change rate are different objects:

$$\begin{aligned} \text{gross executed-unit rate} &= \sum_m \nu_m r_m p_m(\delta_m) z_m, \\ \text{inventory-change rate} &= \sum_m \nu_m r_m p_m(\delta_m) d_m. \end{aligned} \quad (7.4)$$

The second expression is a vector and includes the trade direction. Opposite flows can cancel in that vector while still producing substantial gross turnover. In a general portfolio, units from different instruments may not be economically comparable; our

synthetic example uses a common normalized trading unit so that its gross-unit count is interpretable.

The size distribution in Bergault and Guéant is a probability measure, while their execution kernel also contains a quote-dependent intensity. Their HJB integrates a size-weighted Hamiltonian against that measure. Our discrete sum uses complete mark rates and conditional win probabilities instead. Keeping track of where each probability is included is essential when translating between the two forms. [B, Section 2.1 and Remark 1, p. 3; equation (2), p. 5]

As in Chapter 6, an arrived request is answered and its full execution outcome resolved within one modeled branch. The marked request rate and conditional execution probability are separate inputs; delayed completions would require additional state and a separate physical-execution clock. The size distribution enters the marked rate once, as established in Chapter 3.

7.3 The Same Bellman Logic with More Neighbors

Let S_i be the reference price of the requested instrument and $Q_m = S_i - \eta\delta$ the transaction price. The per-unit execution cost c_m and per-response cost K_m may vary across marks. If I_m is the fill indicator for a response, then

$$\Delta(X + q^\top S) = I_m z_m (\delta - c_m) - K_m, \quad \Delta q = I_m d_m. \quad (7.5)$$

The price-level terms cancel between cash and the newly marked position exactly as in Chapter 6. A response pays K_m whether it wins or loses. Decline incurs no response cost.

We keep an additive expected-wealth objective with running penalty $\psi(q)$ and terminal penalty $\ell(q)$. With martingale reference prices and inputs independent of absolute cash and price levels, the reduced value is $\theta(t, q)$. Its terminal condition is $-\ell(q)$. Correlation matters through the chosen portfolio penalties even though the expected martingale price contribution is zero.

For an assigned continuation vector w , define the mark-specific continuation difference and effective unit cost:

$$\Delta_m w(q) = w(q + d_m) - w(q), \quad c_{\text{dyn},m}(q; w) = c_m - \frac{\Delta_m w(q)}{z_m}. \quad (7.6)$$

The difference is a currency value for the whole trade. Dividing by z_m converts it to a per-unit cost that can be compared with a quoted price offset. Omitting that division would make the quote depend on an inconsistent mixture of total currency and currency per unit.

The best incremental value of an arrived marked request is

$$\mathcal{H}_m(q; w) = \max \left\{ 0, \max_{\substack{\delta \in \mathcal{D}_m \\ q + d_m \in \mathcal{Q}}} [p_m(\delta) \{z_m(\delta - c_m) + \Delta_m w(q)\} - K_m] \right\}. \quad (7.7)$$

Each mark has its own maximization because the instrument, direction, and size are known when the dealer responds. The current request is not a choice between executing several different sizes. The dealer chooses a quote or decline for the size requested.

For constant mark rates, the multi-instrument HJB is

$$0 = \partial_t \theta(t, q) - \psi(q) + \sum_m \nu_m \mathcal{H}_m(q; \theta(t, \cdot)), \quad \theta(T, q) = -\ell(q). \quad (7.8)$$

The mark index replaces the two-direction sum from Chapter 6. Larger sizes reach states farther away in the inventory grid; different instruments reach states along different coordinate directions. The optimization at each request remains a comparison of current contribution and future opportunity.

The categorical time discretization is equally direct:

$$\theta_n(q) = \theta_{n+1}(q) + h \left[-\psi(q) + \sum_m \nu_m \mathcal{H}_m(q; \theta_{n+1}) \right], \quad h\nu \leq 1, \quad \theta_N(q) = -\ell(q). \quad (7.9)$$

As before, this is exact for the model with at most one marked request per interval, beginning-of-interval holding cost, and end-of-interval request resolution. It approximates the continuous-time Poisson model. All states at a time step use the same next-time vector.

Guéant's multi-asset Model B provides the additive finite-shift structure for instrument-specific fixed trade sizes. Allowing several possible sizes within an instrument adds request marks to that structure. It is a substantive extension of the constant-size assumption, rather than a change in the unit used to measure a single fixed lot. [M, equations (5.1)-(5.4), p. 24, and (5.9)-(5.12), p. 26; G, footnote 15, printed p. 222, PDF p. 247]

7.4 Three Ways Size Can Change the Quote

Hold the current request type fixed except for size. Increasing size changes the total quote revenue, may change the conditional win curve, and changes the post-trade inventory. These effects need not move the optimal per-unit offset in the same direction.

At the terminal limit, take the quadratic penalty $\ell(q) = \alpha_T q^\top \Gamma q / 2$, where Γ is a supplied covariance metric. Expanding the quadratic at the full post-trade position gives

$$\ell(q + \eta z e_i) - \ell(q) = \eta \alpha_T z (\Gamma q)_i + \frac{\alpha_T z^2}{2} \Gamma_{ii}. \quad (7.10)$$

The associated inventory charge per unit is

$$u_{i,\eta,z}(q) = \eta \alpha_T (\Gamma q)_i + \frac{\alpha_T z}{2} \Gamma_{ii}. \quad (7.11)$$

The first term prices the interaction with existing portfolio inventory. The second is the finite-size contribution of the request itself. Doubling size doubles the latter per-unit term, even at zero inventory. The effective cost at terminal continuation is $c_m + u_m$.

For the logistic curve $p_m(\delta) = 1/[1 + \exp((\delta - a_m)/b_m)]$ with $b_m > 0$, the continuous interior optimum still satisfies

$$\frac{[\delta_m^* - c_{\text{dyn},m}][1 - p_m(\delta_m^*)]}{b_m} = 1. \quad (7.12)$$

The continuation charge changes $c_{\text{dyn},m}$, while a size-dependent win curve changes a_m or b_m . Holding those inputs fixed and changing only the multiplicative quantity would leave the best offset conditional on responding unchanged. That restricted result does not apply when quantity changes the continuation state.

A fixed response cost has another role. Because K_m is paid once per response, its cost per requested unit is K_m/z_m . It can affect whether the dealer responds. When it is independent of the chosen quote, it does not alter the best quote conditional on responding to that fixed mark.

Our controlled experiment starts from the flat portfolio $q = (0, 0)^\top$ and considers a purchase of instrument 1. It first changes size from 5 to 10 while keeping the win curve fixed, and then changes the large request's win-curve location. The notebook reports both continuous and actual grid choices. The distinction matters because the quote tick can conceal a change in the continuous optimum.

Request (z, a)	Continuous offset	Grid offset	Grid advantage
(5,3)	3.266249	3.25	3.830956
(10,3)	3.567143	3.50	5.663110
(10,2.5)	3.404674	3.50	4.034121

The advantage is the expected trading contribution plus the terminal continuation change for one arrived request, measured in JPY thousands. Moving from the first row to the second raises the inventory charge per unit from 0.5 to 1.0. Moving from the second row to the third leaves that charge unchanged and shifts the win curve. The last two continuous offsets differ, but both select 3.50 on the quote grid.

An observed size effect in data cannot, by itself, identify which of these mechanisms produced it. Larger requests may also come from different clients or at different market times. The calculations here isolate mechanisms by changing explicit model inputs.

7.5 Correlation Connects the Instrument Quotes

The local inventory of the requested instrument is not generally sufficient to determine its risk charge. Equation (7.11) uses $(\Gamma q)_i$, which contains covariances with all other held instruments.

Represent the supplied covariance as a factor component and a positive semidefinite residual:

$$\Gamma = D\Omega D^\top + \Psi, \quad f = D^\top q, \quad \Omega \succeq 0, \quad \Psi \succeq 0. \quad (7.13)$$

The factor inventory f summarizes the exposure to the factor shocks. The quadratic risk metric then splits as

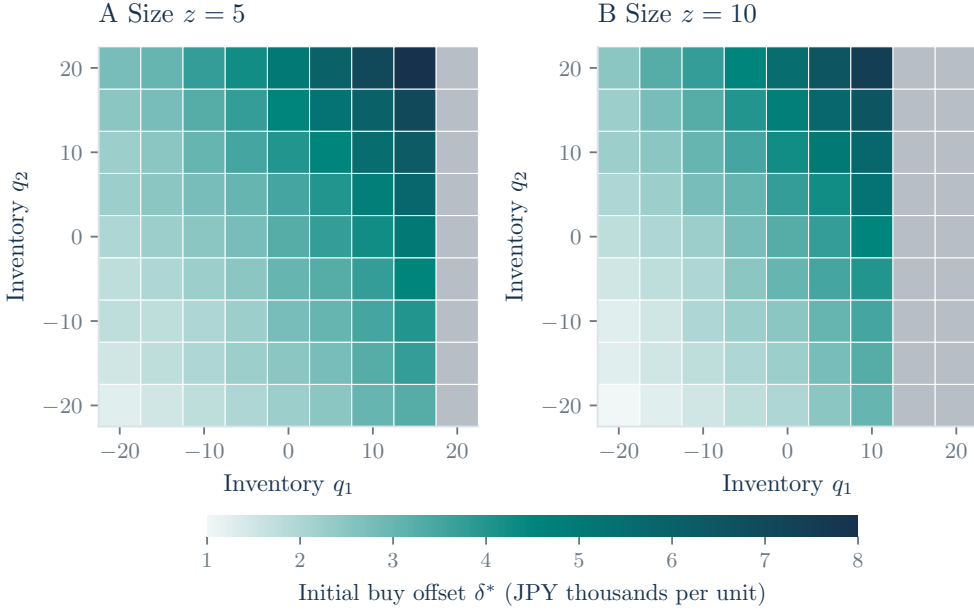
$$q^\top \Gamma q = f^\top \Omega f + q^\top \Psi q. \quad (7.14)$$

In the numerical example, $D = (1, 1)^\top$, $\Omega = 0.8$, and $\Psi = 0.2I$. Thus Γ has diagonal entries 1 and off-diagonal entries 0.8. Two long positions reinforce the common exposure; a long and a short position can offset that common component while retaining residual risk.

For $q = (10, -10)^\top$, factor inventory is zero but the full quadratic risk is 40, so the terminal penalty is 4. The covariance-weighted inventory is $\Gamma q = (2, -2)^\top$. A purchase of instrument 1 adds to the positive residual position, while a purchase of instrument 2 reduces the negative one.

With $\alpha_T = 0.2$, the terminal inventory charges per unit for purchases of sizes 5 and 10 are respectively 0.9 and 1.4 in instrument 1, and 0.1 and 0.6 in instrument 2. These are inventory charges before adding the instruments' execution costs. The same factor inventory has not made the two requests equivalent.

There is no cross-instrument trade in this example: an RFQ changes only one coordinate of q . Cross-instrument effects arise because that coordinate change is valued against a portfolio objective and the future opportunities associated with the whole state.



Instrument 1, $t = 0$, $N = 400$. Gray cells: infeasible full requests. Inventories are trading units. The sizes also use different win curves: $a_5 = 3$, $a_{10} = 2.5$.

Figure 7.1: Full-state optimal buy offsets for instrument 1 use a shared color scale; gray cells indicate infeasible trades

7.6 When Factor Inventory Is a Sufficient State

Two portfolios belong to the same factor class when they have the same value of $D^\top q$:

$$q \sim \tilde{q} \iff D^\top q = D^\top \tilde{q}. \quad (7.15)$$

To solve the entire control problem using only f , the model must assign equivalent future decisions to portfolios in the same class. A factor-only risk penalty is part of a transparent sufficient set of conditions for our proposed reduction, but it is not the only condition.

Require the terminal penalty and running penalty to depend on q only through f . For each mark, require the arrival rate, win probability, execution cost, and response cost to depend on inventory at most through f . Finally, require the same marked quotes to be feasible at every full state with that factor inventory. These mark-by-mark conditions are sufficient; other exact reductions may follow from agreement of aggregate rewards and projected transitions without agreement of every individual input.

Different instruments are still allowed to have different costs, arrival rates, or win curves. The request mark remains known in the reduced problem. The restriction is that a given mark cannot depend on hidden inventory information within a factor class.

The factor transition of a filled request is determined without recovering the full portfolio:

$$f^m = f + D^\top d_m = f + \eta z b_i, \quad b_i = D^\top e_i. \quad (7.16)$$

If the conditions above hold, let $\bar{\psi}(f)$ and $\bar{\ell}(f)$ be the corresponding penalties. Define a reduced request value and HJB by

$$\begin{aligned} \bar{\mathcal{H}}_m(f; \varphi) &= \max \left\{ 0, \max_{\delta \in \bar{\mathcal{A}}_m(f)} [p_m(\delta) \{z_m(\delta - c_m) + \varphi(f^m) - \varphi(f)\} - K_m] \right\}, \\ 0 &= \partial_t \varphi(t, f) - \bar{\psi}(f) + \sum_m \nu_m \bar{\mathcal{H}}_m(f; \varphi(t, \cdot)), \\ \varphi(T, f) &= -\bar{\ell}(f). \end{aligned} \quad (7.17)$$

The reduced action set $\bar{\mathcal{A}}_m(f)$ includes the post-trade factor constraint. It is empty for a blocked request. The notation in (7.17) uses the constant mark parameters of our experiment; the same construction permits factor-dependent inputs when they agree within each class.

Exactness can first be seen in the discrete Bellman recursion. If next-time values are functions of factor inventory, then the terminal data, immediate rewards, continuation transitions, and feasible maximizations all depend only on that same inventory. Applying the Bellman update therefore produces another function of factor inventory:

$$\mathcal{B}_h(\varphi \circ D^\top) = (\bar{\mathcal{B}}_h \varphi) \circ D^\top, \quad \theta_n(q) = \varphi_n(D^\top q). \quad (7.18)$$

Start with the factor-only terminal penalty and repeat this argument backward. It proves equality of the value functions and of the sets of optimal actions on the relevant states. A shared tie convention then selects the same action. Under conditions ensuring uniqueness of the continuous-time value problem, the corresponding HJB reduction follows as well.

This is an information condition, not just a statement about the number of factors in a covariance model. Bergault and Guéant obtain a factor-space equation when the residual covariance is removed and the inventory penalties have the required factor form. Their reduced equation retains instrument-specific intensities and size distributions. [B, Section 3.1, equations (3)-(4), pp. 11-12]

7.7 Residual Risk and Limits Test the Reduction

A nonzero residual matrix does not automatically prevent reduction through a chosen D . What matters is whether it distinguishes portfolios in the same factor class. For a positive semidefinite Ψ on the full inventory space, its quadratic form is a function of $D^\top q$ precisely when

$$\ker(D^\top) \subseteq \ker(\Psi). \quad (7.19)$$

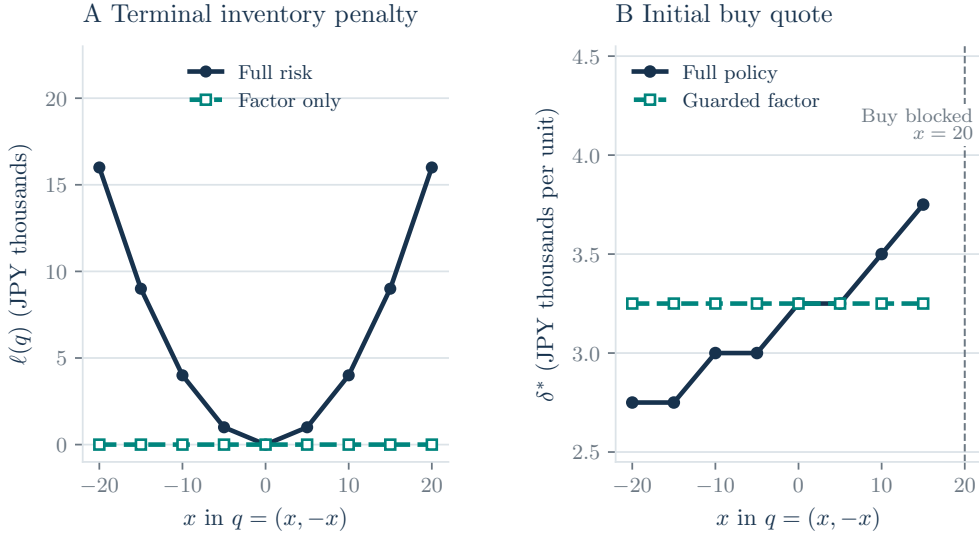
To see necessity, compare zero inventory with a vector h in $\ker(D^\top)$. They have the same factor inventory, so factor-only residual risk would require $h^\top \Psi h = 0$. Positive semidefiniteness implies $\Psi h = 0$. Conversely, if every such hidden direction is annihilated by Ψ , adding h to any portfolio leaves the residual quadratic form unchanged.

In our example, $h = (1, -1)^\top$ has zero common-factor exposure but $\Psi h = 0.2h$ is nonzero. Residual risk therefore distinguishes portfolios in the same factor class. The proof above concerns the full vector space; on a restricted set of reachable states, sufficiency can be assessed directly on the classes that actually occur.

Instrument limits create a separate failure. Under the full limits $|q_1| \leq 20$ and $|q_2| \leq 20$, consider $(20, 0)$ and $(0, 20)$. Both have $f = 20$. A five-unit purchase of instrument 1 is forbidden at the first state and permitted at the second. The proposed factor state cannot determine the action set.

The projected factor range $|f| \leq 40$ does not repair this. Both candidate purchases lead to $f = 25$, within that range, while only one full position passes the instrument limits. A limit on aggregate factor exposure and individual position limits are different restrictions.

These two tests should be kept separate. Removing residual risk would not repair the incompatible instrument action sets. Replacing instrument limits with compatible factor limits would not remove residual risk that varies within a factor class.



All states have $f = q_1 + q_2 = 0$, but full residual risk varies with x .
 Right: instrument 1 buys size 5 at $t = 0$; both policies obey the full inventory limit.
 $N = 400$; x is in trading units. Dots are grid states; lines guide the eye.

Figure 7.2: Portfolios with the same common-factor inventory can have different residual penalties and optimal quotes

7.8 Two Instruments and Two Sizes on a Small Grid

The full numerical model uses $q_1, q_2 \in \{-20, -15, \dots, 20\}$, giving 81 states. Each instrument has both dealer directions and requested sizes 5 and 10, giving eight marks. Quotes lie from -6 to 10 in increments of 0.25 .

Input	Value	Interpretation
$\nu_{i,\eta,5}$	0.5	Small-request rate for each instrument and direction
$\nu_{i,\eta,10}$	0.25	Large-request rate for each instrument and direction
a_5, a_{10}, b	3, 2.5, 1	Size-dependent logistic location and common scale
c_1, c_2, K_m	1, 1.2, 0	Instrument execution costs and response cost
T, N	1, 400	Horizon and baseline number of time steps
$\psi(q)$	$0.01q^\top \Gamma q$	Running penalty
$\ell(q)$	$0.10q^\top \Gamma q$	Terminal penalty

There are four small-mark streams and four large-mark streams. Their rates sum to 3. Equivalently, each instrument-direction pair has rate 0.75 and size probabilities $2/3$ and $1/3$. At the baseline step $h = 1/400$, the no-request probability is 0.9925.

Prices, offsets, and execution costs are in JPY thousands per normalized trading unit. Γ is a supplied price-shock covariance over a fixed reference horizon, in squared price units. Consequently, $q^\top \Gamma q$ is in currency squared. The coefficient $\alpha_T = 0.2$ converts it to a terminal currency penalty; the coefficient $\kappa = 0.02$ converts it to a currency penalty per normalized unit of time, with the factor of one half included in both formulas.

The reference covariance horizon is fixed. It is not automatically the remaining trading horizon, and the implementation does not multiply it again by the remaining time. The penalty coefficients are chosen inputs, not an estimate of an exponential-utility parameter. Their units distinguish the terminal monetary charge from the running monetary charge per unit time; the covariance matrix itself is not a running cost or a covariance rate.

The model solves the full Bellman recursion, stores the selected action for each time, state, and mark, and then evaluates comparison policies independently. It also repeats the calculation on different time grids. The notebook in WSL recomputes the arrays and exercise answers using the same inputs.

Decline is selected when the best response advantage is zero. Among tied quote choices, the smaller offset is selected. The numerical implementation applies a currency tolerance of 10^{-12} to these comparisons.

At the initial state $q = (10, -10)^\top$, the full dynamic policy chooses the following per-unit offsets. All eight requests are feasible here.

Instrument and dealer direction	Size 5 offset	Size 10 offset
1, buy	3.50	3.50
1, sell	3.00	3.00
2, buy	3.00	3.00
2, sell	3.50	3.75

The directions refer to the dealer. A smaller offset makes either a buy or a sell response more competitive under the quote convention. Different offsets across instruments and sizes reflect the full continuation problem, including the instrument costs and the conditional win curves.

Symmetries must be checked against the actual parameters. Reversing all inventory signs and all dealer directions preserves this symmetric buy/sell model. Swapping instruments is not a symmetry of the baseline because their execution costs differ.

7.9 An Exact Reduction on Reachable States

Before assessing approximation error, the experiment verifies a case in which reduction should be exact. Set the residual covariance to zero, so that $\Gamma = 0.8DD^\top$, and impose only the factor limit $|q_1 + q_2| \leq 20$. Keep the instrument- and size-specific request parameters. Penalties and feasible marked actions now depend on inventory through f alone.

The reduced grid has nine factor states, from -20 to 20 in steps of 5 . For a transparent finite calculation, use a horizon of 0.1 and two categorical time steps. The corresponding full-state calculation is stored on a wider grid with individual coordinates from -40 to 40 , intersected with the factor limit.

That storage boundary is not an intended economic instrument limit. Its irrelevance is checked on all reachable state layers from five starting portfolios: $(0, 0)$, $(10, -10)$, $(15, -15)$, $(10, 0)$, and $(15, -5)$. There can be at most two full trades, each of size at most 10 . Starting from an absolute coordinate at most 15 , no such path can exceed 35 in absolute value.

The calculation compares full and reduced values on all those reachable layers, and every marked action and post-trade feasibility on the layers before the terminal time. Equality is asserted only there. It is not asserted at unrelated full-grid states close to the artificial storage boundary, where the truncation could change available actions.

The first three starting states all have factor inventory zero; the last two have factor inventory ten. They therefore have equal values within each group under this compatible model. Different costs for the two instruments remain present and do not invalidate the reduction because those costs are still known from the request mark.

The implementation checks 98 reachable state-time pairs. The largest full-versus-factor value difference is zero at floating-point precision, all compared marked actions and feasibility decisions agree, and the largest absolute reachable coordinate is 35 . At factor inventory zero, each of the three starting portfolios has value 1.203992 . At factor inventory ten, each of the other two has value -6.533846 . These are values for the separate two-step compatible model.

This check does not establish exactness of the baseline model. It verifies the implementation against the sufficiency argument in [Section 7.6](#) after deliberately changing the residual covariance and constraints to satisfy its assumptions.

7.10 Turning a Compressed Model into a Feasible Policy

Return to the full baseline covariance and instrument box. To construct a smaller approximate model, remove the residual component and retain only factor inventory and its projected range:

$$\Gamma_F = 0.8DD^\top, \quad \bar{\psi}(f) = 0.008f^2, \quad \bar{\ell}(f) = 0.08f^2, \quad \mathcal{F} = \{-40, -35, \dots, 40\}. \quad (7.20)$$

This factor model has 17 states. It uses the same eight marks, quote grid, request rates, win curves, costs, horizon, and time step as the full baseline. It has different penalties and a different constraint representation. Its value function is therefore the value of a different model.

Let $\pi_n^F(f, m)$ be its chosen response to a mark. To use that choice in the full portfolio, apply the actual instrument limit before executing the suggested response:

$$\hat{\pi}_n^F(q, m) = \begin{cases} \pi_n^F(D^\top q, m), & q + d_m \in \mathcal{Q}, \\ \text{decline}, & q + d_m \notin \mathcal{Q}. \end{cases} \quad (7.21)$$

The guard makes the deployed policy feasible. It does not make the compressed continuation values correct. Those values still anticipate opportunities using the factor model's penalties and action sets. In particular, preventing an infeasible current trade does not restore information about which future requests will be blocked.

Evaluate this guarded policy, the static-terminal-penalty policy, and the full dynamic policy under the same full-state model. The static policy chooses each response using $w = -\ell$ at every time, without solving for future RFQ opportunities. For a fixed policy π , define its expected immediate reward b_n^π and full-state transition matrix P_n^π . The evaluation is

$$U_n^\pi = b_n^\pi + P_n^\pi U_{n+1}^\pi, \quad U_N^\pi(q) = -\ell(q). \quad (7.22)$$

If the policy responds to mark m at q , write $r_{n,m}(q) = 1$ and use its chosen $\delta_{n,m}(q)$. If it declines, the corresponding contribution is zero. For $q' \neq q$, the transition and immediate reward are

$$\begin{aligned} P_n^\pi(q, q') &= h \sum_{\{m: q+d_m=q'\}} \nu_m r_{n,m}(q) p_m(\delta_{n,m}(q)), \\ P_n^\pi(q, q) &= 1 - \sum_{q' \neq q} P_n^\pi(q, q'), \\ b_n^\pi(q) &= h \left[\sum_m \nu_m r_{n,m}(q) \{p_m(\delta_{n,m}(q)) z_m[\delta_{n,m}(q) - c_m] - K_m\} - \psi(q) \right]. \end{aligned} \quad (7.23)$$

This sums all marked routes to a destination state, rather than assuming there can be only one. No-request events, declines, and lost quotes contribute to the unchanged-state probability. The running and terminal penalties use the full Γ , including residual risk, for every policy.

The statewise policy loss is measured against the full optimizer:

$$L_n^\pi(q) = \theta_n^{\text{full}}(q) - U_n^\pi(q) \geq 0. \quad (7.24)$$

The inequality follows from optimization over the same feasible full-state actions, with the same objective and categorical timing. It is checked numerically, along with equality between Bellman’s value and independent evaluation of the stored maximizing policy.

A difference between the factor model’s own value $\varphi_n(D^\top q)$ and the full value is not this policy loss. Such a difference mixes changed risk penalties, changed constraints, and the resulting policy. The meaningful performance test is to evaluate the implemented guarded policy with the full objective.

Expected trading contribution and expected holding and terminal penalties are retained separately. A policy can earn more trading contribution and still have a lower value under the chosen portfolio objective. Conversely, a higher penalized value is not automatically higher cash income.

Starting inventory	Policy	Full-model value	Loss
(0,0)	Full dynamic	11.639732	0.000000
(0,0)	Static penalty	11.598866	0.040866
(0,0)	Guarded factor	11.600877	0.038855
(10,-10)	Full dynamic	7.382985	0.000000
(10,-10)	Static penalty	7.354166	0.028819
(10,-10)	Guarded factor	7.123196	0.259789

All values and losses are in JPY thousands. The full dynamic policy’s own loss is shown as zero after rounding numerical evaluation error. At (0,0), the guarded factor policy earns 15.687556 of expected trading contribution, compared with 15.403206 for the full optimizer. Its running and terminal penalties are also higher: 0.218019 and 3.868660, versus 0.203273 and 3.560201. Subtracting those penalties reverses the ranking of the trading contributions.

At (10,−10), the guarded policy’s loss is 0.259789, substantially larger than its 0.038855 loss at flat inventory. Those portfolios have the same factor inventory. The comparison shows why performance must be assessed across full portfolios, even if the approximate solver observes only the factor.

7.11 Accuracy and Computation Must Be Reported Together

For an exhaustive quote-grid Bellman solver, a useful work count is

$$C_{\text{full}} = O(NMA|\mathcal{Q}|), \quad C_{\text{factor}} = O(NMA|\mathcal{F}|), \quad (7.25)$$

Here M is the number of marked request types and A the number of quote choices per mark. In the baseline, $M = 8$, $A = 65$, $|\mathcal{Q}| = 81$, and $|\mathcal{F}| = 17$. The state-count ratio is about 4.76. It is not a guaranteed speedup: array operations, setup, memory access, and fixed overhead also affect elapsed time.

The notebook records median solver times from three measured runs after one warmup per model in the existing WSL Python environment. Both solvers use the same time and quote grids. The recorded medians are 46.02 ms for the full model and 14.47 ms for the factor model, a ratio of 3.18. This run used Python 3.12.3 and NumPy 2.4.6 on an Intel Core i7-14700 in WSL2.

Timing includes value and policy array allocation and the full backward recursion on already constructed models. It excludes model and probability-table construction, policy lifting and guarding, full-state evaluation, plotting, notebook rendering, and file operations. These measurements describe this small workload, not production latency. The notebook preserves the recorded measurement used in the figure and provides an explicit option to remeasure.

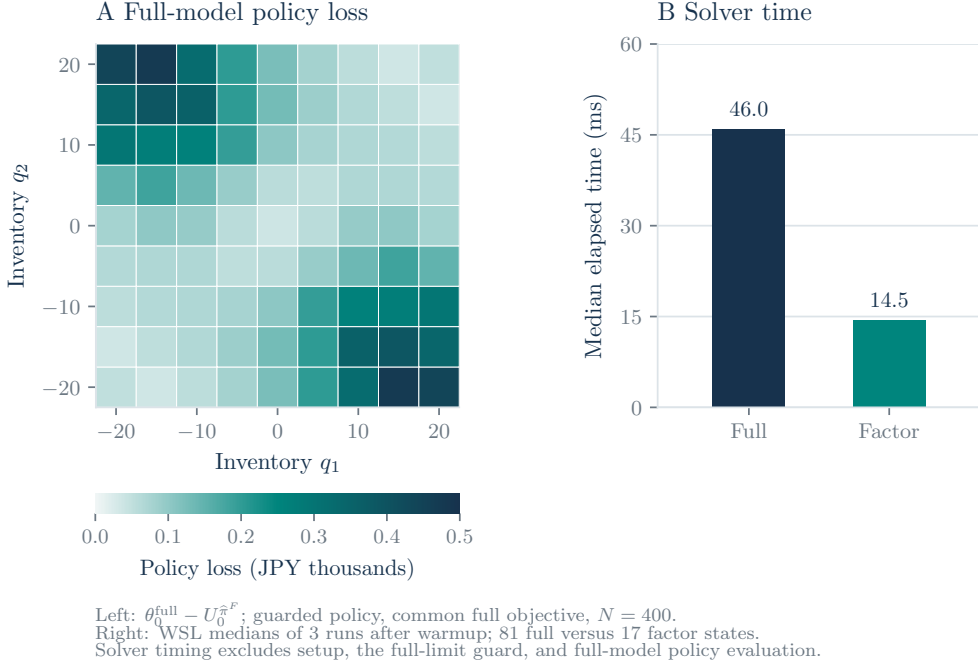


Figure 7.3: Full-model loss of the guarded factor policy and separately measured solver time; timing excludes guarding and policy evaluation

Computation saved by solving the factor model must be assessed together with its statewise decision loss. A small loss at zero inventory does not establish a small loss near a limit or in a portfolio with large offsetting residual positions. The comparison therefore reports the loss over the entire 81-state baseline grid.

Across the initial full-state grid, the guarded factor loss ranges from 0.036761 to 0.475072 JPY thousands. One state attaining the largest loss is (15,-20). The guard overrides 14.20% of the raw factor actions across the enumerated time-state-mark combinations. This is an unweighted grid diagnostic; it is not the probability that the guard will intervene along a realized trading path.

Time discretization is a separate source of error. Keeping the covariance, inventory set, request marks, and quote grid fixed, compare full-state solutions using $N = 100, 200, 400, 800$:

$$E_N = \max_{q \in \mathcal{Q}} \left| \theta_0^{(2N)}(q) - \theta_0^{(N)}(q) \right|. \quad (7.26)$$

This comparison measures changes with time resolution. It does not measure the error from discarding residual risk or changing the constraint representation. We use E_{100} , E_{200} , and E_{400} ; computing E_{800} would require another solution with 1600 steps.

Coarse steps N	E_N	Changed initial actions
100	0.016770	0.309%
200	0.008377	0.617%
400	0.004187	0.000%

The action comparison gives equal weight to all 81 initial states and eight marks, including blocked requests. The value differences approximately halve when the time step halves in this example. Initial action changes need not decrease monotonically: a small value change can cross a quote-grid tie. The 400-step and 800-step solutions choose the same initial actions while retaining a nonzero value difference.

Likewise, agreement between two initial quotes does not imply equality of policies. They may differ at another inventory, for another size, or later in the horizon. Value differences, action differences, and feasibility corrections answer different questions and should be reported separately.

Passing the numerical checks establishes the behavior of the specified models and their implementations. It does not calibrate the covariance, estimate the request-size distribution, or show that a chosen compressed policy remains reliable when the input model is wrong.

7.12 What Carries Forward

The multi-instrument RFQ decision combines marked request rates, conditional win probabilities, trade-sized inventory transitions, and continuation values. A factor state gives an exact reduction when it preserves the information needed for rewards, transitions, terminal values, and feasible actions.

When those conditions fail, a factor policy is an approximation. Apply the full trading limits and evaluate its decisions under the intended portfolio objective. The small-grid benchmark makes both the computational saving and the decision loss visible.

Chapter 8 separates inventory-driven quote adjustments from changes driven by the direction and composition of client flow. Similar-looking quote skews can arise from different states, assumptions, and forecasts.

Exercises

1. Convert an instrument-direction rate of 0.75 and size probabilities $2/3$ and $1/3$ into the two marked rates. Sum across both instruments and directions. At $N = 400$, calculate all categorical event probabilities and the rate of gross requested units. Explain why that quantity rate is not an execution rate.
2. At $q = (10, -10)^\top$, calculate Γq and the terminal inventory charge per unit for a purchase of each instrument at each size. Verify the charges against direct differences of the quadratic penalty.
3. Starting from $q = (0, 0)^\top$, compare purchases of instrument 1 with $(z, a) = (5, 3), (10, 3), (10, 2.5)$ using terminal continuation. Report continuous and grid-optimal offsets and expected response advantages. Separate the effects of size, the win curve, and the finite-trade penalty.
4. Compare the first-instrument five-unit purchase at $q = (20, 0)$ and $q = (0, 20)$. Explain why the portfolios' common factor inventory and the projected factor cap cannot determine the correct full-state feasibility.
5. Reproduce the two-step exact-reduction experiment. Check values, actions, and feasibility on the reachable layers from the five stated starting portfolios. Explain why the individual storage bounds cannot bind there, and why equality should not be claimed over the entire truncated grid.
6. At $q = (0, 0)$ and $q = (10, -10)$, evaluate the full dynamic, static-penalty, and guarded factor policies under the full objective. Separate expected trading contribution, holding penalty, terminal penalty, and value. Compute the guarded factor policy's statewise loss.
7. Compare the state-count ratio with the measured solver-time ratio. State what is included in the timing. Explain why the factor model's own value cannot substitute for the guarded policy's full-model performance.
8. Compute E_{100} , E_{200} , and E_{400} and inspect initial quote changes. Distinguish time-discretization error, quote-grid effects, and information lost by factor compression.

Source Notes

[B] Philippe Bergault and Olivier Guéant, *Size matters for OTC market makers: general results and dimensionality reduction techniques*, arXiv:1907.01225v5, 21 September 2022, supplied 42-page PDF. Printed and PDF page numbers agree. Section 2.1 and Remark 1, p. 3, distinguish RFQ arrival, conditional acceptance, and size probability measures. Equation (1), p. 4, gives the additive inventory-penalty objective; equation (2), p. 5, gives the full inventory HJB with size integration. Section 3.1, equations (3)-(4), pp. 11-12, gives the zero-residual factor model and its reduced equation. Section 3.2, pp. 12-14, discusses residual risk and a formal correction. Available from [arXiv](#).

[M] Olivier Guéant, *Optimal market making*, arXiv:1605.01862v5, 6 May 2017, supplied 43-page PDF. Printed and PDF page numbers agree. Equations (5.1)-(5.4), p. 24, specify multi-asset inventories, reference prices, execution intensities, and cash with fixed instrument-specific lots. Model B on p. 25 and equations (5.9)-(5.12), p. 26, provide the additive multi-asset objective and finite-shift HJB. Equation (5.13), p. 27, normalizes continuation differences by trade size; equation (5.14) gives the terminal condition. Available from [arXiv](#).

[G] Olivier Guéant, *The Financial Mathematics of Market Liquidity: From Optimal Execution to Market Making*, CRC Press, 2016, supplied 302-page PDF. Footnote 15, printed p. 222, PDF p. 247, distinguishes rescaling a fixed trade size from allowing size to vary across trades. Section 11.3 provides multi-instrument context; its CARA formulation is not the additive objective implemented here.

The eight marked RFQ streams, logistic parameters, two covariance regimes, explicit decline and response-cost treatment, finite grids, guarded factor policy, reachable-layer checks, and numerical comparisons are constructions for this book. The sources motivate the model structure and the factor-reduction conditions. The chapter has not reproduced the sources' numerical studies or independently audited all their proofs.

Chapter 8

Flow Imbalance and Client Relationships

8.1 Four Reasons for a Quote to Move

A dealer with no inventory can still have a reason to shade a quote. If most incoming requests would add inventory, taking the next such trade changes the position from which the dealer must handle the remaining flow. If winning a request also improves access to future requests, the next response can have consequences beyond its immediate margin and inventory change.

These mechanisms should be distinguished from two others: a change in the reference-price forecast and a change in the competition for the currently observed request. A movement in the quoted price alone does not reveal which mechanism caused it.

This chapter isolates incoming flow and access to future requests. Reference prices remain supplied martingale inputs. We first vary the direction of incoming RFQs while leaving conditional win probabilities unchanged. We then introduce an access state that changes after a win, a loss, or a decline. The latter extension changes the state transition even when there is no trade. Risk coefficients are held fixed: the separate volatility forecasts in [Chapter 5](#) introduce neither an unannounced price signal nor an adaptive risk state into these examples.

Keep the dealer-direction convention from earlier chapters. A purchase has $\eta = +1$ and a sale has $\eta = -1$. The corresponding offsets are δ_+ and δ_- . Define the full width W and the midpoint skew s by

$$\begin{aligned} Q_{\text{bid}} &= S - \delta_+, & Q_{\text{ask}} &= S + \delta_-, \\ W &= \delta_+ + \delta_-, & s &= \frac{Q_{\text{bid}} + Q_{\text{ask}}}{2} - S = \frac{\delta_- - \delta_+}{2}. \end{aligned} \tag{8.1}$$

A negative s places the quote midpoint below the reference price. Increasing the buy offset while decreasing the sell offset produces that movement. The two offsets describe responses to the corresponding RFQ directions at the same state; they need not be simultaneous streaming quotes. We report a two-sided width or skew only when both full trades are feasible and both directions receive quoted responses.

8.2 Measure Incoming Flow Before Measuring Fills

Let λ_+ and λ_- be the arrival rates of dealer-buy and dealer-sell requests. Their total rate is λ . Define request imbalance by

$$\beta = \frac{\lambda_+ - \lambda_-}{\lambda_+ + \lambda_-}, \quad \lambda_\eta = \frac{\lambda}{2}(1 + \eta\beta), \quad -1 < \beta < 1. \quad (8.2)$$

Positive β means that customers more often ask to sell to the dealer. It is a tendency for incoming requests to add dealer inventory, before the dealer chooses how to respond.

If N_+ and N_- count eligible observed requests over a measurement window, the simple count estimator is

$$\hat{\beta} = \frac{N_+ - N_-}{N_+ + N_-}, \quad N_+ + N_- > 0. \quad (8.3)$$

This is an estimator for the request population represented by those observations. It is not automatically a market-wide imbalance estimate. Invitation rules, operating hours, instrument filters, and missing requests can change the population. Our numerical experiment supplies β directly and does not estimate it from trading records.

Executed flow has an additional selection mechanism: the dealer's response and win probability. For one fixed trade size z , response indicators a_η , and conditional win probabilities p_η , the request and fill drifts are

$$\begin{aligned} \text{signed requested-unit rate} &= z\lambda\beta, \\ \text{expected inventory-change rate} &= z\{\lambda_+a_+p_+(\delta_+) - \lambda_-a_-p_-(\delta_-)\}. \end{aligned} \quad (8.4)$$

The first counts units that clients ask to trade. The second counts expected changes in the dealer's inventory under a policy. Neither is a reference-price forecast. The rate interpretation in (8.4) uses the model's immediate request-resolution convention. With positive execution delay it instead describes completed quantities attached to request-origin exposure unless the separate physical timing is modeled, as in Chapter 3.

For example, 80 dealer-buy requests and 20 dealer-sell requests give $\hat{\beta} = 0.6$. If only 20 and 10 respectively become fills, the fill-count imbalance is $1/3$. Those are different measurements of different processes. If all requests received responses, the corresponding conditional win rates are 25% and 50%. If some were declined, fills divided by all requests instead measures a combined response-and-win rate.

The same distinction matters for client scores. A win rate among responses, a fill rate among all invitations, and a share of the client's total business have different denominators. A model cannot switch between them without changing its transition assumptions.

8.3 Why an Arrival Rate Drops Out of a Single RFQ

Suppose the request has already arrived. Its side, size, costs, and conditional win curve are known. For a fixed continuation function $w(q)$, its response advantage over decline is

$$A_\eta(q, \delta; w) = p_\eta(\delta) \{z(\delta - c_\eta) + w(q + \eta z) - w(q)\} - K_\eta. \quad (8.5)$$

This is a currency value conditional on that arrived request. There is no arrival rate in it. A positive rate only multiplies the same maximization when converting a conditional advantage into a contribution per unit of time:

$$\arg \max_{a \in \mathcal{A}_\eta(q)} \lambda_\eta A_\eta(q, a; w) = \arg \max_{a \in \mathcal{A}_\eta(q)} A_\eta(q, a; w), \quad \lambda_\eta > 0. \quad (8.6)$$

Here the action set includes decline, assigned zero advantage, and all feasible quoted responses. The equality requires that the continuation function and the other conditional inputs are held fixed.

This gives a useful diagnostic. In an immediate-profit model with fixed win curves and costs, changing only how often a request type arrives does not change the best response to that type. It changes how often that decision is encountered. If a proposed single-RFQ formula changes its conditional optimum merely because the same positive rate multiplies every action, the formula needs another mechanism.

The dynamic model supplies such a mechanism. The arrival rates change the future opportunities, which change w . A rate can therefore affect the current quote through continuation value without entering the conditional maximization directly.

At flat inventory with symmetric request economics and fixed total arrival rate, even the sum of the two instantaneous optimized contributions can remain unchanged as β varies. The contributions assigned to the two sides change in opposite directions. That restricted statement says nothing about the value of carrying the resulting inventory over a finite horizon.

8.4 Future Flow Changes Continuation Value

Let $\theta^\beta(t, q)$ be the reduced value under request imbalance β , running penalty $\psi(q)$, and terminal penalty $\ell(q)$. The HJB becomes

$$\begin{aligned} 0 &= \partial_t \theta^\beta(t, q) - \psi(q) \\ &\quad + \sum_{\eta \in \{-1, +1\}} \lambda_\eta \max \left\{ 0, \max_{\delta \in \mathcal{D}_\eta(q)} A_\eta(q, \delta; \theta^\beta(t, \cdot)) \right\}, \\ \theta^\beta(T, q) &= -\ell(q). \end{aligned} \quad (8.7)$$

The set $\mathcal{D}_\eta(q)$ is empty when the entire requested trade would violate a position limit. In that case the request is declined. Arrival rates may be asymmetric even though win curves, costs, and inventory limits are symmetric across dealer directions.

With at most one categorical request per step, the recursion is

$$\begin{aligned}\theta_n^\beta(q) &= \theta_{n+1}^\beta(q) + h \left[-\psi(q) + \sum_\eta \lambda_\eta \max \left\{ 0, \max_\delta A_\eta(q, \delta; \theta_{n+1}^\beta) \right\} \right], \\ h\lambda &\leq 1, \quad \theta_N^\beta(q) = -\ell(q).\end{aligned}\tag{8.8}$$

The inner maximum is over the feasible quote set from (8.7). Holding cost uses beginning-of-step inventory; the request resolves at the end of the step. This recursion is exact for that categorical model and approximates the continuous-time Poisson model as the time step decreases.

For a logistic win curve with location a and scale $b > 0$, the effective unit cost and continuous interior quote condition are

$$\begin{aligned}c_{\text{dyn},\eta}^\beta &= c_\eta - \frac{\theta^\beta(t, q + \eta z) - \theta^\beta(t, q)}{z}, \\ \frac{(\delta_\eta^* - c_{\text{dyn},\eta}^\beta)[1 - p_\eta(\delta_\eta^*)]}{b} &= 1.\end{aligned}\tag{8.9}$$

There is still no direct λ_η multiplier in the second line. The flow dependence enters through the first line.

At terminal continuation, the rate-dependent future has disappeared:

$$c_{\text{dyn},\eta}(T, q) = c_\eta + \frac{\ell(q + \eta z) - \ell(q)}{z}.\tag{8.10}$$

Consequently, the final categorical-step response choices are independent of β when all the other inputs are held fixed. Their arrival-weighted contribution and the earlier dynamic choices can still differ. This is a useful implementation check, not a claim that flow is irrelevant before terminal time.

In the example, an excess of dealer-buy requests makes additional long inventory harder to manage through the remaining client flow. At the same current inventory, the optimizer can therefore quote less competitively to buy and more competitively to sell. The numerical comparison measures that effect; it does not impose a skew formula in advance.

8.5 What an Exponential Benchmark Explains

A decreasing exponential win curve provides a compact view of how continuation value affects width and skew. Take $p(\delta) = \exp(-\delta/\omega)$ on $\delta \geq 0$, with scale $\omega > 0$, common unit cost c , and zero response cost. For a fixed continuation function w , an interior optimum satisfies

$$\delta_\eta^* = c + \omega - \frac{w(q + \eta z) - w(q)}{z}. \quad (8.11)$$

The nonnegative-offset domain matters: extending this exponential to negative offsets would produce probabilities above one. If the interior expression is negative, the constrained optimizer is at zero. Full-trade inventory limits must also be checked.

Where both unconstrained expressions are interior and both trades are feasible, substitute them into the definitions of width and skew:

$$\begin{aligned} W^* &= 2(c + \omega) - \frac{w(q + z) + w(q - z) - 2w(q)}{z}, \\ s^* &= \frac{w(q + z) - w(q - z)}{2z}. \end{aligned} \quad (8.12)$$

The first difference determines the midpoint shift, while the second difference determines the inventory-dependent part of the width. If w is concave in inventory, its second difference is nonpositive, adding to the width.

For $w(q) = -0.1q^2$, $z = 5$, $c = 1$, and $\omega = 1$, the interior offsets are $2.5 + 0.2q$ to buy and $2.5 - 0.2q$ to sell. Width is 5 and midpoint skew is $-0.2q$. This is a particular constant-width, linear-skew construction. It follows from a quadratic continuation function and interior exponential responses; it is not a general identity for dynamic market making.

At $q = 15$, the unconstrained sell offset is -0.5 . Imposing the zero floor gives offsets 5.5 and 0, hence width 5.5 and skew -2.75 . Even this simple benchmark changes when an action boundary binds.

Cotton's inventory-cost formulation gives related first- and second-difference expressions. Its cost slope has the opposite sign from our actual midpoint skew because the reduced continuation value is the negative of an inventory-cost function. Its exponential hazard scale is also distinct from the dealer's full quoted width. [C, Section 2.2, equations (1)-(2), p. 4]

The numerical experiments below use logistic probabilities, finite horizons, and explicit limits. They compute the imbalance effect by solving the corresponding Bellman problem. They do not append a stationary log-odds or width correction to an already optimized logistic quote.

8.6 A Controlled Flow Experiment

To isolate the direction of future flow, return to one instrument and one trade size. The portfolio grid has nine states, $q \in \{-20, -15, \dots, 20\}$. The reference covariance metric is fixed at $\Gamma = 1$ in squared price units.

Input	Value	Interpretation
z, c, K	5, 1, 0	Full trade size, unit cost, response cost
a, b	3, 1	Logistic win-curve location and scale
λ, β	4; $-0.6, 0, 0.6$	Total request rate and direction scenarios
T, N	2, 400	Horizon and baseline step count
$\psi(q)$	$0.01q^2$	Running penalty per time unit
$\ell(q)$	$0.10q^2$	Terminal inventory penalty
$\mathcal{D}$	$-6, -5.75, \dots, 10$	65 per-unit quote offsets

Inventory is in normalized trading units. Offsets and unit costs are in JPY thousands per trading unit. Values and total penalties are in JPY thousands; time is normalized. The covariance horizon is fixed, and the penalty coefficients are chosen inputs. No additional remaining-time covariance factor is applied.

At the baseline time step, the no-request probability is 0.98. In the $\beta = 0.6$ scenario, dealer-buy and dealer-sell request probabilities are 0.016 and 0.004. Only the direction mix changes between scenarios; the total request count rate and conditional win curve remain fixed.

Compare the flow-aware optimizer with a policy planned under balanced arrivals and with a static-terminal-penalty policy. The balanced-flow policy uses the dynamic solution for $\beta = 0$; the static policy uses $w = -\ell$ at every request. Evaluate both under the actual scenario's rates:

$$U_n^\pi = b_n^\pi + P_n^\pi U_{n+1}^\pi, \quad U_N^\pi(q) = -\ell(q). \quad (8.13)$$

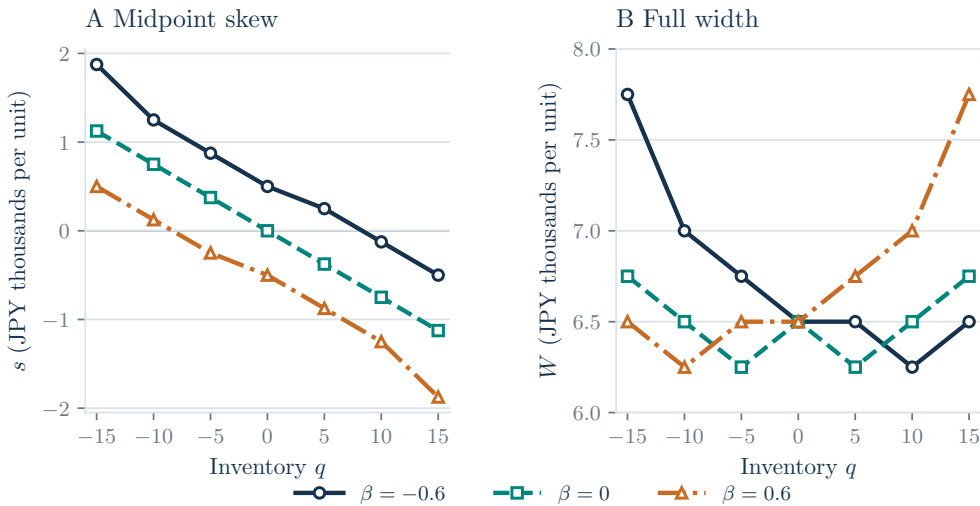
The transition matrix and immediate reward use the actual flow scenario and the stored policy. Expected trading contribution, running penalty, and terminal penalty are recorded separately. This evaluates a feasible policy under the intended model rather than comparing values from different arrival models.

Decline wins a zero-advantage tie; among tied quoted responses, the smaller offset is selected. The numerical comparison tolerance is 10^{-12} in currency units. Quote differences can therefore change by a tick even when values change smoothly.

β	Buy / sell offset	Midpoint skew	Signed fill rate
-0.6	2.75 / 3.75	+0.50	-2.884435
+0.0	3.25 / 3.25	+0.00	+0.000000
+0.6	3.75 / 2.75	-0.50	+2.884435

These initial responses use $q = 0$. Signed fill rate is in units per normalized time; offsets and skew are prices. Requested-unit drift is -12, 0, and +12 respectively. The optimized policy reduces its magnitude without exactly balancing expected fills. A finite-horizon inventory objective does not impose stationary flow balance.

All three quoted widths in this particular table are 6.50. This is a result of these inputs and tick-constrained choices, not a constant-width restriction in the solver. Other states and request mixes can produce different widths.



Initial optimal RFQ offsets; $T = 2$, $N = 400$. Dots are inventory states; lines guide the eye. Both directions are feasible here. At $q = 0$, all three widths equal 6.5 on this quote grid; this coincidence is specific to the example. Inventories are normalized trade units.

Figure 8.1: Controlled changes in request imbalance and inventory separate their effects on quoted width and midpoint skew

Under actual imbalance $\beta = 0.6$ at $q = 0$, the common-model values are:

Policy	Value	Loss
Flow-aware	27.146475	0.000000
Balanced plan	25.814745	1.331731
Static	26.297861	0.848615

Values are in JPY thousands. The balanced-flow planner earns 35.237660 of expected trading contribution, above the flow-aware policy's 33.622058. Its expected running and terminal penalties are also larger: 0.934607 and 8.488308, versus 0.617635 and 5.857948. Subtracting those penalties gives the values shown above.

The static policy performs better than the balanced-flow planner in this scenario. A more elaborate planner can lose value when its assumed future flow is mismatched to the evaluation model.

8.7 Access to Future Requests as a State

Now keep the baseline direction mix balanced and change the model of future access. Let $r \in \{0, 1, 2, 3, 4\}$ be an ordinal access state for one pool of RFQ opportunities. It is a synthetic model state, not an observed win percentage or a measured client rating.

Higher r increases the rate at which the dealer is invited to respond:

$$g(r) = 0.6 + 0.3r, \quad \lambda_\eta(r) = 2g(r), \quad \lambda(r) = 4g(r). \quad (8.14)$$

The gate multiplies request intensity, not the probability of winning an already arrived request. The conditional logistic win curve remains unchanged. This separates access to opportunities from competitiveness within an opportunity.

After an observed request, the model updates the access state according to the outcome:

$$\begin{aligned} r^\uparrow &= \min(r + 1, 4), & r^\downarrow &= \max(r - 1, 0), \\ \text{win: } (q, r) &\longmapsto (q + \eta z, r^\uparrow), \\ \text{loss or decline: } (q, r) &\longmapsto (q, r^\downarrow), \\ \text{no request: } (q, r) &\longmapsto (q, r). \end{aligned} \quad (8.15)$$

The update occurs at request times. A lost or declined RFQ reduces the state even though it produces no inventory change. At the bottom state, a further loss leaves r at zero. This floor is part of the specified model.

This ladder is deliberately small: nine inventory states times five access states gives 45 joint states. Its purpose is to expose the continuation accounting and make exact finite-state comparisons possible. It does not claim that clients update relationships by adding or subtracting a point.

Barzykin models feedback from success scores to request intensity. For the RFQ protocol, both wins and losses update the score, and the HJB includes the continuation change after an unsuccessful request. The source uses an exponentially weighted score and also treats a separate streaming protocol. Our one-pool ordinal ladder is a different educational construction. [B, equation (1), p. 3; equations (3)-(9), pp. 3-4]

The model has no terminal access bonus. At the horizon, the only terminal penalty is for inventory. Any value attached to access must therefore come from opportunities before the horizon.

8.8 A Lost RFQ Can Change Value

Let $w(q, r)$ be the assigned next-time continuation function. Define the continuation change after a decline by

$$D(q, r; w) = w(q, r^\downarrow) - w(q, r). \quad (8.16)$$

This quantity need not be zero. It can be negative when the lower access state removes valuable opportunities. It is not a cash fee paid to a client; it is a change in modeled continuation value.

For a feasible quoted response, the expected incremental value of the arrived request is

$$\begin{aligned} R_\eta(q, r, \delta; w) &= p(\delta)\{z(\delta - c) + w(q + \eta z, r^\uparrow)\} \\ &\quad + [1 - p(\delta)]w(q, r^\downarrow) - w(q, r) - K. \end{aligned} \quad (8.17)$$

A win changes both inventory and access. A loss changes access alone. A response pays K regardless of the outcome, while decline pays no response cost. Reorganizing the two possible outcomes gives

$$R_\eta(q, r, \delta; w) = D(q, r; w) + p(\delta)\{z(\delta - c) + w(q + \eta z, r^\uparrow) - w(q, r^\downarrow)\} - K. \quad (8.18)$$

Thus the response advantage relative to decline compares a winning continuation state with an unsuccessful-request continuation state. It does not compare the winning state with a fictitious unchanged access state.

The corresponding effective unit cost is

$$c_{\text{joint}, \eta}(q, r; w) = c - \frac{w(q + \eta z, r^\uparrow) - w(q, r^\downarrow)}{z}. \quad (8.19)$$

The logistic interior first-order condition from (8.9) applies with this cost. A valuable change in access can make a tighter quote worthwhile in the model. The benefit has not been added as an arbitrary revenue credit: it is computed from the future RFQs available in the continuation problem.

One exact decomposition makes the two mechanisms visible:

$$\begin{aligned}
 w(q + \eta z, r^\uparrow) - w(q, r^\downarrow) = & \underbrace{w(q + \eta z, r^\uparrow) - w(q, r^\uparrow)}_{\text{inventory change at the higher access state}} \\
 & + \underbrace{w(q, r^\uparrow) - w(q, r^\downarrow)}_{\text{access change at fixed inventory}}.
 \end{aligned} \tag{8.20}$$

The decomposition holds access fixed at $r^\uparrow$ for the inventory term. Choosing a different intermediate state would allocate the interaction differently. The total winning-versus-unsuccessful continuation difference is the quantity that enters the decision.

At terminal continuation, $w(q, r) = -\ell(q)$ is independent of r . Then $D = 0$, the access term in (8.20) vanishes, and the inventory charge reduces to (8.10). The access effect is therefore a finite-horizon opportunity effect, not a permanent subsidy to winning.

For the initial dealer-buy request at $q = 0, r = 2$, the next-time continuation values are:

Outcome state	Continuation	Coordinates
Current state	39.451265	$(q, r) = (0, 2)$
Loss or decline	33.290384	$(q, r) = (0, 1)$
Win	42.476308	$(q, r) = (5, 3)$

The decline increment is -6.160881. The winning-versus-unsuccessful continuation difference is 9.185924, giving an effective unit cost of -0.837185 after including the unit execution cost. The grid-optimal offset is 2.25 and its fill probability is 0.679179.

Its improvement over decline is 10.483751, while its complete conditional request increment is 4.322870 after including the negative decline increment. These are different quantities. They are measured using the next-time value vector at the initial decision, as required by the categorical recursion.

At $q = 20, r = 2$, a dealer-buy request is blocked. Decline moves the state to $(20, 1)$ and has a continuation increment of -7.264278. Setting that request kernel to zero would erase a real transition in the specified model.

8.9 Bellman Recursion with Client Feedback

For an arrived marked request, choose between decline and a feasible response. The request kernel and HJB are

$$\begin{aligned}\mathcal{J}_\eta(q, r; w) &= D(q, r; w) + \max \left\{ 0, \max_{\delta \in \mathcal{D}_\eta(q)} [R_\eta(q, r, \delta; w) - D(q, r; w)] \right\}, \\ 0 &= \partial_t \theta(t, q, r) - \psi(q) + \sum_\eta \lambda_\eta(r) \mathcal{J}_\eta(q, r; \theta(t, \cdot, \cdot)), \\ \theta(T, q, r) &= -\ell(q).\end{aligned}\tag{8.21}$$

If a full trade is blocked, only decline is feasible and the kernel is D . In particular, the kernel itself can be negative. Replacing it by the maximum of zero and the response contribution would give the dealer an extra action: ignoring a request without its specified effect on access. That would be a different model.

The categorical recursion uses the rates at the current state:

$$\begin{aligned}\theta_n(q, r) &= \theta_{n+1}(q, r) + h \left[-\psi(q) + \sum_\eta \lambda_\eta(r) \mathcal{J}_\eta(q, r; \theta_{n+1}) \right], \\ h \max_r \lambda(r) &\leq 1.\end{aligned}\tag{8.22}$$

Our largest gate is 1.8, giving a maximum request rate of 7.2 and a maximum per-step request probability of 0.036. Every interval contains at most one request. Its outcome changes the access state for the next interval, not the event rate retrospectively within the current one.

For independent policy evaluation, write $x = (q, r)$, $x_\eta^\uparrow = (q + \eta z, r^\uparrow)$, and $x^\downarrow = (q, r^\downarrow)$. Let $a_\eta(x)$ be the response indicator. The full transition matrix is

$$\begin{aligned}P_n^\pi(x, x') &= [1 - h\lambda(r)] \mathbf{1}_{\{x'=x\}} \\ &\quad + h \sum_\eta \lambda_\eta(r) \left[a_\eta(x) p(\delta_\eta) \mathbf{1}_{\{x'=x_\eta^\uparrow\}} + [1 - a_\eta(x) p(\delta_\eta)] \mathbf{1}_{\{x'=x^\downarrow\}} \right].\end{aligned}\tag{8.23}$$

Terms with a declined response have zero fill probability. A quoted response is permitted only if its full trade is feasible. The matrix sums contributions to the same destination; at the floor $r = 0$, an unsuccessful request and no request both leave the state unchanged.

The immediate trading-contribution rate is the sum over directions of the request rate times the response indicator times $p(\delta)z(\delta - c) - K$. Subtracting $\psi(q)$ and multiplying by h gives the one-step reward. Using this reward with (8.13), now on the joint state space, gives a separate evaluation of the stored policy.

An expected request count accumulates $h\lambda(r)$ at the currently occupied state. An expected fill count accumulates the corresponding response-and-win probabilities. Their

values depend on the state path and the policy. Counting invitations and counting successful responses remain distinct even after access becomes endogenous.

8.10 Compare Policies Under the Same Feedback

Compare the full feedback optimizer with a frozen-access planner and the static-terminal-penalty policy. For each access state r , the frozen planner solves an inventory-only problem using the constant rate $2g(r)$ on each side. When deployed, it looks up the policy associated with the actual current access state:

$$\pi_n^{\text{frozen}}(q, r, \eta) = \pi_n^{[g(r)]}(q, \eta). \quad (8.24)$$

This policy recognizes today's gate but plans as though the gate will stay fixed. It omits the influence of today's outcome on tomorrow's opportunities. That is different from a policy that ignores the access state altogether.

All three deployed policies are evaluated with the actual score updates, gates, full inventory limits, running penalty, and terminal condition. A decline by the frozen or static policy still reduces the access state in the evaluation model. Their planning assumption does not alter the environment used to measure performance.

For each feasible comparison policy, define the common-model loss by

$$L_n^\pi(q, r) = \theta_n^{\text{feedback}}(q, r) - U_n^\pi(q, r) \geq 0. \quad (8.25)$$

The inequality is checked over the complete finite state and time grids, allowing numerical roundoff. The stored optimizing policy must also reproduce the Bellman value under independent Markov evaluation.

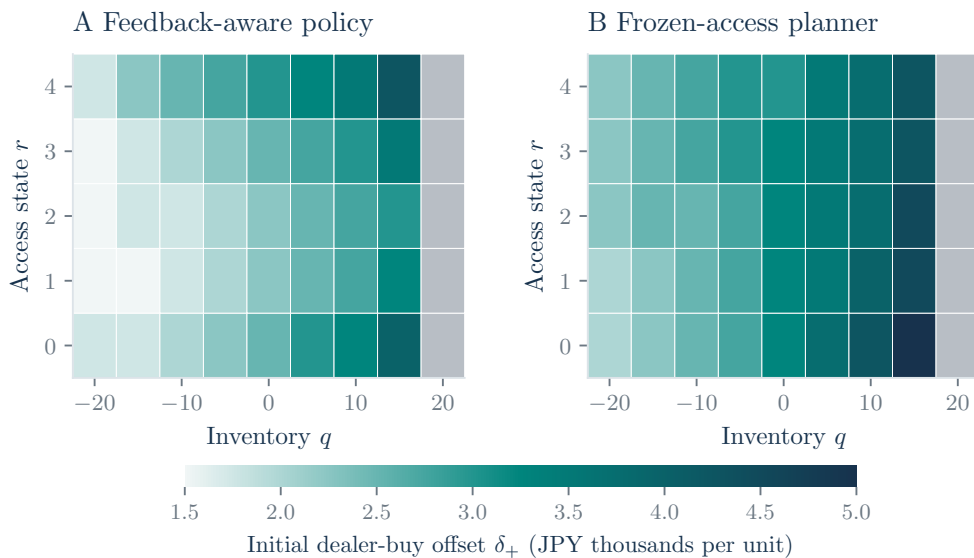
A higher expected terminal access state is not itself the objective. Nor is the largest request count or win rate. The objective is expected trading contribution minus the specified running and terminal inventory penalties. Access has value only through the opportunities and risks it creates before the horizon.

At $q = 0, r = 2$, the full feedback policy quotes 2.25 on both sides, compared with 3.25 for the frozen planner and the static policy. The symmetric access gate changes competitiveness without creating a directional skew at flat inventory.

The value components at this starting state are:

Component	Feedback	Frozen	Static
Trading	46.398221	39.815508	37.741580
Running penalty	0.915118	0.578824	0.453946
Terminal penalty	5.928089	4.122564	3.305104
Value	39.555014	35.114120	33.982530

All entries are in JPY thousands. Penalties are displayed as positive costs; value equals trading contribution minus both penalties. The feedback policy takes more trading opportunities and also incurs more inventory cost. Its advantage is established by the resulting common-objective value, not by the number of wins alone.



$t = 0$, $N = 400$; common color scale. Gray: a full dealer buy is infeasible at $q = 20$. r is an ordinal access state, not a win probability. Inventories are normalized trade units. Frozen-access planning holds the current access level fixed when computing continuation.

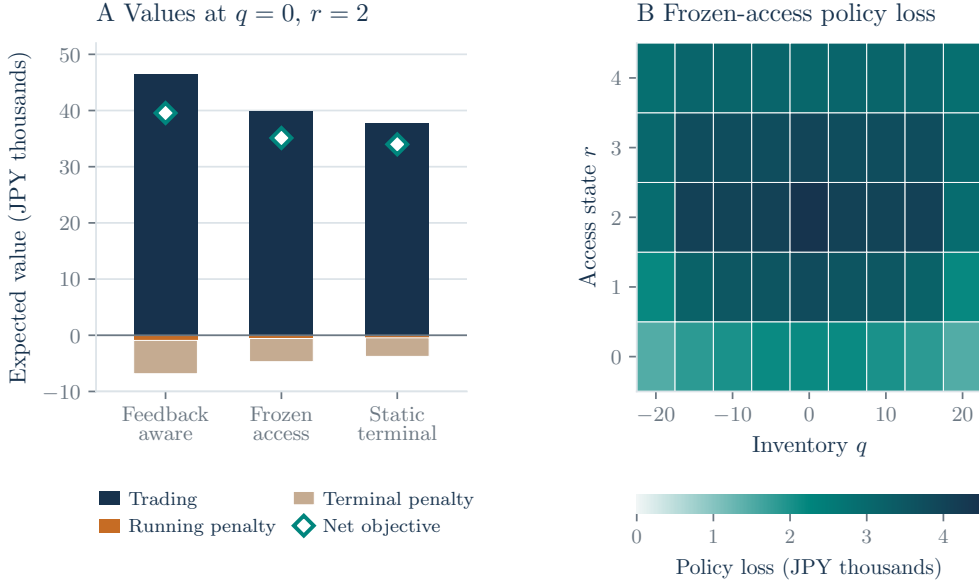
Figure 8.2: The feedback policy and frozen-access planner can choose different quotes at the same inventory and current access state

The numerical outputs retain the three value components alongside expected invitations, fills, and terminal access state. This prevents an increase in one operational metric from being presented as an increase in the modeled trading objective.

Expected metric	Feedback	Frozen	Static
Requests	10.325353	8.477497	8.290999
Fills	5.681184	3.740974	3.526102
Terminal access	1.884601	1.191778	1.110460

Counts are expectations over the full horizon and need not be integers. Terminal access is an expected ordinal state, not a win-rate percentage. The feedback policy's expected terminal state is below its starting state of 2, despite its more competitive initial quotes. The optimizer does not impose a target score.

Across all 45 initial states, the largest frozen-planner loss is 4.440894 JPY thousands, attained at $q = 0, r = 2$. Figure 8.3 shows how this loss varies across the state grid.



All policies are evaluated in the same access-feedback model; $T = 2, N = 400$.
 Left: net objective = trading minus running and terminal penalties; penalties plot below zero.
 Right: optimal full-model value minus frozen-access policy value at each starting state.

Figure 8.3: Comparison policies are evaluated under the actual feedback model with trading contribution and inventory penalties accounted for

8.11 Remove the Mechanism and Refine the Grid

A useful test removes the channel that is supposed to generate an effect. Introduce a gate-strength parameter χ :

$$g_\chi(r) = 1 + \chi[g(r) - 1], \quad \chi \in \{0, 0.5, 1\}. \quad (8.26)$$

At $\chi = 0$, every access state has the same gate of one. The state still updates after outcomes, but no reward, win probability, inventory constraint, or terminal value depends on it. It can be removed from the decision problem:

$$\theta_n^{\chi=0}(q, r) = \theta_n^{\text{balanced}}(q), \quad \pi_n^{\chi=0}(q, r, \eta) = \pi_n^{\text{balanced}}(q, \eta). \quad (8.27)$$

Equality follows by backward induction, starting from the access-independent terminal condition. It holds with the common tie convention. This is a direct check that an access premium has not been inserted into the cash reward or terminal value by mistake.

At $\chi = 0.5$ and 1, the model's actual opportunity process changes. Comparing those scenarios is a mechanism sensitivity test. Within each scenario, policy performance must still be evaluated under that scenario's true gate. A comparison across different gates is not a policy-loss measurement under one unchanged environment.

At $q = 0, r = 2$, the sensitivity calculation gives:

χ	Feedback buy offset	Feedback value	Frozen loss
0.0	3.25	32.842459	0.000000
0.5	2.50	35.786971	1.227471
1.0	2.25	39.555014	4.440894

Both value columns are in JPY thousands. The zero-feedback model has no loss from freezing the access state. As the gate mechanism strengthens in these examples, the full policy selects a tighter initial response and the frozen planner's loss increases. This is a result for the specified gate family and parameters.

Time discretization is checked independently. For a fixed model and state set $\mathcal{X}$, define

$$E_N = \max_{x \in \mathcal{X}} \left| \theta_0^{(2N)}(x) - \theta_0^{(N)}(x) \right|. \quad (8.28)$$

The notebook uses $N = 100, 200, 400, 800$ for the $\beta = 0.6$ flow model and for the full feedback model. Refinement changes the approximation to continuous-time arrivals; it does not validate the assumed access updates or identify a causal client response.

Coarse steps N	Flow model E_N	Feedback model E_N
100	0.069554	0.029258
200	0.034637	0.014609
400	0.017286	0.007295

Both sets of value differences approximately halve when the time step halves. The baseline and 800-step solutions have identical initial quote choices on their respective grids, while their value differences remain nonzero. Agreement of a few discrete actions is therefore a weaker check than agreement of the value functions.

The final-step response policies are also checked. With no terminal access value, changing r cannot change the terminal continuation advantage. With a fixed terminal inventory penalty, changing β cannot change it either. These checks concern the conditional final-step action, not the earlier value of future opportunities.

8.12 What Must Be Observed Before Using the Model

The experiments supply arrival rates, win curves, and a reference-price model. In actual records, request selection, competition, and subsequent price changes can vary together. Filled trades alone do not identify their separate effects.

The access hypothesis is stronger: request outcomes change future invitations. Recent wins can predict future flow because market activity, client needs, or unrecorded dealer information affect both. Correlation alone does not establish the feedback mechanism.

Record invitations, responses, declines, wins, offsets, sizes, times, observed client or protocol identifiers, and subsequent opportunities. Access proxies must be available at decision time, with explicit denominators, update clocks, and treatment of missing requests.

Barzykin's score gates are a modeled feedback channel. The paper's numerical long-run score analysis uses further stationary and discounting approximations. Our finite-horizon experiment establishes neither persistent client regimes nor a causal commercial benefit from a reputation-building campaign. [B, Numerical results, p. 7]

Chapter 9 introduces an external way to reduce inventory. It will compare waiting for offsetting client requests with paying to trade outside the client flow.

Exercises

1. From 80 dealer-buy and 20 dealer-sell requests, estimate request imbalance. If 20 and 10 respectively fill, calculate fill imbalance and the two conditional win rates when every request received a response. Explain what changes in the denominator when some requests were declined.
2. At flat inventory, compare the single-RFQ optimum under $\beta = -0.6, 0, 0.6$, first with immediate-only continuation and then with terminal continuation. Report the side-specific arrival-weighted contributions and their sum. Explain why the conditional quote is unchanged, and why this result does not imply equal dynamic values.
3. At $q = 0$, report the flow-aware initial buy and sell offsets, full width, midpoint skew, and expected inventory-change rate for the three imbalance scenarios. Then hold $\beta = 0$ and compare $q = -10, 0, 10$. Keep requested-unit drift separate from expected filled-unit drift.
4. Verify the exponential benchmark at $q = -5, 0, 5$ with $w(q) = -0.1q^2$, $z = 5$, $c = \omega = 1$. Check the optimum by the first-order condition and nearby quoted offsets. At $q = 15$, enforce the nonnegative quote floor and calculate the resulting width and skew.

5. In the feedback model at $q = 0, r = 2$, calculate the initial decline increment, winning-versus-unsuccessful continuation difference, chosen buy offset, and full request increment. Verify (8.18). Repeat the feasibility reasoning for a dealer-buy request at $q = 20, r = 2$: inventory stays fixed on decline, but the access state need not.
6. Evaluate the full feedback, frozen-access, and static policies at four starting states: $(0, 0)$, $(0, 2)$, $(0, 4)$, and $(10, 2)$, ordered as (q, r) . Report trading contribution, running penalty, terminal penalty, total value, expected terminal access state, request count, and fill count. Distinguish those operational metrics from the objective.
7. Compare $\chi = 0, 0.5, 1$. Verify the complete state-and-time collapse at $\chi = 0$, and evaluate the comparison policies under the actual gate in each scenario. Explain why changing the true gate differs from changing only the planner's assumption.
8. Compute E_{100} , E_{200} , and E_{400} for the positive-imbalance model and the full feedback model. Verify final-step conditional action independence from β or r . Distinguish numerical refinement from evidence about client behavior.

Source Notes

[C] Peter Cotton, *On a Simple Relationship Between Order Imbalance, Skew and Width in Over-The-Counter Trading*, supplied nine-page PDF, arXiv:2608.07690v1. The version stamp and arXiv submission record say 7 August 2026; the title page is dated 11 August 2026. Section 2, pp. 3-4, gives the stationary dealer model and its exponential-response assumptions. Section 2.2, equations (1)-(2), p. 4, links inventory-cost differences to quote offsets, width, and midpoint shift. Our exponential benchmark is derived explicitly with its quote-domain qualification. We do not use the source’s claimed additional stationary width correction as a rule for the logistic finite-horizon model. The reading note records the precise floor, sign, and transformation caveats. Available from [arXiv](#).

[B] Alexander Barzykin, *Strategic OTC market making with reputation feedback*, supplied 11-page PDF, arXiv:2607.11328v2. The version stamp and arXiv revision record say 15 July 2026; the title page is dated 16 July 2026. Equation (1), p. 3, defines request-outcome score updates. Equations (2)-(9), pp. 3-4, give the additive objective, unsuccessful-request and successful-request continuation increments, HJB, and RFQ Hamiltonian. The numerical analysis on p. 7 introduces a discounted stationary approximation. The source’s two scores refer to RFQ and streaming protocols; they are not two named clients. We retain the consistent dealer cash-flow convention of the preceding chapters rather than copying the ambiguous cash-SDE prefactor printed on p. 3. Available from [arXiv](#).

The finite grids, logistic parameters, imbalance scenarios, ordinal access ladder, gates, comparison policies, and exercise outputs are constructions for this book. The source checks cover the passages used here. The notebook does not reproduce either paper’s complete model or numerical study, and its inputs are not estimates of actual client behavior.

PART IV

Execution and evaluation

Manage external transactions, reconcile the cash ledger, and evaluate a complete policy under explicit uncertainty.

- 9 Internalization and External Hedging
- 10 Optimal Execution
- 11 Transaction-Cost Analysis and Profit Attribution
- 12 Simulation and Evaluation

Chapter 9

Internalization and External Hedging

9.1 Two Ways to Reduce the Same Inventory

A dealer holds a positive position after buying from a client. The dealer can encourage a later client to buy some of that position, or sell externally before such a request arrives. Both routes can reduce inventory. They differ in who controls the timing, which price is paid or received, and what opportunities remain afterward.

Internalization means using the client business to absorb and offset inventory. It includes the decision to quote more competitively on the side that reduces a position. It does not guarantee an offsetting request, a successful response, or a particular waiting time. External hedging makes a trade available under the external market's execution conditions. Its cost must be compared with the value of keeping the position and continuing to quote.

Chapters 6–8 made future client flow part of continuation value. This chapter adds an external action to that continuation. We first use the same instrument for both client trades and external inventory reduction. A later section treats an imperfect hedge through supplied risk exposures. To isolate the hedge decision, the baseline fixes the client arrival process and conditional win curve. It does not carry Chapter 8's changing access state: a decline or an unsuccessful quote therefore leaves future access unchanged. Combining both mechanisms would require retaining both inventory and access in the state.

The numerical decision is deliberately concrete: at specified review times, choose one bounded external trade, then quote to arriving clients until the next review. Execution at a review is immediate and certain in this model. Waiting for the next review and waiting for an already-submitted order to execute are different timing problems.

9.2 A Hedge Is a Cash Transaction

Let q be dealer inventory and X cash. An external trade of signed size e is a purchase when $e > 0$ and a sale when $e < 0$. Let S be the supplied reference price and let $C(e)$ be the total execution cost relative to trading e units at S . At execution,

$$\begin{aligned} q^{\text{new}} &= q + e, & X^{\text{new}} &= X - eS - C(e), \\ X^{\text{new}} + q^{\text{new}}S - (X + qS) &= -C(e). \end{aligned} \tag{9.1}$$

The reference-price purchase or sale cancels against the change in marked inventory. The immediate loss of marked wealth is the execution cost. A later reduction in inventory risk is a separate consequence of the new position.

We use a cost schedule with a fixed charge per nonzero trade, a proportional charge, and a quadratic charge:

$$C(e) = F\mathbf{1}_{\{e \neq 0\}} + s_{\text{ext}}|e| + \frac{k_{\text{ext}}}{2}e^2, \quad C(0) = 0. \quad (9.2)$$

Here F is currency, s_{ext} is currency per trading unit, and k_{ext} is currency per squared trading unit. The quadratic term is a supplied cost of executing one trade. It is not a model of the reference price remembering previous orders. Permanent or transient price impact would require a corresponding price or impact state and consistent wealth accounting.

This distinction also separates C from the running inventory penalty ψ and terminal penalty ℓ . Those penalties express the objective's cost of carrying risk. They are not another cash invoice for the same external trade. Guéant's active-order extension makes a related separation between the cost of crossing the spread and the running inventory objective. [G, Section 11.4, pp. 238–240]

9.3 Put Client Quotes and Hedges in One Objective

Keep $\eta = +1$ for a dealer purchase and $\eta = -1$ for a dealer sale, so the inventory change is ηz . A successful response of size z at offset δ contributes $z(\delta - c_\eta)$ relative to the reference price, after the supplied per-unit client-trade cost c_η . A response cost K_η is paid whenever a quote is submitted.

Under a policy π controlling both client responses and external trades, the value is

$$\begin{aligned} \Theta^\pi(t, q) = \mathbb{E}_{t,q}^\pi \bigg[& \sum_{\text{client fills}} z(\delta - c_\eta) - \sum_{\text{quoted responses}} K_\eta \\ & - \sum_{\text{external trades}} C(e) - \int_t^T \psi(q_u) du - \ell(q_T) \bigg]. \end{aligned} \quad (9.3)$$

In the discrete experiment the integral is replaced by the explicitly timed sum below. We use supplied martingale reference prices, so expected marked price changes do not add a directional forecast to this reduced value. Exposure still matters through the chosen inventory penalties. This remains an additive objective, consistent with the preceding chapters.

Client acceptance and execution are represented by one immediate, full-size event. The win curve is conditional on an arriving request and a valid response, using [Chapter 3](#)'s probability contract. A delayed award, execution, or report would require additional event states; an eventual request-cohort win probability alone would not supply their

timing. Chapter 2's conditional reference-price cost would also need to enter the reward if the martingale/selection assumptions used for this reduced value were relaxed.

For fixed continuation w , the conditional client-request kernel is

$$H_\eta(q; w) = \max \left\{ 0, \max_{\delta \in \mathcal{D}_\eta(q)} [p_\eta(\delta) \{z(\delta - c_\eta) + w(q + \eta z) - w(q)\} - K_\eta] \right\}. \quad (9.4)$$

The zero is a decline. The quote set is empty when the full client trade would violate an inventory limit, leaving decline as the only action. In this baseline, an unsuccessful client request changes neither inventory nor future access. This state-transition assumption is why decline has zero increment here; Chapter 8 showed a different case.

An external trade does not earn a client spread, and a client quote does not promise immediate inventory reduction. Both actions are evaluated through the same future value, with their own event probabilities and cash flows.

9.4 Specify the Clock Before Writing the Bellman Equation

Let the horizon have M review intervals, each containing K client-request steps:

$$t_m = m\Delta, \quad \Delta = \frac{T}{M}, \quad h = \frac{\Delta}{K}, \quad m = 0, \dots, M. \quad (9.5)$$

External actions occur at $t_0, \dots, t_{M-1}$. There is no external action at $t_M = T$. At each permitted date the dealer observes inventory, chooses at most one external trade, pays its cost, and then enters the client-request interval. Within that interval, quotes adapt after every observed inventory change.

Here K is the number of client steps per review interval; the subscripted K_η in the request reward is a cash response charge. The two symbols have different units and roles.

The baseline grid is $\mathcal{Q} = \{-20, -15, \dots, 20\}$. External trades are restricted to inventory reduction without crossing zero:

$$\mathcal{E}(q) = \{0\} \cup \{-\text{sgn}(q)d : d \in \{5, 10\}, d \leq |q|\}. \quad (9.6)$$

Thus at $q = 15$ the possible destinations are 15, 10, and 5. At $q = 0$ only waiting is allowed. This is an explicit action restriction. A dealer permitted to build a speculative external position or cross through zero would face a larger problem. Our result is optimal only within the stated action set.

Write $\Theta_m(q)$ for value immediately before review m . For each interval, start from $V_{m,K}(q) = \Theta_{m+1}(q)$ and solve the client steps backward:

$$V_{m,k}(q) = V_{m,k+1}(q) + h \left[-\psi(q) + \sum_{\eta=\pm 1} \lambda_\eta H_\eta(q; V_{m,k+1}) \right], \quad (9.7)$$

$$k = K - 1, \dots, 0, \quad h(\lambda_+ + \lambda_-) \leq 1.$$

The running penalty uses inventory at the beginning of the client step. The step then contains no request or one request, whose response outcome changes inventory for the next step. This is exact for the chosen categorical model and an approximation to Poisson arrivals. A finer client step changes that approximation, without adding external review dates.

Let $W_m = V_{m,0}$ be value after the external action has been settled. The review recursion is

$$\begin{aligned} \Theta_m(q) &= \max_{e \in \mathcal{E}(q)} \{-C(e) + W_m(q + e)\}, \\ \Theta_M(q) &= -\ell(q). \end{aligned} \quad (9.8)$$

The order of these operations matters. First solve the interval conditional on its starting inventory and next review value. Then select the external action using that interval value. Solving the client problem once without future hedging and inserting an external rule afterward generally changes the policy.

With the logistic win curve $p(\delta) = [1 + \exp((\delta - a)/b)]^{-1}$, a differentiable interior quoted optimum has effective cost and first-order condition

$$c_{\text{eff}} = c_\eta + \frac{w(q) - w(q + \eta z)}{z}, \quad (9.9)$$

$$1 - \frac{1 - p(\delta)}{b}(\delta - c_{\text{eff}}) = 0.$$

Here $w = V_{m,k+1}$. The value of future external execution is already in w . Adding a separate hedge benefit to this quote objective would count part of that continuation twice. The actual solver compares all quote ticks and decline; it does not assume that the interior equation has a feasible answer.

9.5 The Region Where Waiting Wins

For any candidate external action, compare its value with leaving inventory unchanged at the same review:

$$A_m(q, e) = W_m(q + e) - W_m(q) - C(e). \quad (9.10)$$

The first difference is the value of the new starting position for the upcoming client business and all later decisions. It can reflect future risk, available client trades, and later external opportunities. The last term is the cash cost paid now.

With a tie convention that keeps the current position, the no-hedge region is

$$\mathcal{N}_m = \left\{ q \in \mathcal{Q} : \max_{e \in \mathcal{E}(q)} A_m(q, e) = 0 \right\}. \quad (9.11)$$

The zero action has $A_m(q, 0) = 0$, including at states where waiting is forced. The region depends on time and on all continuation inputs. The numerical solver does not impose a symmetric interval, a fixed threshold, or a rule to return inventory to zero.

For intuition, replace continuation by a terminal quadratic charge $w(q) = -\alpha q^2/2$. Reducing a positive inventory by d gives the proxy gain

$$w(q - d) - w(q) - C(-d) = \alpha qd - \frac{\alpha}{2}d^2 - C(-d). \quad (9.12)$$

Even this simple comparison uses the finite trade size. A derivative-only approximation omits the d^2 correction, and a fixed transaction charge cannot be recovered from a marginal price alone.

An exact one-interval example is more revealing. Turn off client arrivals, retain $\Delta = 0.05$, $\psi(q) = 0.01q^2$, and $\ell(q) = 0.10q^2$. Then

$$W(q) = -a_\Delta q^2, \quad a_\Delta = 0.10 + 0.01\Delta = 0.1005. \quad (9.13)$$

Use $F = 1$, $s_{\text{ext}} = 0.5$, and $k_{\text{ext}} = 0.04$. Trading 5 units costs 4; trading 10 costs 8. At $q = 10$, the gain from reducing by 5 is 3.5375, while reducing by 10 gives 2.05. The dealer chooses a partial hedge and finishes with inventory 5. At $q = 5$, trading to zero loses 1.4875 relative to waiting.

These calculations include the remaining holding cost and terminal penalty. There is no uncharged terminal transaction afterward. Cartea, Jaimungal, and Penalva's continuous-time liquidation example highlights why the last trading opportunity requires care: allowing interventions arbitrarily near maturity can change the limiting terminal value. Our scheduled model instead specifies a last review before T . [C, Section 8.4, pp. 197–200]

9.6 Evaluate a Stored Policy Independently

Policy evaluation should not reuse the optimizing maximum as evidence that the stored decisions earn the reported value. For one client step, let $a_\eta(q)$ indicate a quoted response, and let $\delta_\eta(q)$ be its offset. With $\lambda = \lambda_+ + \lambda_-$, the explicit transition matrix is

$$P^\pi(q, q') = (1 - h\lambda)\mathbf{1}_{\{q'=q\}} + h \sum_{\eta} \lambda_{\eta} \left[a_{\eta} p_{\eta}(\delta_{\eta}) \mathbf{1}_{\{q'=q+\eta z\}} + (1 - a_{\eta} p_{\eta}(\delta_{\eta})) \mathbf{1}_{\{q'=q\}} \right]. \quad (9.14)$$

Only feasible quoted trades use the first destination. Declines and unsuccessful quotes keep inventory unchanged. Contributions to the same destination are added, so every row sums to one.

The one-step reward and backward policy evaluation are

$$b^\pi(q) = h \left[\sum_{\eta} \lambda_{\eta} a_{\eta}(q) \{p_{\eta}(\delta_{\eta}) z (\delta_{\eta} - c_{\eta}) - K_{\eta}\} - \psi(q) \right], \quad (9.15)$$

$$U_{m,k}^\pi = b^\pi + P^\pi U_{m,k+1}^\pi.$$

At a review, the stored external action is a deterministic inventory transition. Its evaluation is

$$U_m^\pi(q) = -C(e_m^\pi(q)) + U_{m,0}^\pi(q + e_m^\pi(q)), \quad U_M^\pi(q) = -\ell(q). \quad (9.16)$$

The same transition operators propagate separate expected trading contribution, external fixed/proportional/quadratic costs, running penalty, and terminal penalty. They also propagate external volume, external trade count, client fills, and residual inventory. These operational measures help explain the value; they do not replace the objective.

The final accounting identity is

$$\begin{aligned} U^\pi = & \text{client contribution} - \text{external fixed cost} \\ & - \text{external linear cost} - \text{external quadratic cost} \\ & - \text{running penalty} - \text{terminal penalty}. \end{aligned} \quad (9.17)$$

A client request that fails does not pay a filled-trade cost. A quoted response pays K_η whether it wins or loses. A zero external action pays no fixed charge. These event distinctions are verified separately from the total identity.

The client-contribution column includes response charges as well as fill revenue; the baseline sets those charges to zero. Inventory penalties are deductions from the decision objective, so the value in (9.17) is not itself a cash-ledger P&L. Chapter 11 reconciles realized fills, fees and marks separately.

9.7 A Controlled Comparison of Five Policies

The baseline uses $T = 2$, $M = 40$, $K = 10$, $z = 5$, $\lambda_+ = \lambda_- = 2$, and $p(\delta) = [1 + \exp(\delta - 3)]^{-1}$. Client offsets range from -6 to 10 in increments of 0.25. Client cost is $c = 1$ and response cost is $K_\eta = 0$. The running and terminal penalties are $\psi(q) = 0.01q^2$ and $\ell(q) = 0.10q^2$. External costs are those of the preceding example.

Prices and per-unit costs use JPY thousands per normalized trading unit; values and total costs use JPY thousands. Time is normalized. The coefficients and arrival rates are synthetic inputs, not estimates of actual dealer behavior.

Numerical comparisons use a tolerance of 10^{-12} in currency units. External ties choose zero first and then the smallest absolute trade. Client-response ties choose decline first and then the lowest offset among tied quoted responses.

The joint policy optimizes both client quotes and external actions. Four comparison policies make distinct restrictions. The immediate policy trades the maximum feasible inventory-reducing amount at every review. The periodic policy does so only at times 0.5, 1.0, and 1.5, initially waiting. For both of these external rules, client quotes are optimized with full knowledge of the rule. Thus their cost is not artificially inflated by leaving client decisions unoptimized.

The client-only policy never trades externally and optimizes its quotes. The fixed-quote hedger then takes those client-only quotes as a given policy, while optimizing its own external trades:

$$\pi_{\text{RFQ}}^{\text{fixed}}(m, k, q, \eta) = \pi_{\text{RFQ}}^{\text{client only}}(m, k, q, \eta). \quad (9.18)$$

The lookup uses actual inventory after any hedge and intervening client trades. Its hedge decisions still use the actual external costs, timing, and constraints. This is a strong version of separate planning: the hedge layer is optimized, but the client quote layer does not adjust to the hedge opportunity.

Every deployed policy is evaluated under the same actual market model. At each pre-review state, optimality requires

$$\Theta_m^{\text{joint}}(q) - U_m^\pi(q) \geq 0. \quad (9.19)$$

The check covers the complete time and inventory grids, with allowance for numerical roundoff. The stored joint policy must also reproduce its Bellman value under the independent evaluation.

At initial inventory $q = 10$, the common-objective values and first external actions are:

Policy	Value	Loss	First trade
Joint	31.021296	0.000000	0
Immediate	18.289617	12.731680	-10
Periodic	25.004373	6.016923	0
Fixed quotes	30.011565	1.009731	0
Client only	26.778914	4.242382	0

Value and loss are in JPY thousands; trade size is signed inventory units. The immediate policy removes the initial 10 units, but its repeated later hedges also consume execution costs. Its lower terminal inventory does not imply a higher value.

To see where the gain from joint planning comes from, compare its components with those of the fixed-quote hedger at the same initial state:

Component	Joint	Fixed quotes
Client contribution	38.853636	36.204232
External fixed cost	1.107373	0.794345
External linear cost	2.836874	2.007593
External quadratic cost	0.594751	0.410211
Running penalty	1.275096	1.124794
Terminal penalty	2.018245	1.855724
Value	31.021296	30.011565

All component entries are in JPY thousands. Costs and penalties are displayed as positive deductions. The joint policy earns more client contribution while also spending more externally and bearing more inventory cost than this comparison policy. [Equation \(9.17\)](#) determines the net difference.

Policy	External volume	Hedge count	Terminal $ q $
Joint	5.673748	1.107373	3.836642
Immediate	24.122493	3.777363	0.415687
Periodic	14.159396	1.941323	2.748739
Fixed quotes	4.015186	0.794345	3.529918
Client only	0.000000	0.000000	6.325110

These are horizon expectations. Volumes and absolute inventory are trading units; counts need not be integers. A policy that hedges more often can produce less residual inventory and still lose value under the same objective.

At $q = 15$, the joint policy initially waits and has value 27.024482. The fixed-quote hedger initially sells 5 and has value 26.011565. Within the joint problem, waiting exceeds its best immediate nonzero alternative by only 0.003185. The first action is therefore a close decision for these inputs; the larger loss of the separately planned policy also reflects its later decisions.

On the baseline initial grid, the joint policy hedges only at $q = \pm 20$, moving 5 units toward zero. At the last review it also hedges at $|q| = 10$ and 15, with a maximum size of 10. The complete review-time map is shown in Figure 9.1.

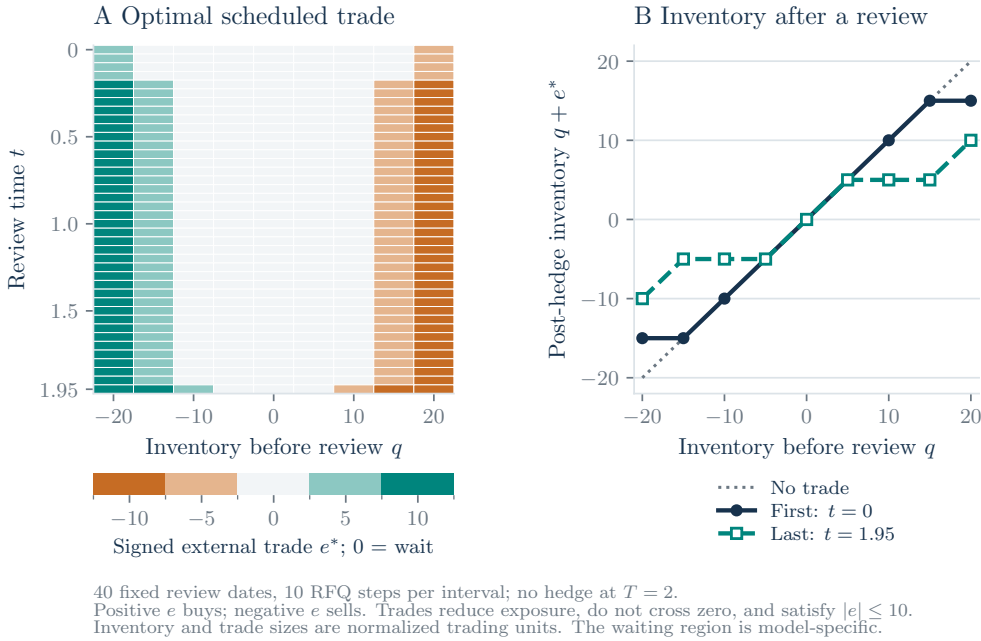


Figure 9.1: The optimal external action depends on inventory and the remaining review opportunities

9.8 Why a Hedge Changes the Client Quote

Suppose two dealers reach the same inventory inside a client interval. One expects to have external execution available at later reviews; the other plans to rely on client flow. Their current cash payoff from a particular fill can be identical, while their continuation increments differ. Consequently, their effective costs in (9.9) and their chosen offsets can differ.

The correct comparison fixes the current post-hedge inventory, the RFQ time within the interval, and the request direction. Comparing the joint policy's quote after reducing inventory with a client-only quote at the original inventory would mix two effects: a changed state and a changed continuation.

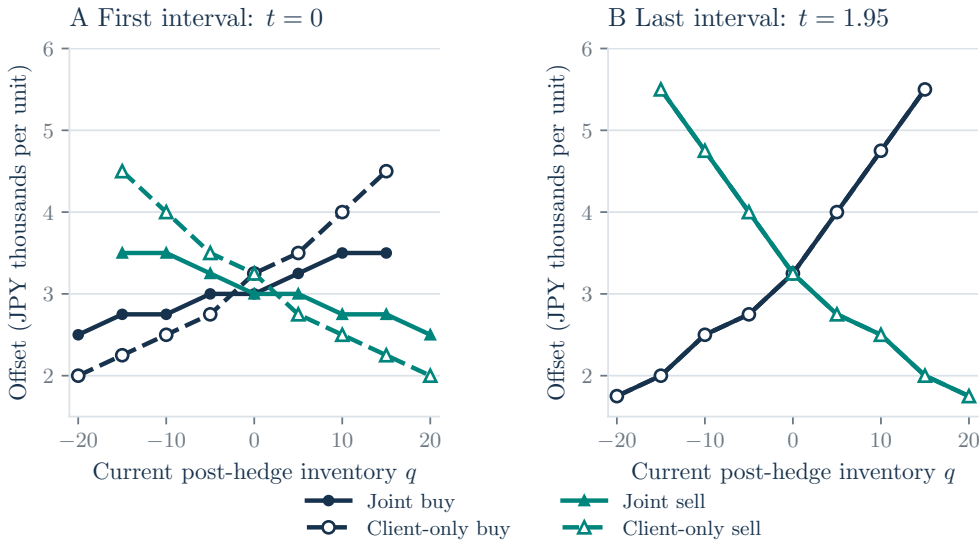
The same-state continuation difference relevant to this comparison is

$$\Delta_{\text{eff},\eta}(q) = \frac{[w^{\text{joint}}(q) - w^{\text{joint}}(q + \eta z)] - [w^{\text{client only}}(q) - w^{\text{client only}}(q + \eta z)]}{z}. \quad (9.20)$$

This is a price-unit difference. Its sign and size are computed from future opportunities and constraints. The availability of hedging does not justify adding an arbitrary tighter spread or smaller skew to every quote.

Time and inventory	Joint buy / sell	Client-only buy / sell
$t = 0.00, q = 10$	3.50 / 2.75	4.00 / 2.50
$t = 0.00, q = 15$	3.50 / 2.75	4.50 / 2.25
$t = 1.95, q = 10$	4.75 / 2.50	4.75 / 2.50
$t = 1.95, q = 15$	5.50 / 2.00	5.50 / 2.00

All entries are price offsets. Time identifies the first RFQ step after the indicated review, with current inventory held fixed. In the last interval the two policies have the same RFQ continuation problem: there is no later external review before the common terminal penalty. Their conditional quotes therefore coincide throughout that interval. Earlier, the future hedge opportunity changes them.



Both policies use the same current inventory and first RFQ step of the indicated interval. No later hedge review remains in the last interval, so the same-state quotes coincide. Infeasible buy/sell directions at $q = 20/-20$ are absent. Inventories are normalized trading units.

Figure 9.2: Client quotes are compared at the same current inventory and RFQ decision time

The direct dealer-market study by Barzykin, Bergault, and Guéant jointly controls client quotes and a continuous external execution rate, including permanent price impact. It supplies a useful connection between quoting and hedging. The present experiment instead uses scheduled, finite trades with a fixed charge and no price-impact state. Its numerical values are not a reproduction of that paper. [B, Sections 2–3]

9.9 Costs, Capacity, and Two Different Clock Experiments

There are useful checks that do not require an estimated market dataset. If one model offers every action of another at the same times and with the same consequences, its optimized value cannot be lower. If an external cost increases pointwise while everything else stays fixed, optimized value cannot increase:

$$\mathcal{E}_1(q) \subseteq \mathcal{E}_2(q) \implies \Theta^{(1)} \leq \Theta^{(2)}, \quad C_1(e) \leq C_2(e) \implies \Theta^{C_1} \geq \Theta^{C_2}. \quad (9.21)$$

These are value statements. They do not prove that the number of hedges, their average size, or each state's action changes monotonically. A fixed charge can alter both the decision to trade and the size chosen when trading. Capacity also affects how much inventory can be removed in one review.

The numerical refinement holds the 40 external review dates fixed and changes only the number of client steps within each interval. For K and $2K$, define

$$E_K = \max_{q \in \mathcal{Q}} \left| \Theta_0^{(M, 2K)}(q) - \Theta_0^{(M, K)}(q) \right|, \quad M = 40. \quad (9.22)$$

This measures a change in the categorical approximation to client arrivals. It does not give the dealer more chances to hedge.

A separate experiment keeps the client step at 0.005 and uses 10, 20, or 40 external reviews over the same horizon. The review dates are nested, so a policy from the coarser schedule can be implemented on the finer one by choosing zero at additional dates:

$$\mathcal{T}_{10} \subset \mathcal{T}_{20} \subset \mathcal{T}_{40} \implies \Theta_0^{10}(q) \leq \Theta_0^{20}(q) \leq \Theta_0^{40}(q). \quad (9.23)$$

This is a change in available opportunities, not merely a more accurate calculation of the same scheduled-trading problem. Conflating the two clocks can make a numerical improvement look like a benefit of a supposedly unchanged policy.

At $q = 10$, changing only the fixed external charge gives:

Fixed charge F	Joint value	Expected hedge count
0	32.228126	1.310025
1	31.021296	1.107373
3	29.630531	0.610245
8	27.906936	0.225753

The optimized value falls as the fixed charge rises. The corresponding trade count is an outcome of the policy, rather than the quantity constrained by the value-ordering argument.

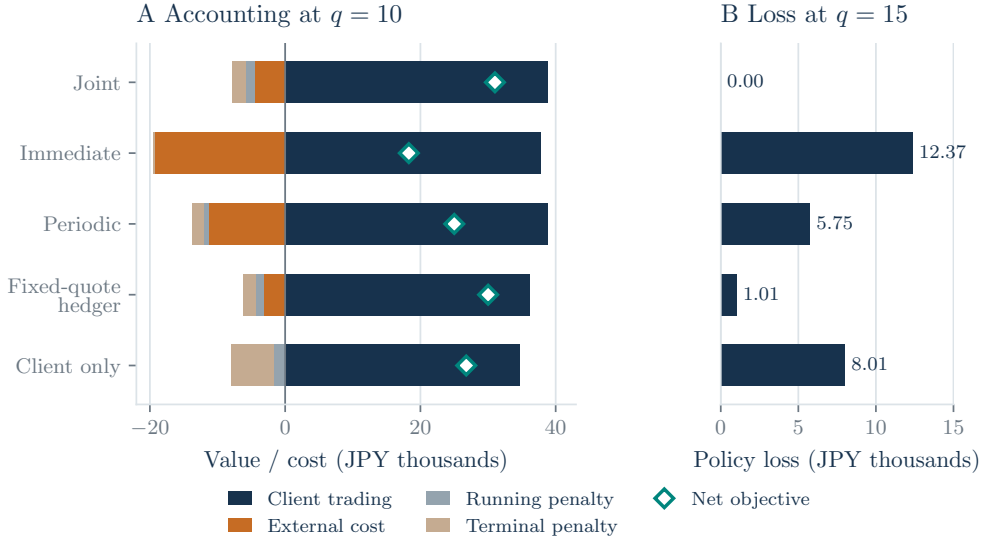
Restricting the maximum external size from 10 to 5 lowers the $q = 10$ value from 31.021296 to 31.000549. At the same positive inventory, request imbalance $\beta = -0.6$ gives value 34.122999 and an initial wait; $\beta = 0.6$ gives 23.974124 and an initial sale of 5. Positive imbalance sends more inventory-adding requests to the dealer. These are different actual arrival environments, not policy losses under one unchanged model.

With 40 review dates held fixed, client-step refinement gives:

Coarse K	Fine K	Maximum value difference
5	10	0.004849
10	20	0.002079
20	40	0.001005

The differences decrease, while initial hedge and post-review quote choices are unchanged across these grids. Agreement of the discrete initial actions does not make the remaining value difference zero.

At a fixed client step of 0.005, the $q = 10$ value is 30.665154, 30.904655, and 31.021296 with 10, 20, and 40 reviews respectively. Additional review opportunities increase value in this nested comparison. [Figure 9.3](#) returns to the single 40-review baseline to compare the five policies' accounting.



All five policies use the same market, review schedule, costs, and inventory penalties.
 Left: net = client trading minus external costs and both penalties; external cost sums all three parts.
 Right: joint value minus policy value; the joint loss rounds to zero. Values start before the first review.

Figure 9.3: Policy values retain client contribution, execution costs, and inventory penalties under one common objective

9.10 When the External Instrument Leaves Residual Risk

The baseline hedge reduces units of the same instrument. With a different hedge instrument, the state transition must use its supplied exposure. Let x be a vector of risk coordinates, Γ a supplied covariance metric over a fixed reference horizon, and b the exposure added by one external trading unit. Then

$$\begin{aligned} R(e) &= (x + be)^\top \Gamma (x + be) \\ &= x^\top \Gamma x + 2e b^\top \Gamma x + e^2 b^\top \Gamma b. \end{aligned} \quad (9.24)$$

If $b^\top \Gamma b > 0$, unconstrained minimization gives

$$e_{\text{risk}}^* = -\frac{b^\top \Gamma x}{b^\top \Gamma b}, \quad R_{\min} = x^\top \Gamma x - \frac{(b^\top \Gamma x)^2}{b^\top \Gamma b}. \quad (9.25)$$

The residual need not be zero. For $x = (10, 5)^\top$, unit variances, correlation 0.6, and $b = (1, 0)^\top$, initial risk is 185. The unconstrained trade is -13 and residual risk is 16. With trade sizes restricted to multiples of 5 and a maximum size of 10, the feasible trade -10 leaves risk 25.

The coordinates matter as much as the covariance. Here x is an exposure vector, rather than the quantity vector used in Chapter 5, and Γ must describe shocks in the matching coordinates. If x contains factor exposures, the corresponding factor-shock covariance plays the role denoted by Ω in Chapter 5. A covariance in instrument-price coordinates cannot be applied unchanged to factor exposures, and residual instrument risk must still be retained if those factors do not span it.

Chapter 5 explains how a forecast $\Gamma_{t,H}$ is constructed for a stated horizon and distinguished from a variance rate. This section uses a fixed supplied metric and does not multiply an already horizon-scaled covariance by time again. Nor does the quadratic form quantify uncertainty about the covariance forecast itself. Updating that forecast during control, or allowing for its estimation error, calls for an explicit information and risk specification; it is not a hidden modification of the fixed-metric example.

This is a risk-minimizing calculation, not the full trading optimum. Execution cost, future client opportunities, hard position limits, and the available trade grid still matter. A one-step cost-and-risk proxy would minimize $C(e) + \gamma R(e)/2$ over feasible trades; the dynamic problem uses continuation differences instead.

A position in a hedge instrument must be retained in the state unless future transitions, costs, constraints, and rewards depend only on the reduced exposure. Chapter 7's criteria for exact state reduction still apply. A small factor residual does not by itself justify discarding a separately constrained or costly hedge position.

9.11 Waiting for a Review Is Not Execution Latency

In the baseline, a chosen hedge executes before the client interval starts. Delaying execution would change which inventory earns subsequent client fills and incurs holding costs. If an order has been submitted but not filled, a state containing only current inventory can no longer distinguish the outstanding commitment.

A minimal delayed-execution extension would carry

$$\text{state} = (q, e_{\text{pending}}, \tau_{\text{remaining}}), \quad (q, e_{\text{pending}}, 0) \longmapsto (q + e_{\text{pending}}, 0, 0). \quad (9.26)$$

The transition needs its own fill price, cost, cancellation rule, and event ordering. A policy must not be allowed to forget a pending order merely because a later client trade offsets the position. An order that reduced inventory when submitted can increase its magnitude by the time it fills.

Order completion and local knowledge of completion are separate clocks. Chapter 1's event and availability records determine what the controller knows; Chapter 3's delayed-outcome distinction determines when the inventory transition actually occurs. A delayed report cannot be treated as a new trade, and a submitted order cannot be treated as an already completed hedge.

For a single certain pending trade, first maintain $q \in \mathcal{Q}$ and $q + e_{\text{pending}} \in \mathcal{Q}$. These conditions cover current inventory and pending execution after no client fill. Before allowing a client fill, additionally require

$$q + \eta z \in \mathcal{Q}, \quad q + \eta z + e_{\text{pending}} \in \mathcal{Q}. \quad (9.27)$$

Together these checks cover no fill, decline, and the two inventory stages after a client fill under the stated event order. They are not a general rule for many partially filled or cancelable orders. A richer execution mechanism requires its reachable intermediate positions to be checked.

The notebook's review-frequency experiment measures delayed access to a decision opportunity. It does not simulate outstanding orders or estimate the cost of real execution latency. Those mechanisms would require the expanded state and appropriately observed execution records.

9.12 What the Experiment Establishes

In this finite model, waiting is a chosen action with continuation value. An external hedge buys a new inventory state at a cash cost. The joint policy compares those alternatives while changing client quotes consistently with the future hedge opportunity.

Volume alone cannot establish successful internalization. Starting inventory, client fills, and external trades obey the pathwise conservation identity

$$q_T = q_0 + z(N_+ - N_-) + \sum_j e_j. \quad (9.28)$$

Gross external volume can exceed gross client volume over a short sample because the dealer started with inventory. An inventory-reducing client trade can offset an old position rather than a trade in the same measurement window. Any internalization ratio therefore needs explicit event definitions, denominators, and treatment of the opening and closing inventory.

The numerical checks establish policy consistency under supplied costs, probabilities, limits, and clocks. They do not establish execution quality in an actual market. Reference-price errors, correlated flow and price changes, venue depth, partial fills, impact persistence, and unobserved opportunities can change the result.

Chapter 10 opens the external-execution action itself. Given a target quantity, it compares immediate execution with distributing trades over time. That changes the question from whether to hedge at this review to how to complete an instruction while paying for liquidity and carrying the remaining exposure.

Exercises

1. At reference price $S = 100$, sell 5 external units using the baseline cost schedule. Calculate the cash change, inventory change, and marked-wealth change. Explain why the sale proceeds are not a trading profit and why a holding penalty is not another execution invoice.
2. Reproduce the last-interval example with zero client arrivals at inventories 5, 10, and 15. Compare every feasible external trade, including zero. Report the selected destination and explain why the optimum need not liquidate completely.
3. At initial inventories 0, 5, 10, and 15, report the joint policy's first external trade and its no-hedge advantage. Compare the first and last review regions. Use the actual interval continuation rather than only the terminal penalty.
4. Compare joint, immediate, periodic, fixed-quote, and client-only policies at initial inventories 10 and 15. Reconcile client contribution, all three external cost components, running penalty, terminal penalty, and value. Also report external volume, trade count, client fills, and expected terminal absolute inventory.
5. Compare joint and client-only RFQ offsets at the same inventory and client-step time, first at the beginning of the first interval and then in the last interval. Relate their differences to effective continuation costs. Explain why comparing different post-hedge inventories answers a different question.
6. Change the fixed external charge to 0, 1, 3, and 8. Then restrict maximum trade size to 5 rather than 10 and vary request imbalance to -0.6 and 0.6. Verify the appropriate value-ordering statements without assuming monotone action counts or a fixed symmetric threshold.
7. Refine the client step with $K = 5, 10, 20, 40$ while retaining 40 review dates. Separately compare 10, 20, and 40 review dates at client step 0.005. Explain which experiment changes an approximation and which changes the decision opportunities.
8. Reproduce the imperfect-hedge risk calculation in [Section 9.10](#), including the constrained trade. For a pending external sale of 10 from current inventory 5, examine subsequent client trades and the state required at execution. Reconcile the inventory conservation identity and explain why an external-volume ratio alone does not measure internalization.

Source Notes

[G] Olivier Guéant, *The Financial Mathematics of Market Liquidity: From Optimal Execution to Market Making* (2016), supplied book. Section 11.4, printed pp. 237–240 / PDF pp. 262–265, adds active market orders in an order-book setting. Printed p. 238 specifies cash and inventory jumps and an additive running-risk objective. Equations (11.27)–(11.28), pp. 239–240, combine continuation with market-order interventions. The preceding quote-driven model explicitly excludes active hedging. Our RFQ arrival model, finite review dates, inventory-reduction restriction, and external cost schedule are constructions for this chapter.

[C] Álvaro Cartea, Sebastian Jaimungal, and José Penalva, *Algorithmic and High-Frequency Trading* (2015), supplied book. Section 8.4, printed pp. 196–205 / PDF pp. 213–222, treats liquidation with limit and market orders. Objective (8.20), the unnumbered full-state QVI on p. 198, reduced system (8.22), intervention equality (8.24), and optimized QVI (8.25) are the relevant bridge. The terminal discussion on p. 200 explains a limiting-value change from last-instant market orders. The reading note records a sign inconsistency in printed (8.22a); our response model is specified and checked independently. Section 8.5 targets an inventory schedule and does not impose our external review calendar.

[B] Alexander Barzykin, Philippe Bergault, and Olivier Guéant, *Algorithmic market making in dealer markets with hedging and market impact*, arXiv:2106.06974v6, revised 11 December 2022. Sections 2–3 combine dealer quotes with a continuous liquidity-pool execution rate and permanent price impact. We use this as the direct connection between client quoting and external trading, not as the source of our scheduled-trade recursion. Available from [arXiv](https://arxiv.org/abs/2106.06974).

The reading notes identify inspected source passages and distinguish them from the numerical constructions. All experiment inputs are synthetic. The source models’ proofs and market calibrations are not reproduced here.

Chapter 10

Optimal Execution

Chapter 9 treated an external hedge as one available action. Here we open that action and ask how its quantity should be distributed through time. Trading faster reduces the inventory exposed to subsequent price changes. Trading more slowly can reduce the price concession paid for consuming liquidity. An execution schedule chooses between those effects, subject to actually completing the target.

The chapter starts with a deterministic target and a fully specified cash ledger. It derives a finite-grid optimum, checks it against an independent numerical optimizer, and then introduces changing liquidity and participation limits. A final, separate example shows why new client fills can make a fixed schedule inadequate. All numerical inputs remain synthetic. Chapter 5 supplies the estimation and forecast-horizon framework; here the chosen volatility and liquidity inputs are held fixed so that the execution decision can be studied on its own.

10.1 A Target, a Clock, and a Completion Rule

Suppose a dealer has decided to sell $Q > 0$ normalized trading units over a horizon T . This is a decision about the execution of an already chosen target. It does not reconsider whether that target was the best hedge of a larger portfolio. Correlated residual exposures from Chapter 7 or Chapter 9 would require an expanded state and risk model.

Quantities in the sale schedule are positive amounts to dispose of. A sale of n_k therefore has signed dealer trade $e = -n_k$ in Chapter 9's convention, and x_k here means remaining sale quantity, not a vector of portfolio risk coordinates. This local notation keeps the completion constraint and the cash-flow sign explicit.

Divide the horizon into N equal intervals of length Δt . Trade n_k executes at the **beginning** of interval k , at time t_{k-1} . Let x_k denote the quantity remaining immediately after that trade. It is this quantity that remains exposed during the following interval.

$$\Delta t = \frac{T}{N}, \quad t_k = k\Delta t, \quad x_0 = Q, \quad x_k = x_{k-1} - n_k. \quad (10.1)$$

The baseline admits continuous quantities and certain, immediate execution of every chosen slice. It is sell-only and must finish by its final trade at $T - \Delta t$:

$$n_k \geq 0, \quad \sum_{k=1}^N n_k = Q, \quad x_N = 0. \quad (10.2)$$

There is no additional execution opportunity at T . The last interval therefore has no residual inventory risk under full completion. Immediate execution means $n_1 = Q$; equal slicing means $n_k = Q/N$. The first comparison does not impose a volume cap, so both are admissible. Later, a participation cap will change that feasible set explicitly.

A deterministic schedule is selected before any price shocks are observed. It can depend on an announced liquidity curve, but it cannot use tomorrow's realized price or client flow. This restriction makes the first derivation transparent and defines precisely what its optimum means.

10.2 From Prices to a Cash Ledger

Write M_k for an unaffected reference price after interval k . Its innovations are independent standard normal variables, with no forecastable drift in the baseline:

$$M_k = M_{k-1} + \sigma\sqrt{\Delta t} \varepsilon_k, \quad \varepsilon_k \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1). \quad (10.3)$$

The word unaffected is a modeling distinction: this process excludes the dealer's own specified price impact. The execution price also depends on cumulative prior sales and on the size of the current slice. We use a block-average convention for linear permanent impact:

$$P_k = M_{k-1} - \gamma \left(Q - x_{k-1} + \frac{n_k}{2} \right) - s - \eta \frac{n_k}{V_k \Delta t}. \quad (10.4)$$

Here $V_k > 0$ is a supplied deterministic market-volume rate. The ratio $n_k/(V_k \Delta t)$ is participation over the interval associated with that slice. The proportional charge s includes an assumed half-spread or per-unit fee. The coefficient $\eta > 0$ converts participation into a temporary price concession. This participation cost is an effective finite-grid model: certain execution at the interval's start is assumed even though the volume input summarizes an interval.

The permanent coefficient $\gamma \geq 0$ moves the impacted reference price down as cumulative sales increase. Equation (10.4) charges the current slice its average own impact, $\gamma n_k/2$, as well as the impact of earlier sales. It does not price the whole block at the pre-block impacted quote. That distinction will matter when comparing formulas with the sources.

For a completed sale, execution shortfall relative to the initial unaffected reference is initial marked value minus actual sale proceeds:

$$C = QM_0 - \sum_{k=1}^N n_k P_k. \quad (10.5)$$

Positive shortfall is a cost. A favorable reference-price move can make a realized shortfall negative. Neither the sale principal nor a risk penalty is itself execution profit. The units used throughout the baseline are as follows.

Input	Meaning	Units
Q, n_k, x_k	Target, slice, remaining quantity	Normalized trading units
$T, \Delta t$	Horizon and interval	Normalized time
M_k, P_k, s, η	Prices and per-unit concessions	Currency per trading unit
V_k	Market-volume rate	Trading units per time
γ	Permanent price slope	Currency per squared trading unit
σ	Reference-price volatility	Currency per trading unit per square root of time
C	Execution shortfall	Currency, measured in JPY thousands

Three further symbols change roles in this execution chapter: η is the temporary-cost slope, not the dealer-side sign; γ is permanent price impact, not an inventory-risk preference; and λ below weights shortfall variance, rather than counting request arrivals. Their definitions and units belong to this model. The numerical figures use these same local conventions.

10.3 Which Costs Depend on the Schedule?

Let $y_k = Q - x_k$ be cumulative sales. Since $y_k - y_{k-1} = n_k$, the permanent-impact contribution telescopes:

$$\sum_{k=1}^N n_k \left(y_{k-1} + \frac{n_k}{2} \right) = \frac{1}{2} \sum_{k=1}^N (y_k^2 - y_{k-1}^2) = \frac{Q^2}{2}. \quad (10.6)$$

This result uses both full completion and the average own-impact price in (10.4). It is exact for every grid in this model. The proportional charge is also fixed at sQ . Temporary cost varies with the sizes and liquidity of individual slices.

A price shock during interval k affects the proceeds of the x_k units still to be sold. Substituting (10.3)-(10.4) into (10.5), and collecting each innovation's remaining quantity, gives the pathwise identity

$$C = sQ + \frac{\gamma Q^2}{2} + \sum_{k=1}^N \frac{\eta n_k^2}{V_k \Delta t} - \sigma \sqrt{\Delta t} \sum_{k=1}^N x_k \varepsilon_k. \quad (10.7)$$

For a deterministic schedule, the last term is a linear combination of independent normal innovations. The mean and variance therefore follow directly:

$$\begin{aligned} E(n) &:= \mathbb{E}[C] = sQ + \frac{\gamma Q^2}{2} + \sum_{k=1}^N \frac{\eta n_k^2}{V_k \Delta t}, \\ R(n) &:= \text{Var}(C) = \sigma^2 \Delta t \sum_{k=1}^N x_k^2. \end{aligned} \quad (10.8)$$

Here R is shortfall variance, not standard deviation or an extra inventory charge, and it has squared-currency units. The scalar σ^2 is price-change variance per unit time; the factor Δt converts it into the variance of one innovation. In Chapter 5's notation, $\Gamma_{t,H} \approx HC_t$ requires an explicit locally constant variance-rate assumption. A fixed-horizon price covariance, or an annualized return-volatility estimate, cannot simply replace σ^2 in (10.8).

For a supplied sequence of independent, zero-drift price innovations with known variances g_k , the same cash derivation would give shortfall variance $\sum_k x_k^2 g_k$ for a deterministic schedule. Serial covariance would add cross terms. Neither extension is part of the constant- σ numerical baseline, and an uncertain forecast of the g_k is a further source of model uncertainty. Chapter 12 distinguishes that uncertainty from sampling more paths under one fixed forecast.

The saved 8 April 1999 Almgren-Chriss preprint instead prices its slice before the slice's permanent impact enters the quoted price. With its constant-volume temporary rate coefficient η_{AC} , its finite-grid mean contains

$$E_{\text{pre}}(n) = sQ + \frac{\gamma Q^2}{2} + \left(\frac{\eta_{AC}}{\Delta t} - \frac{\gamma}{2} \right) \sum_{k=1}^N n_k^2. \quad (10.9)$$

Its effective coefficient must remain positive for the stated strict convexity argument. The corresponding coefficient in (10.8) is $\eta/(V\Delta t)$ when volume is constant. Removing or adding a half-impact term changes the finite model; it is not a harmless relabeling. The notebook separately checks both pricing identities. Neither constant permanent impact nor the absence of drift establishes that actual impact is linear, permanent, or independent of schedule.

10.4 Choose a Risk-Cost Objective

A mean-variance criterion assigns a currency value to uncertainty through a nonnegative risk weight λ :

$$J_\lambda(n) = E(n) + \lambda R(n), \quad \lambda \geq 0. \quad (10.10)$$

Because R is measured in squared currency, λ has units of inverse currency. A value cannot be transferred unchanged between yen and thousands of yen. This chapter uses λR throughout; a convention with $\lambda R/2$ would produce a different numerical weight and a different decay rate.

For each deterministic schedule in the Gaussian baseline, exponential disutility of shortfall has the exact certainty equivalent

$$\frac{1}{A} \log \mathbb{E}[e^{AC}] = E(n) + \frac{A}{2} R(n), \quad A > 0. \quad (10.11)$$

Thus $\lambda = A/2$ gives the same ranking within this deterministic Gaussian family. This identity follows from the normal moment-generating function. It does not, by itself, prove that a deterministic plan is optimal over all possible feedback controls. With adaptive quantities, random costs, signals, or changing targets, the Gaussian distribution used here need not survive.

There is no separate running inventory penalty in the baseline in addition to λR . The post-trade squared-inventory sum is exactly the variance term already being priced. Counting it again as another risk cost would change the objective. Expected spread and permanent cost remain in reported shortfall even though they are constants in this fixed-target optimization.

The direction of optimization differs from the preceding dealer-control chapters: here we minimize cost, whereas [Chapter 9](#) maximizes trading contribution less costs and penalties. For an isolated completed sale with the same opening benchmark and cash coverage, marked wealth change is minus realized shortfall. The risk-adjusted criteria agree only after their variance and penalty conventions are also matched; their numerical weights are not interchangeable.

10.5 Derive the Constant-Liquidity Schedule

For constant V , discard only the two schedule-independent terms while solving. Set $a = \eta/(V\Delta t) > 0$ and $b = \lambda\sigma^2\Delta t$. The quantity to minimize over the intermediate inventories is

$$F(x) = a \sum_{k=1}^N (x_{k-1} - x_k)^2 + b \sum_{k=1}^N x_k^2, \quad x_0 = Q, \quad x_N = 0. \quad (10.12)$$

Changing an interior x_k changes two neighboring trades. Differentiating those two temporary-cost terms and its risk term yields

$$2a(2x_k - x_{k-1} - x_{k+1}) + 2bx_k = 0, \quad k = 1, \dots, N-1. \quad (10.13)$$

The Hessian is positive definite on the free inventories: even when $b = 0$, a nonzero perturbation with both endpoints fixed has a positive sum of squared differences. Consequently this stationary solution is the unique unconstrained optimum. Its decreasing inventory path makes it feasible for the baseline sell-only condition too.

For $b > 0$, write $2 \cosh \kappa = 2 + b/a$. Solving the second-order difference equation and imposing both endpoints gives

$$\begin{aligned} \kappa &= \operatorname{arcosh} \left(1 + \frac{b}{2a} \right), \\ x_k^* &= Q \frac{\sinh\{\kappa(N-k)\}}{\sinh(\kappa N)}, \quad n_k^* = x_{k-1}^* - x_k^*. \end{aligned} \quad (10.14)$$

The parameter κ is dimensionless decay per step. Larger risk relative to temporary cost makes the path bend more strongly toward earlier sales. It does not change the target, create a final trade at T , or remove an imposed volume cap. If $\lambda = 0$, the limit of the hyperbolic-sine ratio is linear:

$$x_k^* = Q \left(1 - \frac{k}{N} \right), \quad n_k^* = \frac{Q}{N}. \quad (10.15)$$

Equal slices are therefore an optimum under specific assumptions: constant liquidity, constant quadratic cost, no risk charge, and no binding additional constraints. The formula is not a reason to divide every real order equally.

For comparison, let Δt shrink while holding $Q, T, V, \eta, \sigma, \lambda$ fixed. The corresponding continuous-inventory solution is

$$x^*(t) = Q \frac{\sinh\{\omega(T-t)\}}{\sinh(\omega T)}, \quad \omega = \sqrt{\frac{\lambda \sigma^2 V}{\eta}}. \quad (10.16)$$

At $\lambda = 0$ it has the same linear limit. Here ω is decay per unit time. Formula (10.16) assumes constant V ; a varying volume curve must enter the first-order equations or the numerical problem explicitly.

10.6 A Complete Numerical Comparison

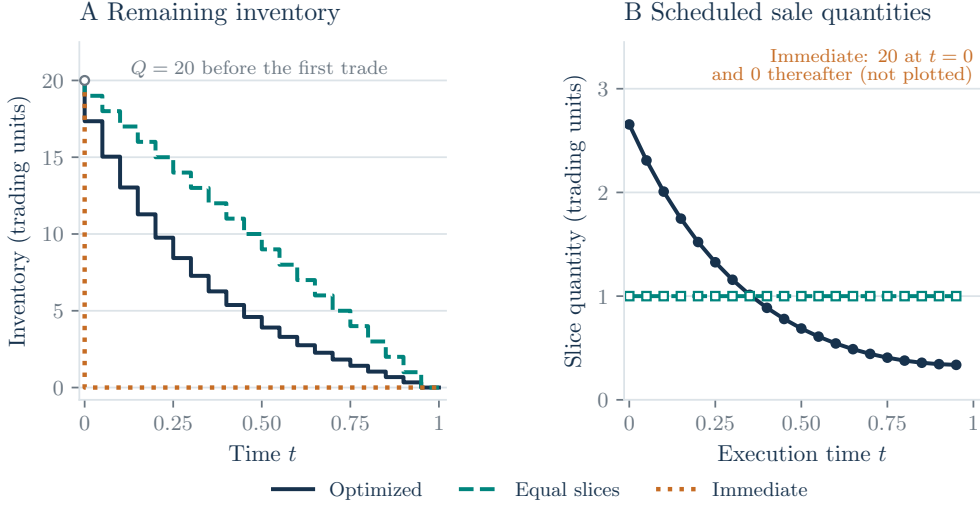
Use $Q = 20$, $T = 1$, $N = 20$, $V = 20$, $\sigma = 2$, $s = 0.05$, $\gamma = 0.02$, $\eta = 0.10$, and $\lambda = 0.01$. Each interval has length 0.05 and reference market volume $V\Delta t = 1$. Quantities are continuous in this example; no lot rounding is hidden in the calculation.

The shared fixed component is $sQ + \gamma Q^2/2 = 5$. The temporary coefficient per squared slice is $a = 0.1$, and the risk coefficient per squared post-trade inventory is $b = 0.002$. The following comparison evaluates all three schedules under (10.8)-(10.10), including the same fixed components.

Schedule	Expected cost	Standard deviation	$J_{0.01}$
Immediate	45.000000	0.000000	45.000000
Equal slices	7.000000	22.226111	11.940000
Optimized	7.954369	15.351089	10.310929

Immediate execution has zero subsequent inventory risk because the entire order trades before the first innovation. It pays a temporary cost of 40. Equal slices pay only 2 in temporary cost, but their declining remaining inventory has variance 494. The optimized schedule exchanges some additional expected cost for lower exposure; it is selected by the complete objective, not by expected cost alone.

The optimized first slice is approximately 2.65546 and its last slice is 0.33716. It pays 2.954369 in temporary cost and 2.356559 in variance charge, giving expected shortfall 7.954369 and objective 10.310929. These are model expectations and a risk-adjusted criterion; the chapter does not label their difference as realized trading profit.



$Q = 20$, $N = 20$, $\lambda = 0.01$; all three schedules complete the same target.
 Each block trades before its interval; the first jump is at $t = 0$, the last trade at $t = 0.95$.
 No trade occurs at $T = 1$. Panel B omits the immediate schedule to show the smaller slices clearly.

Figure 10.1: Execution schedules under the same baseline. Inventory is shown after each beginning-of-interval trade; the first drop occurs at time zero. Slice quantities and remaining inventory describe the same completed target.

The notebook calculates the schedule in three complementary ways: the closed-form inventory path, the stationary linear equations, and constrained numerical optimization. A separate evaluator computes proceeds, expected components, and variance from the stored quantities. Agreement checks both the optimizer and the convention linking a trade to the next price shock.

10.7 Read the Cost-Risk Frontier

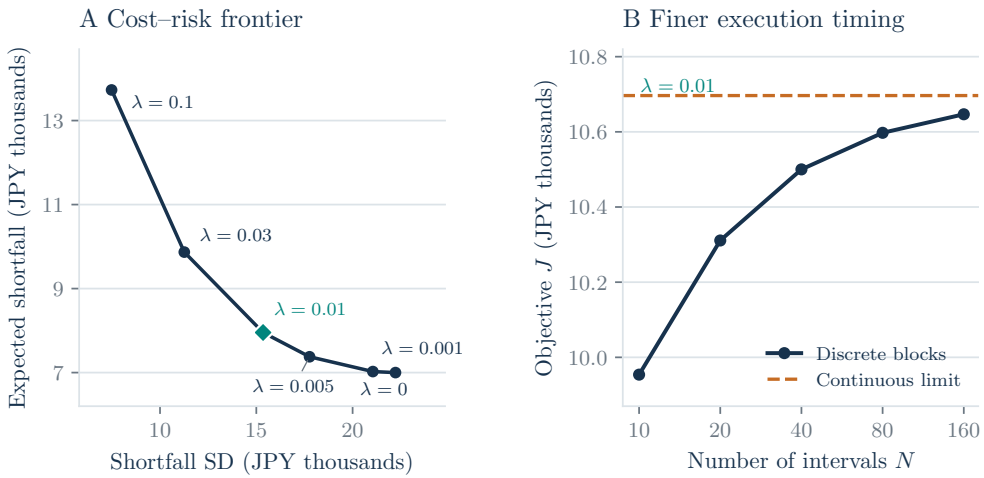
Changing λ produces different optimal schedules from one common feasible set. Plotting expected cost against standard deviation makes their tradeoff visible without treating the risk penalty as a realized cash debit. Each point still solves an objective that uses variance.

Let schedules n_A, n_B minimize (10.10) at $0 \leq \lambda_A < \lambda_B$. Comparing each optimum with the other feasible schedule gives

$$\begin{aligned}
 E_A + \lambda_A R_A &\leq E_B + \lambda_A R_B, \\
 E_B + \lambda_B R_B &\leq E_A + \lambda_B R_A \\
 \implies R_B &\leq R_A, \quad E_A \leq E_B.
 \end{aligned} \tag{10.17}$$

The inequalities follow from optimality, not from a visual pattern in a few simulations. They require the same costs, price model, completion condition, and action constraints. They do not establish a comparable ordering when liquidity or the target changes. Nor do they prove that every individual slice must rise or fall monotonically in a richer constrained model.

λ	Expected cost	Standard deviation	First slice
0	7.000000	22.226111	1.000000
0.001	7.024491	21.051353	1.233825
0.005	7.376781	17.764655	1.977348
0.01	7.954369	15.351089	2.655464
0.03	9.868665	11.260002	4.336147
0.1	13.728723	7.484371	7.165152



A: each point minimizes $\mathbb{E}[C] + \lambda \text{Var}(C)$; the axis shows SD, not variance.
 B: economic inputs stay fixed while Δt shrinks; coarse blocks move inventory earlier.
 Weights use inverse currency units. Synthetic inputs; the continuous limit is a timing benchmark.

Figure 10.2: Left: expected shortfall versus its standard deviation for deterministic optimized schedules on the baseline grid. Right: the discrete objective and its continuous limit as the time grid is refined at fixed economic parameters, examined in Section 10.9. The panels vary different inputs and answer different questions.

There is a useful unit check. If all currency values are multiplied by a positive factor c , the variance is multiplied by c^2 . Preserving the selected schedule requires

$$C' = cC, \quad E' = cE, \quad R' = c^2R, \quad \lambda' = \frac{\lambda}{c}, \quad J'_{\lambda'} = cJ_{\lambda}. \quad (10.18)$$

The notebook verifies this invariance by converting every relevant price coefficient and the risk weight consistently. A copied numerical risk parameter without its currency convention would fail this check.

10.8 Liquidity and Capacity Change the Answer

Suppose more market volume is available late in the horizon. Waiting now offers cheaper temporary execution as well as longer exposure. A supplied participation limit $\rho_{\max}$ adds interval capacities:

$$0 \leq n_k \leq c_k := \rho_{\max} V_k \Delta t, \quad \sum_{k=1}^N c_k \geq Q. \quad (10.19)$$

With continuous nonnegative slices and only these caps plus completion, the final inequality is necessary and sufficient for feasibility. It should be checked before optimization. An optimizer cannot repair a target that exceeds every remaining execution opportunity. Discrete lots, minimum tickets, or additional venue constraints would require stronger checks.

In the risk-neutral problem without binding caps, minimizing $\sum_k \eta n_k^2 / (V_k \Delta t)$ with a quantity multiplier gives $n_k / (V_k \Delta t)$ constant. Thus

$$n_k^* = Q \frac{V_k \Delta t}{\sum_{j=1}^N V_j \Delta t}. \quad (10.20)$$

This is a volume-proportional schedule based on a deterministic input curve. It is not a promise to match a benchmark computed from subsequently realized market volume. At positive risk weight, earlier execution can be preferable even when its temporary cost is higher.

If the volume curve is a forecast, both the expected concession and the projected capacity inherit its uncertainty. The feasibility result above is conditional on the supplied capacities. Revising them after new observations requires checking the still-unexecuted quantity against the remaining capacity; the original deterministic completion claim does not guarantee liquidity that fails to appear.

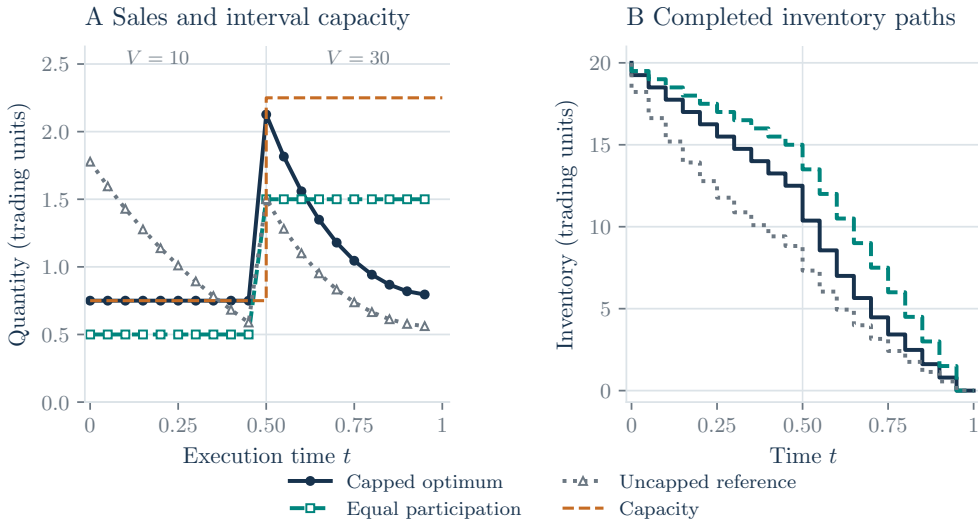
For the general finite problem, let L be the lower-triangular matrix of ones, so $x = Q\mathbf{1} - Ln$, and let D be diagonal with $D_{kk} = \eta / (V_k \Delta t)$. With $b = \lambda \sigma^2 \Delta t$, the constrained stationarity and complementary-slackness conditions are

$$\begin{aligned}
2Dn - 2bL^\top x + \nu \mathbf{1} - \ell + \mu &= 0, \\
\ell_k n_k &= 0, \quad \mu_k(c_k - n_k) = 0, \quad \ell_k, \mu_k \geq 0.
\end{aligned} \tag{10.21}$$

These accompany the feasibility conditions and full completion. Since D is positive definite and $b \geq 0$, the objective is strictly convex and these conditions certify the unique solution. At a capacity bound, an unconstrained stationary equation need not hold: its residual is balanced by the multiplier. The implementation verifies the appropriate conditions instead of demanding a zero unconstrained gradient everywhere.

Set $V_k = 10$ for the first ten intervals and $V_k = 30$ for the last ten. Total reference volume remains 20. With $\rho_{\max} = 1.5$, the slice caps are 0.75 early and 2.25 late; total capacity is 30. An equal-participation schedule sells 0.5 early and 1.5 late, so it is a feasible comparator.

Schedule	Expected cost	Standard deviation	$J_{0.01}$
Uncapped reference	8.365303	19.353850	12.111018
Capped optimum	7.291526	23.955957	13.030405
Equal participation	7.000000	26.972208	14.275000



$Q = 20$, $\lambda = 0.01$, $\rho_{\max} = 1.5$; each capacity is $\rho_{\max} V_k \Delta t$.
Capped and equal-participation schedules respect the limits. The uncapped reference violates them.
All three complete; simply clipping the uncapped quantities leaves 3.90 units unsold.

Figure 10.3: Execution with low early liquidity, high late liquidity, and participation limits. Feasible comparisons complete the same target; the unconstrained reference indicates where caps alter the chosen schedule.

The uncapped reference uses a larger feasible set and is not executable under the displayed caps. The constrained optimum binds all ten early caps, selling 0.75 at each of those reviews. It has temporary cost 2.291526 and variance charge 5.738879. Clipping the uncapped schedule without reallocating leaves 3.904508 units unsold. The comparison between the capped optimum and equal participation uses the same capacity-constrained problem.

Clipping each excessive slice of an unconstrained schedule generally leaves unsold quantity. Reallocating that quantity changes both subsequent risk and temporary cost. The correct constrained solution solves the whole schedule again; it is not a clipped curve with the original value attached. The notebook also rejects an explicitly infeasible capacity configuration.

10.9 Refine Time Without Changing the Model Silently

The finite schedule has trades at the beginning of intervals; the continuous model spreads them through time. Compare them by holding the economic coefficients and horizon fixed while refining the grid. For constant liquidity, the recurrence parameter satisfies

$$\frac{\kappa}{\Delta t} = \frac{1}{\Delta t} \operatorname{arcosh} \left(1 + \frac{\lambda \sigma^2 V (\Delta t)^2}{2\eta} \right) \rightarrow \omega. \quad (10.22)$$

This gives a concrete convergence target for the discrete inventory path. The experiment compares grids with $N = 10, 20, 40, 80, 160$ with (10.16), evaluating each discrete schedule under its own stated grid and evaluating continuous cost and variance by their continuous integrals.

N	Expected cost	Standard deviation	Inventory error
10	7.935284	14.208008	4.953959
20	7.954369	15.351089	2.655464
40	7.959214	15.940230	1.375031
80	7.960430	16.238809	0.699677
160	7.960734	16.389047	0.352921
Continuous	7.960835	16.539896	—

Inventory error is $\max_k |x_k - x^*(t_{k-1})|$: the post-trade finite inventory is compared with the continuous path at the actual execution time. Evaluating the same x_k at t_k would compare the formal recurrence nodes instead and produce a much smaller error. Both are recorded in the notebook, with separate labels. The first trade is a time-zero block whose size vanishes as the grid is refined.

Do not interpret a coarse grid's lower measured exposure as superior trading technology. Beginning-of-interval execution moves quantity before an entire interval's

innovation. Refinement changes that timing approximation. It differs from adding external review opportunities while holding an underlying client-arrival grid fixed, as in Chapter 9.

One indexing error is particularly easy to detect. Charging the same innovations to pre-trade inventories x_{k-1} instead of post-trade inventories x_k changes the reported variance by

$$\sigma^2 \Delta t \sum_{k=1}^N (x_{k-1}^2 - x_k^2) = \sigma^2 \Delta t Q^2. \quad (10.23)$$

For any completed schedule on this grid, the difference is the same constant. It can therefore leave a computed optimizer unchanged while making every risk report wrong for the declared timing. In the baseline the error is 80 squared-currency units. It would even report risk after the entire target had already been sold at time zero.

10.10 A Terminal Penalty Is Not Completion

A completion requirement and a preference against leftover quantity are different mathematical objects. In a minimization problem they can be represented by different terminal functions:

$$\Phi_{\text{hard}}(r) = \begin{cases} 0, & r = 0, \\ +\infty, & r \neq 0, \end{cases} \quad \Phi_B(r) = Br^2, \quad 0 \leq B < \infty. \quad (10.24)$$

Consider a separate one-period, deterministic example with no spread, permanent impact, or price risk. Mark residual inventory at the same reference price at the horizon, rather than treating unsold principal as an execution loss. If the temporary cost is $a(Q - r)^2$ and the residual penalty is Br^2 , the optimum is

$$r_B^* = \frac{a}{a+B}Q, \quad n_B^* = \frac{B}{a+B}Q. \quad (10.25)$$

For positive a, B, Q , a finite penalty leaves a positive residual. With $a = 0.1$, $B = 1$, and $Q = 20$, it leaves 20/11 units. Letting B grow approaches completion in this example, but does not make a finite penalty identical to (10.2). A capacity shortfall can make hard completion impossible at every penalty value.

The notebook reports that example separately. It does not rank its penalized, partially completed trade against the fully completed baseline as if they delivered the same service. When evaluating an incomplete order, both the residual mark and the required completion convention belong in the benchmark definition.

10.11 When Client Fills Change the Target

On a dealer desk, inventory can change while an external order is being worked. A new client purchase or sale changes what remains to be hedged. A deterministic schedule for the original target then becomes a policy input, not a complete control rule.

To isolate information and feedback, use a small, separate two-review example. Let r_k be signed inventory before review k , and let signed $u_k > 0$ mean an external sale. Thus $u = -e$ relative to [Chapter 9](#), while a positive client change still increases dealer inventory. Unlike the nonnegative remaining target x_k , r_k may become negative. An intervening client inventory change ξ_{k+1} produces

$$r_{k+1} = r_k - u_k + \xi_{k+1}. \quad (10.26)$$

Each external trade is an integer in $\{-2, -1, 0, 1, 2\}$. Initially $r_0 = 2$. After the first trade and its holding period, a client shock is $+2$ or -2 with probability one half. It is observed before the second trade. There is no further client shock. The cost of a trade and the following holding period is $g(r, u) = 0.25u^2 + 0.1(r - u)^2$, and the final residual penalty is $\Phi(r) = r^2$.

This is an additive expected-cost tracking problem with synthetic coefficients. It contains neither the baseline permanent-impact cost nor its exponential-utility interpretation. Its finite terminal penalty is intentional. Write V_1 for the value before the second review and V_0 before the first:

$$\begin{aligned} V_1(r) &= \min_{u \in \{-2, \dots, 2\}} \{g(r, u) + \Phi(r - u)\}, \\ V_0(2) &= \min_{u \in \{-2, \dots, 2\}} \left\{ g(2, u) + \frac{1}{2}V_1(4 - u) + \frac{1}{2}V_1(-u) \right\}. \end{aligned} \quad (10.27)$$

An open-loop comparator must choose both trades before observing the client shock. Its expected cost is instead

$$\begin{aligned} J_{\text{open}} &= \min_{u_0, u_1 \in \{-2, \dots, 2\}} \left\{ g(2, u_0) + \frac{1}{2}f(4 - u_0, u_1) + \frac{1}{2}f(-u_0, u_1) \right\}, \\ f(r, u) &= g(r, u) + \Phi(r - u). \end{aligned} \quad (10.28)$$

The feedback policy may use a different second action on each observed branch. For any fixed first trade, this gives the comparison

$$\mathbb{E} \left[\min_u f(r, u) \right] \leq \min_u \mathbb{E} [f(r, u)], \quad V_0(2) \leq J_{\text{open}}. \quad (10.29)$$

Both sides use the same shock probabilities, timing, action bounds, and terminal penalty. The inequality comes from a larger information-dependent set of actions, not from changing the cost of a leftover position. It cannot be transferred to unrestricted mean-variance feedback execution without a new derivation.

Expected component	Feedback	Fixed in advance
Temporary cost	0.875000	0.500000
Holding cost	0.150000	0.500000
Terminal penalty	0.500000	4.000000
Total objective	1.525000	5.000000

The feedback policy first sells one unit. After a client shock of -2 , its observed inventory is -1 and it buys one unit, so its second signed sale is -1 . After a shock of $+2$, its inventory is 3 and it sells two, leaving one unit. The best fixed plan sells one unit at each review, leaving -2 or $+2$. Their objective difference is 3.475000 . The fixed plan's expected residual is zero, yet its terminal squared-inventory cost is 4 ; averaging inventory before squaring would erase precisely the uncertainty that matters.

The notebook enumerates both branches and all allowed actions. It also verifies inventory conservation along each path. A production extension would need a state containing current inventory, the still-required target, outstanding external orders, remaining capacity, and relevant signals. Pending orders must be accounted for before another controller spends the same capacity or inventory limit.

10.12 What an Execution Plan Commits To

An execution model commits to a benchmark, a quantity, an information set, an execution clock, and a completion rule. Its cost forecast is meaningful only when those definitions agree with the trade record. Sending an order is not the same event as filling it. If only part of a slice executes, the remaining quantity must be recomputed from actual signed fills before planning the next action.

The baseline's useful result is a complete chain from prices to proceeds, from proceeds to a cost distribution, and from that distribution to a feasible schedule. Its closed form exposes the role of risk and temporary cost; the constrained experiment shows why liquidity and capacity must enter before selecting the schedule. The feedback example then separates a fixed plan from a rule that reacts to information available at a later review.

The numerical evidence establishes consistency of these specified models. It does not estimate their coefficients from market data, certify a venue's available liquidity, or demonstrate execution quality in a rates market. Signals, correlated price and client flow, uncertain fills, transient impact, minimum lots, and correlated hedge instruments would each require changes to the objective or state.

Chapter 11 connects this decision model to transaction-cost analysis and profit attribution. The predicted shortfall is now confronted with actual fills and a declared benchmark. Expected cost, realized slippage, intervening price movement, and the mark on any residual must remain separate while reconciling to the same cash and inventory ledger.

Exercises

1. **Cash and permanent impact.** For $Q = 20$, compare immediate execution and equal slices using the block-average price. Verify the permanent contribution $\gamma Q^2/2$. Repeat with pre-block pricing and explain the extra finite-grid term.
2. **Which inventory sees the shock?** Derive the coefficient of ε_k in shortfall. Compute the baseline equal-slice variance and the error caused by using pre-trade inventory. Explain why this error need not change the optimizer.
3. **Solve and evaluate.** Derive (10.13), impose both endpoints, and recover (10.14). Compare the resulting slices and objective with the numerical optimizer. Check nonnegativity and full completion.
4. **Change the volume curve.** At zero risk weight, derive (10.20). For the low-early, high-late volume curve, compute the slice quantities and their participation rate. State which conditions make that schedule feasible under the cap.
5. **Capacity and completion.** Re-solve the positive-risk volume-curve example with its participation limits. Identify binding slices and verify the constrained optimality conditions. Show why clipping an unconstrained schedule is not a completed solution, and construct an infeasible target-capacity pair.
6. **Time and units.** Compare the five grids with the continuous solution, including inventory-path error and cost components. Convert currency amounts by a factor of 1,000 and adjust λ so that the chosen schedule remains unchanged.
7. **Penalize a residual.** Derive (10.25), calculate its residual and component costs, and compare its completion condition with the hard constraint. Explain how the residual mark enters the incomplete-order benchmark.
8. **Use new information.** Enumerate the two-review tracking example. Find the optimal first action and both second actions, then compare them with the best pair chosen in advance. Reconcile temporary costs, holding costs, terminal penalties, and inventory on each branch.

Source Notes

Almgren and Chriss, saved 8 April 1999 preprint. The inspected source is the 50-page draft of *Optimal Execution of Portfolio Transactions*, not the later journal pagination. Pages 8-11 define its finite price and shortfall convention; equations (1)-(5) give prices, execution, expectation, and post-trade variance. Equation (8) on page 11 contains the finite-step permanent-impact correction. Equations (16)-(18) on page 14 lead to its difference equation and hyperbolic-sine path. Our chapter uses an explicitly different block-average own-impact price and derives its finite-grid coefficient accordingly. The [journal record](#) identifies the published work; draft page pointers here refer to the stored source.

Guéant, 2016. In the saved book, printed pages 48-53 (PDF pages 73-78), equations (3.5)-(3.15), connect Gaussian execution proceeds, deterministic optimization, and the flat-volume quadratic solution. Printed pages 59-61 (PDF pages 84-86), equations (3.16)-(3.22), make the finite-step timing and permanent-impact correction explicit. The source's deterministic-optimality result for its CARA model has additional assumptions; we do not infer it from the Gaussian moment formula alone. Printed pages 95-97 (PDF pages 120-122) give participation-cap feasibility and the constrained control formulation. The reading note also records a varying-volume differential-equation caution; the chapter uses the general discrete convex problem for nonconstant volume.

Cartea, Jaimungal, and Penalva, 2015. Printed pages 139-141 (PDF pages 156-158), equations (6.4)-(6.12), derive equal-speed execution with a completion constraint. Printed pages 144-147 (PDF pages 161-164), equations (6.20)-(6.30), introduce running inventory cost and terminal liquidation. In that setup the terminal block is actually liquidated at a stated cost; a finite coefficient does not mean that a residual is left indefinitely. Our Section 10.10 explicitly studies a different residual-marking example. Printed pages 150-151 (PDF pages 167-168), equations (6.33) and (6.35), connect CARA risk aversion to the running coefficient. These source passages support the distinctions, not an empirical calibration of the present examples.

Original experiments. The quantities, coefficients, time grids, liquidity curve, participation cap, and two-branch client shock are educational constructions. The finite-grid algebra and computations illustrate the chapter's declared models; they do not reproduce a source's empirical calibration. The companion notebook contains the calculations and worked exercise answers. Reading notes identify the source editions and the passages supporting each comparison.

Chapter 11

Transaction-Cost Analysis and Profit Attribution

The previous chapters chose quotes, hedges, and execution schedules. This chapter asks what their outcomes actually mean. A dealer may earn a favorable execution edge and still lose money while holding inventory. A seller may execute below the contemporaneous reference price and nevertheless outperform a decision-time benchmark. A favorable average markout may describe a small, selected subset of requests. Each statement can be correct; none is a complete account of the others.

Our starting point is a cash and inventory ledger. It gives an exact marked-profit identity, then provides a common foundation for transaction-cost analysis (TCA), execution benchmarks, post-fill markouts, and request-level statistics. The numerical experiment uses one synthetic instrument, supplied reference prices, eight fills, and twelve client requests. All prices are level prices in thousands of JPY per normalized trading unit; all monetary totals are in thousands of JPY. The integer clock is a synthetic observation grid, not a claim about a particular market's event frequency. No parameters are estimated from real transactions.

11.1 Write down the clock before measuring performance

Let M_k be the reference price at time k , q_k the dealer's inventory after events at that time, and X_k its cash account. A signed trade quantity v_k is positive when the dealer buys and negative when it sells. Its execution price is p_k , and its cash fee is f_k . The client has the opposite side. The accounting updates are

$$q_k = q_{k-1} + v_k, \quad X_k = X_{k-1} - v_k p_k - f_k. \quad (11.1)$$

At a time without a fill, $v_k = 0$ and no execution price is observed or required. A separately recorded cash charge can still make f_k positive. In the baseline example, such times have zero fees. Multiple fills at the same mark time can be processed sequentially; the price-movement term occurs once, and every cash event is counted once.

This restores the dealer-inventory convention used for client trades and [Chapter 9](#)'s external trades: $v = \eta z$ for a client fill and $v = e$ for an external fill. [Chapter 10](#)'s positive sale amount corresponds to $v = -n$, or $v = -u$ in its feedback example. A positive cost is unfavorable; a positive marked-wealth increment is favorable.

The mark is updated first, then the trade is booked. Inventory held immediately before the mark update therefore earns the price change into that time. Inventory after the trade earns the next interval's change. This convention prevents using the new position to earn a move that occurred before it existed. It also matters at the final time: stopping the event ledger after the last trade would omit the remaining inventory's exposure to subsequent marks.

This is a ledger of resolved events, not an assumption that their reports reached the policy instantly. Chapter 1's availability clock is still needed when reconstructing a historical decision. The later-known fill belongs at its actual execution time in this account, while a decision replay may still show it as pending until its report becomes available.

Marked wealth is the cash account plus the inventory valued at the supplied mark:

$$W_k = X_k + q_k M_k, \quad \Pi_H = W_H - W_0. \quad (11.2)$$

This is marked P&L over the chosen window, not cash profit alone and not a valuation at executable liquidation prices. The experiment excludes financing, coupons, carry cash flows, currency conversion, and transfers because no such cash events are supplied. If a production account has them, they require explicit entries and an extended reconciliation. A risk penalty from a control objective is not a cash payment and must not be silently inserted into X_k .

The synthetic opening position is $q_0 = 10$, with $X_0 = 0$ and $M_0 = 100$. Opening wealth is therefore 1,000. The reference path at times 0 through 12 is 100, 100.2, 99.9, 100.4, 100.1, 100.0, 100.5, 100.3, 100.6, 100.2, 100.7, 100.4, 100.8. The eight fills are recorded below. Signed quantities are always the dealer's quantities.

Time and origin	Signed quantity	Execution price	Cash fee
1, client	+5	100.05	0.05
2, external	−4	99.82	0.04
3, client	−6	100.65	0.06
4, client	+3	99.95	0.03
5, external	−5	99.88	0.05
7, client	−4	100.50	0.04
8, client	+7	100.45	0.07
10, external	−3	100.60	0.03

An event ID distinguishes a fill from a repeated report of the same fill. Client fills link to request IDs; external fills link to a parent instruction. Request states belong in a separate table. A lost quote, a declined inquiry, or an expired request does not change inventory merely because a hypothetical execution price can be attached to it.

11.2 Derive the P&L bridge from cash

Substitute the cash and inventory updates into marked wealth and subtract the preceding wealth. The exact event identity is

$$W_k - W_{k-1} = q_{k-1}(M_k - M_{k-1}) + v_k(M_k - p_k) - f_k. \quad (11.3)$$

The first term belongs to inventory held through the price change. The second belongs to the actual fill relative to the current reference. A dealer purchase below the mark makes the second term positive; a dealer sale above the mark does too. A fee reduces wealth regardless of trade direction. These conclusions follow from the signed cash update, so they do not depend on an informal choice of what a bid or ask ought to mean.

Partition fills into client fills $\mathcal{C}$ and external fills $\mathcal{E}$. Define gross client execution edge E_C , gross local external execution cost C_E , inventory carry I , and total cash fees F . Summing the event identity gives

$$\begin{aligned} \Pi_H &= E_C - C_E + I - F, \\ E_C &= \sum_{i \in \mathcal{C}} v_i(M_i - p_i), \quad C_E = \sum_{i \in \mathcal{E}} v_i(p_i - M_i), \quad F = \sum_k f_k. \end{aligned} \quad (11.4)$$

Here M_i means the reference mark at fill i 's execution time. Positive C_E is adverse to the dealer. It is measured against each fill's contemporaneous mark, not against the parent order's decision or release price. Neither E_C nor C_E is necessarily positive in a general data set. The word edge describes a signed reference-price difference, not evidence of a profitable strategy.

Inventory carry includes every price interval, including those with no fill:

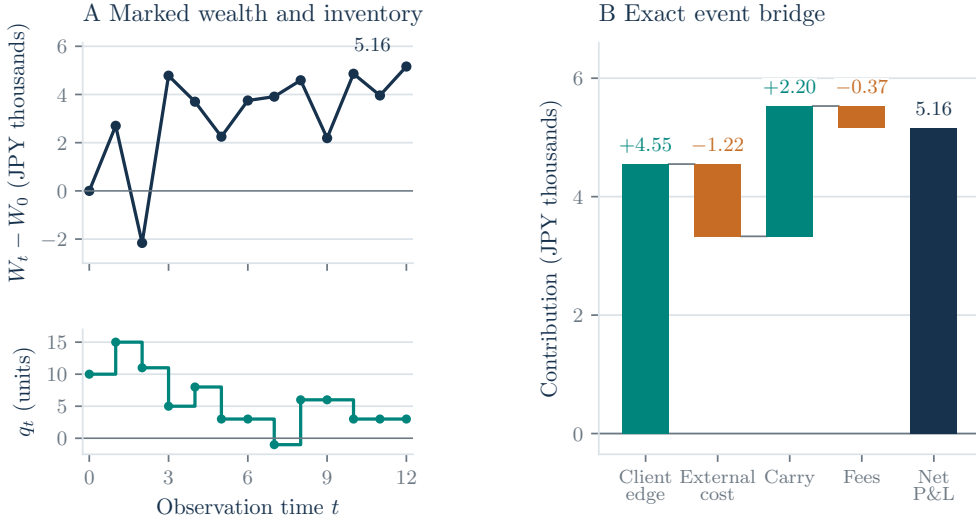
$$I = \sum_{k=1}^H q_{k-1} \Delta M_k, \quad \Delta M_k = M_k - M_{k-1}. \quad (11.5)$$

Here carry means marked price movement on inventory already held. It does not include financing or contractual carry cash flows. Chapter 5's variance forecasts describe uncertainty about such movement before it occurs; they are not additional realized entries in this identity. Changing a risk forecast can change a policy score without changing the booked cash or this realized price-movement sum.

The numerical ledger ends with cash 702.76 and inventory 3, marked at 100.8. Terminal wealth is 1,005.16, so total P&L is 5.16. The independent component calculation gives client edge 4.55, external local cost 1.22, inventory carry 2.20, and fees 0.37. Thus $4.55 - 1.22 + 2.20 - 0.37 = 5.16$. Client fees are 0.25 and external fees are 0.12; the gross components above exclude both.

The inventory path also explains why turnover alone cannot reconstruct carry. After the fill at time 7, the dealer is short one unit; after the client purchase at time 8, it is

long six. The final external sale leaves three units exposed through times 11 and 12. A calculation that looks only at prices attached to fills loses some of these intervals.



$W_0 = 1,000$; the same supplied ledger ends at $W_{12} = 1,005.16$ with $q_{12} = 3$.
 Marks update before executions. Client edges and external costs are gross; fees are subtracted once.
 Observation-clock paths are synthetic. The bridge attributes marked P&L, not causal impact.

Figure 11.1: Exact reconciliation of the synthetic account. The inventory and wealth paths use the stated mark-then-trade clock. The component bridge sums to the change in marked wealth; all monetary amounts are thousands of JPY.

11.3 A second exact attribution uses a common terminal mark

Another useful identity follows by expanding terminal inventory into opening inventory plus all signed fills. Cash proceeds and purchases then telescope into

$$\Pi_H = q_0(M_H - M_0) + \sum_{i \in \mathcal{C} \cup \mathcal{E}} v_i(M_H - p_i) - F. \quad (11.6)$$

The opening position earns its change to the terminal mark. Every subsequent trade is attributed a signed value against that same terminal mark. This is an accounting allocation: an individual fill need not have remained economically unhedged until H . Offsetting fills appear elsewhere in the same sum and cancel the exposure appropriately. A client purchase followed by an external sale can therefore be reconciled without choosing which physical inventory unit was sold.

In the example, the opening inventory contribution is 8.00, the client-fill contribution after client fees is 6.40, and the external-fill contribution after external fees is -9.24 .

Their sum is again 5.16. The external contribution is negative largely because external sales are compared with the later price of 100.8. Calling the entire 9.24 an execution defect would confuse subsequent price movement with the price concession paid at execution.

Equations (11.4) and (11.6) are alternative complete decompositions. Adding a client terminal markout to the client edge and inventory carry in the first decomposition would count some price movement twice. Similarly, matching only profitable client fills to favorable hedges and leaving adverse positions in a residual bucket can change the apparent allocation without changing total wealth. A matching rule may be useful for business reporting, but it needs a documented rule and a reconciliation to the full account.

These identities are our accounting derivations. They do not assert that all gains assigned to a client fill were caused by that client's trade, or that an observed hedge was the optimal action available at the time.

11.4 Execution shortfall depends on its benchmark

Consider the external parent instruction separately from the full desk account. It is an instruction to sell $Q = 15$ units, decided at time 0 when $D = 100$, and released at time 1 when $A = 100.2$. Its three children sell 4, 5, and 3 units at times 2, 5, and 10. Executed quantity is $Q_F = 12$, external fees total $F_E = 0.12$, and 3 units remain unfilled.

For a parent with a common side, write $\eta = +1$ for a buy and $\eta = -1$ for a sell, and let $z_i > 0$ be child quantities. The filled-quantity shortfall against decision price D is

$$C_D^F = \eta \sum_{i \in \mathcal{E}} z_i (p_i - D) + F_E, \quad Q_F = \sum_{i \in \mathcal{E}} z_i. \quad (11.7)$$

Positive shortfall is a cost: buying above the benchmark or selling below it is adverse. The superscript F emphasizes that this measure covers the executed quantity only. It does not pretend the entire instruction was completed.

Chapter 10's completed sell target is the special case $\eta = -1$, $Q_F = Q$, with the same initial benchmark and the same fee coverage. Its modeled execution costs are already reflected in the execution prices. They cannot be subtracted again as a separate estimate when reconciling these realized proceeds.

The arrival-price version C_A^F substitutes A for D . Adding and subtracting A yields

$$C_D^F = C_A^F + \eta Q_F (A - D). \quad (11.8)$$

The second term is the decision-to-release benchmark movement applied to executed quantity. In the example, $C_A^F = 2.04$ and $\eta Q_F (A - D) = -2.40$, so $C_D^F = -0.36$. The same three sales underperform the arrival price but outperform the earlier decision price. Nothing about the cash flows changes when the benchmark changes.

One can instead add and subtract each child's execution-time reference price. With fees included, let $C_M^F = C_E + F_E$ be local execution cost. Then

$$C_D^F = C_M^F + \eta \sum_{i \in \mathcal{E}} z_i (M_i - D). \quad (11.9)$$

Here $C_M^F = 1.34$ and the decision-to-fill reference-price contribution is -1.70 . Their sum is again -0.36 . The reference-price contribution reflects the observed path and selected execution times. An accounting bridge alone does not identify how much movement was caused by trading, how much came from public information, or what would have happened under another schedule.

For completeness, the parent's own gross executed volume-weighted price is

$$\bar{p}_F = \frac{\sum_i z_i p_i}{Q_F}, \quad C_D^F = \eta Q_F (\bar{p}_F - D) + F_E, \quad Q_F > 0. \quad (11.10)$$

Gross proceeds are 1,200.48, so this price is 100.04. Net proceeds are 1,200.36, or 100.03 per sold unit. The fee-adjusted figure is a net sale price, not the actual average execution price. Neither is a market VWAP benchmark: no market-volume series has been supplied. An empty parent has no observed executed VWAP; a reporting system should preserve that missing value rather than report zero.

11.5 Partial completion needs an explicit residual convention

Almgren and Chriss's saved 1999 preprint imposes complete liquidation and defines realized shortfall as the initial marked portfolio value less sale proceeds. Our parent is incomplete. To extend the measurement, specify a common evaluation time H and value the residual $R = Q - Q_F$ at the reference price M_H . The full-target marked shortfall is

$$C_D(H) = C_D^F + \eta R (M_H - D), \quad R = Q - Q_F \geq 0. \quad (11.11)$$

For a sell instruction this follows directly from comparing the initial marked target with actual net proceeds plus the value of retained units:

$$C_D(H) = QD - \left[\sum_i z_i p_i - F_E + R M_H \right], \quad \eta = -1. \quad (11.12)$$

For a buy instruction, the analogous comparison is actual cash spent minus the cost of buying the full target at D , plus the marked cost of the unpurchased units. The buy residual is an unfulfilled acquisition target, not an asset already owned; its marked cost is a comparison amount, not an actual cash debit. Both sides give (11.11). This is an independent marked-residual extension, not an additional source formula attributed

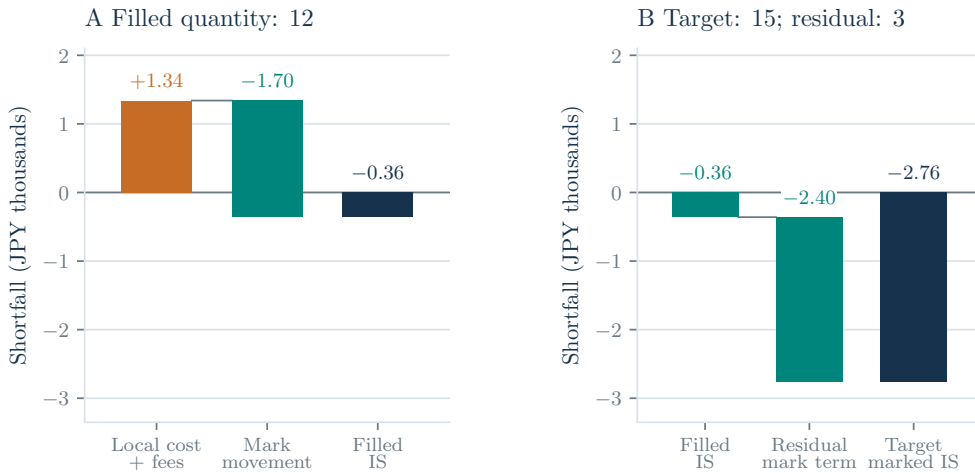
to the complete-liquidation model. It specifies the comparison portfolio; it does not execute the residual at M_H .

The example has residual contribution $-3(100.8 - 100) = -2.40$, so full-target shortfall is -2.76 . Retaining units during a price rise looks favorable for this sell-side comparison. It does not mean the instruction was fulfilled, that those units could actually be sold at the mark, or that leaving the residual was a good decision given the information and mandate at release.

If the full-target measure is computed against arrival price, the benchmark bridge must use the full target quantity:

$$C_D(H) = C_A(H) + \eta Q(A - D). \quad (11.13)$$

Here $C_A(H) = 0.24$, and the full-target benchmark movement is -3.00 . Using the filled quantity 12 in this full-target bridge would leave an unexplained 0.60. The distinction between (11.8) and (11.13) is a useful reconciliation test precisely because both formulas look plausible when their quantity scope is omitted.



SELL: target 15, fills 12, residual 3. Negative shortfall favors the seller.
A: local cost plus decision-to-fill mark movement. B: adds a terminal mark for the residual.
The residual term is not cash or completion; these scopes do not rank policies.

Figure 11.2: Benchmark bridges for the same partial sell instruction. The filled-quantity comparison covers 12 units. The full-target comparison adds three unfilled units marked at the common horizon. These are different quantity scopes, not competing scores to rank without qualification.

A residual penalty in an optimizer is a different object again. It expresses a preference or constraint tradeoff; it is not an executed cleanup fee and need not equal the marked opportunity term above. Comparing strategies fairly requires the same mandate,

horizon, completion convention, reference construction, and cost coverage, alongside their fill quantities. A strategy cannot be declared cheaper solely by excluding difficult units from its measured sample.

11.6 Markouts measure a signed value at a stated horizon

For an actual fill at time t_i , set $z_i = |v_i|$ and $\eta_i = \text{sign}(v_i)$. Define its net dealer markout per executed unit at elapsed horizon $h \geq 0$ by

$$\alpha_i(h) = \eta_i (M_{t_i+h} - p_i) - \frac{f_i}{z_i}, \quad z_i > 0. \quad (11.14)$$

The reference horizon starts at the fill time in this chapter. A buy benefits from a higher future mark, while a sale benefits from a lower one. Fees are allocated once to the executed units in that fill. We include initial execution edge in the word markout; other systems reserve that word for the subsequent reference-price change. The definition must accompany the label.

This is the same net dealer markout denoted by $\text{MO}_i(h)$ in Chapter 2 when its benchmark $B(t_i + h)$ is this chapter's M_{t_i+h} . The symbol $\alpha_i(h)$ is a local label, not a different return measure. Chapter 2 separated decision-reference edge from the movement into execution; the ledger here starts from actual fills and reconciles their combined contribution.

Adding and subtracting the fill-time mark gives the exact decomposition

$$\alpha_i(h) = \eta_i (M_{t_i} - p_i) + \eta_i (M_{t_i+h} - M_{t_i}) - \frac{f_i}{z_i}. \quad (11.15)$$

At zero horizon, this is execution edge less fees per unit. At later horizons, it adds a signed mark movement. A fill-level value can be informative even if the inventory was offset before the horizon, but it is then an attribution or diagnostic convention, not the actual isolated P&L of a position held without interruption.

If every actual fill is instead marked to the same terminal calendar time H , its amount is $z_i \alpha_i(H - t_i)$. Let F_{other} denote cash charges recorded separately from the fill fees. Including opening inventory and these charges gives the exact terminal reconciliation

$$\begin{aligned} \Pi_H &= q_0(M_H - M_0) + \sum_i z_i \alpha_i(H - t_i) - F_{\text{other}}, \\ &= 8.00 + 6.40 - 9.24 = 5.16. \end{aligned} \quad (11.16)$$

The sum here includes both client and external fills and allocates their respective fees once. In the baseline $F_{\text{other}} = 0$; a separate charge without a fill reduces wealth through this additional term. This is equation (11.6) written in markout notation, with total fees split into fill fees and other cash charges. Equal elapsed horizons, such as one minute after each fill, generally lead to different calendar marks; their sum does

not automatically equal desk P&L at any common terminal time. Summing several horizons for the same fill also counts overlapping reference-price movement repeatedly.

Oomen’s accepted manuscript provides a precise reason to resist a causal reading of adverse markouts. Its standard-amount aggregator model generates unfavorable conditional midpoint changes even with uninformed client demand, because an LP tends to receive the request when its quote is most aggressive. The source’s effective spread and last-look results have their own conditioning, units, and clock. In particular, source equation (13) is a lower bound, and the accepted-cohort curves use a request-time origin. They are not direct substitutes for our exact level-price cash ledger or fill-time markouts.

11.7 Keep the cohort fixed when comparing horizons

An average requires both a population and a weighting rule. For an eligible fill set $\mathcal{I}_h$, the executed-quantity-weighted mean is

$$\bar{\alpha}_z(h) = \frac{\sum_{i \in \mathcal{I}_h} z_i \alpha_i(h)}{\sum_{i \in \mathcal{I}_h} z_i}, \quad \sum_{i \in \mathcal{I}_h} z_i > 0. \quad (11.17)$$

An equal-fill mean uses one observation per fill instead. In this example each executed client request produces one fill, so equal-fill and equal-executed-request means coincide. That equivalence fails if requests have different numbers of partial fills. A split execution must not acquire extra statistical weight merely because its reporting granularity changes.

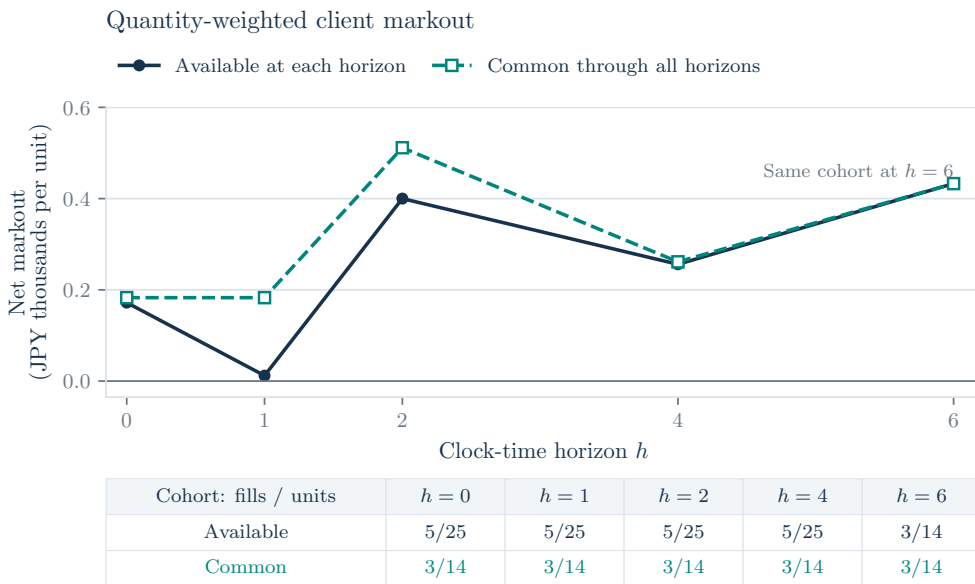
The reference data end at $H = 12$. A requested mark at $t_i + h > 12$ is unavailable. It must not be replaced by the final observed mark and labeled as a complete h -horizon observation. For the client-fill population $\mathcal{C}$, define the fixed common cohort for the largest requested horizon $h_{\max}$ by

$$\mathcal{I}^{\text{common}} = \{i \in \mathcal{C} : t_i + h_{\max} \leq H\}, \quad \mathcal{I}_h^{\text{available}} = \{i \in \mathcal{C} : t_i + h \leq H\}. \quad (11.18)$$

For horizons 0, 1, 2, 4, 6, the available cohort contains all five client fills and 25 units through horizon 4. At horizon 6, only three fills and 14 units remain. The common cohort contains those same three fills at every horizon. Its comparison holds sample membership fixed, although it is still a selected early-fill population rather than the full day’s client flow.

Horizon	Available: fills / units	Available mean	Common-cohort mean
0	5 / 25	0.17200	0.18286
1	5 / 25	0.01200	0.18286
2	5 / 25	0.40000	0.51143
4	5 / 25	0.25600	0.26143
6	3 / 14	0.43286	0.43286

All means in the table are net dealer markouts in thousands of JPY per unit. The common cohort always has three fills and fourteen units. Its particularly different value at horizon 1 is a composition effect. The equal-request means for all five fills at horizons 1 and 2 are 0.01000 and 0.41000, respectively, which also differ from the quantity-weighted results.



Client fees are included; the available cohort loses two fills when their future marks are unavailable. Common-cohort means hold membership fixed. Horizons are supplied clock units, not trade counts. Markouts are overlapping diagnostics: do not add horizons or add these curves to portfolio P&L.

Figure 11.3: Quantity-weighted client markouts with available and common cohorts. The accompanying table reports each point's eligible fill count and executed quantity. The last available-cohort point drops later fills because their six-step marks are unavailable; it does not extend their last observed prices.

Counts and exposure totals are part of the estimate, not optional decoration. A production study also needs uncertainty estimates that respect dependence among observations. These five synthetic fills are an accounting illustration and provide

no statistical evidence about expected profitability. Missing marks, stale marks, or end-of-window censoring can themselves be related to market conditions, so keeping a common cohort does not eliminate every selection problem.

11.8 Requests, quotes, and executions have different denominators

The twelve-request example includes nine quoted requests, two declined requests, and one request canceled before a quote. Among quoted requests, four fill completely, one fills partially, three are lost, and one expires. The partial request asks for six units and executes three. Five requests therefore have an execution, but only four complete.

Chapter 3's binary full-size example deliberately made execution and completion coincide. This new record set relaxes that assumption, so its nine-quote denominator and partial fill are not a revision of the earlier eight-quote example. It shows why the outcome target must be stated before carrying an estimated probability into a decision model.

Count-based execution ratios are

$$r_{\text{invited}} = \frac{5}{12}, \quad r_{\text{quoted}} = \frac{5}{9}, \quad r_{\text{complete|quoted}} = \frac{4}{9}. \quad (11.19)$$

The requested gross quantity is 79 units across all inquiries and 59 units across quoted inquiries. Gross executed client quantity is 25 units. Quantity fulfillment is therefore

$$r_{z|\text{quoted}} = \frac{25}{59}, \quad r_{z|\text{invited}} = \frac{25}{79}. \quad (11.20)$$

These are gross quantities: buy and sell requests are added by absolute size. Netting opposite sides would create an unstable denominator unrelated to service coverage. The partial request contributes one to the any-execution count, three to filled quantity, and six to quoted requested quantity. It does not contribute one to the completed-request count.

We call these execution ratios because the example does not separately observe award and acceptance events. Calling $5/9$ a win rate would imply an event definition not supplied by the records. Oomen distinguishes showing the best quote, receiving a request, rejecting it conditionally, and share among accepted executions. Those denominators cannot be interchanged. A production record should preserve separate request, quote, award, acceptance, and fill events whenever the venue actually exposes them.

The source literature also illustrates why a convenient sample may have a narrow interpretation. Kakuma studies a JGB RFQ sample with five dealers requested and all five responding, including inquiries without a completed trade. Its cross-dealer quote dispersion is not an observed post-fill markout. The quotes are in yield space, where

the investor's preferred direction differs from level-price ranking. Transferring a formula or side convention without its measurement units would corrupt the comparison.

A rejected or lost request can be assigned a hypothetical mark using a subsequently observed reference path. It still has zero actual fill quantity in the ledger. Its hypothetical value does not establish what would have occurred under acceptance, a different quote, or another hedge. It should be shown as a separately defined comparison with its assumptions, never inserted as realized revenue.

11.9 A reference-price change redistributes attribution

The cash account is observed from trade and fee records, while the reference series is a measurement choice. Consider replacing the internal marks by

$$\widetilde{M}_k = M_k + b_k, \quad b_0 = b_H = 0, \quad (11.21)$$

with execution prices, quantities, and cash unchanged. Equal endpoint marks imply equal opening and terminal wealth. Intermediate allocations can nevertheless move. The component changes are

$$\begin{aligned} \Delta E_C &= \sum_{i \in \mathcal{C}} v_i b_i, & \Delta C_E &= - \sum_{i \in \mathcal{E}} v_i b_i, \\ \Delta I &= \sum_{k=1}^H q_{k-1} (b_k - b_{k-1}). \end{aligned} \quad (11.22)$$

Discrete summation by parts gives

$$\Delta I = - \sum_i v_i b_i, \quad \Delta E_C - \Delta C_E + \Delta I = 0. \quad (11.23)$$

The first sum includes every fill. The zero-endpoint condition is essential; it removes the boundary terms. In the numerical experiment, raise marks by 0.10 at times 1 through 11 while keeping the two endpoints fixed. Client edge rises from 4.55 to 5.05, external local cost from 1.22 to 2.42, and inventory carry from 2.20 to 2.90. Total P&L remains 5.16. The larger client edge is not newly created cash profit.

If the terminal mark itself rises by δ with cash and terminal inventory fixed, total marked P&L changes by

$$\Delta \Pi_H = q_H \delta. \quad (11.24)$$

For three remaining units and $\delta = 0.10$, the change is 0.30. Thus a report can reconcile perfectly under a poor terminal mark. Reconciliation tests accounting consistency; independent mark validation addresses whether the valuation input is suitable. Both are needed, and they answer different questions.

11.10 Observation times and units are part of the model

A reference lookup must state which observation was available when a decision or fill occurred. Let τ_j be a mark's source event time and a_j its availability time. Among observations whose source event and availability times do not exceed the decision time, choose the latest source event:

$$\begin{aligned} j^*(t) &= \arg \max_{j: \tau_j \leq t, a_j \leq t} \tau_j, \\ M^{\text{asof}}(t) &= M_{j^*(t)}, \quad \text{age}(t) = t - \tau_{j^*(t)}. \end{aligned} \quad (11.25)$$

The synthetic example has unique source timestamps; a feed that permits ties needs a documented sequence or revision rule. If the eligible set is empty, the as-of reference is unavailable. An older observation arriving later must not overwrite a newer source observation already available. Age is measured from the source event, so delayed delivery does not make stale information fresh. Excessive age should be flagged against a declared tolerance. Source event time, receipt time, and availability time may differ. A reconstructed event-time mark can be useful for retrospective analysis, but should not be described as information available to the quoting policy without evidence of its receipt timing.

An example in the BrokerTec study makes the clock issue concrete. Its transaction-time observations combine an initiating order with its workup, and use the midpoint at the end of that workup. Its long-run price-impact measure accumulates a fitted response over fifty transactions. That object is not a fixed-clock post-fill markout and does not identify a causal cash cost for every individual child in our ledger. Measurement windows must travel with the numerical label.

Unit changes must preserve both cash and exposure. If one new trading unit represents c old units, use

$$q'_k = q_k/c, \quad v'_i = v_i/c, \quad p'_i = cp_i, \quad M'_k = cM_k, \quad f'_i = f_i. \quad (11.26)$$

Then quantities times prices, cash flows, and every currency-valued P&L component are unchanged. Per-unit markouts change scale, as they should. This is a trading-unit conversion with the currency unit fixed. A currency-unit conversion would also rescale cash and fees. A yield difference or a risk exposure cannot simply replace a level-price difference in an exact cash identity; an explicit price mapping or stated approximation and multiplier is required.

For several instruments with all level marks and cash flows expressed consistently in one currency, the same discrete identity is

$$\Pi_H = \sum_k \mathbf{q}_{k-1}^\top \Delta \mathbf{M}_k + \sum_k \mathbf{v}_k^\top (\mathbf{M}_k - \mathbf{p}_k) - F. \quad (11.27)$$

Each instrument’s contract multiplier can be absorbed consistently into its quantity or price convention, but must not appear in both. Foreign-currency cash balances, external cash flows, and changes in conversion rates require additional accounting terms. Equation (11.27) is a conceptual extension under its stated common-currency assumptions; the notebook deliberately tests the single-instrument case rather than implying a full multi-currency implementation.

11.11 Make every metric auditable from its inputs

A useful metric record states its side convention, price and quantity units, event population, benchmark construction, timestamp rule, horizon, fee coverage, weighting, and treatment of missing observations. The minimum ledger reconciliation residual is

$$\varepsilon_{\text{rec}} = (W_H - W_0) - (E_C - C_E + I - F). \quad (11.28)$$

It should vanish up to declared numerical tolerance. That is necessary but not sufficient: duplicated trades may preserve this algebra if both sides of the calculation ingest the same duplicates. The notebook therefore supplements reconciliation with event-ID uniqueness, valid request and parent links, side checks, requested-quantity limits, parent completion limits, finite numerical inputs, valid reference coverage, and time consistency.

The experiment also checks identities using independent constructions. Direct cash accumulation is compared with the component bridge and the common-terminal-mark decomposition. Parent shortfalls are rebuilt from proceeds and residual value. Filled and full-target benchmark bridges use their different quantities. A fill is split into two records with apportioned fees to verify that monetary totals and quantity-weighted markouts are unchanged. A zero-quantity cash-charge case verifies that an actual fee changes wealth without creating inventory or a fictitious execution price.

Reference shifts test attribution dependence, a terminal-mark perturbation tests residual exposure, and trading-unit rescaling tests dimensional consistency. Invalid inputs are deliberately rejected. Unavailable long-horizon marks remain missing, and a future quote cannot serve as an as-of observation. These checks target different failure modes; a large check count is not itself evidence that the data represent the correct economic events.

The companion notebook retains the synthetic requests and events behind the benchmark and cohort tables. Its worked answers reconstruct the reported totals from those records, making the reconciliation rules available for inspection as well as calculation.

11.12 From reconciliation to evaluation

The chapter establishes what was booked, how it was marked, and how several common performance measures relate. Total P&L answers an account-level question. Filled shortfall answers an execution-price question for a specified quantity and benchmark. A marked-residual extension answers a full-target comparison under a declared terminal convention. Markouts answer a signed reference-value question for a stated cohort and horizon.

These measures do not by themselves establish which policy caused a better outcome. A more aggressive quote changes who trades; a different execution schedule changes exposure and may change the observed market path; a stricter acceptance rule changes the retained sample. Reconciliation cannot recover the missing outcomes of actions that were not taken.

Chapter 12 therefore turns to simulation and evaluation. It uses these accounting identities as invariants, then examines information timing, policy comparisons, dependence, uncertainty, and the limits of conclusions drawn from observed or simulated outcomes. The next step is to make policy evidence as explicit as the ledger, while preserving the distinction between exact accounting and assumptions about alternative worlds.

Exercises

1. **Cash, signs, and the final mark.** Rebuild the eight-fill ledger from the raw records. Report terminal cash, inventory, wealth, and P&L. Explain why a dealer sale above the reference contributes positively to immediate wealth. Quantify the error from omitting the final reference update.
2. **Two complete decompositions.** Reproduce both the bridge through client edge, local external cost, and inventory carry and the common-terminal attribution. Identify which fees each component includes. Explain why adding the two client measures together has no accounting interpretation.
3. **Benchmarks and an incomplete instruction.** Compute filled decision, arrival, and local shortfalls for the sell parent. Reconcile the decision-to-arrival and decision-to-fill bridges. Add the residual at H and explain why the full-target bridge uses 15, while the filled bridge uses 12.
4. **Markout weights.** Compute all five client fills' one-step and two-step markouts. Compare equal-request and quantity-weighted means. Split one fill into two proportional records and explain which averaging rules remain invariant.

5. **Horizon eligibility.** Build available and common cohorts for horizons 0, 1, 2, 4, 6. Report counts and quantities with each mean. Explain what false observation is created by replacing an unavailable six-step mark with the final data point.
6. **Request denominators.** Reconstruct the twelve-request funnel and the count- and quantity-based execution ratios. Explain the treatment of the partial fill, lost quotes, declined requests, and the pre-quote cancellation. State what additional event data would be needed to measure award and acceptance rates separately.
7. **Marks and units.** Apply the interior reference shift of 0.10, leaving endpoint marks fixed, and reconcile the changed components. Perturb only the terminal mark by 0.10. Then change trading units by a factor of ten and verify currency totals remain unchanged.
8. **Audit failures.** Introduce a duplicate event ID, a parent overfill, a request overfill, and an invalid future as-of quote. Verify rejection. Add a separate cash charge with zero trade quantity and check its wealth effect. Explain why zero reconciliation residual alone would not catch every corrupted input.

Source Notes

Almgren and Chriss. The local source is the preprint dated 8 April 1999, not the journal layout. Printed/PDF p. 7 states the complete-liquidation condition; p. 10 defines realized shortfall in unnumbered prose, while equation (4) is expected shortfall. Printed/PDF pp. 8-11 establish the execution-price, proceeds, and risk conventions relevant here. Our incomplete-parent marked-residual formula and benchmark bridges are independent accounting extensions. See the [publication record](#) and [chapter reading note](#).

Oomen. The inspected local file is the June 2016 accepted manuscript of *Execution in an aggregator*. Main-text PDF pages equal printed pages plus two. Printed p. 9, equations (6)-(8), separates initial spread capture from a conditional midpoint change; printed pp. 10-14 discuss acceptance and request-time conditioning; printed p. 13, equation (13), is a lower bound. Printed p. 16 distinguishes request-winning probability, conditional rejection, and share among accepted executions. These are standard-amount model results in the source's additive quote/log-price scale, not an exact currency ledger or evidence that adverse marks identify informed flow. See the [LSE source record](#) and [chapter reading note](#).

Kakuma. The JGB RFQ paper provides market-specific examples of sample selection and quote units. Its five-dealer, all-responding sample and yield-quote direction must be retained when interpreting its dispersion and execution-cost measures. We do not substitute its cross-dealer dispersion for a post-fill markout, reproduce its empirical estimates, or apply yield-ordering conventions to the level-price ledger. Exact page locators and local formula qualifications are recorded in the [chapter reading note](#); see also the [Securities Analysts Association of Japan source listing](#).

Fleming, Mizrach, and Nguyen. The revised BrokerTec staff report supplies a concrete observation-clock example: transactions include initiating orders and their workups, and the fitted price-impact response uses transaction time. Its empirical estimates are specific to its sample and construction. They are not reproduced or treated as causal cash attribution in this chapter. The [New York Fed source record](#) and [chapter reading note](#) identify the inspected version, pages, and measurement details.

Scope of originality and verification. The exact ledger bridges, partial-parent extension, reference-change identities, synthetic records, numerical checks, and worked exercises are constructed for this book. The sources motivate distinct measurement and conditioning issues; selected relevant pages were inspected, with page images and unchanged-file hashes retained in the reading records. No whole-book audit, empirical replication, causal calibration, or market-data claim is implied. Every number in the chapter's worked example is synthetic and reproducible in the accompanying notebook.

Chapter 12

Simulation and Evaluation

A quote policy can be mathematically optimal in a model and still perform poorly when the model’s assumptions change. An apparently strong backtest can also use information that was unavailable at the decision time, value an incomplete instruction inconsistently, or compare policies on different selected samples. Chapter 11 made the accounting explicit. This chapter makes the experiment explicit: what information a policy sees, which parts of the environment respond to its actions, how candidates are selected, and what uncertainty a reported advantage actually measures.

We construct a small, reproducible RFQ simulator rather than train a large trading model. Fixed-width and inventory-skew policies share the same constraints and cash accounting. A validation sample selects a candidate before a separate test sample is opened. Further environments alter client-flow feedback, volatility, and reference-price delay. A separate one-step example shows what importance weighting can recover from randomized logs, and why it cannot manufacture outcomes outside their support. Every numerical experiment is synthetic; none establishes live market performance.

12.1 Define the object being evaluated

A policy is more than a formula for a quote. It includes the observations admitted to that formula, the conditions under which it declines, and its behavior when a limit binds. Write S_t for the full simulator state, O_t for the policy’s permitted observation, and A_t for its action. A useful distinction is

$$O_t = g(S_t, \mathcal{H}_t^{\text{available}}), \quad A_t = \pi(O_t). \quad (12.1)$$

The simulator may know the current reference price, latent access state, and future random numbers already allocated in memory. The policy does not automatically receive those quantities. Its observation builder must enforce the intended information boundary. For this chapter’s quote engine, the inputs are the current request’s dealer side and size, inventory after any scheduled hedge, an as-of reference, its age, and the remaining time. The policies below do not infer fresh reference prices from the execution prices of the fixed-schedule hedges.

One formal requirement is that a decision uses information available by its decision time. If $\mathcal{F}_t^{\text{available}}$ denotes that information set, then

$$A_t \text{ is } \mathcal{F}_t^{\text{available}}\text{-measurable.} \quad (12.2)$$

This condition rules out using a later market move merely because it is present in the evaluation file. It does not require using every available field. Restricting a policy to a deliberately smaller observation set can be a meaningful experiment, provided that restriction is stated.

Let K_θ describe the environment's transition and reward law, including the timing of requests, fills, external execution, and marks. Its dependence on an action matters:

$$(S_{t+1}, R_t) \sim K_\theta(\cdot \mid S_t, A_t), \quad V_\theta(\pi) = \mathbb{E}_{\theta, \pi} \left[\sum_t R_t \right]. \quad (12.3)$$

The symbol θ includes both numerical parameters and structural assumptions. Whether clients change their future requests after receiving an unattractive quote is a structural choice, not just a random seed. A score estimated under one K_θ answers a question about that environment. Changing the environment changes the question even when the same policy code is retained.

The notation is local: S_t is the full state, while M_t below is its reference-price component; R_t is reward, rather than Chapter 10's shortfall variance. The transition law K_θ includes the request and execution mechanisms, not a response fee or a number of time steps.

12.2 Reuse the cash ledger with an explicit event order

There are sixty decision opportunities, at times $t = 0, \dots, 59$, followed by a terminal mark at $T = 60$. The initial reference is 100, and opening cash and inventory are zero. At each decision time, the current mark is available to the environment; a scheduled hedge is processed, a request may arrive, the quote engine observes its allowed inputs, and a fill may occur. The resulting inventory is then held until the next mark. Hedges occur at times 10, 20, 30, 40, 50, before that time's RFQ. There is no scheduled hedge at the terminal time.

Hedge executions and successful RFQs settle immediately in this simulator. They have no pending-order or delayed-fill state. The later delay stress changes delivery of reference information, not execution timing. Chapter 9's pending-order extension and Chapter 3's separation of request origin from physical completion would be needed to test those different delays.

Prices are in thousands of JPY per normalized trading unit, and monetary amounts are in thousands of JPY. With normalized time step $\Delta t = 1$, the reference follows $M_{t+1} = M_t + \sigma\sqrt{\Delta t}Z_t$, where the Z_t are independent standard normal shocks, independent of the other exogenous random inputs. The baseline has $\sigma = 0.07$; the

high-volatility stress doubles it. This supplied reference process has no action-dependent price impact.

Let q_t, X_t be inventory and cash just before the actions at time t . Let h_t be the signed external hedge, v_t the signed client fill, and p_t^H, p_t^C their execution prices. Both signed quantities are positive for dealer purchases. With f_t the total separately booked cash fees at that time,

$$\begin{aligned} q_t^+ &= q_t + h_t + v_t, \\ X_t^+ &= X_t - h_t p_t^H - v_t p_t^C - f_t, \\ q_{t+1} &= q_t^+, \quad X_{t+1} = X_t^+. \end{aligned} \quad (12.4)$$

At each scheduled hedge time, the rule flattens the pre-action inventory: $h_t = -q_t$. At other times $h_t = 0$. For a nonzero hedge, its adverse execution price is $p_t^H = M_t + \text{sign}(h_t)\{0.03 + \frac{1}{2}(0.004)|h_t|\}$. Thus its cost relative to the current mark is $h_t(p_t^H - M_t) = 0.03|h_t| + \frac{1}{2}(0.004)h_t^2$. This cost is included once through the hedge execution price. The separately booked fee in this experiment is the client fee $f_t = 0.001|v_t|$.

Absent trades have zero quantity and require no invented execution price. The superscript $+$ denotes the position after both possible trades at the same mark. Terminal marked profit therefore satisfies

$$\Pi_T = \sum_{t=0}^{T-1} \left[q_t^+ (M_{t+1} - M_t) + v_t (M_t - p_t^C) + h_t (M_t - p_t^H) - f_t \right]. \quad (12.5)$$

This is the [Chapter 11](#) identity expressed with the action-then-next-mark clock. It includes the exposure after the last RFQ, from time 59 to time 60. The terminal inventory is marked rather than automatically liquidated. External execution costs enter through the actual hedge prices and any separately recorded fees; they must not be subtracted a second time under a risk-cost label.

Thus q_t^+ plays the role of the inventory held over an interval, just as [Chapter 10](#)'s post-trade x_k sees the following shock. The signed hedge h_t and client fill v_t follow [Chapter 11](#)'s purchase-positive convention; h_t is a quantity here, not the elapsed markout horizon denoted by h there. All three formulations assign each price increment to the position that actually held it.

Candidate selection uses a stated preference for inventory control, separate from marked P&L:

$$J = \Pi_T - \lambda \sum_{t=0}^{T-1} (q_t^+)^2 \Delta t - \phi q_T^2. \quad (12.6)$$

The running coefficient λ has currency per squared quantity per time units, and ϕ has currency per squared quantity units. These are preference parameters, not inferred cash expenses. The notebook reports J , marked P&L, execution costs, and inventory

statistics separately. A policy with a higher J need not have the largest mean marked profit.

Unlike Chapter 10’s inverse-currency weight on shortfall variance, this λ directly prices squared inventory per unit time. It is held fixed when the simulator’s volatility changes. Multiplying it by a new variance estimate would define another objective, not preserve the comparison below. These policies are also compared under a fixed external hedge rule; this chapter does not claim to re-solve Chapter 9’s joint optimum.

When R_t in (12.3) represents this objective, its rewards include the cash-and-mark effects and running charge, with the terminal penalty assigned once at the final transition. Reporting Π_T instead removes those preference deductions; it does not remove actual hedge costs or client fees.

12.3 Start with policies simple enough to audit

Let $\eta_t \in \{+1, -1\}$ denote the dealer’s side of a request and $z_t > 0$ its size. The inventory used for quoting is $\hat{q}_t = q_t + h_t$, after the scheduled hedge. For base offset b and skew coefficient κ , define the offered offset and price by

$$\delta_t = \text{clip}(b + \eta_t \kappa \hat{q}_t, \delta_{\min}, \delta_{\max}), \quad p_t^C = M_t^{\text{asof}} - \eta_t \delta_t. \quad (12.7)$$

A positive inventory raises the offset for further dealer purchases and lowers it for dealer sales. Thus the policy makes inventory-reducing trades more attractive. A fixed-width benchmark sets $\kappa = 0$. The parameter b is a one-sided distance, so an unskewed two-sided quote has total spread $2b$ when clipping is inactive. Before any quote is allowed, the full requested quantity is checked against the same inventory limit. A blocked request is a decline; it is not silently resized into an invented smaller request.

The environment measures competitiveness against the actual contemporaneous reference. If the quote engine’s mark is stale, its intended offset and its true price distance differ:

$$\delta_t^{\text{eff}} = \eta_t(M_t - p_t^C) = \delta_t + \eta_t(M_t - M_t^{\text{asof}}), \quad p_t^{\text{fill}} = \frac{1}{1 + \exp\{(\delta_t^{\text{eff}} - 0.12)/0.04\}}. \quad (12.8)$$

These price differences are in the synthetic currency-price unit. The sigmoid is a supplied model of execution probability conditional on an arriving request and a valid quote. It is not an RFQ-arrival intensity or an empirical estimate. A stale buy quote can become too generous after the true mark falls; measuring fill probability from its intended offset alone would miss that effect.

The policy width b is a one-sided quote offset; the sigmoid’s response scale is the separate constant 0.04. Chapter 3’s conditional probability contract applies to this full-size, immediate-execution event. Chapter 2’s uninformed stale-quote example explains why selected fills can have adverse reference differences without establishing that the customer has private information.

The baseline uses the current mark. Delay experiments separately test an age filter and an unfiltered quote engine. The current reference used internally by the environment to determine price competitiveness is not passed back as a policy feature. Fixed hedging rules and their market-priced executions remain part of the declared environment.

The numerical settings below are deliberately transparent. Prices are in thousands of JPY per trading unit, and marked P&L and objective values are in thousands of JPY. Quantity and time are normalized. Request arrival, side, size, and fill draws use separate random streams; the two dealer sides are equally likely, and sizes 1 and 2 are equally likely independently of side.

Input	Value	Interpretation
Requests and limit	$p = 0.8, q_{\max} = 8$	At most one RFQ per decision time; require $ \hat{q}_t + \eta_t z_t \leq 8$ before quoting
Offset bounds	$0.02 \leq \delta \leq 0.30$	Applied to every quoted request
Client fee	$0.001 v_t $	A separately booked cash expense on fills
Inventory preference	$\lambda = 0.001, \phi = 0.02$	Running and terminal penalties in (12.6), with $\Delta t = 1$
Fixed benchmark	$b = 0.10, \kappa = 0$	Symmetric width subject to the full-size limit
Linear benchmark	$b = 0.10, \kappa = 0.015$	Prespecified inventory skew
Candidate grid	$b \in \{0.06, 0.10, 0.14\};$ $\kappa \in \{0, 0.01, 0.02\}$	Nine rules scored on validation episodes

The linear benchmark is prespecified separately from the nine-rule grid. Reference dynamics do not include price impact from our trades. Client-flow feedback below changes request opportunities, while the Gaussian reference remains exogenous.

The candidate grid varies only width and linear skew. This limits the amount of fitting hidden inside a comparison and makes every selected rule readable. A complex learned policy should earn its place by improving an appropriately controlled evaluation; adding representational capacity is not itself a trading result.

12.4 Distinguish replay, generated paths, and action feedback

In the baseline, request arrival probability is fixed. For a deliberately different environment, introduce an access state $a_t \in [0, 1]$. After a quoted request, it moves toward a response to the quote's effective distance:

$$a_{t+1} = (1 - \rho)a_t + \rho \frac{1}{1 + \exp\{(\delta_t^{\text{eff}} - 0.10)/0.025\}}. \quad (12.9)$$

The corresponding arrival probability is

$$p_t^{\text{request}} = 0.40 + 0.50a_t. \quad (12.10)$$

This is a synthetic response hypothesis. It says that quote competitiveness can affect future access; it does not estimate customer behavior. The experiment starts at $a_0 = 0.8$ and uses $\rho = 0.2$. If no request arrives, $a_{t+1} = a_t$. If a request arrives but is declined, $a_{t+1} = 0.8a_t$. Equation (12.9) applies when a quote is given, whether or not it fills. A quote can therefore affect access without a fill, so replaying the same realized fill tape cannot reproduce all of this feedback.

Chapter 8 used an ordinal access ladder updated by request outcomes. Here the continuous state responds to quoted competitiveness, with ρ an update weight rather than a response probability. The shared idea is action-dependent future opportunity; the two update laws are different hypotheses and are evaluated separately.

There are three distinct evaluation objects. A fixed observation tape is useful for checking timestamps, reproducing decisions, and valuing executed records. A generator of market paths can supply alternative realizations and stress conditions. An action-responsive simulator additionally specifies how the target policy's own actions enter future states. One cannot obtain that last property merely by relabeling a historical path as a simulated one.

M3's inspected appendix makes this distinction concrete: its closed-loop execution experiment matches a submitted child order against the current simulated book, appends the resulting event information, and continues generation. FlowLOB instead generates a window of book snapshots conditional on scenario channels. Its scenario-control test changes a condition; that alone does not define an order submission, a queue rule, or the response to a particular trading action. Both can be useful, but they answer different questions.

ABIDES-MARL is another action-responsive example, with interacting execution and liquidity policies. Its market mechanism, action synchronization, observation timing, completion treatment, and evaluation setup are part of the experimental definition. Its reported simulator behavior should not be substituted for the behavior of a rates RFQ venue. This chapter's access-state experiment is an independent, much smaller construction, not a reproduction of any of those systems.

12.5 Use common random inputs without forcing common outcomes

For episode e , preallocate independent random inputs by time and purpose: reference shocks, request uniforms, side draws, size draws, and fill uniforms. Denote that complete input array by U_e . A policy run is then a deterministic transformation of its inputs, parameters, and initial state:

$$\mathcal{T}_e^\pi = \Phi(\pi, \theta, U_e). \quad (12.11)$$

Every policy receives the same indexed random inputs in a paired comparison. A policy that declines still consumes the same conceptual time slot: subsequent uniforms are not shifted because one branch made fewer calls to a random-number generator. This prevents accidental changes in future shocks from contaminating the comparison.

Common random inputs do not require identical states, requests, or fills. In the access-state environment, the same request uniform can be below one policy's arrival probability and above another's. That is the intended response of different histories to a shared exogenous draw. Forcing the request outcomes to remain identical would remove part of the action feedback the experiment is supposed to measure.

The reference process here is exogenous, so policies within the same volatility scenario share its path. In a model with action-dependent price dynamics, even prices could differ under a shared innovation sequence. The resulting paired potential outcomes are defined by the simulator's transition equations; the coupling does not identify corresponding counterfactual outcomes from real historical data.

12.6 Select on validation data and freeze before testing

Let $\mathcal{P}$ be a candidate set fixed before final evaluation. If $\hat{V}_{\text{val}}$ is the average objective on validation episodes, candidate selection is

$$\hat{\pi} = \arg \max_{\pi \in \mathcal{P}} \hat{V}_{\text{val}}(\pi). \quad (12.12)$$

The selection score is optimistic as an estimate of the selected candidate's future performance: the candidate is chosen partly because its validation noise was favorable. Report it as the selection score. A separate test set evaluates the frozen selection once, alongside prespecified benchmarks, with no retuning to make the final table more attractive.

In the synthetic experiment, design, validation, and test episode IDs and random seeds are disjoint:

$$\mathcal{D}_{\text{design}} \cap \mathcal{D}_{\text{val}} = \mathcal{D}_{\text{design}} \cap \mathcal{D}_{\text{test}} = \mathcal{D}_{\text{val}} \cap \mathcal{D}_{\text{test}} = \emptyset. \quad (12.13)$$

The design stage checks the model and experiment contract; the known synthetic transition parameters are supplied rather than estimated from final-test outcomes. The notebook records the candidate grid, objective coefficients, seed namespaces, selected rule, and test configuration. Stress scenarios evaluate that same frozen selection. Selecting a new winner separately within each stress table would answer a different, explicitly retuned-policy question.

Chapter 3 separated parameter fitting, candidate selection, probability calibration and testing for an estimated win model. The present simulator supplies its transition parameters, so it needs no fitted probability-calibration stage. If Chapter 5’s volatility estimators or a fitted win curve were introduced, their fitting windows and model choices would also belong before the test boundary. They would become part of the evaluated procedure, rather than invisible preprocessing.

There are 256 design episodes, 512 validation episodes, and 2,048 test episodes. The master seed is 20260912; a stored seed plan separates the split, episode, and purpose of every stream. Selection chooses the maximum validation mean J , with exact ties resolved by lower width and then lower skew. It selects $b = 0.10$, $\kappa = 0.01$, whose validation mean objective is 3.37138. The selection record is written before test draws are generated. Its baseline test mean, reported next, is a new estimate rather than the selection score.

For real market data, disjoint row numbers are insufficient. Features and labels must respect chronological availability, overlapping response horizons must not leak across boundaries, and fitted transformations must use only their permitted training period. A future mark used as a target belongs to the label, not to the quote-time feature vector. A practical split needs a boundary gap at least large enough to remove the overlapping label or position windows actually used; there is no universal gap length independent of the experiment.

A final test set also ceases to be untouched when it is repeatedly consulted to choose features, thresholds, models, or seeds. The remedy is to disclose that use and establish a new independent evaluation boundary, rather than continue calling the reused sample a final holdout.

12.7 Measure differences at the independent-episode level

For two frozen policies A and B , define their paired objective difference in episode e as

$$D_e = J_e^A - J_e^B, \quad \bar{D} = \frac{1}{N} \sum_{e=1}^N D_e. \quad (12.14)$$

With independent simulated episodes, an estimated Monte Carlo standard error is

$$\widehat{\text{SE}}(\bar{D}) = \frac{s_D}{\sqrt{N}}, \quad s_D^2 = \frac{1}{N-1} \sum_{e=1}^N (D_e - \bar{D})^2. \quad (12.15)$$

For a sufficiently large episode sample with finite variance, the notebook displays the approximate normal interval

$$\bar{D} \pm 1.96 \frac{s_D}{\sqrt{N}}. \quad (12.16)$$

This interval describes simulation sampling uncertainty conditional on the fixed environment, the realized frozen policies, and the chosen objective. It does not include variability from repeating policy selection on new validation samples. It is not a confidence interval for real-market profitability, for the correctness of the simulator, or for an arbitrary policy selected after viewing the test results. The reported comparisons are individual comparisons, not a simultaneous guarantee across every policy and stress scenario.

In particular, a covariance forecast from [Chapter 5](#) would be fixed inside this conditional calculation. Uncertainty about that forecast, its estimation window, or its model choice is not the same as price shocks generated conditional on it. Studying the full forecasting-and-control procedure requires repeating the permitted estimation and selection steps at historical origins or under an explicitly specified outer experiment; more inner simulation paths alone do not measure that uncertainty.

Pairing can improve precision because

$$\text{Var}(J^A - J^B) = \text{Var}(J^A) + \text{Var}(J^B) - 2 \text{Cov}(J^A, J^B). \quad (12.17)$$

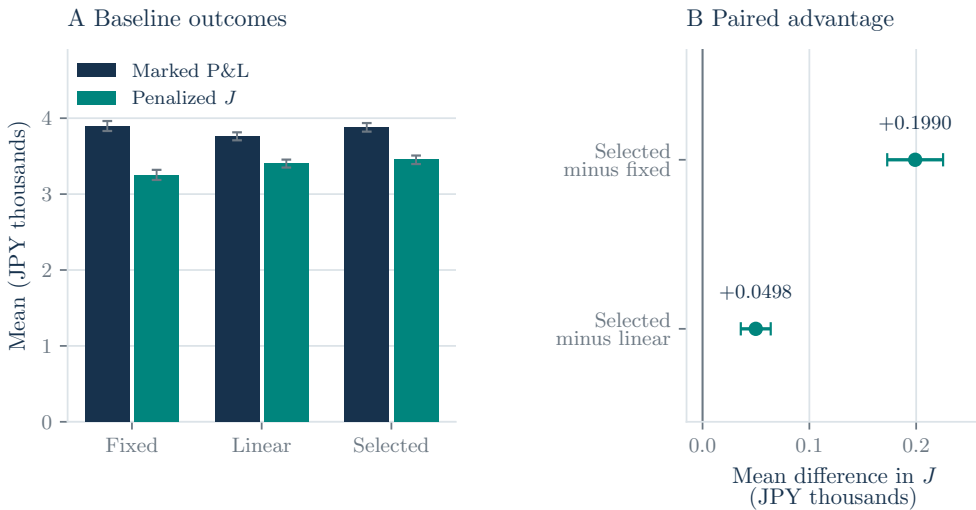
Positive covariance removes some shared environmental noise. It is not guaranteed, so the paired standard error is measured rather than assumed to be smaller. The notebook reports the observed pairing effect and retains one episode-level outcome per run.

Individual RFQs inside an episode are not independent profit observations. They share inventory, marks, access state, and subsequent hedges. Dividing a trade-level sample standard deviation by the square root of the number of fills would generally target the wrong variance. For real data, the independent or approximately independent sampling unit must be justified from the dependence structure; labeling each row an observation does not establish that justification.

The baseline test estimates show why the scoring rule matters. Marked P&L is computed from the cash-and-inventory ledger, and inventory RMS is $\sqrt{(NT)^{-1} \sum_{e,t} (q_{e,t}^+)^2}$ in quantity units.

Policy	Mean marked P&L	Mean J	Inventory RMS
Fixed	3.89707	3.25290	2.58846
Linear	3.76121	3.40214	1.99675
Selected	3.87955	3.45195	2.15528

The selected rule has a lower mean marked P&L than the fixed benchmark, but sufficiently smaller inventory penalties to have a higher mean J . Its paired advantage over the fixed rule is 0.19905, with standard error 0.01334 and an approximate 95-percent interval $[0.17290, 0.22520]$. Against the linear benchmark, the advantage is 0.04981, with standard error 0.00714 and interval $[0.03580, 0.06381]$. Discarding the pairing would instead give standard error 0.03916 for that last comparison. These figures quantify this simulator’s conditional comparison, not a universal superiority of the selected coefficients.



$N = 2,048$ independent test episodes; identical indexed innovations pair policies within each episode. Selected on 512 validation episodes: width 0.10, skew 0.01; then frozen. Intervals are individual approximate 95% Monte Carlo intervals. J subtracts inventory penalties. Terminal inventory is marked. Sampling uncertainty does not measure simulator or market validity.

Figure 12.1: Frozen policies compared on independent paired episodes. Point estimates and intervals describe Monte Carlo uncertainty conditional on the stated synthetic environment and objective; they do not measure uncertainty about real-market validity.

12.8 Read the entire outcome distribution and the stress table

A mean objective is not a complete performance description. The notebook also reports mean marked P&L, external execution costs, fees, inventory exposure, terminal inventory, and constraint or age-filter declines. As a simple lower-tail diagnostic for marked P&L, define

$$Q_{0.05}(\Pi) = \inf\{x : \Pr(\Pi_T \leq x) \geq 0.05\}, \quad p_{\text{loss}} = \Pr(\Pi_T < 0). \quad (12.18)$$

The notebook's sample quantile uses its documented interpolation convention. A sample quantile and an estimated loss frequency are descriptors of the finite simulated sample; they do not establish a reliable extreme-loss limit. In particular, two thousand episodes contain only about one hundred observations in a five-percent tail, with fewer observations available for more remote events.

For the selected policy in the baseline, the marked-P&L five-percent sample quantile is 1.78066, and 3 of 2,048 episodes have negative marked P&L. The corresponding objective quantile is 1.35050, with 8 negative-objective episodes. These are different events: a positive trading result can fall below zero after the stated inventory penalties. The sample quantiles use linear interpolation between adjacent order statistics.

The selected rule's mean marked P&L also reconciles transparently: client gross edge 4.28469, less client fees 0.04446 and external cost 0.38111, plus inventory carry 0.02043, gives 3.87955, up to the displayed rounding. Running and terminal penalties of 0.27871 and 0.14889 then produce mean $J = 3.45195$. The external cost has already entered the cash ledger.

Reference delay illustrates why operating controls need their own measurements. If the latest admissible reference has source timestamp τ_t , its age and quote-eligibility rule are

$$\text{age}_t = t - \tau_t, \quad \text{quote allowed} \implies \text{age}_t \leq L_{\text{max}}. \quad (12.19)$$

The age-filtered delay stress can reduce trading sharply. That does not make it a high-performing quoting policy merely because it avoids some losses; service coverage and executed quantity change too. The unfiltered delay stress instead exposes the economics of quotes based on stale marks. Comparing both reveals the cost of stale information and the behavior of the specified control, without describing absent trades as zero-cost completed service.

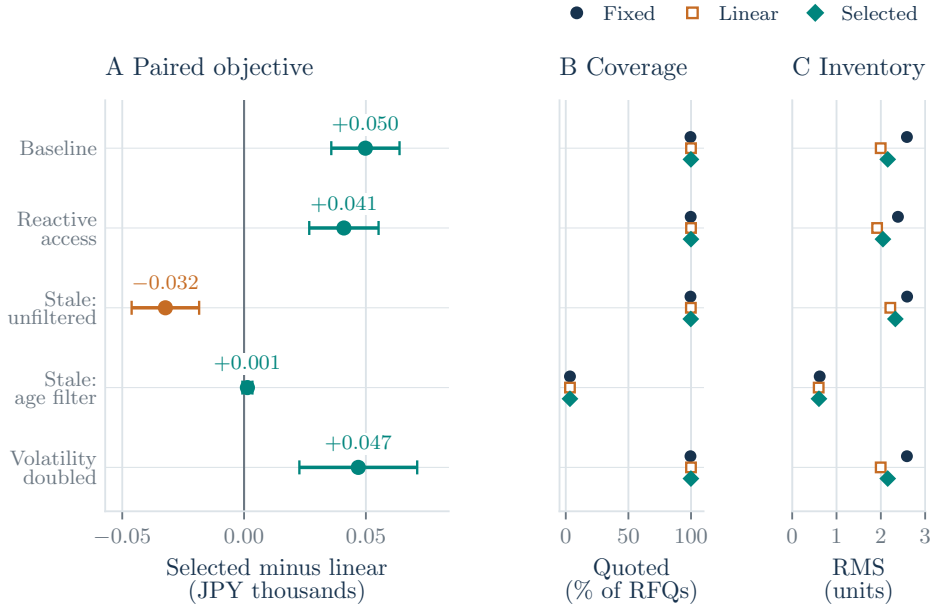
Both delay environments use $M_t^{\text{asof}} = M_{\max(0, t-2)}$. The age-filtered version requires $L_{\max} = 1$, so it can quote during the first two decision times but declines later requests. Scheduled hedges remain active. The reactive environment starts at $a_0 = 0.8$ and uses $\rho = 0.2$ in (12.9). The high-volatility environment doubles σ from 0.07 to 0.14 without changing policy coefficients, costs, or inventory preferences.

Environment	Mean J : fixed	Mean J : linear	Mean J : selected
Baseline	3.25290	3.40214	3.45195
Reactive access	2.69416	2.78619	2.82717
Delayed, unfiltered	0.66340	0.92394	0.89164
Delayed, age filtered	0.06091	0.06095	0.06232
Double volatility	3.26268	3.42549	3.47237

Using delayed marks without an age filter reverses the selected-versus-linear ranking: the paired mean difference is -0.03230 , with interval $[-0.04615, -0.01844]$. No policy is retuned after this result. The age filter instead reduces the selected rule's average quoted requests from 47.81885 to 1.60303 per episode, and filled units from 44.46289 to 1.45361. It records 46.25781 stale declines per episode on average. Its small objective cannot be compared as if it provided the original service volume.

Likewise, increasing volatility changes the distribution of inventory gains and losses while leaving the declared policies frozen. Adding client-flow feedback changes future opportunities. A ranking that persists across these scenarios is more informative than a single favorable mean, but the scenario set is still selected by the experimenter. It cannot certify robustness to an unspecified real environment.

With a fresh reference, the declared policies use offsets rather than price levels, and their parameters do not respond to σ . Under the paired draws, doubling volatility therefore leaves requests, fills, and inventory paths unchanged while scaling their reference-price carry. Mean carry is zero in this exogenous zero-drift model before sampling. A slightly higher simulated mean under doubled volatility is not a new expected return source: the selected rule's negative-objective frequency increases from 8/2048 to 144/2048, and its five-percent objective quantile falls to -0.44403 .



All three policies remain frozen; $N = 2,048$ episodes per environment. A uses paired 95% MC intervals. B and C show point diagnostics for each policy, with small vertical offsets for legibility. The age filter quotes only 3.35% of arrivals and fills 1.45 units per episode for Selected. Quote coverage is service availability, not completion. Lower inventory can accompany less service.

Figure 12.2: The same frozen policies under declared changes in client response, volatility, and reference delay. Paired selected-minus-linear objective differences, quote coverage, and inventory RMS must be read together; policies are not reselected within each stress scenario.

12.9 Historical logs need action support

Simulation specifies outcomes of actions through a model. Logged data reveal only the action actually taken and its observed reward. To see the distinction cleanly, set aside sequential inventory dynamics and consider a single-decision contextual bandit. Context X has a fixed distribution, action A is sampled by a known randomized logging policy $\mu(a | X)$, and only reward $Y(A)$ is observed. The target policy is $\pi(a | X)$.

Identifying its value from those logs requires target-action support:

$$\pi(a | x) > 0 \implies \mu(a | x) > 0 \quad \text{for relevant contexts } x. \quad (12.20)$$

The example assumes randomized logging conditional on context, so the recorded propensity is the actual assignment probability. With observational decisions influenced by unrecorded information, supplying a fitted action-frequency model is not enough to establish that condition or eliminate confounding.

Under the stated single-decision assumptions, the inverse-propensity estimator is

$$\hat{V}_{\text{IPS}} = \frac{1}{n} \sum_{i=1}^n \frac{\pi(A_i | X_i)}{\mu(A_i | X_i)} Y_i. \quad (12.21)$$

Write $m(x, a) = \mathbb{E}[Y(a) | X = x]$. Conditional randomization and positive support give the elementary identity

$$\mathbb{E}_\mu \left[\frac{\pi(A | X)}{\mu(A | X)} Y \right] = \mathbb{E}_X \left[\sum_a \pi(a | X) m(X, a) \right]. \quad (12.22)$$

The proof sums over logging actions and cancels their probabilities. It does not predict the missing reward of an individual request. It reconstructs an average under the specified context distribution and assignment mechanism. The denominator is the logging probability, not the target's action probability, its observed win rate, or a convenient aggregate frequency across unlike contexts.

For a deterministic target action $\pi(x)$, merely averaging rewards when the logged action matches the target instead converges to

$$\frac{\mathbb{E}_X[\mu(\pi(X) | X) m(X, \pi(X))]}{\mathbb{E}_X[\mu(\pi(X) | X)]}. \quad (12.23)$$

This weights contexts by how likely the logger was to match the target. It is generally a different population. A narrow exception occurs when the match probability is constant across relevant contexts; that exception should be established rather than assumed.

12.10 A small example exposes selection and missing support

There are two contexts, with probabilities 0.8 and 0.2, and two actions. The reward is a unitless Bernoulli outcome, separate from the trading simulator's currency ledger. The target chooses action 1 in context 0 and action 0 in context 1. Its reward means and logging probabilities are

Context	Reward means: actions 0, 1	Logging probabilities: actions 0, 1
0, probability 0.8	0.2, 0.4	0.9, 0.1
1, probability 0.2	0.9, 0.1	0.2, 0.8

The exact target value is $0.8(0.4) + 0.2(0.9) = 0.50$. The logger matches the target with probability $0.8(0.1) + 0.2(0.2) = 0.12$. Conditional on a match, the exact average reward is $(0.08(0.4) + 0.04(0.9))/0.12 = 0.56667$. That favorable matched-sample value comes from a changed context mixture, not from a better target policy.

IPS uses weights 10 for the target action in context 0, 5 for the target action in context 1, and zero for nonmatching actions. Its expectation is 0.50. The independently generated logging sample illustrates finite-sample fluctuations around these exact quantities. Large weights also make uncertainty important. A common weight-concentration diagnostic is

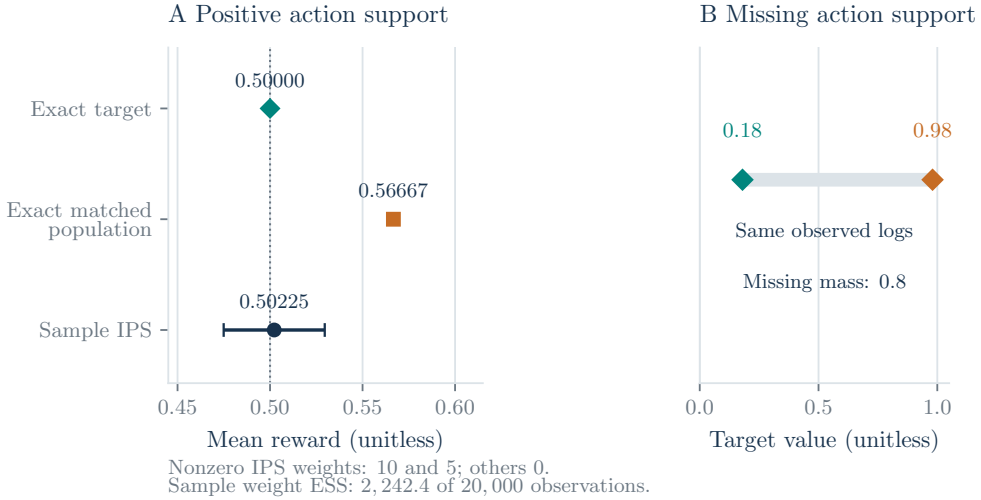
$$n_{\text{eff}} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}, \quad w_i = \frac{\pi(A_i | X_i)}{\mu(A_i | X_i)}. \quad (12.24)$$

It is defined when the denominator is positive. Here $\mathbb{E}[w] = 1$ and $\mathbb{E}[w^2] = 9$, so the large-sample fraction n_{eff}/n approaches $1/9$. This number diagnoses concentrated weights; it is not literally a count of independent trading episodes and does not by itself validate a confidence interval. If all weights are zero, the estimate has no matched support in that sample and the diagnostic is undefined, not evidence of a zero-valued policy.

The independent sample contains 20,000 logged decisions, of which 2,417 match the target. The matched-sample mean is 0.56227, whereas IPS gives 0.50225, standard error 0.01394, and an approximate 95-percent interval $[0.47494, 0.52956]$. Its weight-concentration diagnostic is $n_{\text{eff}} = 2242.38$, and the largest weight is 10. The finite-sample target comparison uses the known exact value, not the naive matching estimate as its reference.

Now change the logger so that action 1 is never taken in context 0. The missing target-support mass is 0.8. Two possible reward models can agree on every action the logger ever takes while assigning different rewards to that missing action. They then imply different target values despite indistinguishable logging data. No importance-weighting formula removes this identification failure.

For example, assign the missing action reward mean 0 in one world and 1 in another, leaving all logged-action reward laws unchanged. The target values are then $0.8(0) + 0.2(0.9) = 0.18$ and $0.8(1) + 0.2(0.9) = 0.98$. The data-generating law of the observed log is identical in the two worlds. This is a lack of identification, not a standard error that disappears with more logged observations.



A: exact values have no sampling interval. IPS uses 20,000 randomized logged observations. Only sampled IPS has a 95% normal interval. Weight ESS is a concentration diagnostic.
B: the unobserved target-action reward mean is 0 or 1 in two otherwise identical worlds. The segment is identification ambiguity, not a confidence interval. This is a one-step example.

Figure 12.3: A separate contextual-bandit illustration of logging selection. Exact target value, exact matched-population value, finite-sample IPS, and support diagnostics use known randomized propensities. These are unitless one-step rewards, not the sequential trading simulator’s P&L.

12.11 Do not transfer one-step guarantees to a sequential desk

Dudik, Langford, and Li study contextual-bandit policy evaluation and combine reward prediction with propensity correction. For fixed nuisance estimates obtained independently of the evaluation observations, the corresponding doubly robust form can be written

$$\hat{V}_{\text{DR}} = \frac{1}{n} \sum_i \left[\sum_a \pi(a | X_i) \hat{m}(X_i, a) + \frac{\pi(A_i | X_i)}{\hat{\mu}(A_i | X_i)} \{Y_i - \hat{m}(X_i, A_i)\} \right]. \quad (12.25)$$

Under the stated bandit assignment, support, and independent-estimation assumptions, correctness of either the reward model or the logging-probability model removes the estimator’s bias; the error interaction and variance still matter. The notebook implements the simpler known-propensity IPS example and exact enumeration, not a learned doubly robust estimator. Equation (12.25) is a source-guided extension, with its assumptions shown rather than an additional claimed empirical result.

In a sequential inventory problem, earlier actions change later inventory, access, and observations. Weighting only the current quote does not correct the distribution of those states. If full histories and actual behavior-policy probabilities are observed, a trajectory likelihood ratio has the schematic form

$$W_{0:T-1} = \prod_{t=0}^{T-1} \frac{\pi(A_t | \mathcal{H}_t)}{\mu(A_t | \mathcal{H}_t)}. \quad (12.26)$$

This expression requires common initial and environment laws, appropriate sequential randomization, and support at every relevant action-history pair. The transition factors cancel only under those assumptions. Products can have severe variance, and an unobserved information variable can invalidate the assignment model. A deterministic historical quote engine also usually provides no positive probability for most alternative continuous quotes. The chapter does not treat the one-step demonstration as an implemented sequential off-policy evaluator.

There is a related distinction in generative simulation. Conditioning on a future volatility regime or a future path statistic is not the same operation as intervening with a trading action:

$$\mathcal{L}(S_{t+1:T} | C = c) \quad \text{and} \quad \mathcal{L}(S_{t+1:T} | \text{do}(A_t = a)) \quad \text{are different targets.} \quad (12.27)$$

They can coincide only under additional assumptions connecting the condition and the intervention. Realistic marginal spreads, volumes, or return distributions are useful generator diagnostics, but they do not alone verify the action-response law needed to rank strategies. Feedback, execution feasibility, accounting, and information timing require their own checks.

12.12 Turn an experiment into a reviewable decision

The experiment retains configuration, seed allocation, candidate scores, the frozen selected policy, per-episode outcomes, event traces, and the accounting checks. These records separate a reproducibility claim from a performance claim. Reproducing a table means that the declared procedure gives the same table; it does not mean that the procedure describes the real market sufficiently well.

The gap between a simulated estimated value and an unknown real-environment value can be separated conceptually as

$$\widehat{V}_\theta(\pi) - V_{\text{real}}(\pi) = \{\widehat{V}_\theta(\pi) - V_\theta(\pi)\} + \{V_\theta(\pi) - V_{\text{real}}(\pi)\}. \quad (12.28)$$

More independent simulated episodes can reduce the first term's sampling uncertainty. They do not automatically reduce the second term's model discrepancy. A very narrow simulation interval can therefore coexist with a badly misspecified action-response

mechanism. Calibration evidence, independent data, and targeted failure tests address different parts of that problem.

A reviewable policy decision states the objective and cash consequences, its information boundary, feasible actions, evaluation population, selection procedure, uncertainty unit, failure conditions, and assumptions that remain unverified. Operating checks should record missing or stale inputs, rejected actions, response deadlines, executions, and any fallback behavior in the same event vocabulary used for evaluation. Timing a small Python example is not evidence that a production quote system meets its deadline; that requires measurements in the actual deployment setting.

The book's decision chain now returns to its starting point. Chapters 1–3 define the observable events, reference values and conditional probabilities. Chapters 4–8 turn those inputs and explicit inventory preferences into quote decisions; Chapters 9–10 connect those decisions to external execution. [Chapter 11](#) reconciles the resulting cash and inventory, and this chapter tests what the evidence can support. A useful next model is one that addresses a documented failure of that chain while preserving its event definitions, units, information boundary and accounting.

Exercises

1. **Execution clock and accounting.** Reconstruct one complete episode from the retained trace. Identify the inventory exposed to each price interval and the five scheduled hedge times. Reconcile terminal marked wealth to client edge, external cost, and inventory carry. Explain why the terminal mark still matters when no terminal hedge is allowed.
2. **Observation boundaries.** Reconstruct a quote from only its permitted inputs. Compare intended and effective price distance under a two-step reference delay. Show how the age filter changes quote coverage. Verify that changing future shocks cannot change earlier decisions.
3. **Action feedback.** Use identical indexed innovations with fixed and reactive arrival environments. Find an example where two policies receive different later request outcomes despite the same request uniform. Explain why imposing the original realized request tape would remove part of the feedback.
4. **Selection and final testing.** Reproduce the validation selection from the declared candidate grid. Verify design, validation, and test episode IDs are disjoint. Report the selected parameters and distinguish the validation selection score from the untouched test score. Explain why choosing another width after reading the stress table would require a new evaluation claim.
5. **Paired uncertainty.** Recalculate one objective-difference interval from the per-episode differences. Compare its standard error with the variance calculation that

discards pairing. Explain why treating all fills as independent observations is inappropriate for the same desk trajectory.

6. **Cash, risk, and stress.** Compare mean P&L, mean objective, inventory exposure, tail diagnostics, and execution or decline counts for the frozen policies. Explain why a stale-data filter that eliminates most trades cannot be assessed from marked P&L alone. Keep terminal marking and inventory penalties distinct.
7. **Logging selection.** Derive the exact bandit target value 0.50, match probability 0.12, and matched-sample reward 0.56667. Recompute sample IPS, the nonzero weights, and the weight-concentration diagnostic. Explain why the matched observations have a different context mixture.
8. **Missing support and valid conclusions.** Construct two reward models with identical observed rewards under the logger that never chooses action 1 in context 0, but different target values. State why one-step weighting does not evaluate a sequential inventory policy. Separate Monte Carlo error from model discrepancy and describe one check for each.

Source Notes

Dudik, Langford, and Li. The source is *Doubly Robust Policy Evaluation and Learning*, ICML 2011, available from the [official proceedings](#). The paper’s contextual-bandit setup, IPS discussion, doubly robust equation, independent-estimation assumptions, and bias/variance qualifications guide Sections 12.9-12.11. The toy reward table and its exact enumeration are constructed for this book. They do not reproduce the paper’s experiments or establish a sequential inventory-policy guarantee. See the [reading note](#) for the inspected pages and source mapping.

M3. The saved source is *M3: A State-Event Generative Foundation Model for Market Microstructure Dynamics*, arXiv 2608.19227v1. The relevant distinction is its explicit feedback of matched intervention events into continued generation, separate from mechanically replaying a stress intervention. We do not train the model, replicate its empirical evaluation, or transfer its stock-market results to rates RFQs. The [source record](#) and [reading note](#) identify the local edition, exact passage locations, and scope of inspection.

FlowLOB. The saved source is *FlowLOB: Efficient and Controllable Limit Order Book Generation with Flow Matching*, arXiv 2608.13096v1. Its jointly generated snapshot windows, scenario conditioning, and data partitions illustrate why the forecasting, scenario-generation, and intervention targets must be distinguished. Its distributional control tests are not treated here as a trader-specific execution model. See the [source record](#) and [reading note](#).

ABIDES-MARL. The saved source is arXiv 2511.02016v2 of *ABIDES-MARL: A Multi-Agent Reinforcement Learning Environment for Optimal Execution with Endogenous Liquidity*. Its interacting policies and synchronized learning decisions motivate inspection of the full environment contract. Clearing rules, completion treatment, benchmark construction, and holdout setup are source-specific; they are not silently imported into the RFQ experiment. The paper does not establish formal Nash convergence or real-market policy validity. See the [source record](#) and [reading note](#).

Construction and validation scope. The simulator, access response, policy grid, objective coefficients, common-random-input coupling, stress definitions, and logging example are independent educational constructions. The companion notebook checks their accounting and experimental consistency. Source claims rest on the inspected original pages, while the numerical results rest on the retained synthetic configuration and episode records. Neither source inspection nor numerical reproducibility is a claim to have validated a production trading strategy.

Research and notebook guide

The chapter source notes distinguish published results from the book's derivations and simulations. The links below connect the integrated argument to those notes and to the canonical executed notebooks. Local notebook and reading-note links require this library and the WSL workspace to remain available on the reading machine. The book's contents, equations, and chapter links work within this PDF.

Chapter 1: RFQ, Streaming, Order Books, and Available Information

[Chapter 1](#) | [Source notes](#) | [Exercises](#) | [Working notebook](#)

- [Brokertec Protocols — reading note](#)
- [Gould Protocols — reading note](#)
- [Kakuma Protocols — reading note](#)
- [Oomen Protocols — reading note](#)

Chapter 2: Reference Prices, Adverse Selection, and Markouts

[Chapter 2](#) | [Source notes](#) | [Exercises](#) | [Working notebook](#)

- [Brokertec Reference — reading note](#)
- [Glosten Milgrom — reading note](#)
- [Oomen Reference — reading note](#)

Chapter 3: Arrivals and Conditional Execution

[Chapter 3](#) | [Source notes](#) | [Exercises](#) | [Working notebook](#)

- [Bergault Arrivals — reading note](#)
- [Gould Arrivals — reading note](#)
- [Kakuma Arrivals — reading note](#)

Chapter 4: Optimizing a Single RFQ

[Chapter 4](#) | [Source notes](#) | [Exercises](#) | [Working notebook](#)

- [Cartea Book](#) — reading note
- [Guéant Book](#) — reading note

Chapter 5: Inventory Risk and Factor Representation

[Chapter 5](#) | [Source notes](#) | [Exercises](#) | [Working notebook](#)

- [Bergault Guéant](#) — reading note
- [Guéant Book](#) — reading note

Chapter 6: Bellman and HJB

[Chapter 6](#) | [Source notes](#) | [Exercises](#) | [Working notebook](#)

- [Cartea Book](#) — reading note
- [Guéant Book](#) — reading note

Chapter 7: Multiple Instruments and Sizes

[Chapter 7](#) | [Source notes](#) | [Exercises](#) | [Working notebook](#)

- [Bergault Guéant](#) — reading note
- [Guéant Multi Asset](#) — reading note

Chapter 8: Flow Imbalance and Client Relationships

[Chapter 8](#) | [Source notes](#) | [Exercises](#) | [Working notebook](#)

- [Barzykin](#) — reading note
- [Cotton](#) — reading note

Chapter 9: Internalization and External Hedging

[Chapter 9](#) | [Source notes](#) | [Exercises](#) | [Working notebook](#)

- [Barzykin Bergault Guéant](#) — reading note
- [Cartea Book](#) — reading note
- [Guéant Hedging](#) — reading note

Chapter 10: Optimal Execution

[Chapter 10](#) | [Source notes](#) | [Exercises](#) | [Working notebook](#)

- [Almgren Chriss](#) — reading note
- [Cartea Book](#) — reading note
- [Guéant Execution](#) — reading note

Chapter 11: Transaction-Cost Analysis and Profit Attribution

[Chapter 11](#) | [Source notes](#) | [Exercises](#) | [Working notebook](#)

- [Almgren Chriss TCA](#) — reading note
- [Brokertec TCA](#) — reading note
- [Kakuma TCA](#) — reading note
- [Oomen TCA](#) — reading note

Chapter 12: Simulation and Evaluation

[Chapter 12](#) | [Source notes](#) | [Exercises](#) | [Working notebook](#)

- [ABIDES Marl](#) — reading note
- [Dudik Offpolicy](#) — reading note
- [FlowLOB](#) — reading note
- [M3](#) — reading note

Volatility-estimation supplement

[Forecasting models and their risk connection](#) | [Source reading note](#) | [Working notebook](#)

The supplement compares rolling variance, EWMA, GARCH(1,1), and log-HAR forecasts on one declared synthetic population. Its frozen chronological comparison is a numerical illustration, with separate training, selection, and testing blocks.

Primary source access

These links lead to the recorded original source or its institutional host. The chapter notes identify the exact saved editions and passages used. Public access does not imply an open redistribution license.

- [A Simple Approximate Long-Memory Model of Realized Volatility](#)
- [ABIDES-MARL: A Multi-Agent Reinforcement Learning Environment for Optimal Execution with Endogenous Liquidity](#)
- [Algorithmic and High-Frequency Trading](#)
- [Algorithmic market making in dealer markets with hedging and market impact](#)
- [Bid, ask and transaction prices in a specialist market with heterogeneously informed traders](#)
- [Doubly Robust Policy Evaluation and Learning](#)
- [Execution in an aggregator](#)
- [FlowLOB: Efficient and Controllable Limit Order Book Generation with Flow Matching](#)
- [Generalized autoregressive conditional heteroskedasticity](#)
- [Limit order books](#)
- [Liquidity Structure and Execution Cost in the Japanese Government Bond Market: Analysis and Consideration of Trading Inquiry Data on an Electronic Trading Platform](#)
- [M3: A State-Event Generative Foundation Model for Market Microstructure Dynamics](#)
- [On a Simple Relationship Between Order Imbalance, Skew and Width in Over-The-Counter Trading](#)
- [Optimal execution of portfolio transactions](#)
- [Optimal market making](#)
- [RiskMetrics Technical Document, Fourth Edition](#)
- [Size matters for OTC market makers: general results and dimensionality reduction techniques](#)
- [Strategic OTC market making with reputation feedback](#)
- [The Financial Mathematics of Market Liquidity: From Optimal Execution to Market Making](#)
- [The Microstructure of a U.S. Treasury ECN: The BrokerTec Platform](#)
- [Volatility forecast comparison using imperfect volatility proxies](#)